



SHIPFLOW[®]

8.0

USERS MANUAL

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1. INTRODUCTION

The following section gives an overview of the contents and conventions of the users manual, the capabilities of the various modules of SHIPFLOW and the commands inherent to the system.

1.1. Manuals

There are two manuals that describe the SHIPFLOW system. The SHIPFLOW Users Manual and the SHIPFLOW Tutorial. The Tutorial contains a tutorial and examples with complete input files and comments to several cases. The Users Manual is the reference manual and describes in detail all the input commands and the input files.

The first time user of the SHIPFLOW system is recommended to start by studying the tutorials and use the Users manual when more detailed information is needed.

Sections 1.2 and 1.3 give an overview of the capabilities and commands for the six SHIPFLOW modules XMESH, XPAN, XBOUND, XGRID, XCHAP and MOTIONS. The command syntax and manual conventions are explained in Section 1.4. XFLOW commands can be found in Chapter 2. These are the commands that are common for all modules and Chapters 3-8 describe the commands for the modules XMESH, XPAN, XBOUND, XGRID and XCHAP, respectively. The format of the offset file that contains the hull geometry is described in Chapter 9. Files etc. are described in Chapter 10. Chapter 11 contains a description of the "Standard Case Mode"; the simplest way of using SHIPFLOW.

1.2. Capabilities

In this section the various capabilities and options available in SHIPFLOW are summarized.

1.2.1. XMESH

XMESH is a panel generator for the potential flow module XPAN. XMESH can be executed as a separate program to check the panelization of the body and free-surface before the potential flow computation is executed. The XMESH module is also executed during the potential flow computation when sinkage/trim or non-linear iterations are performed and the panelization is updated in each iteration. XMESH generates the panels used for a sink-disk representation of a propeller in the potential flow. Off-body points can also be generated using the XMESH module. Off-body points are used in potential flow computations when the result is to be displayed at points in the flow field outside the hull surface.

1.2.2. XPAN

XPAN is the flow solver for the potential flow around three dimensional bodies based on a surface singularity panel method. A wide range of problems may be analysed. These include

- Flows with or without a free surface
- Ship flows with or without a transom stern
- Ship flows with or without sinkage and trim
- Multiple ship speeds
- Multiple onset flow directions
- Influence of the propeller (for more detailed studies XCHAP is recommended)
- Lifting surfaces
- Shallow water
- Ship in a canal

With the following options

- First or higher order panel method
- Linear or non-linear free surface boundary condition
- Neumann-Kelvin, double-model or single-model solution as base flow
- Symmetry feature
- Initial ship position specification
- Velocity and pressure computations at specified off-body points

Using the XPAN program, the following features of the flow around the hull may be computed

- Wave resistance from pressure integration and from transverse wave cuts
- Wave pattern
- Wave profile along the waterline
- Wave profile along longitudinal and transverse wave cuts
- Far-field waves in deep water
- Potential streamlines (traced in XBOUND)
- Pressure contours
- Velocity vectors
- Sinkage and trim
- Lift and induced drag

XPAN creates a data base file, id_XPDB, used by XBOUND, XGRID and XCHAP. The data base file contains all the results from the potential flow computation that are needed for the execution of XBOUND and XCHAP (zonal approach).

1.2.3. XBOUND

XBOUND is a program for thin turbulent boundary layer computations. The momentum integral equations for boundary layers are solved along streamlines traced from a potential-flow computation. XBOUND is also capable of computing the laminar boundary layer and the transition to the turbulent boundary layer for simpler cases with a well defined stagnation point or line. The computations can be carried out for a smooth surface or for a specified surface roughness.

The following boundary layer quantities can be computed in XBOUND:

- Boundary layer thickness
- Displacement thickness
- Momentum thickness
- Shape factor
- Cross-flow angle
- Skin friction coefficient
- Transition between laminar and turbulent flow
- Limiting streamlines

The friction is integrated over a specified region of the hull.

XBOUND creates a data base file, id_XBDB, used by XCHAP (zonal approach). The data base file contains all the results from the boundary layer computation needed for the execution of XCHAP.

1.2.4. XGRID

XGRID generates the grid used for the viscous computations in XCHAP

- around a ship or submarine hull
- with constant x-surfaces
- with proper concentration close to the hull surface regardless of humps and hollows.
- where the position of the parametric edges leaving the keel/water line can be specified to avoid singularity problems in the XCHAP module.
- with a concentration of x-planes in the stern region where it is needed the most.
- with a strong concentration of points close to the singularity lines (the parametric edges between the continuation of the hull surface, the wake plane, and the adjacent parametric surfaces)
- which includes sinkage and trim
- create a transom grid for XCHAP
- mirror grid for unsymmetrical cases
- handle twin skeg hulls
- adopt horizontal plane to a prescribed free surface from XPAN

XGRID cannot handle

- appendages

1.2.5. XCHAP

XCHAP is a finite volume code that solves the Reynolds averaged Navier-Stokes equations. It uses several turbulence models (*EASM*, *k- ω BSL*, *k- ω SST*). The solver can be used in a zonal or a global approach. The solver can handle overlapping grids. Several parametrised models of appendices are available in the system, e.g. rudder, shafts, brackets and vortex generators. Grids can also be imported from external grid generators. There are also two actuator disk models available, a simple force model and a lifting line model. The flow can be computed with a double model or with a free-surface.

The following quantities are computed

- Velocity field
- Pressure
- Turbulent kinetic energy and specific turbulent kinetic energy.
- Local skin friction coefficient
- Friction and pressure resistance coefficients for the hull part covered by the grid
- Total resistance and its components using the results from XPAN, XBOUND and XCHAP.

XCHAP can use the grid provided by XGRID. Inflow boundary conditions are generated from results provided by XPAN and XBOUND when the zonal approach is used.

XCHAP can also use block structured grids generated in an external grid generator. These grids can be either the entire hull so that no XGRID grid is needed or appendage grids that are added to the XGRID grid.

1.2.6. XPOST

XPOST module is a tool for generating comprehensive report for the computed cases. It contains the integrated results as well as the most common pictures and graphs showing the flow fields, wave patterns and pressure distributions.

1.3. Overview of the Commands

The information that governs the behavior of SHIPFLOW is given in the command file in the form of a series of commands.

Only a minority of the commands will be used by all modules of SHIPFLOW. They are described briefly in the chapter “XFLOW” below.

The vast majority of the commands however are only used by one of the SHIPFLOW modules. They are summarized in the other chapters below.

Note that the commands must be grouped and placed between delimiters in the command file, e.g. all commands used by XGRID must be given between the commands XGRID and END.

In the following description a (M) indicates that the command is mandatory.

Module delimiter commands

In order to run a SHIPFLOW computation a command file must first be created. The command file tells SHIPFLOW which modules should be executed. Each module has a specific set of commands which are unique to its operation. When specifying these commands it is necessary to tell SHIPFLOW

- Which module you have chosen
- Where you are beginning to define the commands
- Where you have ended your command definition

This is accomplished by the module delimiter commands. In the example below the delimiter commands are **XPAN** and **END**. The information found between the delimiters is the command definitions for the XPAN module.

```
xpan
      cont      ( free, nonl )
      iter      ( maxi=10 )
end
```

1.3.1. XFLOW

Delimiter commands

<i>command</i>	<i>description</i>
XFLOW	Marks the beginning of commands to the XFLOW level.
END	Marks the end of commands to the XFLOW level.

Master commands

<i>command</i>	<i>description</i>
TITLE	Title.
PROGRAM	Which modules of SHIPFLOW to run (M).
OFFSETFILE	Name of the file containing hull offsets or iges, orientation and size of the hull in the offset file (M).
HULLTYPE	Specify the type of ship and specify input for the Standard Case option (M).
IPOSITION	Initial position of ship.
OSFLOW	Onset flow direction.
TURN	Turning radius and center for XCHAP.
FLUID	Fluid properties.
FMREF	Ship dimensions for nondimensionalised integrated forces and moments.
VSHIP	Speed of ship (M).
PRTOPT	Extra output files.
SYMMETRY	Used to specify the symmetry plane, if any.
PROPELLER	Geometry data and thrust coefficient for a propeller.
SELFPROP	Control self-propulsion.
SPAUTO	Control full propulsion analysis.
ITTC78	Extrapolate XCHAP results from model scale to full scale.
ROUGHNESS	Distribute the roughness parameter KS over the hull.
RUDDER	A parametric rudder model for XCHAP.
SHAFT	A parametric shaft model for XCHAP
BRACKET	A parametric model of bracket for XCHAP
BOX	A rectangular grid for XCHAP.
ENVBOX	The XGRID grid will be embedded in a BOX grid.
CYLINDER	A cylinder grid for XCHAP.
FILE	Overrides the name conventions for some of the files used by XPAN,

	XBOUND, XGRID and XCHAP.
OPTIM	Output and input for optimization.
HYPSURF	Creates a 2D hyperbolic grid for XCHAP.
WSECTION	Creates a 2D grid around a wing section for XCHAP.
ROTBODY	Uses the 2D grids from HYPSURF and WSECTION to make 3D grids.
VORTEXGEN	Creates a vortex generator for XCHAP.
APPENDAGE	Adapts appendage grids to the hull surface for XCHAP.
RECESS	Adapts hole cutting grids for XCHAP.

1.3.2. XMESH

Delimiter commands

<i>command</i>	<i>description</i>
XMESH	Marks the beginning of commands to XMESH.
END	Marks the end of commands to XMESH.

Master commands

<i>command</i>	<i>description</i>
PROPELLER	This command is used to specify panels for the propeller.
ENVIRONMENT	This command is used to specify panels on geometries that are not included in the description of the hull. A typical example is the bottom and the sides of a canal.
LIFT	This command is used to specify panels for lifting surfaces.
BODY	This command is used to specify panels for hull surfaces or other general body surfaces.
FREE	This command is used to specify panels on the free-surface.
TRANSOM	This command is used to specify panels for the free-surface downstream of a transom stern hull.
FSFAR	This command is used to specify panels for far-field waves. Note that the

	command TWCUT on XPAN must be included as well.
STRIP	This command is used to specify the dipole distribution on a lifting surface and its trailing wake.
OBPOINT	This command is used to specify additional points in the flow field where velocity vectors and pressure contour lines are to be plotted.
PLOT	This command is used to specify panels used for plotting purposes only. The panels are not active in the computations.

1.3.3. XPAN

Delimiter commands

<i>command</i>	<i>description</i>
XPAN	Marks the beginning of commands to XPAN.
END	Marks the end of commands to XPAN.

Master commands

<i>command</i>	<i>description</i>
CONTROL	This command is used to specify the type of execution to be performed in XPAN. The command is also used to over-write default values for some numerical parameters.
CONVERG	Specify convergence criteria.
EXFORCE	Specify external forces acting on the ship.
EXMOMENT	Specify the towing point position and the position of external forces acting on the ship.
ITERATION	Specify the maximum number of iterations in sinkage/trim and non-linear computations and to specify at which iterations the geometry is updated.
RELAXATION	Specify relaxation factor.
TWCUT	Input for wave resistance computations based on multiple transverse wave cuts. This command must be included in order to use FSFAR on XMESH.
WAVECUT	Used to specify the positions of longitudinal and transverse wave cuts. The wave cuts are printed in the files id_LWAVECUT and id_TWAVECUT.

1.3.4. XBOUND

Delimiter commands

<i>command</i>	<i>description</i>
XBOUND	Marks the beginning of commands to XBOUND.
END	Marks the end of commands to XBOUND.

Master commands

<i>command</i>	<i>description</i>
CONTROL	This command is used to specify the mode of execution of XBOUND.
TRACE	Streamline tracing parameters.
INICON	Specification of upstream boundary layer quantities for the integration of the boundary layer equations.
RESISTANCE	Sets the limits for the skin friction integration.
LIMIT	Parameters for limiting streamline tracing.
ROUGHNE	Parameters for surface roughness.

1.3.5. XGRID

Brief summary of steps taken

This brief summary of steps taken by XGRID is explained further in the "XGRID theoretical manual". Here it is used to illustrate the connection between different commands and events in the program.

- i) Start. XGRID takes control of SHIPFLOW.
- ii) Distribution of points on the longitudinal boundary surfaces (all boundaries except the inflow and outflow planes).
- iii) Generation of initial grid as needed for first sweep in Poisson solver.
- iv) Generation of initial guess of source terms.
- v) One sweep in Poisson solver.
- vi) Source term improvement.

- vii) Boundary point movement.
- viii) If "convergence" has not been achieved and we haven't run short of iterations, repeat steps v) - vii).
- ix) Copy wake planes and interpolate in radial direction to specified density¹.
- x) Write to file(s).
- xi) Return control to SHIPFLOW.

Delimiter commands

<i>command</i>	<i>description</i>
XGRID	Marks the beginning of commands to XGRID.
END	Marks the end of commands to XGRID.

Master commands

<i>command</i>	<i>description</i>
COARSE	The relation between the preliminary and the final (interpolated) grid.
CONTROL	This command is used to specify the mode of execution of XGRID.
IMPROVE	Extent of area of source change, direction of coordinate lines leaving the surfaces under treatment.
NEUMAN	A Neumann boundary condition at the η -boundaries can be explicitly implemented by movement of the boundary points in between sweeps in the Poisson solver.
OFFSETFILE	Name of offset file, LPP in offset scale, reflections, position of origin.
OUTPUT	This command gives the user control over what grids will be written to files and if any iteration history should be given.
POISSON	This command controls the Poisson solver.
RADIUS	The location of the outermost boundary of the domain and the location of the center of the domain.
SINGUL	Location of the singularity lines (parametric edges between the wake surface and the two η -boundaries).
SIZE	Number of clusters in the final grid.

¹ This interpolation is the step that makes a "final"/"fine" grid out of the "preliminary"/"coarse" grid.

SKIN	The distance between the first and second ζ plane is specified with this command.
YPLUS	An alternative to SKIN. The program estimates the cell height closest to the hull based on the Reynolds number.
XDISTR	Distribution of x-planes.
ETASMOOTH	Smoothing of the longitudinal lines on the η -boundaries.
FEEDBACK	This command allows the user to tune the improvement of the Poisson equation source terms.
TUNE	Feedback amplification and saturation factors.

Notes

- 1) Some of the information in the SHIPFLOW commands is passed to XGRID:
- 2) The additional sinkage and trim as calculated by XPAN for a ship that is free to move is also used by XGRID if these programs are run in succession.
- 3) Since this step in the grid generation process describes the size of the domain and the distribution of points on four out of six surfaces, it uses a lot of commands:
- 4) This action is however fully automatic and only the information contained in the boundary point distribution (produced in step ii) is needed.
- 5) The only information that we supply directly to the Poisson solver is the relaxation factors entered in POISSON and the iteration history output flag in OUTPUT.
- 6) Boundary point movement is nothing else than an explicit implementation of the Neumann boundary condition, it is therefore controlled by the command NEUMANN.
- 7) Convergence criteria and maximum number of sweeps in the Poisson solver is given in the POISSON command.
- 8) Which type of grids that should be written to which file (interpolated or not, id_XVGRID or id_XGPOST) is specified in the OUTPUT command².

Steps 5-7 can be excluded from the execution if the Poisson solver is turned off by the keyword OFF in the POISSON command. This is extremely useful in the initial search for input parameters that give a good boundary point distribution.

1.3.6. XCHAP

² As a matter of fact, this information also tells XGRID when and if to do the above substeps.

Delimiter commands

<i>command</i>	<i>description</i>
XCHAP	Marks the beginning of commands to XCHAP.
END	Marks the end of commands to XCHAP.

Master commands

<i>command</i>	<i>description</i>
ACTUATOR	Manage the force actuator model.
CONTROL	Specify execution mode and maximum number of iterations.
CONVERGENCE	Control convergence criteria.
EXTPROPELLER	Interface for external propeller programs.
EXTRACT	Interpolate data to an external grid.
FREE	Free sinkage and trim
IMPORT	Import of external grids.
KRYLOV	Control the krylov solver.
LLINE	Manage the lifting line actuator model
OVERLAP	Tuning of the overlapping grid algorithm.
PARALLEL	Control the number of threads XCHAP is run with.
POW	Control the Propeller Open Water calculation.
PRIORITY	Control flagging of the non-fluid points in overlap regions.
SAVE	Save XCHAP data in Tecplot or Paraview format.
REFINE	Refine a region in the XGRID grid.
UNION	Union of grids to extend computational domain.
VOLUME	Volume grid from surface grid.
WAKE	Wake integration
VOF	Control Volume of Fluid (VOF) settings.
XGREFINE	Auto-refine XGRID grid.

XGRID	Modify XGRID grid.
-------	--------------------

1.3.7. XPOST

Delimiter commands

<i>command</i>	<i>description</i>
XPOST	Marks the beginning of commands to XPOST.
END	Marks the end of commands to XPOST.

Master commands

<i>command</i>	<i>description</i>
CONTROL	Specify execution mode.

1.4. Command file Syntax

The following syntax rules apply to the SHIPFLOW command file.

The command file may consist of

- blank lines
- comments
- commands

Blank lines are ignored.

Any line that begins with the character "/" will be treated as a comment and will be ignored.

The command syntax rules are explained below.

1.4.1. Command Syntax Rules

Since it is crucial to understand the syntax rules of the commands in order to avoid a lot of frustration in the beginning and irritating errors later on, the following description of the syntactic rules will be illuminated by a brief explanation of how SHIPFLOW reads a command and some examples of syntactically correct commands.

Command components:

<i>note</i>	<i>part</i>	<i>description</i>
(1)	CONTROL WORD	This is the label by which the command will be recognized by SHIPFLOW.
(2)	"("	If any additional information apart from the mere existence of the control word is to be given in the command, this information must be separated from the control word with "(".
(3)	KEYWORD	Just as control words help SHIPFLOW identify different commands in the command file, keywords are used to help SHIPFLOW identify information given in the command.
(4)	"="	An equal sign must separate the keyword and its associated information.
(5)	INFORMATION	The information following the "=" can be integers, real numbers, character strings or real vectors.
	DELIMITER	Keywords (with associated information) must be separated by commas.
	)"	The command must end with a ")".

Notes

- (1) The control word will be identified correctly only if the command is used in the right section of the command file. An XCHAP command, for instance, will not be recognized if used in the XPAN section of the command file. Blanks preceding the control word on the line are ignored. Misspelled or undefined control words will be found and treated as such. Also, SHIPFLOW only uses the first four characters of any command. For example it is entirely legal to specify the command PROPELLER by simply entering PROP.
- (2) Anything, except left brackets and line breaks, between the part of the control word that SHIPFLOW uses for its identification and the "(", will be ignored.
- (3) Keywords must be separated by a ", " .
- (4) Blanks may be used between keywords the "=" sign and numerical values.
- (5) The information is read with "free format" Fortran input statements. This allows some freedom:

The real number 0.0234 may be given as 0.0234, 2.34E- 02.

Integers are allowed to look like real numbers but the value used will be the truncated value of the real number. 3.75 will thus be interpreted as 3 by the program and -3.75 as -3. That is of course not useful for our needs and users interested in this are recommended to look in the Fortran manual of their computer.

Vectors must be enclosed by square brackets [x, y, z] and components separated by commas.

Character strings must be enclosed by quotation marks e.g.

"character string"

The length of the character strings is usually limited. (See detailed description of each command for the limit of the string in question.)

In some cases, the keyword itself is the information. Take for instance the SHIPFLOW master command PROGRAM. The presence of the keyword XPAN in it tells SHIPFLOW that XPAN should be activated in that run of the system. Some commands use sets of mutually exclusive keywords to enable the user to choose between mutually exclusive options. The XPAN command CONTROL is a good example of this.

General rules

SHIPFLOW is not case sensitive except when dealing with file names.

Commands must be shorter than 800 characters long.

A command may be continued on a new line by placing a comma "," between the

keywords. The ", " is the last character on the line.

Some command examples

```
SINGUL (keel, xyzfwd = [123, 0.0, 1.2], xyzaft = [160.0, 0.0, 7.0])
```

This example illustrates that commas are required between keywords (together with their associated information) and between vector components. KEEL belongs to a set of mutually exclusive keywords.

```
/  TRAcE( group = 4, stream = 18, istart = 1,
    TRAcE      ( group = 4, stream = 8, istart = 7,
    IDISTR = 0, s1 = 0.1, sn = 0.9 ,
    p1 = 1.d-04, Station= 15, dp1 = 0.01,
    jdistr = 1, pn = 0.9)
```

This example shows what a command could look like in real life. The first line is a fragment that the user has chosen to save to remember what parameters he/she used in a previous but not quite satisfying run of XBOUND. Note also the disregard for the difference between upper/ lower case letters. This command is legal but not that easy to read, compare it with this version of the same data:

```
    trace (group = 4, stream = 8, station= 15, istart = 7,
          idistr = 0, s1 = 0.1, sn = 0.9,
          jdistr = 1, p1 = 1.d-04, dp1 = 0.01, pn = 0.9 )
```

Note that this command is structured by its information. Even if we don't know the exact meaning of all the keyword values it is easy guess that IDISTR and JDISTR have similar function in SHIPFLOW. The same seems to hold true for the pairs S1-P1 and SN-PN. Note also that JDISTR can no longer be mistaken for a control word and STATION doesn't clutter the set JDISTR, P1, DP1 and PN that are related.

The following is an example of a typical command file that one would set up to execute a SHIPFLOW computation.

```
/
/ shipflow 8.0
/
xflow
  title      ( title="Ferry T = 5.0 m" )
  program    ( xmesh, xpan, xbound, xgrid, xchap )
  hulltype   ( mono, hlgr="xyzhull", fbgr="xyzbulb",fsflow )
  offset     ( file="off_ferry", lpp=100.00 )
  ipos       ( tm=5.0 )
  vship      ( fn=0.275, rn=0.1208e8 )
  prtopt     ( str1 )
end
```

1.4.2. Syntax check

The program has a syntax check for the input commands before any actual computations are started. It is able to catch most error and prints an error message for each error. An error message from the syntax check may look as follows:

```
*** Syntax ERROR: Unknown keyword
    Last: module XFLO ( 1), command OFFS ( 1), keyword FILE ( 1)
    Unknown keyword: LPPO
```

The output is printed after checking the command in the XFLOW section:

```
offs ( file="off_s60", lppoff=1.0 )
```

The last line in the error message informs us that the keyword `lppoff` is not a valid keyword for the command `offs`. The second line tells where to find the error in the command file. The last correctly interpreted module, command and keyword is printed. The numbers inside parenthesis are the number of times the module delimiter command have been found in the file so far (must not be more than one), the number of times the command have occurred, and the number of times the keyword have occurred in the command. The syntax check continue to check the following commands in the command file after an error is found. However when the end of the file is found or after 10 detected errors the program always stop the execution.

1.4.3. Manual Conventions

This section explains how the Users Manual displays SHIPFLOW commands. Each chapter command description is typically broken up into sections of "General Form of Command", "Defaults" and "Keywords".

General Form of Command

This section lists all the valid keywords of the command, e.g:

```
COMMAND (      key1, key2 = n, key3 = v, key4 = c,
               key5 = [n1, .., nkey2],
                   key6,
                   key7,
                   key8 )
```

In this example, `key1 .. key9` are the keywords that SHIPFLOW will look for in the command `COMMAND`. Additional information is associated with `key2 .. key6` whereas `key7 .. key8` and `key1` control SHIPFLOW just by being found in the command.

`Key1` may be present or absent in the command and can thus trigger two different responses from SHIPFLOW.

`n` after `key2` shows that `key2` must be given an integer value.

`v` -"- `key3` --- " --- `key3` ----- " ----- a real value.

"c" -"- `key4` --- " --- `key4` ----- " ----- a character string.

`Key5` gives an example of input of vectors. In this case, `key5` is an integer valued vector with `key2` elements.

`Key6 .. key8` are three mutually exclusive keywords. By giving one of such a group of keywords, the user can for example choose between different numerical operators.

Defaults

This section shows which default values exist for the keywords shown in the previous section of the command description. Keywords not shown in this section don't have any default values and must be given in the command.

If a set of keywords does not have a default listed, then one of the keywords in the set must appear on the command if the command is used. Unless otherwise noted, the default command listed is also the default if the command is not present in the command file.

Keywords

This section gives a detailed description of the meaning and use of each keyword of the command.

Name inside hard brackets [] gives the group name in the SHIPFLOW GUI.

2. XFLOW commands

This section describes the commands that are common for all the SHIPFLOW modules: XMESH, XPAN, XBOUND, XGRID and XCHAP. They are necessary for setting up the input files and for specifying which module(s) should be executed.

In addition, the XFLOW commands are used to define the general physical properties of the underlying problem. These include; initial ship position and speed, onset flow, hull type, propeller geometry, fluid characteristics and symmetry.

2.1. MODULE SECTION DELIMITER COMMANDS

The following commands mark the beginning and end of sections of the command file that are specific for the sub-modules of SHIPFLOW.

XFLOW / END beginning and end of XFLOW section.

XMESH / END beginning and end of XMESH section.

XPAN / END beginning and end of the XPAN section.

XBOUND / END beginning and end of the XBOUND section.

XGRID / END beginning and end of the XGRID section.

XCHAP / END beginning and end of the XCHAP section.

2.2. APPENDAGE

This command specifies that the object ID is adapted to the grids in ADAPT. ADIR specifies the coordinate direction in which ID is adapted. ASIDE is the grid face of adaption. To adapt more than one side of a grid the APPENDAGE command should be repeated for each side.

General form of command

APPENDAGE	(	ID = "c"	,	ADAPT = ["c"]	,	ADIR = n	,
		START=v	,	MORPH	,	DBOX = [v1,v2,v3]	
		ASIDE = i	,	COSVMIN = v	,	BCNOSLIP	,
		CUT					)

Default values

APPENDAGE (START={0.3 for RUDDER objects on the centerline, otherwise 1 if
ASIDE=1 and 0 otherwise.}, DBOX=[0,0,0])

Keywords

<i>keyword</i>	<i>description</i>
ID	Specifies the object with matching ID is an appendage.
ADAPT	Which grid(s) to append (adapt to):
ADIR	The coordinate direction used for adapting ID.
ASIDE	The side(s) that are adapted. Value should be 1 (low side) or 2 (high side).
COSVMIN	Minimum of the cosine between the coordinate line and surface normal where the adaption occurs.
START	Parametric position from where to start the search for an intersection with the surface to adapt to. 0 means start at the parametric low side ,1 at the parametric high side and 0.5 in the middle. Starting from the opposite side, i.e. 1 when ASIDE=1 and 0 when ASIDE=2 is usually the right thing to do and this is the default (except for rudders), but if there are multiple intersections using START can help selecting the one you want.
MORPH	The default behavior is to not change the shape of the adapted grid, only move the points along the grid lines to fit it to the chosen surface. If the keyword MORPH is given, it will instead change the shape of the grid to fit it. This can be useful for example for bilge keels.
DBOX	Extension of the bounding box used for the MORPH version. This is especially useful for long, thin objects. The purpose is to extend the box

so that the base surface to be fitted is completely inside the hull.

BCNOSLIP Force the boundary condition on the adapted surface to NOSLIP, even if a coordinate line failed to intersect the adaption surface.

CUT Allow the adapted and adaption surfaces to cut each other. This is very rarely a good idea.

2.3. ASSEMBLY

This command controls the generation of data from the integration of forces on the ship. It also allows positioning of a group of grids to a specified position and orientation or along the normal at a point on the ship hull.

The force and moment for each assembly are logged in the file <id>_FORCELOG and the average of the force coefficients are written to the file <id>_FORCEAVERAGE. By default an assembly is created for each component grid. In <id>_FORCEAVERAGE the forces are written as coefficients using the ship reference area S as given in the file <id>_OUTPUT.

The idea is to group component grids together in an assembly and save the integrated values at the integration times to get a log and/or the final averaged values. The results are printed to tables in tab separated text files.

The integration surface is the union of boundary cell faces of the component grids in the assembly with boundary condition set to NOSLIP, except for grid faces that are adapted to another grid surface. Those cell faces are instead regarded as belonging to the grid they were adapted to.

The command can also be used to position the specified grids so that they are translated and rotated consistently. For general positioning the parameters TRANSLATE, ROTCENTER, ROTAXIS and ANGLE are used.

For positioning an assembly on the hull the additional parameters POSGRID, POSDIRECTION, POSREFERENCE, POSORI, POSRAY and POSSHIFT are used. POSORI and POSRAY defines a ray. The point POSORI should be inside the hull and the direction ROTAXIS should point out of the hull. The program will find the intersection of the ray and the hull and transform the assembly so that its local coordinate system origin is in the intersection point and the vector POSDIRECTION in the assembly coordinate system is aligned with the hull surface normal. POSREFERENCE is a reference vector for the rotation of the assembly around the normal vector. The default value usually does not need to be changed. Finally, POSSHIFT translates the object in the positive direction of ROTAXIS. As an example, the following commands

```
rudder (      id="fin1", s=[0,1],c=[3,2], section=["NACA4412"],
             xle=[1,0], origin=[0,0,0], span=5, angle=0, tp2r )

append (      id="fin1", adapt=["Xgrid"], adir=3, aside=1 )

assembly (    id="fin1a", grids=["fin1"], posori=[9,0,3], posray=[0,1,0],
             posref=[1,0,0.3], posshift=-0.2 )
```

gives the result in the picture below. The APPEND command is not necessary for positioning with ASSEMBLY but it illustrates the need to position the RUDDER object into the hull with POSSHIFT so that it can be adapted. Changes to the object positioning in its own command like CANT, TILT and ANGLE for RUDDER are preserved.

The function of POSREFERENCE is somewhat complicated. Before the transformation, the design (offset) and assembly coordinate systems coincide and POSREFERENCE and POSDIRECTION defines a plane. In addition to aligning the assembly with the normal vector, the assembly will be rotated so that the reference vector lies in this plane. If

POSREFERENCE is parallel to the undisturbed flow direction this means that the assembly will have the same angle of attack after the transformation as it had before.

For an assembly of more than one grid, the ASSEMBLY parameters ROTCENTER, ROTDIRECTION and ANGLE can similarly be used to adjust the position.

General form of command

```
ASSEMBLY ( ID = "c"           , GRIDS = ["c1",...]   , GROUP = n           ,
           TRANS = [v1,v2,v3] , ROTAX = [v1,v2,v3] , ROTC = [v1,v2,v3]   ,
           ANGLE = v          , SCALE = v         , POSGRID = "c"       ,
           POSDIR = [v1,v2,v3] , POSREF = [v1,v2,v3], POSORI = [v1,v2,v3],
           INTRAY = [v1,v2,v3] , POSSHIFT = v      ,
           INTDIR = i         , INTSIDE = i         , AREF = V             )
```

Default values

```
ASSEMBLY ( TRANS=[0,0,0], ROTAX=[0,0,-1], ROTC=TRANS, ANGLE=0, SCALE=1,
           POSDIR=[0,0,-1], POSREF=[1,0,0], POSORI=[0,0,0], INTRAY=[0,1,0],
           POSSHIFT=0, INTDIR=1, INTSIDE=1)
```

Keywords

<i>keyword</i>	<i>description</i>
ID	Gives the assembly a name. The name is used in general to identify the object and to associate results like integrated forces with it.
GRIDS	Names of the component grids in the assembly. The “_1” suffix in the name should not be given, local refinements will be included.
GROUP	Alternative to GRIDS, all grids with group number equal to GROUP will be included in the assembly.
TRANS	Translation vector for the assembly. Specified in the design (offset) coordinate system.
ROTAX	Direction of axis of rotation.Specified in the design (offset) coordinate system.
ROTC	Centre of rotation. Specified in the design (offset) coordinate system.
ANGLE	Angle of rotation in degrees. Clockwise as seen along ROTAX.
SCALE	Scale factor for the assembly.

POSGRID	Name of grid to position the assembly on. If this parameter is present, TRANSLATION and ROTAXIS are interpreted as a ray for finding an intersection point where the assembly is positioned and aligned with the surface normal.
INTDIR	Integer that specifies the parameter direction of the grid POSGRID to use. Values can be 1..3.
INTSIDE	Integer that specifies which side of the grid POSGRID to use. Values can be 1..2.
POSDIR	Vector in the assembly coordinate system that will be aligned with the hull normal vector.
POSREF	Reference vector for rotation around the hull normal vector, given in the design (a.k.a. offset) coordinate system. See above for explanation.
POSORI	Origin of the ray intersecting the hull.
INTRAY	Direction of the ray intersecting the hull.
POSSHIFT	The assembly is shifted this distance in the direction of ROTAXIS.
AREF	Reference area for the assembly.

2.4. BOX

Creates a rectilinear grid for XCHAP that can be used as a tunnel for inserting other objects like rudders or shafts into. The grid will be stretched towards NOSLIP boundaries (unless that has been switched off with the XCHAP>CONTROL>NOSTRETCH keyword) to get a y^+ value of approximately 1 for the first grid node. The stretching cannot be done for opposing NOSLIP boundaries and doing it for more than one side may cause stability problems for the solver.

No transformations are applied to the box.

General form of command

```
BOX      (  ID = "c"                      , LOW = [v1,v2,v3]  , HIGH = [v1,v2,v3]  ,  
          DIMENSION = [v1,v2,v3]  , BC11 = "c"          , BC12 = "c"          ,  
          BC21 = "c"              , BC22 = "c"          , BC31 = "c"          ,  
          BC32 = "c"              , GROUP = n            , NOTRIM              ,  
          NOTFM                    , PRIO = n            , TRANS = [v1,v2,v3]  ,  
          ROTATE = [v1,v2,v3]      , SKEW = v            , BVEL = [v1,v2,v3]  ,  
          SCALE = v                                     )
```

Default values

```
BOX  (  ID="Box", LOW=[0,0,0], HIGH=[1,1,1], DIMENSION=[4,4,4],  
        BC11="INFLOW", BC12="OUTFLOW", BC21="NOSLIP", BC22="SLIP",  
        BC31="SLIP", BC32="SLIP", PRIO=-1, TRANS=[0,0,0], ROTATE=[0,0,0],  
        SKEW=[0,0,0], BVEL=[0,0,0], SCALE=1)
```

Keywords

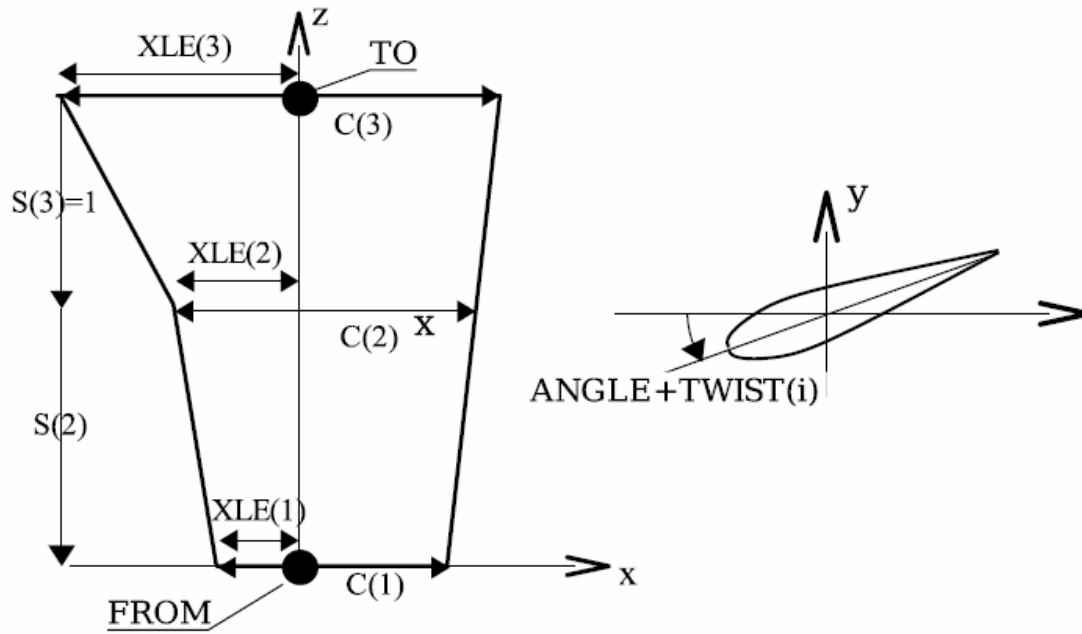
<i>keyword</i>	<i>description</i>
ID	Gives the box a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different grids may have the same name, but then only the total force on all of those objects will be calculated and output.
LOW	x,y,z coordinates of the first corner of the box.
HIGH	x,y,z coordinates of the second corner of the box. For every component it must be true that LOW < HIGH.
DIMENSION	Number of grid nodes along the x,y,z axes respectively.

BC11	Specify the boundary condition on side $l=1$ of the grid. Possible values are “NOSLIP”, “SLIP”, “INFLOW”, “OUTFLOW”, “INOUT” and “INTERIOR”. The INTERIOR condition means that the boundary values are taken by interpolation from other grids.
BC12	Specify the boundary condition on side $l=L$ of the grid.
BC21	Specify the boundary condition on side $m=1$ of the grid.
BC22	Specify the boundary condition on side $m=M$ of the grid.
BC31	Specify the boundary condition on side $n=1$ of the grid.
BC32	Specify the boundary condition on side $n=N$ of the grid.
GROUP	Group number in the overlapping grid algorithm. If not specified the grid will be the only grid in the group.
NOTRIM	Trim rotation will not be applied to the box grid. This is typically used when you define the whole water domain with the box grid.
NOTFM	No transformations applied to the box, it will appear in the computational coordinate system as specified in the command.
PRIO	Set the classification priority for the bracket grid.
TRANS	Translation vector for the box. Specified in the design (offset) coordinate system.
ROTATE	Rotation given in Euler angles of the morphed grid. Unit is degrees.
SKEW	Skew the grid in the x, y and z directions.
BVEL	Velocity applied on NOSLIP boundaries. Most likely use case is to avoid (or create) a boundary layer by specifying a velocity tangential to the surface.
SCALE	Scale factor for the box grid.

2.5. BRACKET

This command adds a bracket (strut) to the ship geometry in XCHAP. The command may be repeated to add more brackets. The geometry is specified in a bracket-fixed coordinate system according to the figure below.

The bracket position and orientation in the offset file coordinate system is given as two points that are the start and end points of the bracket, and a rotations about the brackets z-axis.



The section of the bracket is given as a set of points specified in an offset file. The format of the file is very simple, on every line should be the x and y coordinates of a points, separated by whitespace (space and/or tabs). The points should be ordered starting at the trailing edge, going forward on the upper side, around the leading edge and back on lower side to the trailing edge, i.e. last point must be the same as the first. The trailing edge must be at a point $(x > 0, 0)$ and the leading edge at $(0, 0)$. If the section is symmetrical about $y=0$ it is sufficient to give the points from the trailing edge to and including the point on the leading edge. Several symmetrical and non-symmetrical wing sections are supplied in the SHIPFLOW installation.

Except at the trailing edge the program tries to fit a smooth curve to the set of points in the offset file. This is done with a Theodorsen-Garrick transform, which is a conformal mapping technique. The obtained mapping is also used to make 2D grids for uniformly spaced stations of constant z .

When the bracket grid is added to the overlapping grid that describes the complete geometry, it goes through the transformation to the SHIPFLOW computational coordinate system, including scaling by $1/LPP$, sinkage and trim.

Since the bracket surface is not closed at either end, the bracket must intersect other boundary surfaces at both ends. A typical situation is that one end of the bracket sticks into the hull and the other into the supported structure such as the propeller shaft. The curves at the ends of the bracket must be completely outside of the fluid domain.

General form of command

```

BRACKET ( ID = "c"           , FROM = [v1,v2,v3] , TO = [v1,v2,v3]           ,
          ANGLE = v           , S = [v1,v2,...,vn]   , C = [v1,v2,...,vn]       ,
          SECTION = ["c"]     , XLE = [v1,v2,...,vn] , T = [v1,v2,...,vn]       ,
          BC22 = "c"          , OGRID                ,
          RMAX = v            , GROUP = n            , TWIST = [v1,v2,...,vn]   ,
          BC31 = "c"          , BC32 = "c"           , DIMENSION = [v1,v2,v3]  ,
          YTARGET = v         , TGTRANS              , EXTFROM = v              ,
                               HYPWING              , EXTTO = v                ,
                               HYPPT
          GROW = v            , DTAU = v             , NITER                    ,
          TP1O                , TP2O                 , DN = v                   ,
          TP1R                , TP2R                 ,
          TP1F                , TP2F
          SCALE = v           , STRETCH = v           , PRIO = n                 ,
          STR1 = "c"          )

```

Default values

```

BRACKET( ID = "Bracket", ANGLE = 0, SECTION = "", XLE = C*0.25, TWIST=[],
          RMAX=1, TP1O, TP2O, SCALE=1, STRETCH=1, PRIO=-1)

```

Keywords

<i>keyword</i>	<i>description</i>
ID	Gives the bracket object a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different objects may have the same name, but then only the total force on all of those objects will be calculated and output.
FROM	The starting point of the bracket.
TO	The ending point of the bracket.
ANGLE	Bracket rotation angle in degrees. The axis of rotation is the z-axis in the bracket coordinate system.
S	Bracket section stations along the z-axis, non-dimensionalized by bracket length. s=0 corresponds to the starting point and s=1 to the ending point. The first value should always be 0 and the last 1.

C	Bracket chord lengths. The length of this vector must be equal to the length of S.
XLE	Distance from the leading edge to the z-axis for each station. Positive numbers places the leading edge on the negative side of the axis. The length of this vector, if given, must be equal to the length of S. If not given the default value $0.25 \cdot C$ is used.
T	Bracket thickness. The length of this vector must be equal to the length of S.
TWIST	Rotation of each section in degrees around the z-axis. The length of this vector, if given, must be equal to the length of S. If not given the default value 0 is used for each station.
SECTION	The name of a file containing offsets for a 2D section. See above for a description of the file format. If an empty string is given, the NACA0012 section is used.
TP1O	The bracket is open at the first station.
TP1R	The bracket is closed with a rounded tip at the first station.
TP1F	The bracket is closed with a flat (square) tip at the first station.
TP2O	The bracket is open at the last station.
TP2R	The bracket is closed with a rounded tip at the last station.
TP2F	The bracket is closed with a flat (square) tip at the last station.
RMAX	Change the position of the outer surface of the grid. $RMAX > 1$ increases the distance from the bracket to the outer surface and vice versa.
GROUP	Group number in the overlapping grid algorithm. If not specified the grid will be the only grid in the group.
BC22	Specify the boundary condition at outer radius of the grid. Possible values are "NOSLIP", "SLIP", "INFLOW", "OUTFLOW", "INOUT" and "INTERIOR". The INTERIOR condition means that the boundary values are taken by interpolation from other grids.
BC31	Specify the boundary condition at start of bracket grid.
BC32	Specify the boundary condition at end of bracket grid.
DIMENSION	Number of grid nodes along circumferential, radial and axial directions respectively.
YTARGET	y^+ target value at noslip boundaries.
TGTRANS	Use Theodorsen-Garrick transform. Gives high quality orthogonal planar grids for normal wing sections.

HYPWSECT	Use 2D hyperbolic grid generator. The same format of offset file used by TGTRANS is used to specify the starting curve. This generator can handle more arbitrary shaped sections than TGTRANS.
HYPPT	Use 2D hyperbolic grid generator. The same format of offset file used by TGTRANS but with an additional point type column is used to specify the starting curve. This generator can handle more arbitrary shaped sections than TGTRANS.
OGRID	A zero-thickness section at the trailing edge will be interpreted by HYPWSECT and HYPPT as a C-type grid. If this is not correct, OGRID forces the generation of an O-type grid.
GROW	[HYPSECT only] Growth factor in the normal direction.
DTAU	[HYPSECT only] Artificial time step for solving the hyperbolic equations. Lower the value to increase stability.
NITER	[HYPSECT only] Maximum number of iterations for solving the hyperbolic equations. May need to be increased if DTAU is decreased.
DN	[HYPSECT only] Thickness of first layer of cells. DN is relative to the chord length.
EXTFROM	Extension distance from starting point.
EXTTO	Extension distance from end point.
SCALE	Scale factor for the bracket.
STRETCH	[HYPSECT only] Stretch factor towards regions with high curvature. Higher value gives stronger concentration.
PRIOR	Set the classification priority for the bracket grid. Should be >0.
STR1	Stretching function in the chord-wise direction. The value should be the ID of a STRETCHING.

2.6. CYLINDER

Creates a right circular cylindrical grid for XCHAP. The grid will be stretched towards NOSLIP boundaries (unless that has been switched off with the XCHAP>CONTROL>NOSTRETCH keyword) to get a y^+ value of approximately 1 for the first grid node.

The CYLINDER geometry is specified in the offset coordinate system and the same transformations that are applied on the hull are applied to the cylinder.

To make an interior grid that can be used as a thruster tunnel, set the minimum radius to 0, the maximum radius to the desired value and then BC12 to NOSLIP.

General form of command

```
CYLINDER ( ID = "c" , FROM = [v1,v2,v3] , TO = [v1,v2,v3] ,  
          R = [v1,v2] , DIMENSION = [v1,v2,v3] , GROUP = n  
          BC11 = "c" , BC12 = "c" , BC31 = "c" ,  
          BC32 = "c" , NOCUT . ANGLE = [v1,v2] ,  
          SKEW = v , ANGROT = v )
```

Default values

```
CYLINDER ( ID="Cylinder", FROM=[0,0,0], TO=[0,1,0], DIMENSION=[10,20,20],  
          BC11="NOSLIP", BC12="INTERIOR", BC31="SLIP", BC32="SLIP",  
          ANGLE=[0,360], SKEW=0, ANGROT=0)
```

Keywords

<i>keyword</i>	<i>description</i>
ID	Gives the cylinder a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different grids may have the same name, but then only the total force on all of those objects will be calculated and output.
FROM	Start point of the axis of the cylinder.
TO	End point of the axis of the cylinder.
DIMENSION	Number of grid nodes. Dimension 1 is radial, dimension 2 is circumferential and dimension 3 is axial.
BC11	Specify the boundary condition on side l=1 of the grid. Possible values are "NOSLIP", "SLIP", "INFLOW", "OUTFLOW", "INOUT" and

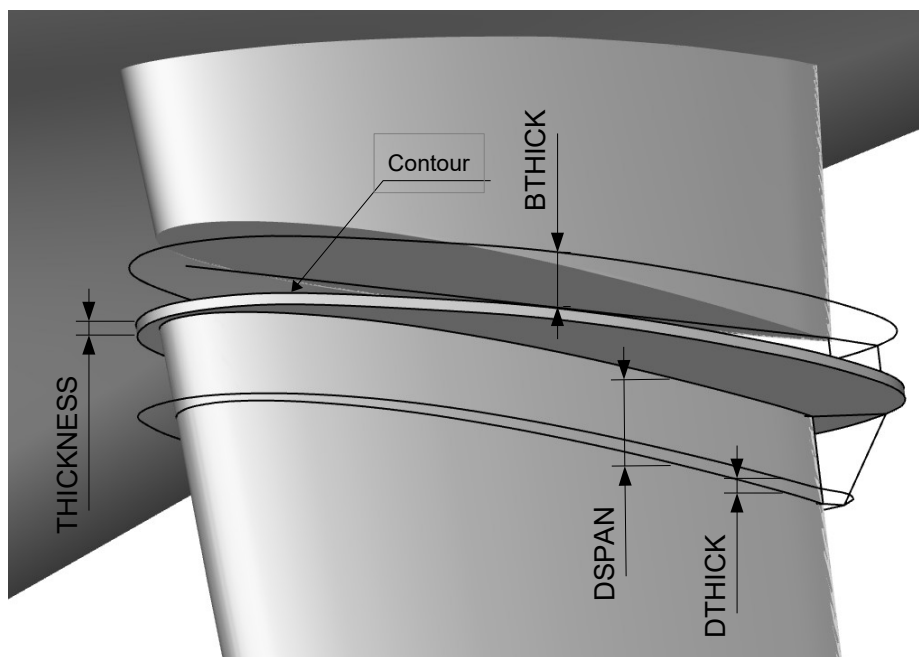
“INTERIOR”. The INTERIOR condition means that the boundary values are taken by interpolation from other grids.

BC12	Specify the boundary condition on side $l=L$ of the grid.
BC31	Specify the boundary condition on side $n=1$ of the grid.
BC32	Specify the boundary condition on side $n=N$ of the grid.
GROUP	Group number in the overlapping grid algorithm. If not specified the grid will be the only grid in the group.
NOCUT	The physical boundaries will not cut holes in other grids.
ANGLE	Start and end angles. Default is $[0, 360]$ which is a closed cylinder.
SKEW	Tangential skew. Used mainly to test discretization.
ANGROT	Rotation angle around cylinder axis.

2.7. ENDPLATE

This command makes an endplate on a RUDDER object. Given a rudder object ID and a curve representing the contour of the plate, grids are generated that conform to both the rudder tip and the plate.

The contour is either given in the command with the XPLATE, YPLATE and PTYPE parameters or imported from an offset file with the SECTION parameter. If the latter option is used, the file should be in the wing section offset format specified for the WINGSECTION command. The size of the contour is assumed to be normalized by the chord length of the rudder at the tip where the endplate is fitted. If needed the size can be adjusted with the SCALE parameter and the chordwise position by XPOS. It is required that the rudder tip contour is completely enclosed by the endplate contour.



General form of command

```

ENDPLATE (ID = "c"           , RUDDER = "c"           , XPOS = v           ,
          SIDE= i            , THICKNESS = v          , SECTION = "c"       ,
          XPLATE = [v1,...] , YPLATE = [v1,...] , PTYPE = [i1,...] ,
          GROW = v           , DN = v                 , DSPAN = v           ,
          DTHICK = v         , BTHICK = v             , DIME = [i1,i2,i3] ,
          YTARG = v          , SCALE = v              , GROUP = i           ,
          PRIO = i           )

```

Default values

```

ASSEMBLY ( XPOS=0, SIDE=0, PRIO=10, THICKNESS=0.03, GROW=1.05, DN=
          -0.001, DSPAN=0.05, DTHICK=0.1, BTHICK=0.1, DIME=, YTARG=1,
          SCALE=1, GROUP=10, PRIO=10, )

```

Keywords

<i>keyword</i>	<i>description</i>
ID	Gives the endplate a name. The name is used in general to identify the object and to associate results like integrated forces with it.
RUDDER	Name of the associated RUDDER object. The position and direction of the endplate is derived from values in RUDDER.
XPOS	Translation of the endplate along the rudder chord line. Positive towards the leading edge.
SIDE	Specify which end of the rudder the endplate is fitted to. 0 → root and 1 → tip.
THICKNESS	Thickness of the plate in the rudder coordinate system.
SECTION	Name of offset file for the endplate contour curve. The file should be in the wing section offset format specified for the WINGSECTION command.
XPLATE	List of x-coordinate values for the contour curve. Do not use this if the contour is specified with SECTION.
YPLATE	List of y-coordinate values for the contour curve. Do not use this if the contour is specified with SECTION.
PTYPE	List of point types for the contour curve, 0 → smooth and 1 → corner. Do not use this if the contour is specified with SECTION.
GROW	Growth rate for the hyperbolic grid generator used for the grid outside the contour.

DN	Initial cell thickness for the hyperbolic grid generator used for the grid outside the contour. The value is negative due to the orientation of the contour curve.
DSPAN	Spanwise thickness of the grid on the inside of the endplate. This value is given in the spanwise parameter of the rudder, so 1 means cover the whole rudder.
DTHICK	Normal thickness of the grid on the inside of the endplate. This value is given in the normal parameter of the rudder, so 1 means from the rudder surface to the outer surface of the rudder grid. This value should be chosen so that the curve is approximately the same distance from the rudder surface as the endplate contour.
BTHICK	Thickness of the outside (bottom) grid, given in the rudder coordinate system.
DIMENSIONS	Dimensions of the grids, first component is along the contour, second is out from the contour and third is spanwise.
YTARG	Target y^+ for the grids.
SCALE	Scale factor for the endplate. The endplate is automatically scaled by the rudder chord at the selected end and that scale factor is then multiplied by SCALE.
GROUP	Set the group number for the endplate grids.
PRIOR	Set the classification priority for the endplate grids.

2.8. ENVBOX

When this command is present the grid generated by XGRID is shrunk and embedded in a BOX grid. Use this command for VOF cases with drift angle (OSFLOW) and/or turn radius (TURN). For such cases the hull grid is rotated by the angle specified in OSFLOW rather than rotating the onset flow. In general cases with VOF free surface and OSFLOW angle and/or TURN radius will not work.

ENVBOX will replace the computational domain filled by the specified XGRID. It does it by reading and modifying the parameters in the XGRID module and generating an XFLOW|BOX command. Parameters in the XFLOW|HULLTYPE and XCHAP|VOF commands are also used. The modified module and command can be read in the generated command file <id>_RUN_DIR/<id>. The sequestered parameter values may be overridden by setting values directly in the ENVBOX command.

For VOF cases the BOX grid is automatically stretched towards the undisturbed position of the free surface as specified by XCHAP|VOF|Z0.

General form of command

```
ENVBOX ( ID = "c" , XSTART = v , XEND = v ,
        YWIDTH = v , ZBOTTOM = v , ZTOP = v ,
        DIMENSION = [v1,v2,v3] , BC11 = "c" , BC12 = "c" ,
        BC21 = "c" , BC22 = "c" , BC31 = "c" ,
        BC32 = "c" , GROUP = n )
```

Default values

```
ENVBOX ( ID="EnvBox", XSTART from XGRID, XEND from XGRID, YWIDTH from
        XGRID, ZBOTTOM from XGRID, ZTOP from XFLOW|OFFSET,
        DIMENSION from XGRID, BC11="INFLOW", BC12="OUTFLOW" or
        "TOPOUT", BC21="SLIP", BC22="SLIP", BC31="SLIP", BC32="SLIP" or
        "TOP")
```

Keywords

<i>keyword</i>	<i>description</i>
ID	Gives the envbox a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different grids may have the same name, but then only the total force on all of those objects will be calculated and output.
XSTART	Position in the Shipflow computational system of the inflow (upstream) grid face.

XEND	Position in the Shipflow computational system of the outflow (downstream) grid face.
YWIDTH	Y-direction half width of the box in the Shipflow computational system. For symmetric cases the extent of the box is [-YWIDTH..0] and for unsymmetric cases [-YWIDTH..YWIDTH]. The default value is XGRID RADIUS RADIUS.
ZBOTTOM	Position in the Shipflow computational system of the lower grid face. The default value is -YWIDTH.
ZTOP	Position in the Shipflow computational system of the upper grid face. The default value is 0 for double model cases and XFLOW OFFSET ZTOP for VOF cases.
DIMENSION	Number of grid nodes along the x,y,z axes respectively.
BC11	Specify the boundary condition on side l=1 of the grid. Possible values are “NOSLIP”, “SLIP”, “INFLOW”, “OUTFLOW”, “INOUT” and “INTERIOR”. The INTERIOR condition means that the boundary values are taken by interpolation from other grids.
BC12	Specify the boundary condition on side l=L of the grid.
BC21	Specify the boundary condition on side m=1 of the grid.
BC22	Specify the boundary condition on side m=M of the grid.
BC31	Specify the boundary condition on side n=1 of the grid.
BC32	Specify the boundary condition on side n=N of the grid.
GROUP	Group number in the overlapping grid algorithm. If not specified the grid will be the only grid in the group.

2.9. EXTRUSION

This command creates a volume grid by extruding a 2D-grid in the z-direction.

General form of command

```
EXTRUSION ( ID = "c" , SECT = "c" , SCALE = v ,
            TRANSLAT = [v1,v2,v3] , DIR = [v1,v2,v3] , DIM3 = n ,
            YTARGET = v , ANGLE = v )
```

Default values

```
ROTBODY ( ID="Rotbody", SCALE=1, TRANSLATE=[0,0,0],
           DIR=[1,0,0], DIM3=20, YTARGET = 1.0,
           ANGLE=360 )
```

Keyword

<i>keyword</i>	<i>description</i>
ID	Gives the extrusion a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different objects may have the same name, but then only the total force on all of those objects will be calculated and output.
SECTION	Name of the 2D grid to extrude. If not overridden the dimensions and boundary conditions are carried over to the 3D grid.
SCALE	Scale the grid.
TRANSLATE	Position of the local coordinate system origin in the offset file (ship) coordinate system.
DIR	Direction of the x-axis in the offset file coordinate system.
DIM3	Number of grid planes generated by the rotation.
YTARGET	Sets the target value for y^+ .
ANGLE	Sets the rotation angle for the object.

2.10. FILE

This command can be used to override the name conventions for some of the files used by XPAN, XGRID and XCHAP.

General form of command

```
FILE (  XPAN = "c"           ,  XBOUND = "c"           ,  PROFILES = "c"           ,  
        MEASUREMENT = "c"    ,  GEOMETRY = "c"           )
```

Default values

```
FILE (          PROFILES = id_XVPROF, MEASUREMENT = id_XVMEAS,  
              GEOMETRY = id_XVGRID, XBOUND = id_XBDB, XPAN = id_XPDB )
```

where "id" is the name of the command file

Keywords

<i>keyword</i>	<i>description</i>
PROFILES	File containing profiles of velocity and turbulent quantities at the inlet plane and velocity and pressure at the outer boundary.
MEAS	File containing boundary layer data at the inlet station for profile calculation.
GEOMETRY	File containing the coordinates generated by XGRID.
XBOUND	Data base file containing results from XBOUND.
XPAN	Data base file containing results from XPAN. To be used for restart and to transfer data to XBOUND and XCHAP.

2.11. FLUID

This command is used to specify the fluid properties and the gravity.

General form of command

```
FLUID ( DENSITY = v      ,  VISCOSITY = v      ,  GRAVITY = v      ,
        TEMPERATURE = v  ,  SEA
                           FRESH
      )
```

Default values

```
FLUID (SEA, TEMPERATURE = 15, GRAVITY = 9.80665)
```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	DENSITY	Density of the fluid [kg/m ³].
(2)	VISCOSITY	Kinematic viscosity of the fluid [m ² /s].
(3)	GRAVITY	Gravitational acceleration [m/s ²].
	TEMP	Water temprature in degrees Celcius. Will give the default values for density and viscosity.
	SEA	Sea water
	FRESH	Fresh water

Notes

- (1) Default density is the density of sea water at 15 C°, only used for calculation of forces for output.
- (2) The default viscosity is the viscosity of sea water at 15 C°, used for conversion of speed to Reynolds number.
- (3) Standard gravitational acceleration, used for conversion of speed to Froude number.

2.12. FMREF

This command is used to specify the ship dimensions for an additional output of the non-dimensionalized integrated forces and moments. The data is written to id_INTEGRALS.

General form of command

FMREF (LENGTH = v, DRAFT= v, BEAM= v, XTRA= v)

Default values

None

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	LENGTH	Length of ship used for non-dimensionalisation only.
(2)	DRAFT	Draft of ship used for non-dimensionalisation only.
(3)	BEAM	Beam of ship used for non-dimensionalisation only.
(3)	XTRA	Longitudinal of the transformed coordinate system. Hence XTRA=0 is at FP and XTRA=1 is at AP.

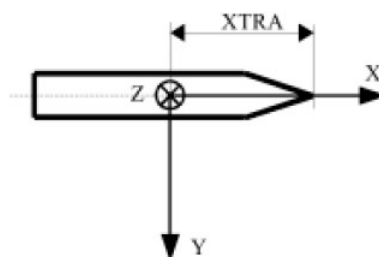
Notes

- (1) Computational forces and moments are non-dimensionalised in the following way:

$$X = \frac{2X}{\rho V^2 L_{PP} T} \quad Y = \frac{2Y}{\rho V^2 L_{PP} T} \quad Z = \frac{2Z}{\rho V^2 L_{PP} B}$$

$$K = \frac{2K}{\rho V^2 L_{PP} T^2} \quad M = \frac{2M}{\rho V^2 L_{PP} B} \quad N = \frac{2N}{\rho V^2 L_{PP} T}$$

- (2) Center of reference the moments at intersection of the water and the center plane at position XTRA from the forward perpendicular. The x-axis points forward, y-axis to starboard and z-axis down.



2.13. HULLTYPE

This command helps XMESH, XPAN and XGRID carry out the grid generation for different types of hulls. This command is also where the information for the Standard Case mode is to be specified. See “Standard Cases” on page 325. The Standard Case mode is triggered by the presence of the keyword H1GR.

General form of command

```

HULLTYPE ( MONO          , NOFSFLOW      , XMAUTO          ,
            CATAMARAN    , FSFLOW       , XMMANU          ,
            SUBMARINE     , VFSFLOW      ,
            TRIMARAN      ,
            TWINSKEG      ,
            YACHT         ,
            POW           ,

            H1GR = "c"    , H2GR = "c"    , H3GR = "c"    ,
            FBGR = "c"    , OGRP = "c"    , ABGR = "c"    ,
            H1GI = "c"    , H2GI = "c"    , H3GI = "c"    ,
            OGRI = "c"    , ABGI = "c"    , FBGI = "c"    ,
            H1GO = "c"    , H2GO = "c"    , H3GO = "c"    ,
            OGRO = "c"    , ABGO = "c"    , FBGO = "c"    ,
            BDENS = v     , FDENS = v     , NOWSING        ,
                               WSING
            WSDY = v      , XWLP = v      ,
            NOKSINGCORR   , TRXDIR        , COARSE          ,
                               TRXYZDIR
                               TRHBUT        , MEDIUM          ,
                               FINE
            TRANSOM       , WTRANSOM      , VOF             )

```

Default values

HULLTYPE (MONO, FSFLOW,XMAUTO, BDENS = 1.0, FDENS = 1.0, TRXDIR)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	MONO	[SHIPTYPE] XGRID will generate a grid that is a quarter of a circular cylinder. XPAN will assume a surface piercing mono hull configuration. A wave profile plot is generated for the read macro.
	CATAMARAN	[SHIPTYPE] XMESH and XPAN will assume a catamaran configuration.
	TRIMARAN	[SHIPTYPE] XMESH and XPAN will assume a trimaran configuration.
	SUBMARINE	[SHIPTYPE] XGRID will generate a grid that is a half of a cylinder. XMESH and XPAN will assume a fully submerged body.
	XMAUTO	[MESHMOD] Automatic mesh generation in XMESH/XPAN. Requires offset group specification in HULLTYPE. Replaces “Standard case”.
	XMMANU	[MESHMOD] Manual mesh generation in XMESH/XPAN. Requires full description of panelization in XMESH section. Replaces “Normal input mode”.
(2)	TWINSKEG	[SHIPTYPE] XGRID will generate a grid for twin skeg hull type.
(3)	YACHT	[SHIPTYPE] XMESH and XPAN will assume a sailing yacht configuration. The panelization at the stern on the free- surface for heeled sailing yachts will be influenced.
(4)	POW	[SHIPTYPE] Run Propeller Open Water calculations. This keyword will override most commands in the command file and just use a few. Propeller geometry in XFLOW:PROPELLER and the corresponding propeller model in XCHAP needs to be set.
	H1GR	Character string that gives the label of the offset group for hull group number one when the Standard Case mode is used.
	H2GR	Character string that gives the label of the offset group for hull group number two when the Standard Case mode is used.
	H3GR	Character string that gives the label of the offset group for hull group number one when the Standard Case mode is used.
	OGRP	Character string that gives the label of the offset group for the stern overhang when the Standard Case mode is used.

	FBGR	Character string that gives the label of the offset group for the fore bulb when the Standard Case mode is used.
	ABGR	Character string that gives the label of the offset group for the aft bulb when the Standard Case mode is used.
(2)	H1GI	Name of an offset group that describes the inner part of the main hull.
(2)	H2GI	Second offset group that describes the inner part of the main hull.
(2)	H3GI	Third offset group that describes the inner part of the main hull.
(2)	OGRI	Name of offset group that describes the inner part of the overhang in the stern.
(2)	ABGI	Name of offset group that describes the inner part of the stern bulb.
(2)	FBGI	Name of offset group that describes the inner part of the bulb.
(2)	H1GO	Name of an offset group that describes the outer part of the main hull.
(2)	H2GO	Second offset group that describes the outer part of the main hull.
(2)	H3GO	Third offset group that describes the outer part of the main hull.
(2)	OGRO	Name of offset group that describes the outer part of the overhang in the stern.
(2)	ABGO	Name of offset group that describes the outer part of the stern bulb.
(2)	FBGO	Name of offset group that describes the outer part of the bulb.
	FSFLOW	Specifies that a free-surface is included in the potential flow computation when the Standard Case mode is used.
	NOFSFLOW	Specifies that a free-surface is not included in the potential flow computation when the Standard Case mode is used.
	VFSFLOW	Specifies that the viscous flow should be calculated under the prescribed free-surface computed by XPAN. It implies that FSCFLOW is set. XGRID will fit the grid to the free-surface computed by XPAN using linear or non-linear boundary conditions. Supported by XCHAP.

(5)	TRANSOM	Specifies that a transom stern case is to be computed in the potential flow solution when the Standard Case mode is used.
	WTRANSOM	Wet transom block in XGRID. Supported by XCHAP.
	BDENS	XMESH panel density factor for the H1GR, H2GR, H3GR, OGRP, FBGR and ABGR groups when the Automatic mode is used.
	FDENS	XMESH panel density factor for the free-surface and transom groups when the Automatic mode is used.
(6)	NOWSING	[SINGULARITY] XGRID will create a grid without a singularity line in the water-line plane.
	WSING	[SINGULARITY] XGRID will create a grid with a singularity line in the water-line plane.
	WSDY	[SINGULARITY] Fraction of max breath used for default WSING.
	XWLP	Start of WL projecion.
(7)	NOKSINGCORR	The keel singularity correction is turned off.
	TRXDIR	[WEDMODE] The wave profile will continue horizontally after the transom. Only used here in Automatic mode. In Manual mode this option can be specified in the TRANSOM group.
	TRXYZDIR	[WEDMODE] The wave profile will follow the slope of the wedge after the transom. Only used here in Automatic mode. In Manual mode this option can be specified in the TRANSOM group.
	TRHBUT	[WEDMODE] The wave profile can take any direction after the transom Only used here in Automatic mode. In Manual mode this option can be specified in the TRANSOM group.
	COARSE	[XMDENS] Corresponds to BDENS=0.5 and FDENS=0.5.
	MEDIUM	[XMDENS] Corresponds to BDENS=0.8 and FDENS=0.8.
	FINE	[XMDENS] Corresponds to BDENS=1.0 and FDENS=1.0.
	VOF	Volume of Fluid solver is executed.

Notes

- (1) XBOUND assumes for the Standard Case mono-hull that a turbulent boundary layer on panel group H1GR should be computed. It automatically creates a streamline group number one. Variables can be altered from the default values in the XBOUND data section. More streamline groups can be added manually.

XGRID creates a default grid based on the information in HULLTYPE for the mono-hull case. Variables can be altered from the default values in the XGRID data section.

- (2) Added support for twin skeg hulls. More information on grid generation for this type of ship can be found in “Offset file format for twin skeg hulls” on page 313
- (3) The free-surface downstream of a sailing yacht may be modeled with or without a TRANSOM group depending on the geometry of the stern. If a TRANSOM group is used it is recommended to extend the hull above the free-surface. See also EXPANEL on the BODY command in XMESH and ZFACT on the CONTROL command in XPAN.

For a heeled sailing yacht where both sides of the hull are modeled it may be necessary to search for two intersections with the free-surface, TWOINT, on the BODY group that goes into the water. The TRANSOM group may be attached to both sides of the yacht or to one side only if the other side is lifted out of the water in the stern region.

The keyword IBDE = [n₁], may be introduced on the FREE group attached the side that goes out of the water in order to attach the FREE group to the second free-surface intersection of the BODY group, [n₁], on the other side of the hull.

The temporary additional draught described above is recommended also for heeled sailing yachts.

The initial draught ZORI must be changed for heeled yachts in order to have the same displacement as in the upright position.

- (4) The keyword POW overrides all of the settings in the command file, except for LPP, XFLOW:PROPELLER, XCHAP:ACTUATOR, XCHAP:LLINE and XCHAP:PARALLEL. This means that your regular command file that contains a hull can be used for an open water simulation by just adding the keyword POW.

The propeller grid is put inside a box grid. For each advance coefficient (J) there is a cgns file created (naming convention: <id>_POW_J<J>.cgns). At the end of the computations a data file is produced which contains a table of J, KT, KQ and Eta_o.

The list of advance coefficients that should be computed and the number of iterations per calculation can be modified with the XCHAP:POW command.

- (5) In the longitudinal direction the hull will be cut at the last offset station that has a point below the wavy free-surface. A TRANSOM group will be generated on the free-surface downstream of the transom. The TRANSOM keyword can be used for high speed vessels, tunnel sterns and for flat overhang sterns. For the latter cases it is recommended to model the hull also above the free-surface.

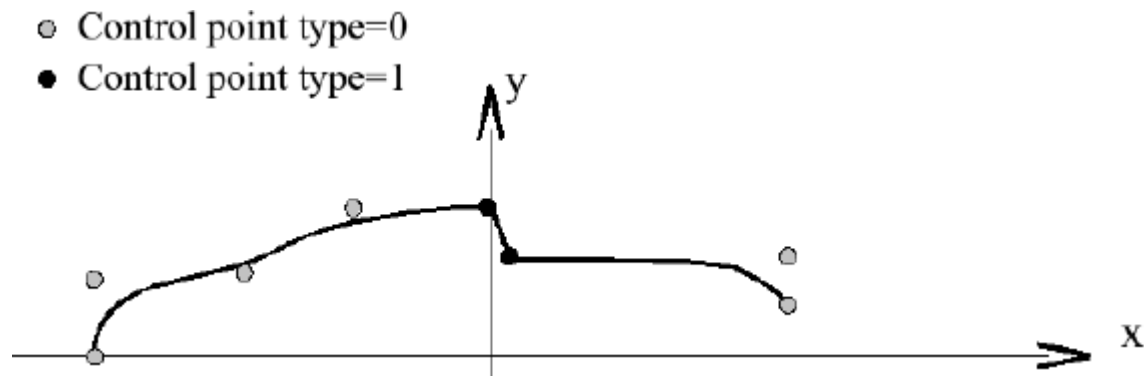
See also EXPANEL on the BODY command in XMESH and ZFACT on the CONTROL command in XPAN.

- (6) XGRID will try to make a singularity line in the free-surface plane starting at the end of the ship in the case neither NOWSING or WSING is given. Specifying NOWSING can be better for ships without a transom or with a small transom.
- (7) For the ships where the transom block option was used together with the prescribed free surface a correction of the keel singularity line is active by default to improve robustness of the grid generation. It is advised to keep this correction active especially when the stern wave has a large amplitude.

2.14. HYP SURF

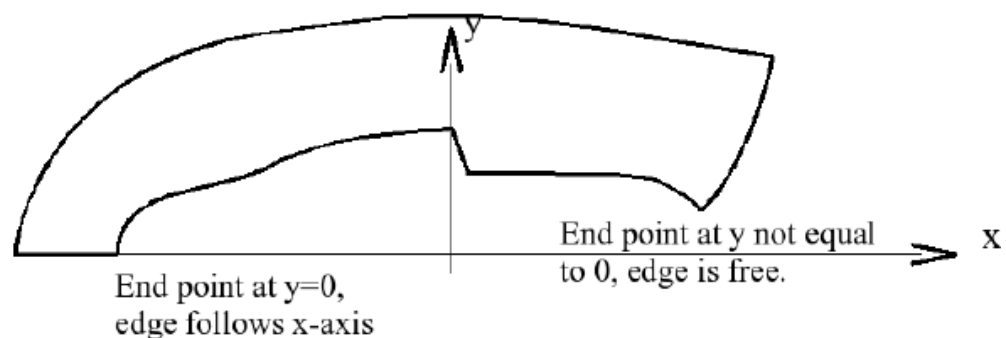
This command creates a hyperbolic two-dimensional grid in XCHAP. The command may be repeated to add more grids.

Hyperbolic grid generation is a technique where a grid is generated from a starting curve by time stepping a hyperbolic equation from a starting curve. Several parameters may be used to control the grid quality. Boundary conditions may be applied at the ends of the starting curve to control the surface shape, but the shape of the outer curve cannot be controlled. This makes the technique especially suited for generating component grids for overlapping grids, since the outer grid boundary does not need to be fitted to any other grid boundary.



The geometry of the starting curve is specified with control points in a local coordinate system according to the figure below. Each point is given a type 0 or 1 where 0 means that the curve is smooth and 1 that the curve has a corner (discontinuous derivative). A curve is constructed from the control points using Chaikin subdivision. This subdivision is approximating, like a B-spline, so the curve will in general not pass through the control points except the end points and the corner points. The type of the end points are ignored except if the first and last points coincide. Then the curve is closed and the point type is taken from the first control point.

The starting curve, or polygon rather, is generated by distributing points approximately equidistantly along the subdivision curve, but grid points are forced to coincide with corner points.



Finally the surface grid is generated by repeatedly generating new curves from the previous with the hyperbolic method. The boundary conditions at the ends of the starting curve are specified implicitly by the y-coordinate of the corresponding control point. If it is zero, then

the surface follows the x-axis, otherwise the boundary is free.

Limitations

The algorithm is not very good at smoothing the grid in concave corners. Increasing CONCAVE, NSMOOTH or the number of cells along the starting curve (the first dimension) may help. Increasing TSMOOTH can make things worse. Interior corners with angle less than 90 degrees tend to give bad grid quality.

General form of command

```
HYPSURF  (  ID = "c"                ,  X = [v1,v2,...,vn]    ,  Y = [v1,v2,...,vn]    ,
           PTYPE = [v1,v2,...,vn]    ,  DN = v                ,  GROW = v                ,
           CAREA = v                  ,  STRETCH = v           ,  NITER = n              ,
           DTAU = v                   ,                               ,
           ORIGIN = [v1,v2]           ,  ANGLE=v               ,  PRESCALE=v            ,
           SCALE = v                  ,  DIME = [v1,v2]       ,  GROUP = n              ,
           BC11 = "c"                 ,  BC12 = "c"           ,  BC21 = "c"            ,
           BC22 = "c"                 ,  YTARGET = v          ,  OFFSET = "c"          ,
           YSCALE = v                 ,  CURVE = "c"          ,  BC1X,                 ,
           BC1Y                       ,  BC2X                 ,  BC2Y                   )
```

Default values

```
HYPSURF(  ID="Hypsurf",  DN=0.01,      GROW=1.1,    CAREA=1,
           STRETCH=0,    NITER=5,      ORIGIN=[0,0],  SCALE=1,
           PRESCALE=1,  ANGLE=0,      DIME=[20,10],  GROUP=unique,
           BC11="INTERIOR",  BC12="INTERIOR",
           BC21="NOSLIP",    BC22="INTERIOR",
           YSCALE=1, DTAU=0.5          )
```

Keyword

<i>keyword</i>	<i>description</i>
----------------	--------------------

ID	Gives the hypsurf object a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different objects may have the same name, but then only the total force on all of those objects will be calculated and output.
----	---

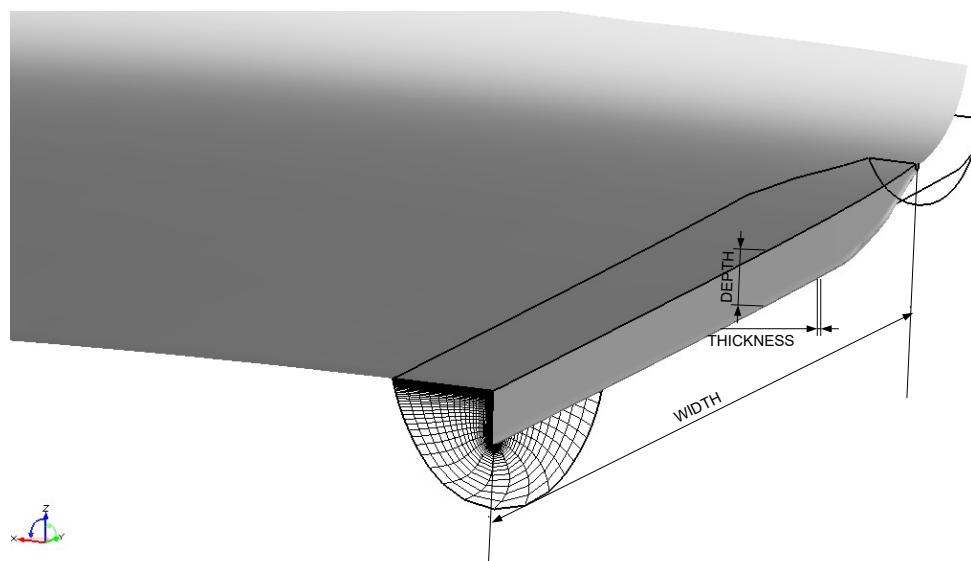
X	Vector of x coordinates for the control points. At least 2 values required.
---	---

Y	Vector of y coordinates for the control points. Same number of values as X.
PTYPE	Vector of control type points. 0 gives a smooth curve and 1 a corner. The end point values are ignored except if the curve is closed, then the first value is used. Same number of values as X.
DN	Thickness of the first layer of cells.
GROW	Growth rate of the cell thickness, typically 1-1.2.
CAREA	Area growth factor. Increasing the value makes small cells grow faster. Tends to smooth the grid in concave regions.
NITER	Maximum number of iterations for solving the hyperbolic equations.
DTAU	Artificial timestep in the update. Default value is 0.5, lower values typically increases stability but larger NITER may be needed to smooth the grid.
STRETCH	Stretch factor towards regions with high curvature. Higher value gives stronger concentration.
ORIGIN	Translate the surface.
SCALE	Scale the surface.
PRESCALE	Grid scale factor applied before rotation and translation.
DIME	Grid dimensions. The first dimension is along the starting curve.
GROUP	Group number used in the overlapping grid algorithm.
BC11	Boundary condition on the edge that starts at the first point of the starting curve. Possible values are "NOSLIP", "SLIP", "INFLOW", "OUTFLOW", "INOUT" and "INTERIOR". The INTERIOR condition means that the boundary values are taken by interpolation from other grids.
BC12	Boundary condition on the edge that starts at the last point of the starting curve. Possible values same as BC11.
BC21	Boundary condition on the starting curve. Possible values same as BC11.
BC22	Boundary condition on the last generated grid line out from the starting curve. Possible values same as BC11.
ANGLE	Rotate the grid around origin in clockwise direction. ANGLE should be given in degrees.
YTARGET	y+ target value at noslip boundaries.
OFFSET	Read x, y, ptype from a text file.

YSCALE	Scale factor for the y coordinate.
CURVE	Name of a curve object to grow the grid on.
BC1X	Grid follows $x=\text{const}$ at boundary curve start.
BC1Y	Grid follows $y=\text{const}$ at boundary curve start.
BC2X	Grid follows $x=\text{const}$ at boundary curve end.
BC2Y	Grid follows $y=\text{const}$ at boundary curve end.

2.15. INTERCEPTOR

This command generates an interceptor appendage at the transom. The geometry is defined by the width, the depth under the hull and the thickness. The contour of the device is parallel to the transom-hull intersection.



General form of command

```
INTERCEPTOR( ID = "c"           , WIDTH = v           , DEPTH = v           ,
                THICKNESS = v      , YTARGET = v        , DIME = [i1,i2,i3] ,
                GROW = v           , DN = v             )
```

Default values

```
ASSEMBLY ( WIDTH=1.0, DEPTH=0.05, THICKNESS=0.01, YTARGET=1, GROW=1.1,
            DN=0.001, DIME=[50,35,60] )
```

Keywords

<i>keyword</i>	<i>description</i>
ID	Gives the interceptor a name. The name is used in general to identify the object and to associate results like integrated forces with it.
WIDTH	Interceptor length measured from the centre plane $y=0$. Given in design (offset) coordinates.

DEPTH	Interceptor height measured from the transom corner vertically to its edge. Given in design (offset) coordinates.
THICKNESS	Interceptor constant thickness. Given in design (offset) coordinates.
YTARGET	For cell thickness given in y^+ units. Typically ≤ 1 .
GROW	Growth factor for the cell thickness for the grid. See the documentation for HYP SURF GROW for more information.
DN	Thickness of the first cell layer in the fairing grid. See the documentation for HYP SURF DN for more information.
DIMENSIONS	Dimensions of the grid. The dimension are along the section, out from the surface and along the edge respectively.

2.16. IPOSITION

This command is used to specify the initial position of the ship. This makes it possible to use the same offset point coordinates for different roll and trim conditions

General form of command

```
IPosition (  ROLL = v   ,  TRIM = v   ,  XCOF = v   ,
            TM = v       ,  TA = v       ,  TF = v       ,
            XLCG = v   ,  ZVCG = v           )
```

Default values

IPosition (ROLL = 0.0, TRIM = [0.0 or from TF and TA], XCOF = Lpp/2, TM = 0.0)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	ROLL	Roll-angle [°].
(3)	TRIM	Trim-angle [°].
(1)	XCOF	The initial position of the centre of flotation.
(2)	XLCG	Longitudinal position of centre of gravity.
(2)	ZVCG	Vertical position of centre of gravity.
(3)	TM	Draught at midship.
(3)	TF	Draught at forward perpendicular.
(3)	TA	Draught at aft perpendicular.

Notes

(1) XCOF is used to specify the axis for the trim rotation. The XCOF value is only used for the model-fixed option and for the first iteration in model-free option. The XCOF value is re-computed in the following iterations according to the new sinkage and trim condition when the model is free to move. XCOF must be specified in the offset coordinate system.

(2) XLCG is used for the calculation of the dynamic trim. This value is usually not necessary to specify, the program will estimate the XLCG by assuming equilibrium in the initial position. The variable ZVCG is normally not used, it will only effect the balance when

external forces are applied in XPAN. XLCG and ZVCG must be specified in the offset coordinate system.

- (3) Use TRIM and TM or TF and TA to set hull position.

2.17. ITTC78

This command is used to specify the data needed to extrapolate model scale computations to full scale and do a performance prediction according to the ITTC 78 procedure. The resulting full scale proeller RPM, efficiency, thrust, torque and delivered power is written to the output file.

Prerequisites for using this command is that the SELPROPULSION command is also given, that a nominal, unpropelled case is computed and that model scale propeller open water results (POW) are available. The POW results may come either from a SHIPFLOW computation, see the POW command, or from another source, such as experiments.

General form of command

```
ITTC78      (  ON          ,  LM = v      ,  LS = v          ,
              OFF
              DS = v      ,  S = v        ,  S_BK = v      ,
              A_T = v     ,  KS = v       ,  NPOW = v     ,
              KP = v      ,  RHOS = v    ,  NY_M          ,
              NY_S        ,  ESD          ,  K_BH = v      ,
              T_BH = v    ,  W_BH = v          )
```

Default values

ITTC78 (S=[from XPAN], S_BK=0, A_T=0, KS=150, KP=30, RHOS=1025, NPOW=17, NY_M=1.048e-6, NY_S=1.188e-6, LS=LPP, LM [from REFL, SCALE or RN])

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	ON	[ONOFF] Command is switched on.
	OFF	[ONOFF] Command is switched off.
	LM	Model reference length [m].
	LS	Ship reference length [m].
	DS	Ship propeller diameter [m].
	S	Ship reference area [m ²]. If the default is accepted, LS must be equal tp Lpp.

S_BK	Area of bilge keels [m ²].
A_T	Maximum transverse area above the water line [m ²].
KS	Ship surface roughness [μm].
NPOW	Model propeller open water RPS [s ⁻¹]. An open water test is normally run with a constant rotational speed.
KP	Propeller model surface roughness [μm].
RHOS	Ship water density [kg m ⁻³].
NY_M	Model water kinematic viscosity [m ² s ⁻¹].
NY_S	Ship water kinematic viscosity [m ² s ⁻¹].
ESD	Case run with an Energy Saving Device. The full scale wake is calculated with a modified ITTC78 method. With ESD set, you also need to set K_BH, T_BH and W_BH which you get from the OUTPUT files of computations without ESD.
K_BH	Form factor from resistance calculation without ESD.
T_BH	Thrust deduction from self propulsion calculation without ESD.
W_BH	Effective mean wake in model scale (WTM) from self propulsion calculation without ESD.

2.18. KNUCKLE

This command puts a knuckle in the contour of an object. It is quite specialized and intended to adapt part of a rudder to the hull. It is different from the general APPEND in that the rudder root should be closed with FLAT and the rudder positioned so that there is a gap between it and the hull. This allows studying the influence of the gap on the rudder lift and drag at different angles.

General form of command

```

KNUCKLE  (ID = "c"                , X0 = v                , Z0 = v                ,
          Z1 = v                    , SLOPE = v           , SIDE = n              )

```

Default values

```

KNUCKLE  ( X0=0, Z0=0, Z1=1, SLOPE=0, SIDE=1)

```

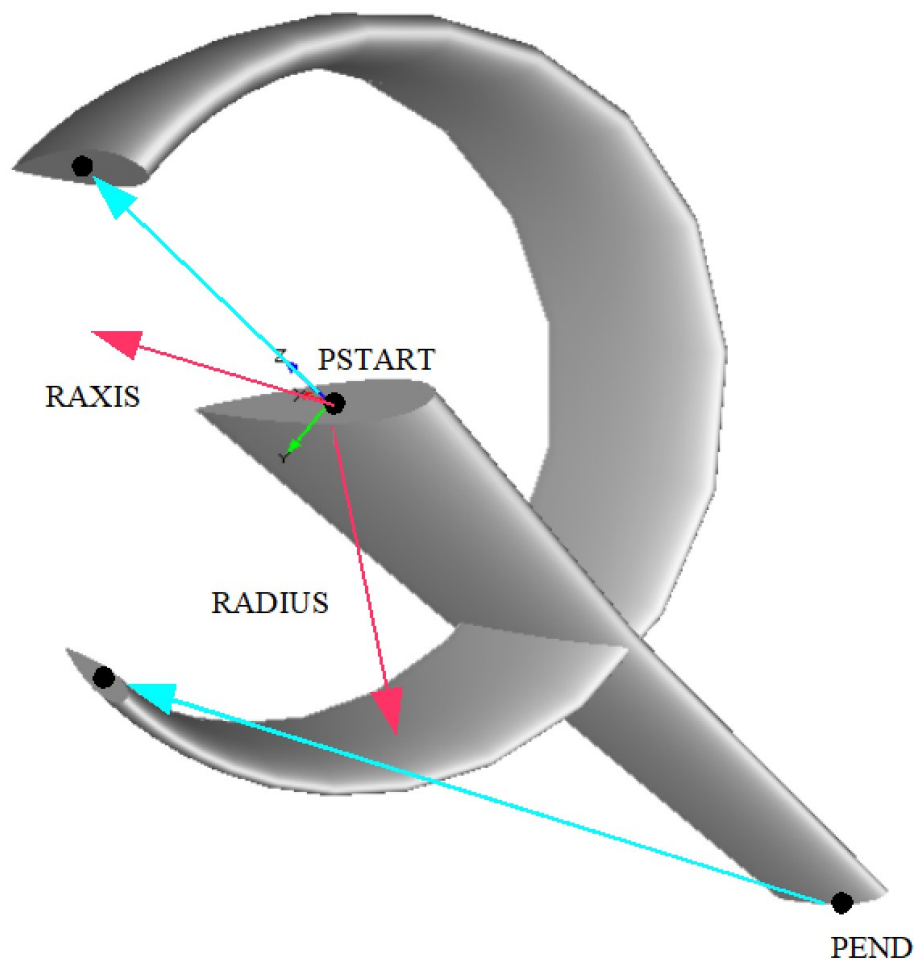
Keywords

<i>keyword</i>	<i>description</i>
ID	Name of the grid to modify.
X0	Knuckle starts at x0 and slope goes in negative x dir.
Z0	Knuckle start level, should coincide with the z-value of the grid surface to be modified.
Z1	Knuckle end level, this is where the knuckle ends.
SLOPE	Slope of the knuckle. The value 0 means no change.
SIDE	Side at which the knuckle is applied. The value 1 means root, the value 2 means tip.

2.19. MCIRCLE

This command morphs a volume grid by transforming a line segment into a circular segment. The command has no information about the grid to be transformed so the user has to specify the two endpoints of a line segment in the grid coordinate system. To specify the transformation, the radius, axis of rotation and start and end angles are given. It is strongly recommended to choose the axis of rotation RAXIS so that it is perpendicular to the line segment PEND-PSTART as this gives more predictable results.

The source grid is by default disabled so it does not appear in the computation. It will however remain available as a source for other morphed volumes.



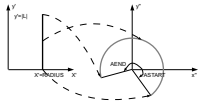
Transformation details:

Let $R = \text{RAXIS}$ and $O = (\text{PEND} + \text{PSTART})/2$. Together they define a plane through O with normal vector R . Let $L = \text{PEND} - \text{PSTART}$ and define a vector $D = R \times L$ which, since it is perpendicular to R , lies in the plane. The transformation to the circle is performed in this plane according to the following:

1. Transform to the plane so that the new coordinate x' is along the D axis, z' is along the R axis and y' is along $R \times D$. If R is perpendicular to L , $L = R \times D$ so that y' is along the user defined line segment.

2. Translate $x' \rightarrow x' + \text{RADIUS}$, $y' \rightarrow y' + |L|/2$. This puts the start point of the line segment at (RADIUS, 0) in the plane coordinate system.
3. Assign the polar coordinates
 $r = x'$
 $a = \text{ASTART} + y' (\text{AEND} - \text{ASTART}) / |L|$
4. Transform $x'' = r \cos(a)$, $y'' = r \sin(a)$. This transforms from a line in the plane to a circle in the plane.
5. Transform (x'', y'', z') back to the coordinate system of the source grid.

If R was chosen perpendicular to L then points on L are transformed to a circle with $r = \text{RADIUS}$ in a plane with normal RAXIS and the point PSTART goes to the angular position ASTART and PEND goes to AEND. If R and L are not perpendicular then L is projected to the plane and then transformed to the circle.



General form of command

```
MCIRCLE  (ID = "c"                , SOURCE = "c"                , RADIUS = v                ,  
          RAXIS = [v1,v2,v3]      , ASTART=v                , AEND=v                    ,  
          PSTART = [v1,v2,v3]    , PEND = [v1,v2,v3]      , TRANS= [v1,v2,v3]      ,  
          ROTATE = [v1,v2,v3]    , KEEPSRC                , SCALE=v                    )
```

Default values

```
MCIRCLE  ( RADIUS=1, RAXIS=[1,0,0], ASTART=0, AEND=90, PSTART=[0,0,0],  
          PEND=[0,0,1], TRANS=[0,0,0], ROTATE=[0,0,0], SCALE=1)
```

Keywords

<i>keyword</i>	<i>description</i>
ID	Gives the MCIRCLE a name. The name is used in general to identify the object and to associate results like integrated forces with it.
SOURCE	Name of the source grid object. This will be disabled unless KEEPSRC is given. More than one morphing commands can use a source. NB: KEEPSRC is disabled in the current release.
RADIUS	Radius of the circle.
RAXIS	Direction of axis of rotation (normal to plane of circle). The centre position is derived from RADIUS, PSTART and PEND.
ASTART	Start angle of the circle. Unit is degrees.
AEND	End angle of the circle. Unit is degrees.
PSTART	Start point of line segment to be transformed into the circle.
PEND	End point of line segment to be transformed into the circle.
TRANS	Translation vector of the morphed grid.
ROTATE	Rotation given in Euler angles of the morphed grid. Unit is degrees.
SCALE	Scale factor for the grid.
KEEPSRC	Generate the SOURCE grid also. This can be helpful to check that it is OK.

2.20. OFFSETFILE

This command is used to give the address to the file that contains the hull offsets, stl or iges, to non- dimensionalize the offset points in this file and to specify the origin of the computational coordinate system.

General form of command

```
OFFSET (  FILE = "c"   ,  LPP = v      ,  XAXDIR = v   ,  YSIGN = v   ,
          STL = "c"
          IGES = "c"

          STLT = v     ,  XORI = v     ,  YORI = v      ,  ZORI = v      ,

          ZTEM = v     ,  ITTE = n     ,  TOTOL = v      ZTOP = v      ,

          FFWD = v     ,  FAFT = v     ,  THIN = v      ,  REFI = v      ,

          BLEN = v     ,  NSTA = n     ,  NORC           ,  UNIF

          XTWS = v     ,  YTWS = v     ,  ZTWS = v              )
```

Default value

```
OFFSETFILE (  LPP = 1.0, XAXDIR = -1.0, YSIGN = 1.0, XORI = LPP, YORI = 0.0,
              ZORI = [TM or (TF+TA)/2], ZTEM = 0.0, ITTE = 4, TOTOL = 1e-4,
              ZTOP = 0.15LPP )
```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	FILE	Address to the file that contains the hull offsets, must be shorter than 256 characters.
	STL	Address to the file that contains the hull geometry in STL format, must be shorter than 256 characters. If STL is specified the hull groups in HULLTYPE command will be set automatically.
	IGES	Address to the file that contains the hull geometry in IGES format, must be shorter than 256 characters. If IGES is specified the hull groups in HULLTYPE command will be set automatically.

	LPP	LPP is the reference length used to non-dimensionalize the coordinates in the offset file. The reference length is normally specified as the length between perpendiculars of the ship measured in the offset coordinate system but other definitions of the reference length may be used. The Froude number specified in the VSHIP command must be based on the same reference length.
	XAXDIR	The x-axis of the computational coordinate system is pointing from the bow towards the stern. XAXDIR must be set to -1.0 if the x-axis of the offset coordinate system is pointing in the opposite direction.
	YSIGN	YSIGN is used to specify if the y-coordinate of the offset points is input using positive (ysign=1.0) or negative (ysign=-1.0) values.
	STLT	STLT is used to specify deflection tolerance for IGES surface input. Default value is 0.0001.
	TOTOL	Tolerance used when creating the twin skeg offset file.
(1)	X-Y-ZORI	XORI, YORI and ZORI specifies the location of the origin of the computational coordinate system in the offset coordinate system. The origin is normally specified at the fore perpendicular.
(2)	ZTEMP	ZTEMP is used to specify an additional draught that is gradually removed during ITTEMP iterations in a non-linear computation.
(2)	ITTEMP	ITTEMP is the number of iterations used to remove the initial extra draught ZTEMP.
	ZTOP	ZTOP is used to specify the height of the XGRID domain above undisturbed free surface level for VOF computations.
	FFWD	Freeboard height at the bow.
	FAFT	Freeboard height at the stern.
	BLN	Bulb length (STL or IGES to OFFSET converting parameter).
	NSTA	Initial number of stations in the offset file.
	THIN	Offset file coarsening after converting from STL or IGES. Input is non-dimensionalized by Lpp.
	REFI	Offset file refining after converting from STL or IGES. Input is non-dimensionalized by Lpp.
	NORC	Switches off ridge correction for twin skeg hull offset file

	generation.
UNIF	Uniform distribution of sections in the generated offset file.
XTWS	X-coordinates of the twinskeg skeg contour (3 points) used for definition of the splitting plane.
YTWS	X-coordinates of the twinskeg skeg contour (3 points) used for definition of the splitting plane.
ZTWS	X-coordinates of the twinskeg skeg contour (3 points) used for definition of the splitting plane.

Notes

- (1) It is assumed that the computational coordinate system is located in the undisturbed free surface at the fore perpendicular. The initial draught at centre of flotation (XCOF) for the hull is therefore specified by ZORI.
- (2) ZTEMP and ITTEMP are used to introduce an additional draught that is gradually removed during the first ITTEMP iteration in a non-linear computation. ZTEMP is in the first iteration added to the original draught specified by ZORI. This option can be used to give a shape of the waterline in the first iteration that is similar to the waterline in the final converged solution. The geometry for large flat sterns, tunnel sterns and sailing yachts may be better represented in the first iteration when the additional draught is introduced and a better convergence can be expected. The distance ZTEMP is specified in the same scale as in the offset coordinate system.

The longitudinal position of the centre of gravity is estimated at the first iteration. This introduces an error in the calculation of the dynamic trim which can be important to correct for some types of hulls, for example hulls with large flare. In this case the longitudinal centre of gravity (XLCG) can be specified in the IPOS command. The value can be computed with a separate XPAN run. Note that the reference area will differ when ZTEMP is used, but this effects only the coefficients and can be corrected afterwards.

2.21. OPENSHAFT

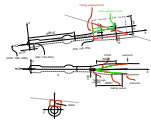
This command generates a propeller shaft with a hull-shaft fairing. The indata position and direction of the shaft is taken from an associated PROPELLER and the shape of the shaft and fairing is given by contour curves. The shaft and fairing are adapted to the hull and partial collar grids represent the shaft-fairing intersection.

The geometry is made with HYPERSURF, ROTBODY and EXTRUDE grids and adaption and transformation are done with APPEND and ASSEMBLY commands, reading the documentation for these provides more in-depth information.

The shaft is divided in two parts, the second of which intersects the hull and the fairing.

It is important that the partial collars go as far forward as possible while not intersecting the hull. A typical symptom of too short collar grids is that the boundary layer thickens too much upstream of the collar grids due to the low fidelity in the shaft-fairing intersection.

The origin of the fairing grid, specified by the lower point of the trailing edge XTE, YTE and ZTE. It is important to select these parameters and LENFAIRING, the fairing length along the trailing edge, so that the lower edge of the fairing is completely inside the second shaft grid and the upper edge is completely inside the hull.



General form of command

```

OPENSHAFT (ID = "c"           , PROP = "c"           , XPOS = v           ,
           YTSHAFT = v         , PRIO = i         , FROLL = v         ,
           FYAW = v           , FYSCALE = v      , FXTWIST = v        ,
           FTW0ANG = v         , FTW1ANG         ,                   ,
           XTE = v           , YTE = v         , ZTE = v           ,
           XSHAFT = [v1,...]   , YSSHAFT = [v1,...] , PTSHAFT = [i1,...] ,
           XSPLIT = v         ,                   ,                   ,
           GRS1 = v           , DNS1 = v         , DIS1 = [i1,i2,i3] ,
           GRS2 = v           , DNS2 = v         , DIS2 = [i1,i2,i3] ,
           XFAIR = [v1,...]   , YFAIR = [v1,...] , PTFAIR = [i1,...] ,
           GRFAIR = v         , DNFAIR = v       , DIFAIR = [i1,i2,i3] ,
           LENFAIR = v        , LENCOLLAR = v    , ZCOLLAR = v       ,
           GRCOLL = v         ,                   ,                   ,
           XCO1 = v           , XCO2             , NOSA              ,
           COPRIO = i         , SHOFF = "c"      , FAIOFF = "c"      ,
           HULL = "c"         ,                   ,                   )

```

Default values

```

ASSEMBLY ( XPOS=0, YTSHAFT=0, PRIO=10, FROLL=0, FYAW=0, XTE=0, YTE=0,
           ZTE=0, GRS1=1.02, DNS1=0.01, GRS2=1.02, DNS2=0.01, GRFAIR=1.02,
           DNFAIR=0.01, HULL="Xgrid")

```

Keywords

<i>keyword</i>	<i>description</i>
ID	Gives the open shaft a name. The name is used in general to identify the object and to associate results like integrated forces with it.
PROP	Name of the associated propeller object. The position and direction of the shaft is derived from values in PROP.
XPOS	Translation of the shaft along its axis. Positive towards the bow.
YTSHAFT	YTARGET parameter for the shaft grids.
PRIORITY	PRIORITY parameter for the second shaft grid and the faring grid. Has to be larger than the hull grid priority (default 0). The collar grids are assigned a priority of PRIORITY+5.
FROLL	Roll rotation angle in degrees of the fairing. Positive counter-clockwise seen from the stern.
FYAW	Yaw rotation angle in degrees of the fairing. Positive counter-clockwise seen from above.
FYSCALE	Fairing scaling in y direction. Default is 1, <1 gives a thinner fairing and

>1 a thicker one.

FXTWIST	X position of fairing twist axis. Measured from the faring trailing edge.
FTW0ANG	Fairing twist angle at shaft end.
FTW1ANG	Fairing twist angle at hull end.
XTE	X coordinate of the fairing relative to the shaft.
YTE	Y coordinate of the fairing relative to the shaft.
ZTE	Z coordinate of the fairing relative to the shaft.
XSHAFT	X coordinate values for the shaft contour. Start with the x value for the end of the shaft and go forward with increasing values.
YSHAFT	Y coordinate values for the shaft contour. Start at 0 and then give only positive values.
PTSHAFT	Point types for the shaft contour. 0 means that the curve is smooth at the point while 1 is a corner point.
XSPLIT	X position of shaft grid split. The grid is split in two parts so that the adaption to the hull is confined to the second shaft grid.
GRS1	Growth factor for the cell thickness for the first shaft grid. See the documentation for HYP SURF GROW for more information.
DNS1	Thickness of the first cell layer in the first shaft grid. See the documentation for HYP SURF DN for more information.
DIS1	Dimensions of the first shaft grid. The dimension are along the shaft, out from the shaft and tangential to the shaft respectively.
GRS2	Growth factor for the cell thickness for the second shaft grid. See the documentation for HYP SURF GROW for more information.
DNS2	Thickness of the first cell layer in the second shaft grid. See the documentation for HYP SURF DN for more information.
DIS2	Dimensions of the second shaft grid. The dimension are along the shaft, out from the shaft and tangential to the shaft respectively.
XFAIR	X coordinate values for the fairing section. Start inboard end (negative y) and go aft with decreasing values until the trailing edge is reached at x=0. Then go forward with increasing x until the end.
YFAIR	Y coordinate values for the fairing section. Start with negative values, change sign at the trailing edge (x=0) and then continue with positive values.
PTFAIR	Point types for the fairing section. 0 means that the curve is smooth at the point while 1 is a corner point. Typically 1 or 2 points at the trailing edge

	should be corners.
GRFAIR	Growth factor for the cell thickness for the fairing grid. See the documentation for HYP SURF GROW for more information.
DNFAIR	Thickness of the first cell layer in the fairing grid. See the documentation for HYP SURF DN for more information.
DIFAIR	Dimensions of the fairing grid. The dimension are along the section, out from the fairing and along the trailing edge respectively.
LENFAIRING	Length of the faring grid along the trailing edge before adaption to the hull.
LENCOLLAR	Length of the collar grids along the trailing edge before adaption to the second shaft grid.
ZCOLLAR	Z coordinate of the collar grids lower sides relative to the shaft. Must be inside the second shaft grid.
XCO1	Collar starting (low) x distance from fairing TE. Should be close the hull intersection but outside of the hull. Measured from the fairing trailing edge.
XCO2	Collar end (high) x distance from fairing TE. Should be close the hull intersection but outside of the hull. Measured from the fairing trailing edge.
COSFAIR	COSVMIN for fairing. See XFLOW APPEND.
COSSHAFT	COSVMIN for fairing. See XFLOW APPEND.
NOSA	NOShaftAdaption. Turn off adaption of the second shaft grid to the hull. Debugging option.
COPRIO	Set the collar grids priority offset. Debugging option.
SHOFF	Shaft offset file. ASCII text file with one point per line: X, Y, POINTTYPE.
FAIOFF	Fairing offset file. ASCII text file with one point per line: X, Y, POINTTYPE.
HULL	Name of the grid to append the open shaft to. This is normally the hull grid from XGRID.

2.22. OPTIM

The file id_OPTRES will be generated at the end of the SHIPFLOW execution when the OPTIM command is included. The file contains the computed resistance, displacement, LCB and wetted surface area. The purpose of this file is to transfer the SHIPFLOW results to the optimizer when SHIPFLOW is included in an optimization loop. The contribution from wave resistance can be based on pressure integration or transverse wave cuts.

General form of command

```
OPTIM  (  ON          ,  FULL      ,  LNONDIM  ,  
        CW           ,  SNONDIM   ,  
        CWTWC       ,  TFORMF    ,  READ          )
```

Default value

OPTIM (FULL, TFORMF = 0.0)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	ON	The file id_OPTRES will be generated using the default values.
(1)	FULL	[OMAGN] Four resistance values, displacement, LCB and wetted surface area at even keel will be printed in the file id_OPTRES.
(1,2)	CW	[OMAGN] Only the wave resistance will be printed in the file id_OPTRES.
(2)	LNONDIM	[ONDIM] A fixed reference length that will be used to non- dimensionalize the wave resistance value printed at the first position in the id_OPTRES file.
(2)	SNONDIM	[ONDIM] A fixed reference surface area that will be used to non- dimensionalize the wave resistance value printed at the first position in the id_OPTRES file.
	CWTWC	The wave resistance value will be computed from a transverse wave cut method when CWTWC is specified. The default wave resistance is computed from transverse wave cut method.
(3)	TFORMF	A total resistance based on the computed wave resistance and the friction resistance from ITTC78 will be printed in

the id_OPTRES file.

- | | | |
|-----|------|---|
| (4) | READ | Values for the XMESH keywords XSCA, YSCA, ZSCA, XTRA and YTRA will be read from the file temp1.dat. |
|-----|------|---|

Notes

- (1) The optres.dat file will contain 2 rows when FULL (default) is specified. On the first row four resistance values are printed and the second row includes the non-dimensionalized displacement, LCB and wetted surface area (WSA) at even keel.

The first resistance value is a measure of the wave resistance computed from:

$C_{wout} = C_w$ wave resistance coefficient (default)

$C_{wout} = C_w * WSA * L_{pp}^2 / L_{NONDIM}^2$ if LNONDIM is specified

$C_{wout} = C_w * WSA * L_{pp}^2 / S_{NONDIM}$ if SNONDIM is specified

The second resistance value is computed from the wave resistance and ITTC78:

$$R_t = 0.5 * \rho * U^2 * WSA * L_{pp}^2 * (C_w + (1 + k) * C_{f_{ITTC57}})$$

$(1 + k) = T_{FORMF}$, see note (3)

The third resistance value is computed from:

$$R_t = 0.5 * \rho * U^2 * WSA * L_{pp}^2 * (C_w + C_{ftot})$$

C_{ftot} is the friction resistance computed in XBOUND and in XCHAP if executed.

The fourth resistance value is computed from:

$$R_t = 0.5 * \rho * U^2 * WSA * L_{pp}^2 * (C_w + C_{vtot})$$

C_{vtot} is the friction resistance computed in XBOUND and the friction and viscous pressure resistance in XCHAP if executed.

- (2) Only the wave resistance will be included in id_OPTRES when CW is specified.
- (3) TFORMF is the total form factor $(1 + k)$. See also note (1).
- (4) Simple geometrical changes can be applied to the geometry specified in the offset file during an optimization. The geometry for one group will be scaled and translated according to the information in the temp1.dat file when READ is specified. The file temp1.dat must contain the values for the XMESH keywords XSCA, YSCA, ZSCA, XTRA and YTRA on the first row. These values will override the values specified for the BODY group in XMESH. Note that initial values must be specified on the BODY group. A typical application for this option is the scaling and positioning of the side hull of a trimaran.

2.23. OSFLOW

This command is used to specify the onset flow direction.

General form of command

OSFLOW (FLOW = [α_1 , , , , α_n])

Default values

OSFLOW (FLOW = [0.0])

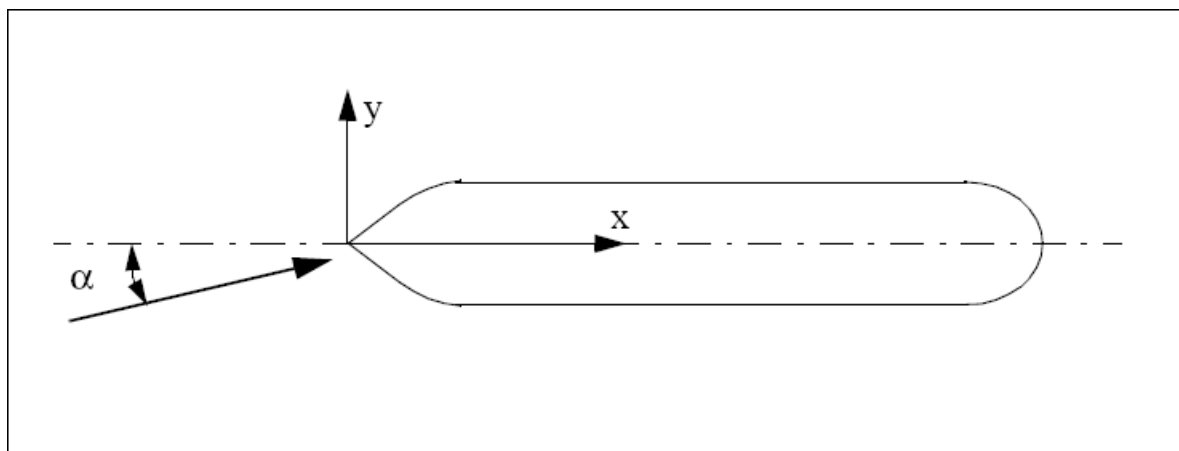
Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	FLOW	Onset flow angle specified in degrees.

Notes

- (1) More than one onset flow angle can only be handled by XPAN. A maximum of 25 values may be specified. No xz-plane symmetry is allowed for angles other than zero. Multiple onset flows cannot be stored in the TECPLOT file.

Definition of positive onset flow angle



Definition of positive onset flow angle

2.24. PROGRAM

MANDATORY

This command is used to specify which module(s) of SHIPFLOW that should be executed.

General form of command

PROGRAM (XMESH, XPAN, XBOUND, XGRID, XCHAP, ALL, XPTD, XPOST)

Default values

None

Keywords

XMESH, XPAN, XBOUND, XGRID, XCHAP:

The SHIPFLOW modules corresponding to the keywords given will be executed. The modules will be run in the order shown above regardless of the order they appear in the command. The XPAN keyword instructs SHIPFLOW to run both the XMESH and the XPAN module.

ALL:

All modules (except MOTIONS) are executed after each other in one go. The information needed by subsequent modules is always saved, so it is possible to run SHIPFLOW module by module and adjust the input for each module during the process.

XPTD:

Run MOTIONS

XPOST:

Run only post-processing features (Automatic reporting and Python scripts)

2.25. PROPELLER

This command is used to specify the geometry and position of a propeller. The propeller thrust coefficient used in the potential flow computation including a propeller is also specified on this command. Several (max 100) propeller commands can be given, but in the XPAN module only the last one is used. For use with XCHAP see the XCHAP ACTUATOR and LLINE commands.

General form of command

```
PROPELLER ( ID = "c" ,
            DPRO = v , DHUB = v , XSH = v , YSH = v ,
            XDIR = v , YDIR = v , ZDIR = v , ZSH = v ,
            NBLA = n , JV = v , EAR = v ,
            CTS = v , CMO = v , ROTDIR = n , FRACTION=v,
            BETA = v , PMOD = v , MODE = n , FTHRUST ,
            KQFIX = v , CAMB = [v1, ..., vNUMB] ,
            R_RT = [v1, ..., vNUMB] , P_D = [v1, ..., vNUMB] ,
            THIC = [v1, ..., vNUMB] , LENG = [v1, ..., vNUMB] ,
            C_D = [v1, ..., vNUMB] , T_D = [v1, ..., vNUMB] ,
            F_C = [v1, ..., vNUMB] , GEOFILE = "c" ,
            WAGENINGEN , WPD = v ,
            BB )
```

Default values

```
PROPELLER ( DPRO = 0.0, DHUB = 0.0, XSH = 0.0, YSH = 0.0, ZSH = 0.0,
            XDIR = XFLOW:OFFSET:XAXDIR, YDIR = 0.0, ZDIR = 0.0,
            CTS = 0.0, FRAC = 1.0, FTHRUST, JV = 0.85,
            ROTDIR=0 for XZPLANE set in SYMMETRI and YSH=0.0
            – otherwise ROTDIR=1)
```


Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	ID	Name used to identify the propeller by XCHAP
(1)	DPRO	Propeller diameter.
(1)	DHUB	Hub diameter.
(1)	XSH	x-coordinate for the propeller centre.
(1)	YSH	y-coordinate for the propeller centre.
(1)	<u>ZSH</u>	z-coordinate for the propeller centre.
	XDIR	Direction cosine in the x-direction for the propeller axis.
	YDIR	Direction cosine in the y-direction for the propeller axis.
	ZDIR	Direction cosine in the z-direction for the propeller axis.
(2)	CTS	Propeller thrust coefficient. For XPAN and XCHAP force actuator disk.
(3)	CMO	Propeller moment coefficient. For XCHAP force actuator disk.
(4)	ROTDIR	Propeller rotation direction. For XCHAP lifting line actuator disk.
	FRACTION	Propeller load fraction. Specifies the fraction of the propeller load for the propeller. The value is 1.0 by default corresponding to a single propeller. The value should be set to 0.5 in both propeller commands for a twin screw. The sum of the fractions specified in all propeller commands should be 1.0.
	FTHRUST	Distribute propeller load by thrust.
	FPOWER	Distribute propeller load by delivered power.
	NBLA	Number of propeller blades.
	JV	Propeller advance ratio $VS/(DPROP*N)$, where VS is the ship speed and N the rate of rotation.
	EAR	Expanded area ratio.
	KQFIX	Specify desired Kq and vary J or BETA depending on MODE.
	MODE	Parameter to vary during self propulsion. 0=JV(default), 1=blade pitch angle, 2=pitch.

	BETA	Blade pitch angle in degrees. Positive->higher pitch.
	PMOD	Additional blade pitch, normalized by diameter.
	R_RT	Propeller radii made dimensionless by the radii of the propeller tip.
	P_D	Pitch ratio.
(1)	LENG	Length of blade section.
(1)	THICK	Maximum blade thickness.
(1)	CAMB	Camber.
	C_D	Length of blade section divided by propeller diameter. This can be entered instead of LENG.
	T_D	Maximum blade thickness divided by propeller diameter. This can be entered instead of THICK.
	F_C	Camber divided by the length of the corresponding blade section. This can be entered instead of CAMB.
	GEOFILE	Read NBLA, EAR, R_RT, P_D, C_D, T_D and F_C from a file with name GEOFILE. The file name can be given as a full path, otherwise SHIPFLOW will look for it first in your working directory and then in shipflow/data/propellers. The file should be an ASCII text file with 5 lines. On each line should be a space-separated list of floating point numbers, followed by an optional comment. The first line is the R_RT values etc. A small example with three stations: 4 !NBLA 0.55 !EAR 0.0 0.5 1.0 !R_RT 0.6 0.65 0.6 !P_D 0.25 0.35 0.1 !C_D 0.03 0.02 0.005 !T_D 0.04 0.02 0.01 !F_C
	WAGENINGEN	Keyword that triggers the inbuilt Wageningen propeller series. In addition to the propeller size and position the following input is required: NBLA, EAR, WPD
	WPD	P/D for Wageningen propeller .
	BB	Wageningen BB-Series.

Notes

- (1) All geometrical data for the propeller is given in the same coordinate system as in the offset file.
- (2) The propeller thrust coefficient is computed from

$$CTS = \frac{T}{\frac{1}{2}\rho U^2 A}$$

Where T is the propeller thrust, computed or estimated, U is the ship speed and A is the propeller disk area.

- (3) The propeller moment coefficient is computed from

$$CMO = \frac{Q}{\frac{1}{2}\rho U^2 AD}$$

Where Q is the torque, computed or estimated, U is the ship speed, A is the propeller disk area and D is the propeller diameter.

- (4) Integer values equal to 1 or -1 specifies clockwise or counter clockwise direction. The default value is 0 which sets the tangential force component to zero. This keyword applies only to the lifting line method in XCHAP.

Requirements for different modules

The following data is required by XPAN: DPRO, DHUB, XSH, YSH, ZSH, XDIR, YDIR, ZDIR and CTS.

2.26. PRTOPT

1. This command is used to specify additional output from XMESH, XPAN and XBOUND.

General form of command

```
PRTOPT ( PGEOM , FDOP , PCON , STRLRES ,  
         CFRES ,  
         PANCO , NLTERM , BMRES , FSRES ,  
         OBPRES , NAPA , XYPLOT , TRACE )
```

Default values

None

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	PGEOM	Panel geometry information.
	FDOP	Finite difference operator terms from XPAN.
	PCON	Panel connectivities.
	STRLRES	Streamline coordinates and detailed results from XBOUND.
	CFRES	Results from boundary layer computation. Control point coordinates, local skin friction and panel surface area.
	PANCO	Panel corner coordinates from XPAN.
	NLTERM	Non-linear terms and velocity derivatives on the free-surface.
	BMRES	Results from basic model calculation double or single model information: control point coordinates, velocities and pressures, panel source strengths and normal vectors.
	FSRES	Results from free surface calculation, control point coordinates, velocities and pressures, panel source strengths and normal vectors, panel area, wave heights.

OBPRES	Results from offbody point calculation, offbody point coordinates, velocities and pressures.
NAPA	Output for NAPA interface. FSRES must also be specified.
XYPLOT	Diagram information for XPOST.
TRACE	Trace file for XMESH/XPAN.

Note

The additional output is printed out to files that are named id_keyword. Where id is the name of the command file and keyword is the keyword that trigger the creation of the output file. For example, the file hamb_STRLRES will be created by SHIPFLOW if the command file hamb contain the command PRTOPT(STRLRES).

2.27. RECESS

This command specifies that the object ID is adapted to the grids in ADAPT from the outside, so that the fluid region is extended. ADIR can presently only have one component and specifies the coordinate direction in which ID is adapted. ASIDE are the grid faces of adaption.

General form of command

RECESS	(	ID = "c"	,	ADAPT = "c1"	,	ADIR = i	,
		ASIDE = i	,	COSVMIN			)

Default values

RECESS ()

Keywords

<i>keyword</i>	<i>description</i>
ID	Specifies the object with matching ID is an appendage.
ADAPT	Which grid(s) to append (adapt to):
ADIR	The coordinate direction used for adapting ID.
ASIDE	The side(s) that are adapted.
COSVMIN	Minimum of the cosine between the coordinate line and surface normal where the adaption occurs.

2.28. RESCALE

If you want to run your computations at Rescale.com's cluster you add this command and then run it like any local computation. Do the following:

- 1) Sign up for an account at rescale.com.
- 2) At rescale.com get you API key and save it in a file on your computer.
- 3) Set the APIKEY keyword to point at the file with your api key.
- 4) Run your case in one of the three possible ways.
 1. Run from Design/CAESES by specifying shipflow_rescale.bat instead of shipflow.bat as your program to execute.
 2. Or from a terminal type "shipflow_rescale <name of configuration>"
 3. Or (only on Windows) from Notepad++ select to run at Rescale from the run menu.

Please notice that there is a cost involved in using these services.

General form of command

RESCALE	(	APIKEY = "c"	,	VERSION = "c"	,	INSECURE	,
		MARBLE	,	ALL	,	PLATFORM = "c"	)
		NICKEL		OUTPUT			
		ONYX		CGNS			

Default values

RESCALE (VERSION= "6.1.00", NICKEL, ALL,
PLATFORM = "platform.rescale.com")

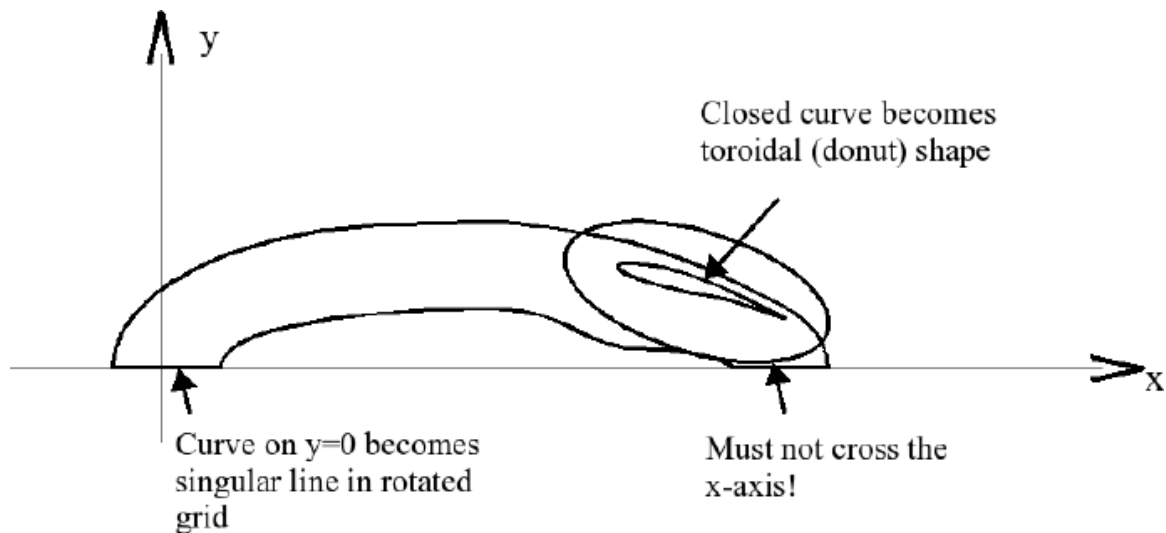
Keywords

<i>keyword</i>	<i>description</i>
APIKEY	File with your API key. Do not show the key to others, since it is basically your password at rescale.com.
VERSION	Version of SHIPFLOW you want to run at rescale. Not all versions are installed.

INSECURE	Adds -k to all the cURL commands.
MARBLE	Computational nodes that supports 1, 2, 4, 8, 12, 16, 20, 24, ... processes. On node has 4 cores.
NICKEL	Computational nodes that supports 1, 2, 4, 8, 16, 32, 48, 64... processes. One node has 16 cores.
ONYX	Computational nodes that supports 1, 2, 4, 8, 18, 36, 54, 72... processes. One node has 18 cores.
TITANIUM	Computational nodes that supports 1, 2, 4, 8, 20, 40, 60, 80... processes. One node has 20 cores.
ALL	Copy all files back after the run is completed. All files will be deleted from your rescale account.
OUTPUT	Just copy the OUTPUT file back.
CGNS	Copy cgns and OUTPUT file back.
PLATFORM	Enter alternative Rescale platform name here. E.g. "eu.rescale.com"

2.29. ROTBODY

This command creates a body of rotation in XCHAP by rotating a two-dimensional grid around the x-axis, see the figure below. The two-dimensional grid is generated by a separate command and its ID is referenced in the ROTBODY command. When generating the 2-dimensional grid it is important that it does not have any nodes with $y < 0$ since that will give degenerate cells with negative volume.



The two-dimensional grids available for generating bodies of rotation are WSECTION that represents a wing section, and HYP SURF that can represent a general shape.

The first two dimensions and the boundary conditions of the generated grid are taken from the two-dimensional grid.

General form of command

```
ROTBODY ( ID = "c" , SECT = "c" , SCALE = v ,
          TRANSLAT = [v1,v2,v3] , DIR = [v1,v2,v3] , DIM3 = n ,
          ROTATE = [v1,v2,v3] , YTARGET = v ANGLE = v ,
          GROUP = n , BC11 = "c" , BC12 = "c" ,
          BC21 = "c" , BC22 = "c" , BC31 = "c" ,
          BC32 = "c" , PRIO = n )
```

Default values

```
ROTBODY ( ID="Rotbody", SCALE=1, TRANSLATE=[0,0,0], DIR=[1,0,0],  
          DIM3=20, YTARGET = 1.0,  
          ANGLE=360, ROTATE=[0,0,0], GROUP=unique,  
          BC11="NOSLIP", BC12="INTERIOR",  
          BC21="INTERIOR", BC22="INTERIOR",  
          BC31="INTERIOR", BC32="INTERIOR",  
          PRIO=-1)
```

Keyword

<i>keyword</i>	<i>description</i>
ID	Gives the rotation body a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different objects may have the same name, but then only the total force on all of those objects will be calculated and output.
SCALE	Scale the grid.
TRANSLATE	Position of the local coordinate system origin in the offset file (ship) coordinate system.
ROTATE	Rotation in Euler angles.
DIR	Direction of the x-axis in the offset file coordinate system.
DIM3	Number of grid planes generated by the rotation.
YTARGET	Sets the target value for y^+ .
ANGLE	Sets the rotation angle for the object.
GROUP	Assign to group.
BC11	Specify the boundary condition on the rotation surface. Possible values are "NOSLIP", "SLIP", "INFLOW", "OUTFLOW", "INOUT" and "INTERIOR". The INTERIOR condition means that the boundary values are taken by interpolation from other grids.
BC12	Specify the boundary condition on the outer surface.
BC12	Specify the boundary condition on the face at the start of the generating curve. Not needed if the curve is periodic or the grid follows $y=0$.
BC22	Specify the boundary condition on the face at the end of the generating curve. Not needed if the curve is periodic or the grid follows $y=0$.
PRIO	Set overlapping grid classification priority,

2.30. ROUGHNESS

This command sets the roughness parameter KS on no-slip boundaries. The distribution is piecewise constant in x-intervals. The command can be repeated to specify KS on different regions. It can be given in four forms:

1. A single value for KS. This is the default value used if no other region matches.
2. N_x x-values, N_z z- and $(N_x-1) \times (N_z-1)$ $KS_{<j>}$ values. Let (x_f, z_f) be the x and z coordinates of the boundary face element to be assigned a KS value. If there is an interval $X[i+1] < x_f \leq X[i]$, $Z[j+1] < z_f \leq Z[j]$ then $KS_f = KS_{X_{<j>}[i]}$, otherwise $KS_f = KS$. If no Z values are given then implicitly $Z[1] = \infty, Z[2] = -\infty$.

The notation $KS_{<j>}$ means vektors named KS01 for the first interval in z, KS02 for the second interval etc.

Note that X and Z are given in decreasing magnitude, so that the KS values are distributed starting from the bow and going aft with increasing i-index and down with increasing j-index. For $N_x=5$ and $N_z=4$ there should be 3 vectors KS01, KS02, KS03 with 4 values each. The distribution over the hull is, with the bow to the left,

KS01(1)	KS01(2)	KS01(3)	KS01(4)
KS02(1)	KS02(2)	KS02(3)	KS02(4)
KS03(1)	KS03(2)	KS03(3)	KS03(4)

3. An assembly name and a single KS value. All grids belonging to the assembly will get the constant KS value.
4. An assembly name, N_x x-values, N_z z- and $(N_x-1) \times (N_z-1)$ $KS_{<j>}$ values. All grids belonging to the assembly will get KS values according to item 2 above.

The KS values are assigned with criterion 4 given the highest priority, then 3 etc. For a description how to define assemblies, see XFLOW|ASSEMBLY.

X values are given from bow to stern, so if the x-axis is pointing forward then the values should be ordered largest to smallest.

Z values are given from waterline to keel, so if the z-axis is pointing upward then the values should be ordered largest to smallest.

If no ROUGHNESS command is given then the default value $KS=0$ is used. See also XCHAP|CONTROL|KS.

General form of command

ROUGHNESS(KS = v , X=[v₁,...,v_N] , KS<j>=[v₁,...,v_{N-1}] ,
ASSEMBLY="c", , Z=[v₁,...,v_M])

Default values

ROUGHNESS ()

Keywords

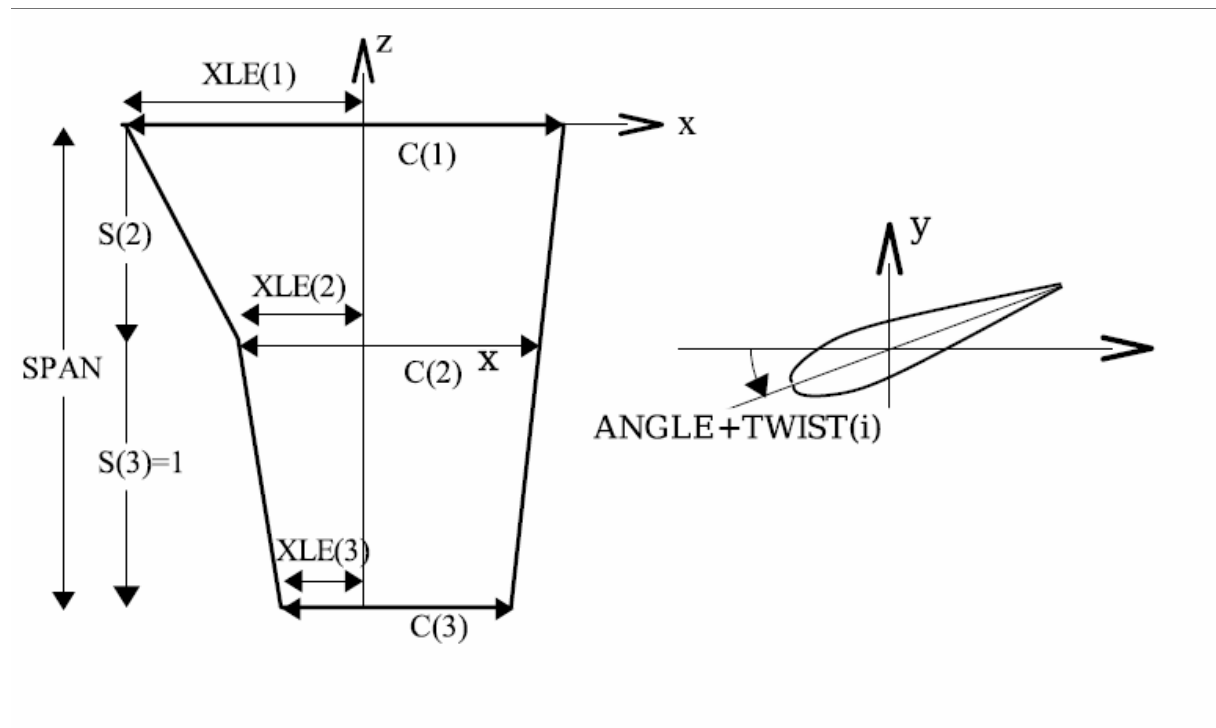
<i>keyword</i>	<i>description</i>
KS	Default value for regions not otherwise specified. Default is KS=0. If the KS parameter is given, X and KSX should be not used.
X	A vector of X values specifying contiguous intervals with constant KS. At least two X values must be specified, giving one interval. The values are given bow to stern.
Z	A vector of Z values specifying contiguous intervals with constant KS. At least two Z values must be specified, giving one interval. The values are given waterline to keel.
KS<j>	Vector with one KS value for each X interval. The length of KS<j> must be the length of X minus one. The number of KS<j> vectors must be the length of Z minus one. The actual names are KS01, KS02, KS03 etc.
ASSEMBLY	Name of the assembly on which to apply the KS value or KS<j> values.

2.31. RUDDER

This command adds a rudder to the ship geometry in XCHAP. The command may be repeated to add more rudders. The geometry is specified in a rudder-fixed coordinate system according to the figure below

The rudder position and orientation in the offset file coordinate system is given as a translation of the origin and two rotations, the rudder angle about the z-axis and the cant angle about the x- axis, in that order.

The tip of the rudder is closed with a half body of rotation with the same section as the rudder. If the case is symmetric about $y=0$ and the y-component of ORIGIN=0, then only one side of the rudder is modelled. ANGLE, CANT and TWIST must be zero for this case.



The section of the rudder is given as a set of points specified in an offset file. The format of the file is very simple, on every line should be the x and y coordinates of a points, separated by white space (space and/or tabs). The points should be ordered starting at the trailing edge, going forward on the upper side, around the leading edge and back on lower side to the trailing edge, i.e. last point must be the same as the first. The trailing edge must be at a point $(x>0, 0)$ and the leading edge at $(0,0)$. If the section is symmetrical about $y=0$ it is sufficient to give the points from the trailing edge to and including the point on the leading edge. Several symmetrical and non-symmetrical wing sections are supplied in the SHIPFLOW installation.

Except at the trailing edge the program tries to fit a smooth curve to the set of points in the offset file. This is done with a Theodorsen-Garrick transform, which is a conformal mapping technique. The obtained mapping is also used to make 2D grids for uniformly spaced stations of constant z . The complete 3D grid has an O-O structure.

When the rudder grid is added to the overlapping grid that describes the complete geometry, it goes through the transformation to the SHIPFLOW computational coordinate system, including scaling by $1/LPP$, sinkage and trim.

If the rudder surface is not closed, the rudder must intersect another boundary surface such as the hull and/or the free surface so that the curve ending the surface at the root of the rudder is completely outside of the fluid domain.

General form of command

```

RUDDER ( ID = "c"                , SPAN=v                , ANGLE = v                ,
        CANT = v                  , ORIGIN=[v1,v2,v3]    , SECTION=["c",...]       ,
        S =[v1,v2,...,vn]         , C =[v1,v2,...,vn]    , XLE =[v1,v2,...,vn]     ,
        T = [v1,v2,...,vn]        , TWIST = [v1,v2,...,vn], FLIP                      ,
        DIMENSION=[v1,v2,v3] , TILT = v                , ROTATE=[v1,v2,v3]
        YSIGN = n                  , SCALE = v                , RMAX=v                ,
        BC22 = "c"                 , BC31 = "c"                 , BC32 = "c"                 ,
        TP1O                       , TP2O                       , IGES = "c"                 ,
        TP1R                       , TP2R                       , STLF = "c"                 ,
        TP1F                       , TP2F
        TPCELLS = n                , YTARGET = v                , STLT = v                ,
        TGTRANS                     , TOP                         , OGRID
        HYPWSECT                     , BOTTOM
        HYPPT                        , BOTH
        FLAP = [v1,v2,...,vn]      , FANGL = v                , SCALE = v
        GROW = v                   , DTAU = v                  , NITER = n                ,
        DN = v                     , STRETCH = v               , PRIO = i                 ,
        HALF                       , NOLESTR                   , CTIP = v                 ,
        STR1 = "c"                  )

```

Default values

```

RUDDER( ID = "Rudder", SPAN = 1, ANGLE = 0, CANT = 0, ORIGIN = [0, 0, 0],
        SECTION = [""], XLE = C*0.25, TWIST=[], T=[], TILT=0, TP1O, TP2R,
        BC22="INTERIOR", BC31="INTERIOR", BC32="INTERIOR", RMAX=1,
        SCALE=1, YTARGET=1, TGTRANS, FANGL=0, GROW=1.1, DTAU=0.5,
        NITER=5, DN=0.01, STRETCH=0, PRIO=-1, BOTH, CTIP=0.1, SCALE=1)

```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	ID	Gives the rudder object a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different objects may have the same name, but then only the total force on all of those objects will be calculated and output.
	SPAN	The total length (depth) of the rudder.
	ANGLE	Rudder rotation angle in degrees. The axis of rotation is the z-axis in the rudder coordinate system.
	CANT	Rotation in degrees around the x-axis in the rudder coordinate system.
	ROTATE	Rotation given in Euler angles (degrees). Do not use together with ANGLE, CANT or TILT.
	TILT	Rotation in degrees around the y-axis in the rudder coordinate system.
	ORIGIN	Position of the rudder coordinate system origin in the offset (ship) coordinate system.
	SECTION	The name of a file containing offsets for a 2D section. The file name can be given as a full path, otherwise SHIPFLOW will look for it first in your working directory and then in shipflow/data/wing_sections. See above for a description of the file format. Either only one section is given, which is then used for all stations, or the number of sections should be the same as the number of sections. If an empty string is given, the NACA0012 section is used. Alternatively SECTION can be the ID of a 2D grid specified in the command file.
	S	Rudder section stations along the z-axis, non-dimensionalized by SPAN. $s=0$ corresponds to the root and $s=1$ to the tip. The first value should always be 0 and the last 1.
	C	Rudder chord lengths. The length of this vector must be equal to the length of S.
	T	Rudder thickness. The length of this vector must be equal to the length of S.
	XLE	Distance from the leading edge to the z-axis for each station. Positive numbers places the leading edge on the negative side of the axis. The length of this vector, if given, must be equal to the length of S. If not given the default value $0.25 \cdot C$ is used.

	TWIST	Rotation of each section in degrees around the z-axis. The length of this vector, if given, must be equal to the length of S. If not given the default value 0 is used for each station.
	TP1O	The rudder is open at the first station.
	TP1R	The rudder is closed with a rounded tip at the first station.
	TP1F	The rudder is closed with a flat (square) tip at the first station.
	TP2O	The rudder is open at the last station. (Same as NOTIP)
	TP2R	The rudder is closed with a rounded tip at the last station. (Same as ROUND)
	TP2F	The rudder is closed with a flat (square) tip at the last station. (Same as FLAT)
	TPCELLS	Number of cells in spanwise direction on each closed tip. The total number of tip cells can of course not exceed the number of axial cells (see DIMENSION keyword).
	BC22	Specify the boundary condition at the outer grid surface. Possible values are "NOSLIP", "SLIP", "INFLOW", "OUTFLOW", "INOUT" and "INTERIOR". The INTERIOR condition means that the boundary values are taken by interpolation from other grids.
	BC31	Specify the boundary condition at S=0. Possible values the same as BC22.
	BC32	Specify the boundary condition at S=0. Possible values the same as BC22.
	DIMENSION	Number of grid nodes in circumferential, radial and axial direction respectively.
	RMAX	Change the position of the outer surface of the grid. RMAX>1 increases the distance from the shaft to the outer surface and vice versa. Only applies to the default section grid type.
	YSIGN	Sign of y-coordinate.
	FLIP	Flip top → bottom and vice versa.
	SCALE	Scale the rudder.
	YTARGET	y+ target value at noslip boundaries.
(1)	STLF	Address to the file that contains the rudder geometry in STL format, must be shorter than 256 characters.

(1)	IGES	Address to the file that contains the rudder geometry in IGES format, must be shorter than 256 characters.
	STLT	STLT is used to specify deflection tolerance for IGES surface input. Default value is 0.0001.
	TGTRANS	Use Theodorsen-Garrick transform. Gives high quality orthogonal planar grids for normal wing sections.
	HYPWSECT	Use 2D hyperbolic grid generator. The same format of offset file used by TGTRANS is used to specify the starting curve. This generator can handle more arbitrary shaped sections than TGTRANS.
	HYPPT	Use 2D hyperbolic grid generator. The same format of offset file used by TGTRANS but with an additional point type column is used to specify the starting curve. This generator can handle more arbitrary shaped sections than TGTRANS.
	OGRID	A zero-thickness section at the trailing edge will be interpreted by HYPWSECT and HYPPT as a C-type grid. If this is not correct, OGRID forces the generation of an O-type grid.
	FLAP	[HYPSECT only] Flap chords. If only 1 value is given it will be used for all stations.
	FANGL	[HYPSECT only] Flap angle in degrees, positive is counter-clockwise seen along the rudder axis.
	GROW	[HYPSECT only] Growth factor in the normal direction.
	DTAU	[HYPSECT only] Artificial time step for solving the hyperbolic equations. Lower the value to increase stability.
	NITER	[HYPSECT only] Maximum number of iterations for solving the hyperbolic equations. May need to be increased if DTAU is decreased.
	DN	[HYPSECT only] Thickness of first layer of cells. DN is relative to the chord length.
	STRETCH	[HYPSECT only] Stretch factor towards regions with high curvature. Higher value gives stronger concentration.
	PRIOR	Set the classification priority for the rudder grid. Should be >0.
	HALF	The section input is one side only and the grid should also be one side. The default behaviour is to mirror the section offsets to make a closed, symmetric section curve.
	NOLESTR	Disable leading edge stretching. The leading edge

	stretching is done to place the parametric value 0.5 at the leading edge. This usually makes the folding that closes the tip better, but in some cases it causes grid generation to fail.
BOTH	Generate a grid over both sides of the rudder. This is the default unless the rudder is at $y=0$ and ANGLE, CANT, TILT are all zero. In that case only one side is generated and the boundary conditions are set for symmetric flow.
TOP	Generate a grid over the top side of the rudder only.
BOTTOM	Generate a grid over the bottom side of the rudder only.
CTIP	Flat tip grid thickness chord scaling factor, default=0.1. This may have to be reduced for extremely low aspect ratio, for example if using a RUDDER as a bilge keel. Reducing CTIP will reduce the thickness of the grid over a flat tip in the spanwise direction.
SCALE	Scale factor for the rudder.
STR1	Stretching function in the chord-wise direction. The value should be the ID of a STRETCHING.

Notes

- (1) Feature for a symmetric rudder. The geometry has to be specified on one side only.

2.32. SELFPROPULSION

This command runs Shipflow in self propulsion mode. Basically this means that the propeller RPM is adjusted so that the drag and thrust are in balance. For a full scale computation this is usually sufficient, but to mimic a model scale experiment there is usually an external towing force added that compensates for the difference in frictional drag. This can also be modeled with SELFPROPULSION, as well as specifying a constant propeller thrust.

In order to run a self propulsion simulation, the computational model has to include a hull, possibly with appendages, and one or more propeller models such that thrust is a function of the advance ratio J . The lifting line mode LLINE fulfills this but not the actuator disc ACTUATOR. External propeller models integrated with the EXTPROP command are also likely to fulfill this property. Since the ship speed and propeller diameter are not varied in the computation, changing J is the same as changing the rate of revolution.

The results are presented in the OUTPUT file. If propeller open water characteristics are given, more results such as effective mean wake, relative rotative efficiency, hull efficiency and propulsive efficiency.

The self propulsion computation can be done alone, but to get the full results from the analysis two or three computation have to be done:

1. A nominal, unpropelled resistance calculation.
2. A Propeller Open Water (POW) computation of the propeller open water characteristics. Data from other sources can also be used.
3. A computation of the self propelled case.

General form of command

```
SELFPROP ( ON , CTOW = v , POW = "c" ,
           OFF
           CTTO = v , CTEFD = v , AEFD = v,
           NOMINAL="c" , FORMFAC = v )
```

Default values

```
SELFPROP ( CTOW=[from SCALE, REFL or RN], CTTO=0.0 )
```

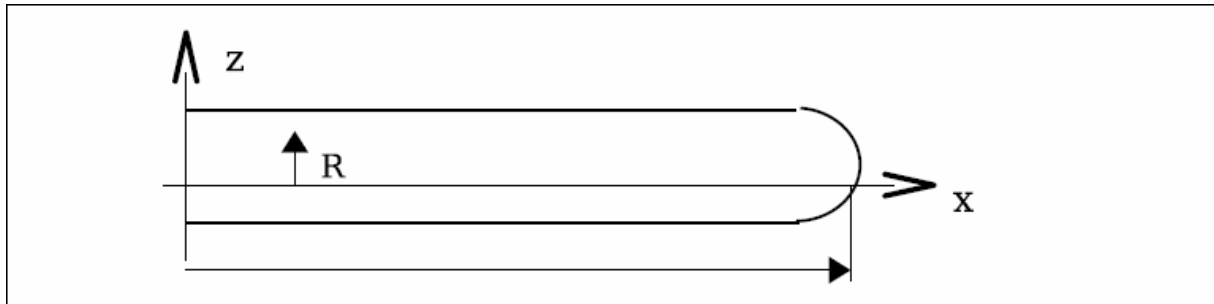
Keywords

<i>keyword</i>	<i>description</i>
ON	[ONOFF] Selpropulsion ON, i.e. J is adjusted to achieve balance between

	thrust, drag and tow force.
OFF	[ONOFF] Computation is run with constant propeller speed.
POW	File containing table with POW data.
CTOW	Coefficient of towing force, same normalization as the drag coefficients. A positive CTOW pulls the ship forwards and reduces the propeller load.
CTTO	Total drag coefficient for the nominal, un-propelled case.
NOMINAL	Name of the CGNS file holding the results of the nominal case. If this parameter is given then CTTO is read from the CGNS file.
CTEFD	Total drag coefficient from the towing tank resistance test.
AEFD	Area of non-dimensionalization for CTEFD.
FORMFAC	Form factor. If not given it is computed from the result in NOMINAL.

2.33. SHAFT

This command adds a propeller shaft to the ship geometry in XCHAP. The geometry is a circular cylinder ended by a hemisphere. The geometry is specified in a shaft-fixed coordinate system according to the figure below.



The position and orientation of the shaft in the offset file coordinate system is given as a translation of the origin and a rotation that aligns the x-axis of the shaft with a given vector.

When the shaft grid is added to the overlapping grid that describes the complete geometry, it goes through the transformation to the SHIPFLOW computational coordinate system, including scaling by 1/LPP, sinkage and trim.

Since the shaft surface is not closed, the shaft must intersect another boundary surface such as the hull and/or the free surface so that the curve that ends the surface at the root of the shaft is completely outside of the fluid domain.

General form of command

```
SHAFT ( ID = "c" , R = v , LENGTH = v ,
        ORIGIN = [v1,v2,...,vn] , DIR = [v1,v2,...,vn] , RMAX = v ,
        DIMENSION = [v1,v2,v3] , GROUP = n , YTARGET = v )
```

Default values

```
SHAFT( ID="Shaft", R=0.1, LENGTH=1, ORIGIN=[0,0,0], DIR=[1,0,0], RMAX=1,
        PRIO=4, BC11="INTERIOR", BC21="NOSLIP", BC22="INTERIOR")
```

Keywords

<i>keyword</i>	<i>description</i>
ID	Gives the shaft object a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different

objects may have the same name, but then only the total force on all of those objects will be calculated and output.

R	The radius of the shaft.
LENGTH	The length of the shaft.
ORIGIN	Position of the shaft coordinate system origin in the offset file (ship) coordinate system.
DIR	Direction of the shaft in the offset file coordinate system.
RMAX	Change the position of the outer surface of the grid. RMAX>1 increases the distance from the shaft to the outer surface and vice versa.
GROUP	Group number in the overlapping grid algorithm. If not specified the grid will be the only grid in the group.
DIMENSION	Number of nodes in each direction.
YTARGET	y+ target value at noslip boundaries.
BC11	Boundary condition at the start (open) end, typically inside the hull.
BC21	Boundary condition on the shaft.
BC22	Boundary condition on the outer radius.

2.34. SPAUTO

To do a full propulsion analysis according to ITTC78 you need results from a resistance computation, propeller open water results and a self-propulsion computation. While this can be done as three separate Shipflow runs, SPAUTO can do all three computations from a single command file. This is especially handy if the design is being optimized with respect to the extrapolated delivered power.

General form of command

```
SPAUTO ( ALL , MITRES = i , MITPOW = i )
RESISTANCE
POW
SELFPROPULSION
EXTRAPOLATION

MITSELF = i
```

Default values

```
SPAUTO ( ALL )
```

Keywords

<i>keyword</i>	<i>description</i>
ALL	[RUN_] All three computations described below required for the propulsion analysis are run.
RESISTANCE	[RUN_] Only resistance computation will be executed.
POW	[RUN_] Only propeller open water computation will be executed.
SELFPROP	[RUN_] Only self-propulsion computation will be executed.
EXTRAPOL	Only the ITTC78 extrapolation will be executed
MITRES	Override the maximum number of solver iterations in the resistance computation.
MITPOW	Override the maximum number of solver iterations in the POW computations.
MITSELF	Override the maximum number of solver iterations in the self propulsion computation.

2.35. SPHERE

Creates a spherical grid for XCHAP. Useful for modeling for example bouys or planets. The grid will be stretched towards NOSLIP boundaries (unless that has been switched off with the XCHAP>CONTROL>NOSTRETCH keyword) to get a y^+ value of approximately 1 for the first grid node.

The SPHERE geometry is specified in the offset coordinate system and the same transformations that are applied on the hull are applied to the cylinder.

General form of command

```
SPHERE  ( ID = "c"                , R = v                , THICK = v                ,  
        PHI = [v1,v2]            , THETA = [v1,v2]        , ORIGIN = [v1,v2,v3]    ,  
        SKEW = v                , DIMENSION = [n1,n2,n3] , GROUP = n                ,  
        BC11 = "c"              , BC12 = "c"              , BC31 = "c"              ,  
        BC32 = "c"              , GROUP                    . PRIO                    )
```

Default values

```
CYLINDER ( ID="Sphere", R=0.5, THICKNESS=0.5, DIMENSION=[10,30,30], PHI=[-  
90,90], THETA[0,360], BC11="NOSLIP", BC12="INTERIOR",  
BC21="INTERIOR", BC22="INTERIOR", BC31="INTERIOR",  
BC32="INTERIOR", GROUP=-1, PRIO=-1, SKEW=0, )
```

Keywords

<i>keyword</i>	<i>description</i>
ID	Gives the sphere a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different grids may have the same name, but then only the total force on all of those objects will be calculated and output.
R	Radius of the sphere.
THICKNESS	Radial thickness of the grid.
PHI	Start and end polar angles (latitude) in degrees.
THETA	Start and end circumferential angles (longitude) in degrees.
ORIGIN	Position of the spheres center.
DIMENSION	Number of grid nodes. Dimension 1 is radial, dimension 2 is

circumferential and dimension 3 is axial.

BC11	Specify the boundary condition on the sphere surface. Possible values are “NOSLIP”, “SLIP”, “INFLOW”, “OUTFLOW”, “INOUT” and “INTERIOR”. The INTERIOR condition means that the boundary values are taken by interpolation from other grids.
BC12	Specify the boundary condition on the outer radius.
BC21	Specify the boundary condition on the start polar angle. Not needed if angle is -90 degrees.
BC22	Specify the boundary condition on the end polar angle. Not needed if angle is -90 degrees.
BC31	Specify the boundary condition on the start circumferential angle. Not needed if $\text{THETA}(2) - \text{THETA}(1) = 360$.
BC32	Specify the boundary condition on the end circumferential angle. Not needed if $\text{THETA}(2) - \text{THETA}(1) = 360$.
GROUP	Group number in the overlapping grid algorithm. If not specified the grid will be the only grid in the group.
SKEW	Circumferential skew. Used mainly to test discretization.

2.36. SPLVOLUME

Split a volume (component) grid at a point. The point is given in the design (offset) coordinate system and it has to be inside the grid to be split. After finding the parametric position in the grid it is split in the specified direction, creating two SPLVOLUMES.

Boundary conditions on the split surfaces are set to INTERIOR, for the rest of the grid faces the boundary conditions of the source grid are used.

General form of command

```
SPLVOL    ( ID = ["c1","c2"]      , SOURCE = "c"          , SPOINT = [v1,v2,v3]  ,  
            SPLDIR = i             , KEEPSRC              )
```

Default values

```
SPLVOL    ( )
```

Keywords

<i>keyword</i>	<i>description</i>
ID	Names of the two new grids made by the split.
SOURCE	Name of the source grid to split.
SPOINT	Point in the design (offset) coordinate system. The point has to be inside the source grid.
SPLDIR	Split direction.
KEEPSRS	Keep the source grid visible. Default is to hide it. This is mainly for debugging.

2.37. STRETCHING

Define a 1-dimensional stretching function. These are used to change the node distribution in a component grid. The stretching is applied in a specified coordinate direction and the node positions are changed in this direction along the grid lines, leaving the distribution in the remaining directions unchanged. One stretching can be applied in each direction.

For all stretching functions $f(r)$: $f(0) = 0$, $f(1) = 1$ and $f'(r) > 0$ for r in $[0,1]$.

Note that not all of the parameters S0, S1, DU, DU1, DS0, DS1 apply to all stretchings.

General form of command

```
STRETCHING ( ID = "c1"           , EXP           , FLIP
              TANH                 NOFLIP
              OUTLET
              MIDDLE
              FS
              MIDPOINT
              S0 = v               , S1 = v               , DU = v               ,
              DU1 = v             , DS0 = v             , DS1 = v             )
```

Default values

STRETCHING()

Keywords

<i>keyword</i>	<i>description</i>
ID	Name of stretching.
FLIP	Reverse the parameter direction so that $f(r) \rightarrow f(1-r)$.
NOFLIP	Do not reverse the parameter direction.
EXP	Exponential stretching towards $r=0$. DS0 specifies stretching at $r=0$. Default is DS0=0.5, DS1=1 gives no stretching.
TANH	Hyperbolic tangent stretching towards $r=0$ and $r=1$. DS0 specifies stretching at $r=0$ and DS1 at $r=1$. Values <1 gives smaller spacing than the original distribution and values >1 gives larger spacing than the original distribution. Defaults are DS0=0.5 and DS1=0.5.
OUTLET	Uniform spacing followed by exponential stretching. The exponential stretching parameter is set so that the spacing is smooth. S0 specifies where the change from uniform to exponential occurs and DS0 the

spacing in the uniform region. Defaults are $S0=0.5$, $DS0=1$.

MIDDLE A uniform stretching between $0 < S0 < S1 < 1$ with matching exponential stretchings on either side. $DS0$ specifies the spacing in the uniform region. Defaults are $S0=0.25$, $S1=0.75$ and $DS0=1$. The parameters are not intuitive, so if you are thinking about using MIDDLE it is better to use FS.

FS Similar to MIDDLE except that the flanking stretchings are $TANH$. $0 < S0 < S1 < 1$ specifies the region of uniform spacing. DU specifies the fraction of cells inside the uniform region, $0 < DU < 1$. The function of $DU0$ is unclear at the time of writing, best leave it alone. Defaults are $S0=0.25$, $s1=0.75$, $DU=0.8$, $DU0=0$.

MIDPOINT Adds a sinusoidal shift to a uniform stretching. $DS0$ is the amplitude of the shift. $DS0=0$ gives no stretching, $DS0 < 0$ shifts towards $r=0$ and $DS0 > 0$ towards $r=1$. Default is $DS0=0$.

S0 Multi use parameter, see above.

S1 Multi use parameter, see above.

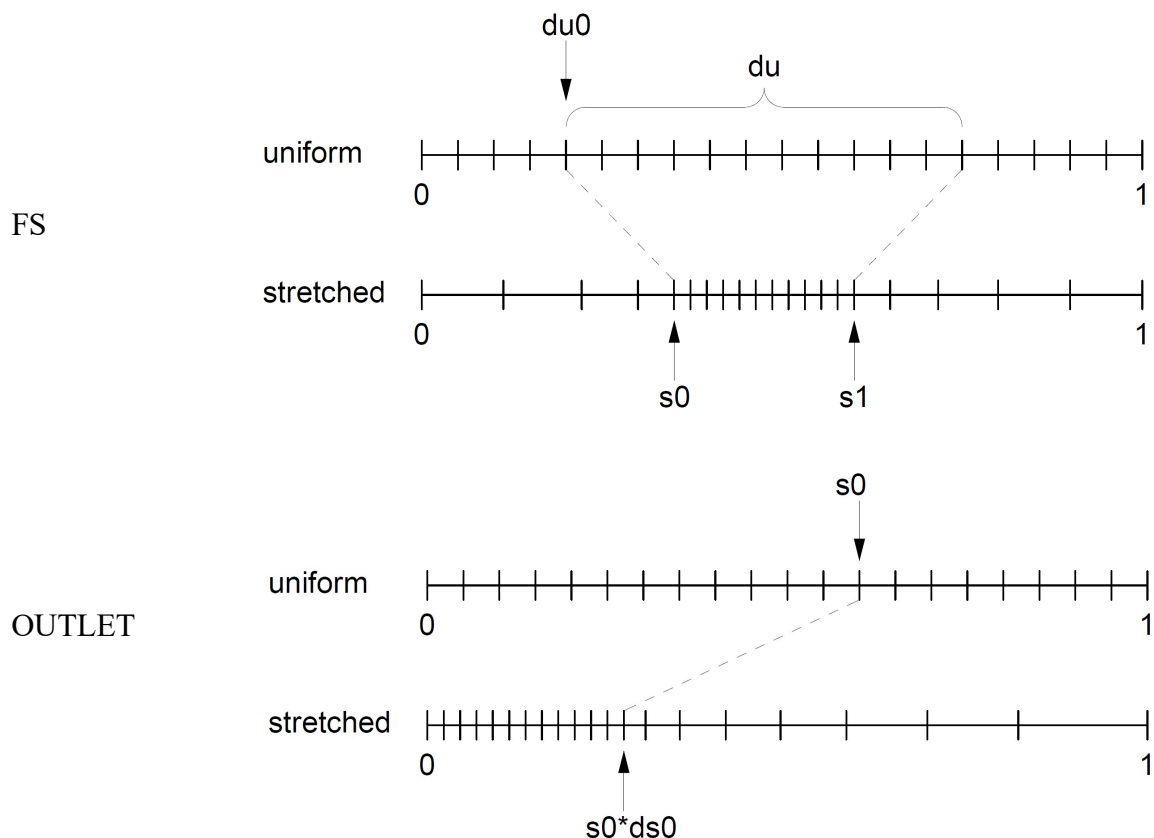
DU Multi use parameter, see above.

DU1 Multi use parameter, see above.

DS0 Multi use parameter, see above.

DS1 Multi use parameter, see above.

Example stretching types:



2.38. SUBVOLUME

Create a smaller grid from a source by specifying a range in parameter space [1..0] in each direction.

Setting LOW=[0,0,0] and HIGH=[1,1,1] makes a copy of the source grid.

General form of command

```
SUBVOL  ( ID = "c"           , SOURCE = "c"           , LOW = [v1,v2,v3]   ,
          HIGH = [v1,v2,v3]   , DIMENSION = [n1,n2,n3] , KEEPSRC           ,
          BC11 = "c"          , BC12 = "c"          , BC21 = "c"          ,
          BC22 = "c"          , BC31 = "c"          , BC32 = "c"          ,
          PRIO                                     )
```

Default values

```
SUBVOL  ( DIMENSION=<from SOURCE>, BC11=<from SOURCE>, BC12=<from
          SOURCE>, BC21=<from SOURCE>, BC22=<from SOURCE>
          BC31=<from SOURCE>, BC32=<from SOURCE>, PRIO=-1)
```

Keywords

<i>keyword</i>	<i>description</i>
ID	Name of the sub-volume.
SOURCE	Radius of the sphere.
LOW	Start corner of range in parameter space.
HIGH	End corner of range in parameter space.
DIMENSION	Number of grid nodes in each parameter direction.
BC11	Specify the boundary condition on the low surface in the first parameter direction. Possible values are "NOSLIP", "SLIP", "INFLOW", "OUTFLOW", "INOUT" and "INTERIOR". The INTERIOR condition means that the boundary values are taken by interpolation from other grids.
BC12	Specify the boundary condition on the high surface in the first parameter direction.
BC21	Specify the boundary condition on the low surface in the second

parameter direction.

BC22 Specify the boundary condition on the high surface in the second parameter direction.

BC31 Specify the boundary condition on the low surface in the third parameter direction.

BC32 Specify the boundary condition on the high surface in the third parameter direction.

2.39. SYMMETRY

This command is used to specify the symmetry condition, if any.

General form of command

```
SYMMETRY (  NOSYM      )
           XZPLANE
```

Default values

SYMMETRY (NOSYM [if runing with PROP at YSH=0], otherwise XZPLANE)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	NOSYM	[SYMMETRY] No symmetry planes are included in the computation.
(1)	XZPLANE	[SYMMETRY] The xz-plane is treated as a symmetry plane.

Notes

- (1) The xz-plane is the only symmetry plane that can be specified on the SYMMETRY command. The xy-plane is automatically treated as a symmetry plane for the body groups when the double-model solution is used as a basic-model.

NOSYM must be used onset flow angels other than zero.

2.40. TITLE

OPTIONAL

This command is used to give title information to SHIPFLOW. It will be used for headers in various files and plots.

General form of the command

TITLE (TITLE = "c")

Default values

None

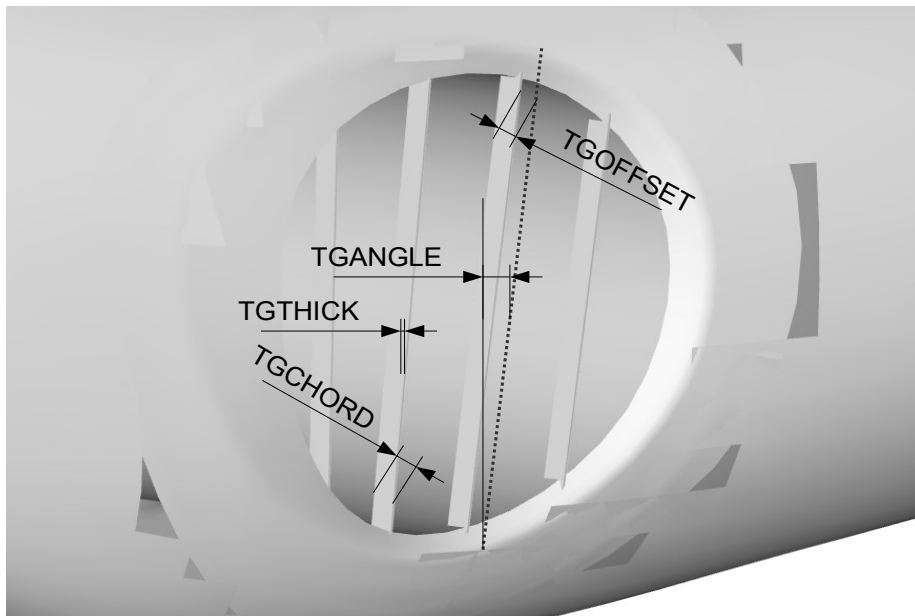
Keywords

TITLE	The maximum length of the string is 80 characters. Only the first 53 characters of the title will be shown in the plots produced by FIPOST.
-------	---

2.41. TUNNEL

Make a thruster tunnel grid. The tunnel is parallel to the y axis. The y-coordinate of the tunnel should always be zero. For cases with XFLOW|SYMMETRY|NOSYMMETRY set, the tunnel is automatically mirrored.

The tunnel opening may be fitted with a protective grid, that may also reduce drag. The parameters that control this function are prefixed with TG (for TunnelGrid).



All geometric dimensions are given in the design (offset file) coordinate system.

General form of command

TUNNEL	(	ID = "c"	,	RADIUS = v	,	ORADIUS = v	,
		LENGTH = v	,	ORIGIN = [v1,v2,v3]	,	ADAPT = "c"	,
		DIME=[i1,i2,i3]	,	CROSSFLOW		ANGLE = v	,
		CWIDTH = [v1,v2,v3]	,	CDEPTH = [v1,v2,v3]			,
		TGRID	,	TGNBAR = i	,	TGCHORD = v	,
		NOTGRID					
		THFGRID					
		TGTHICK = v	,	TGOFFSET = v	,	TGANGLE = v	,
			,	YTARGET = v	,	YTBAR = v	,

		NOADAPT	,	TGNOADAPT		)

Default values

TUNNEL (ID="Tunnel", RADIUS=1, ORADIUS=1.5*RADIUS, LENGTH=10,
ORIGIN=[0,0,0], ADAPT="Xgrid", DIME=[10,20,20], ANGLE=0,
YTARGET=1, YTBAR=4)

Keywords

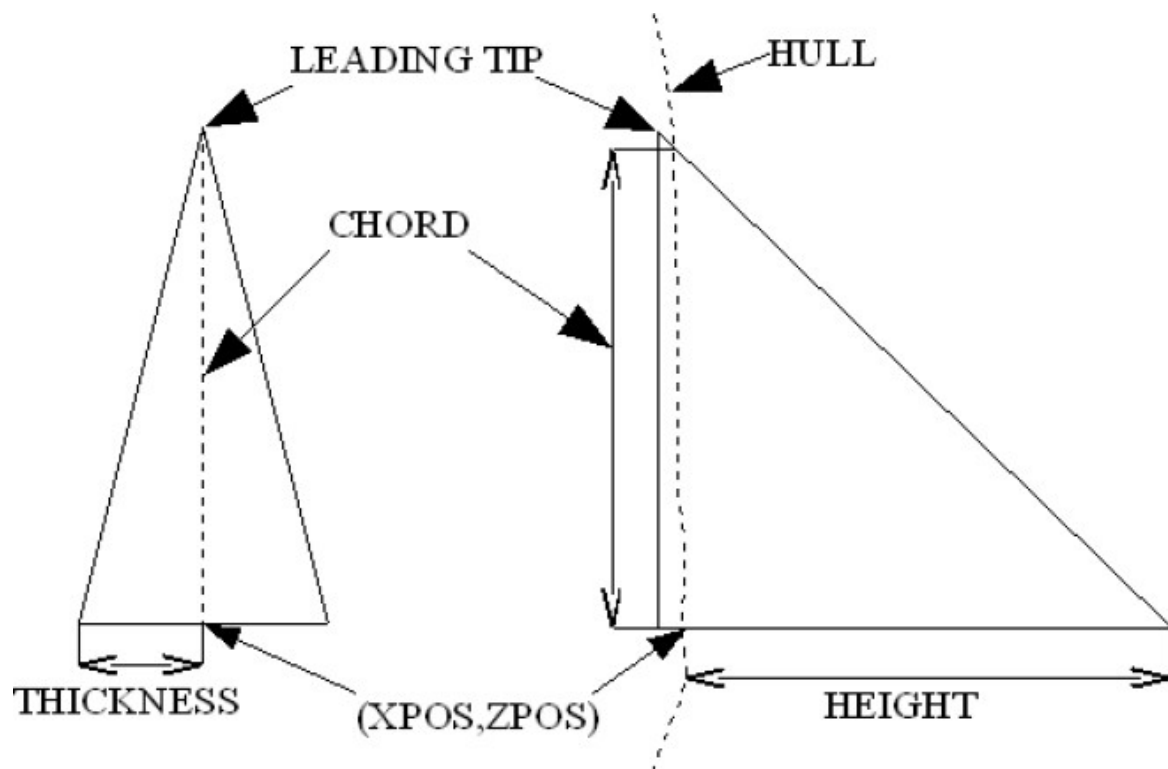
<i>keyword</i>	<i>description</i>
ID	Gives the tunnel a name. The name is used in general to identify the object and to associate results like integrated forces with it.
RADIUS	Radius of the tunnel.
ORADIUS	Outer radius, i.e. the radius of the flange grid that adapts to the hull.
LENGTH	Length of the grid object, from the center plane and including the part the overlaps the hull grid. Half of the cells are distributed inside the tunnel and half in the overlap region.
ORIGIN	Coordinates of the tunnel origin. The y-coordinate should always be zero.
ADAPT	Name of the grid that the tunnel is fitted to. Shipflow appends <n> to grid names and this can be seen while post processing. In this context only the base name should be used, for instance "Xgrid" and not "Xgrid_1".
DIME	Node dimensions of the grid inside the tunnel. The first dimension is radial, the second tangential and the third axial. The axial dimension is doubled so that there is an equal amount of cells in the part of the tunnel grid that is outside the hull, i.e. inside the hull grid. A toroidal flange grid with dimension DIME is also added, so the total number of grid nodes is $i1*i2*(i3*3-1)$. If XFLOW SYMMETRY NOSYMMETRY is set the grids are mirrored so the number of nodes is doubled.
CROSSFLOW	A tunnel through the hull can produce oscillations in the flow if both sides of the hull is modeled. In a real case there is an impeller in the tunnel that damps the cross flow, and in model experiments some kind of obstruction is usually mounted in the tunnel. To avoid the oscillations in the computation the tunnel grid is split at $y=0$ and the SLIP boundary condition is applied to the two split faces. If the CROSSFLOW keyword is given then the split is not done and the tunnel is open for cross flow. This is required if for instance a propeller model is used to pump fluid through the tunnel.
ANGLE	Angle of rotation in degrees. The geometry is only affected if the CWIDTH and CDEPTH parameters are given.

CWIDTH	Width of the region around the tunnel edge where the hull is chamfered. The width is given at three angular positions, 0, 90 and 180 degrees. The width at 270 degrees is the same as that at 90 degrees. The chamfered region is represented by two elliptical curves. From 270 to 90 degrees the half axes are $RADIUS + CWIDTH[1]$ and $RADIUS + CWIDTH[2]$ and from 90 to 270 degrees they are $RADIUS + CWIDTH[3]$ and $RADIUS + CWIDTH[2]$.
CDEPTH	The depth of the chamfer is specified at 0, 90 and 180 degrees with the depth at 270 degrees the same as at 90 degrees. The variation between points is linear.
TGRID	Turn on tunnel protective grid.
NOTGRID	Turn off tunnel protective grid.
THFGRID	Turn on hull following tunnel protective grid.
TGNBAR	Number of bars in the grid.
TGCHORD	Chord length, i.e. the length in the tunnel axial direction, of the bars. The bars have a rectangular cross section.
TGTHICK	Thickness of the grid bars.
TGOFFSET	Inward offset from the hull surface of the bar endpoints.
TGANGLE	Rotation angle around the y-axis in degrees for the grid.
TGDIME	Dimensions of tunnel grid bar
YTARGET	y+ target value at noslip boundaries.
YTBAR	y+ target value at tunnel grid bar noslip boundaries. This may have to be reduced for very high Re cases, i.e. full scale.
NOADAPT	Do not adapt the tunnel to the hull. This is for checking that the tunnel is properly placed and sized. If a tunnel grid is specified, it will be disabled since it requires the intersection curve between hull and tunnel.
TGNOADAPT	Do not adapt the tunnel grid to the tunnel. This is for checking that the bars are properly placed and sized.

2.42. VORTEXGEN

This command adds a delta wing vortex generator to the ship geometry in XCHAP. The command may be repeated to add more vortex generators. Each vortex generator consists of 4 butt-joined component grids.

The position and size of the vortex generator is given in the offset file coordinate system. The user specifies the x-position and z-position and the y-position is automatically calculated by SHIPFLOW so that it is appended on the hull. The vortex generator is also automatically rotated so that it points in the normal direction of the hull surface and with the leading tip in the offset file axial direction. Note: If trim is used the original direction of the leading tip will still point in the axial direction in the offset file coordinate system, which then differs from SHIPFLOW's coordinate system.



For a non-symmetrical case the vortex generator will be placed at the starboard side unless the PORT keyword is specified.

Restrictions:

The vortex generator does only work together with an XGRID-grid. If only imported grids are used SHIPFLOW will stop if the user tries to add a vortex generator object. Risks:

1. If the curvature of the hull surface is large compared to the size of the vortex generator, the vortex generator might stick out somewhere which leads to a “leak of non-fluid points”. If a large part of the total number of cells are flagged as outside (non-fluid) it is a sign of such a leak has occurred. Results from a calculation with leaks can not be considered valid.
2. If the shape of the vortex generator is changed a lot compared to the original shape some cells might be very skewed which will give the solver problems.

If the default delta-shaped vortex generator is not applicable, another shape may be imported through the GRID parameter. The value of GRID should be the full pathname to a CGNS file that contains a block structured grid. The file should also contain the connectivity information for the butt-joined grid faces, and the boundary conditions for all faces. Since there are several ways to store a grid in a CGNS file, the following should be noted:

1. Use CGNS version 2.
2. There should be only 1 base in the file.
3. The domains should be located directly under the base.
4. Each domain should be of ZoneType Structured
5. The ZoneGridConnectivity data should have the Node Label GridConnectivity1to1_t.
6. The ZoneBC data should have the Node Label BC_t and have the Node Data equal to "BCWall" or "BCWallViscous", both of these map to the Shipflow boundary condition NOSLIP. Grid faces that should have the INTERIOR boundary condition should not appear at all in the CGNS file.
7. Under each BC_t node should be
 1. PointRange with Node Label equal to IndexRange_t.
 2. InwardNormalIndex.
 3. Any other nodes are ignored.

Use a browser such as adfviewer to check that the grid data is correctly stored in the CGNS file.

The orientation of the geometry is important if the VORTEXGEN command shall be able to position the appendix correctly. The origin must be at that base and trailing edges of the appendix, the x-axis must point towards the front and the z-axis towards the tip, see the figure below.

General form of command

```
VORTEXGEN ( ID="c"           , XPOS=v     , ZPOS=v     ,
            PITCH=v         , CHORD=v   , HEIGHT=v   ,
            POST            , AXIAL     , PORT      ,
            POSL            , PROJ     , STARBOARD ,
            POSC
            THICKNESS=v    , SCALE=v   , DIMS=v     ,
            GRID = "c"    )
```

Default Values

VORTEXGEN (PITCH=0 , CHORD=1 , HEIGHT=1 ,
THICKNESS=0.1 , DIMS=1 , PORT ,
POST , AXIAL , SCALE=1)

Keywords

<i>keyword</i>	<i>description</i>
ID	Gives the vortex generator object a name. The name is used in general to identify the object and to associate results like integrated forces with it.
XPOS	The x-position of the vortex generator.
ZPOS	The z-position of the vortex generator.
PITCH	Pitch angle in degrees in the local coordinate system for vortex generator after it has been placed normal to the hull. Tip turns up for positive values.
CHORD	Chord length of the vortex generator at the hull surface.
HEIGHT	Height of the vortex generator from the hull surface.
THICKNESS	Thickness of the vortex generator from its center plane.
SCALE	Linear scaling factor.
DIMS	Used for scaling number of cells. 0.5 gives half as many cells in each parameter direction and 2 gives twice as many cells in each parameter direction. Changing this value will require interpolations to create the grid. The most robust way is to leave DIMS unchanged.
POST	[POSITION] The position given by XPOS and ZPOS is for the intersection between the hull surface and the trailing side.
POSL	[POSITION] The position given by XPOS and ZPOS is for the intersection between the hull surface and the leading edge.
POSC	[POSITION] The position given by XPOS and ZPOS is for the intersection between the hull surface and the center of the vortex generator.
AXIAL	[PTYPE] Rotate vortex generator PITCH degrees around its own axis.
PROJ	[PTYPE] Rotate vortex generator so that the projection of the intersection between the hull surface and the vortex generator on the symmetry plane is rotated PITCH degrees.
PORT	[SIDE] Use this keyword if the vortex generator is on the port side.
STARBOARD	[SIDE] Use this keyword if the vortex generator is on the starboard side.

GRID

CGNS file with grids and boundary conditions. If this is specified the parameters that defines the shape of the vortex generator are not needed.

2.43. VSHIP

MANDATORY

This command is used to specify the speed of the ship.

General form of command

```
VSHIP (  FN = v                ,  RN = v                )  
        VKNOT = v              REFLLEN = v  
        VM_S = v              SCALE = v
```

Default values

VSHIP (SCALE = 1)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	FN	Ship speed in Froude number.
(1)	RN	Reynolds number.
(1)	VKNOT	Velocity in knots.
(1)	VM_S	Velocity in meter per second.
(1)	REFLEN	Reference length
(1)	SCALE	Scale factor

Notes

- (1) Geometry in XFLOW:OFFSET should be given in ship scale. Speed in ship scale has to be given by VKNOT, VM_S or FN. If you want to run in model scale then set the appropriate REFLLEN, SCALE or RN.

2.44. TURN

This command is used to specify the turning radius and center position. The ship will always turn left when turning motion is applied.

General form of command

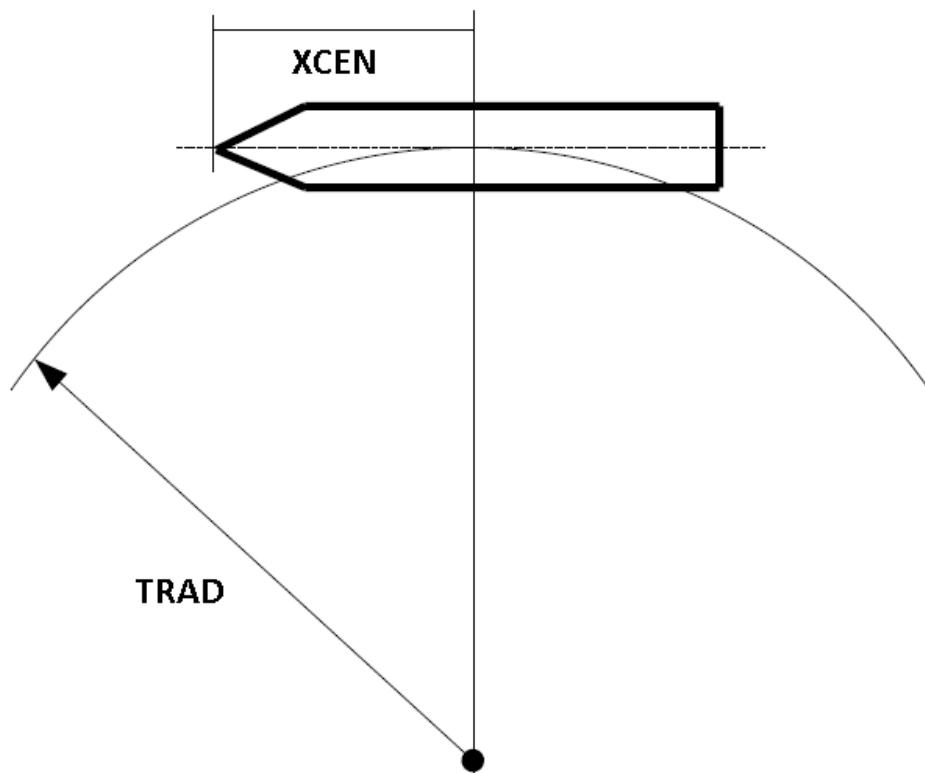
TURN (XCEN = v , TRAD = v)

Default values

The ship moves straight forward as a default.

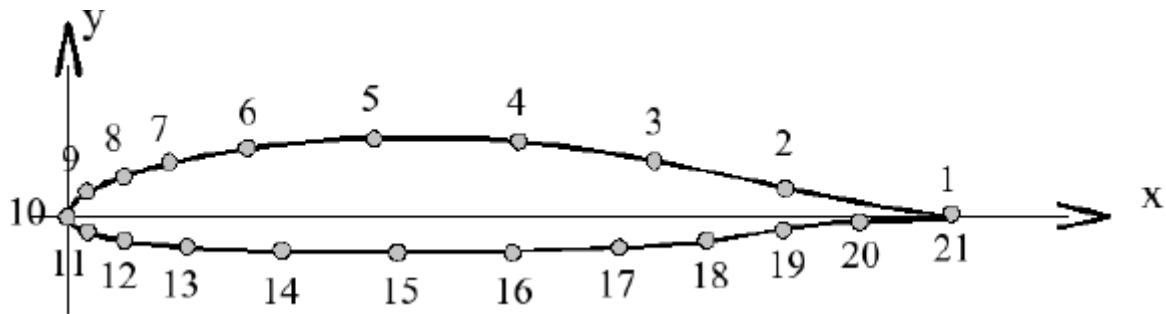
Keywords

<i>keyword</i>	<i>description</i>
XCEN	longitudinal position of the turning center in computational coordinate system
TRAD	turning radius in computational system



2.45. WSECTION

This command creates a two-dimensional grid around a wing section in XCHAP. The command may be repeated to add more grids.



The shape of the wing section is given as a set of points specified in an offset file. The format of the file is very simple, on every line should be the x and y coordinates of a point, separated by white space (space and/or tabs). The points should be ordered starting at the trailing edge, going forward on the upper side, around the leading edge and back on lower side to the trailing edge, i.e. last point must be the same as the first. The trailing edge must be at a point ($x > 0$, 0) and the leading edge at (0,0). If the section is symmetrical about $y=0$ it is sufficient to give the points from the trailing edge to and including the point on the leading edge. Several symmetrical and non-symmetrical wing sections are supplied in the SHIPFLOW installation.

The grid is generated with a conformal mapping technique that gives a high quality orthogonal grid. It is however restricted to wing section shapes. Inputting offsets for other shapes may give unsatisfactory results. Trying to introduce corners with double points will specifically cause the algorithm to fail.

Alternatively a hyperbolic grid generator can be chosen with the HYPERBOLIC keyword. This is more robust but generates lower quality grids that are not orthogonal.

General form of command

```
WSECTION    (  ID = "c"           ,  OFFSETS = "c"       ,  ANGLE = v      ,
               SCALE = v          ,  ORIGIN = [v1,v2]    ,  RMAX = v       ,
               BC21 = "c"         ,  BC22 = "c"         ,  GROUP = n      ,
               PRESCALE=v         ,  DIME = [v1,v2]     ,  CONFORMAL      )
                                   HYPERBOLIC
```

Default values

```
WSECTION (  ID="Wsect",    ANGLE=0,    SCALE=1,
            ORIGIN=[0,0],  RMAX=1,      BC21="NOSLIP",
            BC22="INTERIOR", GROUP=unique,
```

PRESALE=1, DIME = [50,20], CONFORMAL)

Keyword

<i>keyword</i>	<i>description</i>
ID	Gives the wsect object a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different objects may have the same name, but then only the total force on all of those objects will be calculated and output.
ANGLE	Clockwise rotation in degrees.
SCALE	Grid scale factor.
PRESALE	Grid scale factor applied before rotation and translation.
ORIGIN	Translation vector of the grid. The translation is applied before the rotation ANGLE, so the translation vector will be rotated too.
RMAX	RMAX>1 makes the grid thicker and 0<RMAX<1 makes it thinner. The default thickness is approximately half the chord length of the section, but it depends on the actual shape of the section.
GROUP	Group number used in the overlapping grid algorithm.
BC21	Boundary condition on the wing section. Possible values are "NOSLIP", "SLIP", "INFLOW", "OUTFLOW", "INOUT" and "INTERIOR". The INTERIOR condition means that the boundary values are taken by interpolation from other grids.
BC22	Boundary condition on the outer boundary. Possible values same as BC21.
DIMENSION	Dimension of grid in chord-wise and radial direction respectively.
CONFORMAL	Use the conformal grid generator.
HYPERBOLIC	Use the hyperbolic grid generator.

3. XMesh commands

The following section describes the commands that are used to generate panels on offset groups, free-surface, propeller and wakes behind lifting surfaces. A grid for visualization (off-body points) of the solution in the flow-field outside the panels can also be generated.

There are 10 types of commands for the XMesh module, PROPELLER, ENVIRONMENT, LIFT, BODY, FREE, TRANSOM, FSFAR, STRIP, OBPOINT and PLOT.

3.1. PROPELLER

This command is used to generate a panel group for the propeller. Only one propeller group is allowed.

General form of command

```
PROPELLER ( GRNO = n      ,  GENE      ,  HIGHER      ,  
              READ        FIRST  
              STATION = n  ,  POINT = n  ,  OFFSETGROUP = "c" ,  
              ANGLE = v      )
```

Default values

PROPELLER(GENE, HIGHER, STATION = 9 (if ANGLE = 180),
STATION = 17 (if ANGLE = 360), POINT = 11)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	GRNO	Group number. The panel groups must be numbered from 1 to the maximum number of groups, starting with the propeller group followed by the environment groups, the lift groups, the body groups, the free-surface groups, the transom free-surface groups, the strip groups, the offbody point group and last the plot groups. The group number must always be specified.
	GENE	[GENMODE] The panels are generated according to the information on the present propeller command.
	READ	[GENMODE] The offset points are directly used as panel corner points.
	HIGHER	[PTYPE] Parabolic panels with a linearly varying source strength are generated.
	FIRST	[PTYPE] Flat panels with constant source strength are generated.
(1)	OFFSETGR	Label name for the points in the offset file.
(1)	STATION	Number of stations in the circumferential direction.
(1)	POINT	Number of points on each station in the radial direction.

ANGLE	The circumferential extension in degrees of the propeller disc. ANGLE must be set to 180 for a single propeller arrangement if xz-symmetry is used. 360 must be specified for all other cases.
-------	--

Notes

- (1) The offset group label name, the number of stations and points are specified in MANUAL input mode only. The specified number of stations and points must then correspond exactly to the number of stations and points in the offset file. The default number of stations and points are always used in AUTO input mode.

3.2. ENVIRONMENT

This command is used to specify panels on geometries that are not included in the description of the hull. A typical example is the bottom and sides of a canal, see “Order of points and stations” on page 307

General form of command

```
ENVI  (  FSINCL      ,  GENE      ,  HIGHER      ,
        NFSINCL     ,  READ      ,  FIRST      ,

        FIXED       ,
        FOLLOW      ,

        GRNO = n    ,  OFFSETGROUP = "c" ,  XTRA = v    ,
        YTRA = v    ,  ZTRA = v      ,  ZROT = v    ,
        XROT = v    ,  YROT = v      ,  XSCA = v    ,
        YSCA = v    ,  ZSCA = v      ,  YMIR      ,
        STATION = n ,  POINT = n     ,  VELBC = v  ,
        STR1 = n    ,  DF1 = v       ,  DL1 = v    ,
        STR2 = n    ,  DF2 = v       ,  DL2 = v    ,
        STR3 = n    ,  DF3 = v       ,  DL3 = v    ,
        STR4 = n    ,  DF4 = v       ,  DL4 = v    )
```

Default values

```
ENVI  (GENE, HIGHER, FSINCLUDE, FIXED, VELBC = 0, XTRA = 0.0,
        YTRA = 0.0, ZTRA = 0.0, XROT = 0.0, YROT = 0.0, ZROT = 0.0, STR1 = 0,
        DF1 = 0.0, DL1 = 0.0, STR2 = 0, DF2 = 0.0, DL2 = 0.0, STR3 = STR1, DF3
        = DF1, DL3 = DL1, STR4 = STR2, DF4 = DF2, DL4 = DL2, XSCA = 1.0,
        YSCA = 1.0, ZSCA = 1.0)
```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	GRNO	Group number. The panel groups must be numbered from 1 to the maximum number of groups, starting with the propeller group followed by the environment groups, the lift groups, the body groups, the free-surface groups, the transom free-surface groups, the strip groups, the offbody

		point group and last the plot groups. The group number must always be specified.
	GENE	[GENMODE] The panels are generated according to the information on the present ENVIRONMENT command.
	READ	[GENMODE] The offset points are directly used as panel corner points.
	HIGHER	[PTYPE] Parabolic panels with a linearly varying source strength are generated.
	FIRST	[PTYPE] Flat panels with constant source strength are generated.
	OFFSETGR	Label name for the points in the offset file.
(2)	FSINCLUDE	[INCLUDE] The group is included when the intersection with the free surface is generated.
(2)	NFSINCLUDE	[INCLUDE] The group is not included when the intersection with the free surface is generated.
	FIXED	[FIXFRE] The environment panels are kept in a fixed position during sinkage and trim.
	FOLLOW	[FIXFRE] The environment panels moves with the hull if sinkage and trim is computed.
	XTRA	The offset points are moved the distance XTRA in the x-direction for this group. XTRA is given in the offset scale.
	YTRA	The offset points are moved the distance YTRA in the y-direction for this group. YTRA is given in the offset scale.
	ZTRA	The offset points are moved the distance ZTRA in the z-direction for this group. ZTRA is given in the offset scale.
	XROT	The offset points are rotated XROT degrees about the x-axis for this group.
	YROT	The offset points are rotated YROT degrees about the y-axis for this group.
	ZROT	The offset points are rotated ZROT degrees about the z-axis for this group.
	XSCA	The x-coordinates in the offset file are scaled a factor XSCA.
	YSCA	The y-coordinates in the offset file are scaled a factor YSCA.
	ZSCA	The z-coordinates in the offset file are scaled a factor ZSCA.

	YMIR	The offset points are mirrored in the plane $y = 0.0$.
(1)	STATION	Number of stations.
(1)	POINT	Number of points on a station.
	STR1	Stretch function for side 1 of the environment groups.
	STR2	Stretch function for side 2 of the environment groups.
	STR3	Stretch function for side 3 of the environment groups.
	STR4	Stretch function for side 4 of the environment groups.
		= 0 uniform distribution
		= 1 hyperbolic tangent stretching, one end specified
		= 2 exponential stretching, one end specified
		= 3 hyperbolic sine stretching, one end specified
		= 4 geometric stretching, one end specified
		= 5 hyperbolic tangent stretching with spacing specified at two ends.
(3)	DF1,DF2, DF3,DF4	Size of the first panel on side 1, 2, 3 and 4 in percent of Lpp.
(3)	DL1,DL2, DL3,DL4	Size of the last panel on side 1, 2, 3 and 4 in percent of Lpp.
	VELBC	The velocity through the panel in the panel normal direction. VELBC must be normalized by the free stream velocity, a negative value means inflow through the surface and a positive value means outflow from the surface.

Notes

- (1) GRNO, STATION and POINT must always be specified.
- (2) A surface-piercing ENVI group can be excluded from the generation of the intersection line between the group and the free-surface. The keyword NFSINCLUDE must then be included on the ENVI command. The default value for all groups is FSINCLUDE
- (3) DF1 and DL1 must be 0.0 or left out if STR1 = 0.

Only one of DF1 and DL1 must be specified if STR1 = 1, 2, 3 or 4. The mesh is

clustered towards the first station if DF1 is specified and towards the last station if DL1 is specified.

Both DF1 and DL1 must be specified if IGRID = 5.

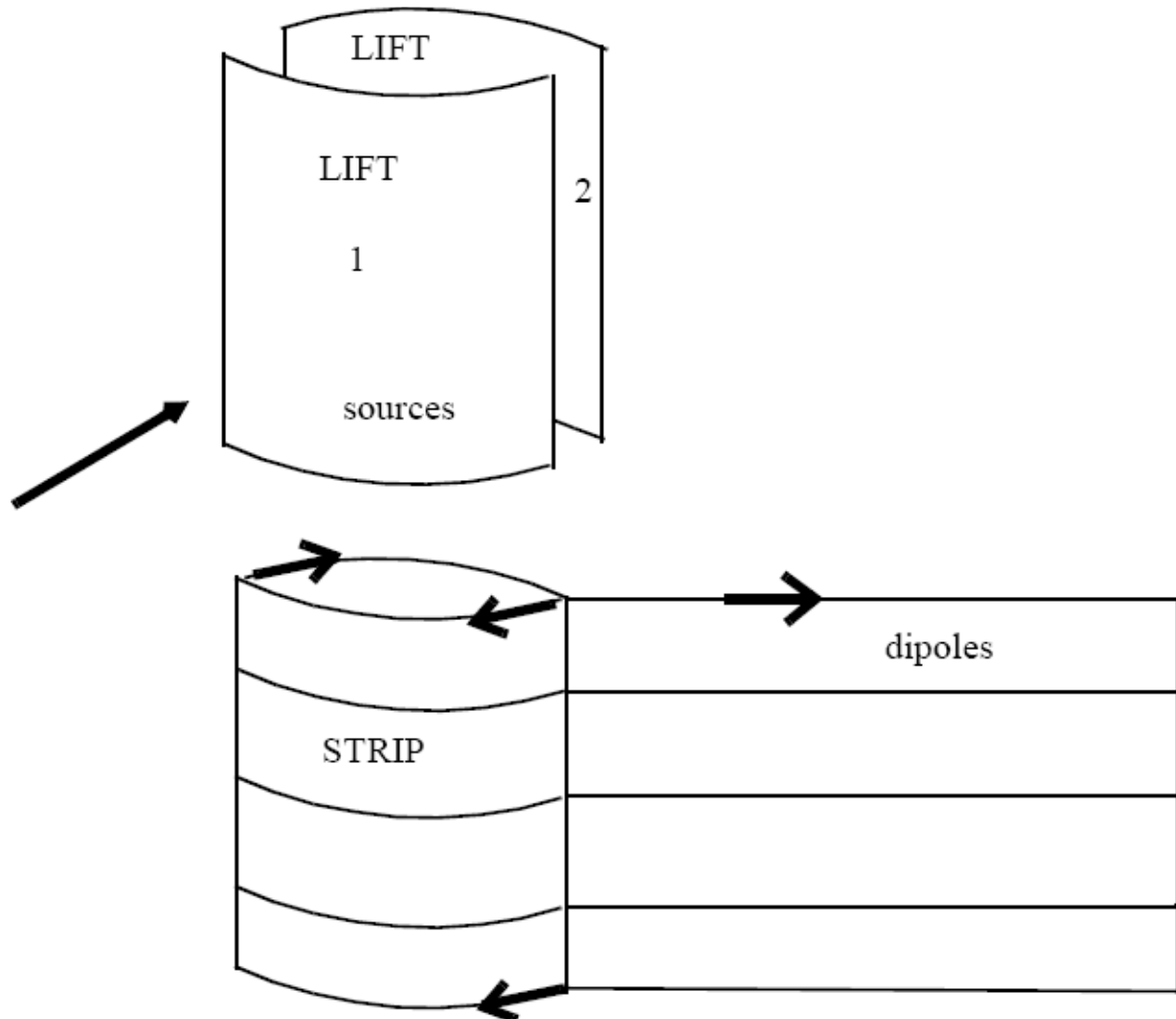
The above is valid also along side 2, 3, 4.

The input for side 1 is used as default for side 3.

The input for side 2 is used as default for side 4.

3.3. LIFT

This command is used to specify source panels for lifting surfaces. A lifting surface is specified by two LIFT groups, one for the pressure and one for the suction side, and one STRIP group. The LIFT groups can be used without an attached STRIP group but the lift force will then not be taken into account. See also the STRIP command and “Order of points and stations” on page 307



General form of command

```
LIFT ( FSINCL      , GENE      , HIGHER  ,
      NFSINCL      READ      FIRST
      GRNO = n      , OFFSETGROUP = "c" , XTRA = v ,
      YTRA = v      , ZTRA = v      , ZROT = v ,
      XROT = v      , YROT = v      , XSCA = v ,
      YSCA = v      , ZSCA = v      , YMIR      ,
      STATION = n   , POINT = n      , DL1 = v   ,
```

```

STR1 = n      ,  DF1 = v      ,  DL2 = v      ,
STR2 = n      ,  DF2 = v      ,  DL3 = v      ,
STR3 = n      ,  DF3 = v      ,  DL4 = v      ,
STR4 = n      ,  DF4 = v      )

```

Default values

LIFT (GENE, HIGHER, FSINCLUDE, XTRA = 0.0, YTRA = 0.0, ZTRA = 0.0, XROT = 0.0, YROT = 0.0, ZROT = 0.0, STR1 = 0, DF1 = 0.0, DL1 = 0.0, STR2 = 0, DF2 = 0.0, DL2 = 0.0, STR3 = STR1, DF3 = DF1, DL3 = DL1, STR4 = STR2, DF4 = DF2, DL4 = DL2, XSCA = 1.0, YSCA = 1.0, ZSCA = 1.0)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	GRNO	Group number. The panel groups must be numbered from 1 to the maximum number of groups, starting with the propeller group followed by the environment groups, the lift groups, the body groups, the free-surface groups, the transom free-surface groups, the strip groups, the offbody point group and last the plot groups. The group number must always be specified.
	GENE	[GENMODE] The panels are generated according to the information on the present lift command.
	READ	[GENMODE] The offset points are directly used as panel corner points.
	HIGHER	[PTYPE] Parabolic panels with a linearly varying source strength are generated.
	FIRST	[PTYPE] Flat panels with constant source strength are generated.
(2)	OFFSETGR	Label name for the points in the offset file.
(3)	FSINCLUDE	[INCLUDE] The group is included when the intersection with the free surface is generated.
(3)	NFSINCLUDE	[INCLUDE] The group is not included when the intersection with the free surface is generated.
	XTRA	The offset points are moved the distance XTRA in the x-direction for this group. XTRA is given in the offset scale.

	YTRA	The offset points are moved the distance YTRA in the y-direction for this group. YTRA is given in the offset scale.
	ZTRA	The offset points are moved the distance ZTRA in the y-direction for this group. ZTRA is given in the offset scale.
	XROT	The offset points are rotated XROT degrees for this group.
	YROT	The offset points are rotated YROT degrees for this group.
	ZROT	The offset points are rotated ZROT degrees for this group.
	XSCA	The x-coordinates in the offset file are scaled a factor XSCA.
	YSCA	The y-coordinates in the offset file are scaled a factor YSCA.
	ZSCA	The z-coordinates in the offset file are scaled a factor ZSCA.
(4)	YMIR	The offset points are mirrored in the plane $y = 0.0$.
(1), (5)	STATION	Number of stations.
(1), (5)	POINT	Number of points on a station.
	STR1	Stretch function for side 1 of the lifting surface groups.
	STR2	Stretch function for side 2 of the lifting surface groups.
	STR3	Stretch function for side 3 of the lifting surface groups.
	STR4	Stretch function for side 4 of the lifting surface groups.
		= 0 uniform distribution
		= 1 hyperbolic tangent stretching, one end specified
		= 2 exponential stretching, one end specified
		= 3 hyperbolic sine stretching, one end specified
		= 4 geometric stretching, one end specified
		= 5 hyperbolic tangent stretching with spacing specified at two ends.
(6)	DF1,DF2, DF3,DF4	Size of the first panel on side 1, 2, 3 and 4 in percent of LPP.
(6)	DL1,DL2, DL3,DL4	Size of the last panel on side 1, 2, 3 and 4 in percent of LPP.

Notes

- (1) GRNO, STATION and POINT must always be specified.
- (2) The major difference between a body and a lift group is the ordering of data in the offset file. Points are ordered along lines in the cord direction on a lift group, e.g. along lines almost parallel to the flow direction. Points in a body group on the other hand is order along frames, which are nearly perpendicular to the flow. See “Offset file format” on page 305
- (3) A surface-piercing LIFT group can be excluded from the generation of the intersection line between the body and the free-surface. The keyword NFSINCLUDE must then be included on the LIFT command. The default value for all groups is FSINCLUDE.
- (4) The suction side of a symmetric profile can easily be generated from the coordinates of the pressure side by the YMIR command.
- (5) The two LIFT groups that form the lifting body must have the same number of points and stations.
- (6) DF1 and DL1 must be 0.0 or left out if STR1 = 0.

Only one of DF1 and DL1 must be specified if STR1 = 1, 2, 3 or 4. The mesh is clustered towards the first station if DF1 is specified and towards the last station if DL1 is specified.

Both DF1 and DL1 must be specified if IGRID = 5.

The above is valid also along side 2, 3, 4 and in the three intervals in the x-direction on the free-surface groups.

The input for side 1 is used as default for side 3.

The input for side 2 is used as default for side 4.

3.4. BODY

This command is used to specify panels for hull surfaces.

General form of command

```

BODY (  FSINCL      ,  GENE      ,  HIGHER      ,
        NFSINCL     ,  READ      ,  FIRST      ,

        ONEINT       ,  EXPANEL = n      ,
        TWOINT

        GRNO = n      ,  OFFSETGROUP = "c" ,  XTRA = v      ,
        YTRA = v      ,  ZTRA = v      ,  ZROT = v      ,
        XROT = v      ,  YROT = v      ,  XSCA = v      ,
        YSCA = v      ,  ZSCA = v      ,  YMIR      ,
        STATION = n   ,  POINT = n      ,  VELBC = v      ,
        STR1 = n      ,  DF1 = v      ,  DL1 = v      ,
        STR2 = n      ,  DF2 = v      ,  DL2 = v      ,
        STR3 = n      ,  DF3 = v      ,  DL3 = v      ,
        STR4 = n      ,  DF4 = v      ,  DL4 = v      )

```

Default values

BODY (GENE, HIGHER, FSINCLUDE, ONEINT, EXPANEL = 2 or 0, VELBC = 0, XTRA = 0.0, YTRA = 0.0, ZTRA = 0.0, XROT = 0.0, YROT = 0.0, ZROT = 0.0, STR1 = 0, DF1 = 0.0, DL1 = 0.0, STR2 = 0, DF2 = 0.0, DL2 = 0.0, STR3 = STR1, DF3 = DF1, DL3 = DL1, STR4 = STR2, DF4 = DF2, DL4 = DL2, XSCA = 1.0, YSCA = 1.0, ZSCA = 1.0)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	GRNO	Group number. The panel groups must be numbered from 1 to the maximum number of groups, starting with the propeller group followed by the environment groups, the lift groups, the body groups, the free-surface groups, the transom free-surface groups, the strip groups, the offbody point group and last the plot groups. The group number must always be specified.

	GENE	[GENMODE] The panels are generated according to the information on the present body command.
	READ	[GENMODE] The offset points are directly used as panel corner points.
	HIGHER	[PTYPE] Parabolic panels with a linearly varying source strength are generated.
	FIRST	[PTYPE] Flat panels with constant source strength are generated.
	OFFSETGR	Label name for the points in the offset file.
(2)	FSINCLUDE	[INCLUDE] The group is included when the intersection with the free surface is generated.
(2)	NFSINCLUDE	[INCLUDE] The group is not included when the intersection with the free surface is generated.
(3)	ONEINT	[INTERSECT] XMESH will search for one intersection with the free- surface on each offset station starting above the free- surface.
(3)	TWOINT	[INTERSECT] XMESH will search for two intersections with the free- surface on each offset station starting above the free- surface.
(4)	EXPANEL	Number of additional panels above the free-surface in the girthwise direction.
	XTRA	The offset points are moved the distance XTRA in the x-direction for this group. XTRA is given in the offset scale.
	YTRA	The offset points are moved the distance YTRA in the y-direction for this group. YTRA is given in the offset scale.
	ZTRA	The offset points are moved the distance ZTRA in the y-direction for this group. ZTRA is given in the offset scale.
	XROT	The offset points are rotated XROT degrees about the x-axis for this group.
	YROT	The offset points are rotated YROT degrees about the y-axis for this group.
	ZROT	The offset points are rotated ZROT degrees about the z-axis for this group.
	XSCA	The x-coordinates in the offset file are scaled a factor XSCA.
	YSCA	The y-coordinates in the offset file are scaled a factor YSCA.

	ZSCA	The z-coordinates in the offset file are scaled a factor ZSCA.
	YMIR	The offset points are mirrored in the plane $y = 0.0$.
(1)	STATION	Number of stations.
(1)	POINT	Number of points on a station.
	STR1	Stretch function for side 1 of the body groups.
	STR2	Stretch function for side 2 of the body groups.
	STR3	Stretch function for side 3 of the body groups.
	STR4	Stretch function for side 4 of the body groups. = 0 uniform distribution = 1 hyperbolic tangent stretching, one end specified = 2 exponential stretching, one end specified = 3 hyperbolic sine stretching, one end specified = 4 geometric stretching, one end specified = 5 hyperbolic tangent stretching with spacing specified at two ends.
(5)	DF1,DF2, DF3,DF4	Size of the first panel on side 1, 2, 3 and 4 in percent of Lpp.
(5)	DL1,DL2, DL3,DL4 VELBC	Size of the last panel on side 1, 2, 3 and 4 in percent of Lpp. The velocity through the panel in the panel normal direction. VELBC must be normalized by the free stream velocity, a negative value means inflow through the surface and a positive value means out flow from the surface.

Notes

- (1) GRNO, STATION and POINT must always be specified in normal input mode.
- (2) A surface-piercing BODY group can be excluded from the generation of the intersection line between the body and the free-surface. The keyword NFSINCLUDE must then be included on the BODY command. The default value for all groups is FSINCLUDE.

The NFSINCLUDE keyword may be used for the inner part of the hull when a tunnel- stern configuration is computed. A dense distribution of offset-stations on the stern is required for a tunnel-stern configuration. No interpolation is performed between offset stations in the x-direction. An offset station is either included or totally excluded (above the free-surface) when the hull panels are generated.

- (3) It is sometimes necessary to find two intersections with the free-surface in order to have a good panelization of the stern region for twin screw tunnel sterns and for heeled sailing yachts. Two intersections will searched for if TWOINT is included,

ONEINT is default.

- (4) The BODY groups may be extended a distance above the wavy free-surface or they may be cut at the intersection between the BODY group and the free-surface level. Hulls with a flat transom stern may be better modelled if the hull is continued to a distance above the free-surface. The part above the free-surface will also ensure that there is a width of the transom stern if a TRANSOM group is specified behind a flat stern. It is also possible to compute the wave profile along the waterline from the hull surface pressure if the hull is extended above the free-surface. A procedure that is expected to be more accurate than to extrapolate the wave height from the free-surface panels to the hull.

BODY groups extended above the free-surface can be used in non-linear computations, but not in linear computations based on a double-model solution. The extended BODY groups are included as default in Standard Case input mode but the number of panels above the free-surface in the girthwise direction must be specified for each BODY group in Normal input mode to include the part above the free-surface.

The BODY groups are extended to a distance $ZFACT \cdot ZRAISE$ above the free-surface. The default value for ZFACT is 0.75. The BODY groups will be cut at the free-surface level if $ZFACT \cdot ZRAISE$ is smaller than 0.001. ZFACT can be specified on the CONTROL command in XPAN. ZRAISE is the distance between the free-surface and the free-surface source panels. The default value of ZRAISE is computed by the program.

The keyword EXPANEL on the BODY command specifies the number of panels in the girthwise direction above the free-surface. If EXPANEL = 0 the BODY group is cut at the free-surface level. The default value of EXPANEL is 2 in Standard Case input mode and 0 in Normal input mode.

The hull should not be extended above the free-surface for high speed vessels, $Fn > 0.6$.

The free-surface panels are attached to the intersection between the free-surface and the BODY groups specified on IBD2 and IBD4 unless the keyword NFSINCLUDE is specified on the BODY command. If the BODY groups are extended above the free-surface level the free-surface panels will be attached to the intersection between the free-surface and the BODY groups for $x/L_{pp} < 0.6$ and to the vertical projection of upper edge of the extended panels aft of $x/L_{pp} = 0.6$. This approach will give a gap between the hull and the free-surface for sterns with a flare but this is necessary to avoid singularity problems for the potential-flow method.

See also ZFACT and ZRAISE on the CONTROL command for XPAN.

- (5) DF1 and DL1 must be 0.0 or left out if STR1 = 0.

Only one of DF1 and DL1 must be specified if STR1 = 1, 2, 3 or 4. The mesh is clustered towards the first station if DF1 is specified and towards the last station if DL1 is specified.

Both DF1 and DL1 must be specified if IGRID = 5.

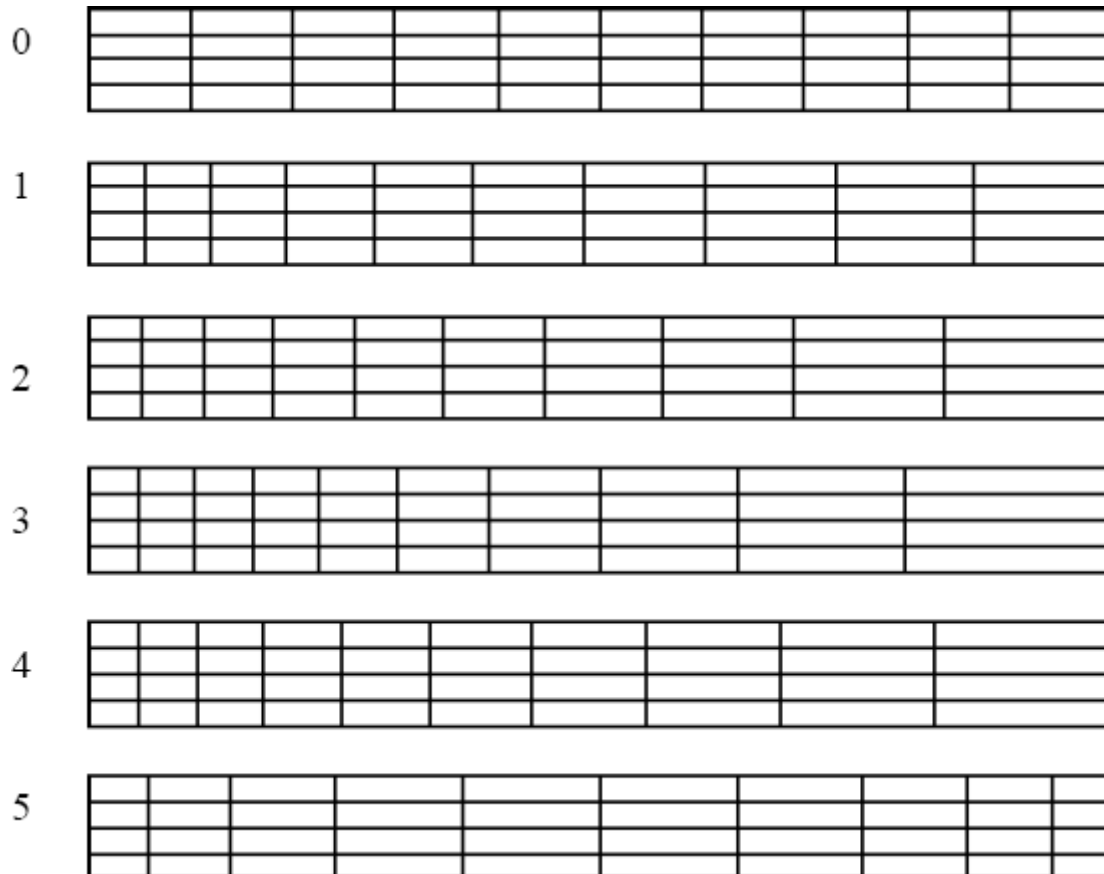
The above is valid also along side 2, 3, 4 and in the three intervals in the x-direction

on the free-surface groups.

The input for side 1 is used as default for side 3.

The input for side 2 is used as default for side 4.

The effect of selecting different stretch functions for distributing station or points with the STR1 to STR4 keys is illustrated in the figure below. The number of the stretch function is shown to the left of the corresponding mesh. DF2 is 0.05 in mesh 1 to 5 and DL2 is 0.05 for mesh 5.



3.5. FREE

This command is used to specify panels on the free-surface.

General form of command

```

FREE  (  GRNO = n           ,  GENE
      READ                  ,  HIGHER
                              FIRST
      ,
      OFFSETGROUP = "c"    ,  POINT = n      ,  STR1 = n      ,
      DF1 = v              ,  DL1 = v        ,  IBD2 = [n1, ... ,nn] ,
      IBD4 = [1, ... ,nn]  ,  IBDE = [n1]    ,
      Y2SIDE = v          ,  Y4SIDE = v      ,  SMOOTH = v      ,
      XU1 = v             ,  YU1 = v        ,  XU2 = v        ,
      YU2 = v             ,  XU3 = v        ,  YU3 = v        ,
      XU4 = v             ,  YU3 = v        ,  XD1 = v        ,
      YD1 = v             ,  XD2 = v        ,  YD2 = v        ,
      XD3 = v             ,  YD3 = v        ,  XD4 = v        ,
      YD4 = v             ,  XUPS = v        ,  XBOW = v        ,
      XSTE = v            ,  XDOW = v        ,  STAU = n      ,
      STAM = n            ,  STAD = n        ,  STRU = n      ,
      STRM = n            ,  STRD = n        ,  DFU = v        ,
      DFM = v             ,  DFD = v        ,  DLU = v        ,
      DLM = v             ,  DLD = v

```

Default values

```

FREE  (GENE, FIRST, SMOOTH = 10, Y2SIDE = 0.0, Y4SIDE = -0.7, XUPS = -0.5,
      XBOW = 0.0, XSTE = 1.0, XDOW = 2.0 STR1 = 1, DF1 = 0.02, DL1 = 0.0,
      IBD2 = 0, IBD4 = 0, STRU = 0, DFU = 0.0, DLU = 0.0,   STRM = 0,
      DFM = 0.0, DLM = 0.0, STRD = 0, DFD = 0.0, DLD = 0.0)

```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1), (2)	GRNO	Group number. The panel groups must be numbered from 1 to the maximum number of groups, starting with the propeller group followed by the environment groups, the lift groups, the body groups, the free-surface groups, the transom free-surface groups, the strip groups, the offbody point group and last the plot groups. The group number must always be specified.
	GENE	[GENMODE] The panels are generated according to the information on the present free command.
	READ	[GENMODE] The offset points are directly used as panel corner points.
	HIGHER	[PTYPE] Parabolic panels with a linearly varying source strength are generated.
	FIRST	[PTYPE] Flat panels with constant source strength are generated.
	OFFSETGR	Label name for the points in the offset file. Specified in manual input mode, MANU.
(1)	POINT	Number of points on a station.
	STR1	Stretch function for side 1 of the free groups. = 0 uniform distribution = 1 hyperbolic tangent stretching, one end specified = 2 exponential stretching, one end specified = 3 hyperbolic sine stretching, one end specified = 4 geometric stretching, one end specified = 5 hyperbolic tangent stretching with spacing specified at two ends.
(3)	DF1	Size of the first panel on side 1 in percent of Lpp.
(3)	DL1	Size of the last panel on side 1 in percent of Lpp.
(4)	IBD2	The group numbers for the body groups that are cut by the free-surface at side 2 of this free-surface.
(4)	IBD4	The group numbers for the body groups that are cut by the free-surface at side 4 of this free-surface.

IBDE	Used for hulltype YACHT. Group number of an additional BODY group where the free-surface side 2 or 4 are to be attached at the stern of a heeled yacht having its max length aside of the centre line. The FREE group is attached to the second intersection of the BODY group.
Y2SIDE	y-coordinate of side 2 if NBD2 and IBD2 are not specified.
Y4SIDE	y-coordinate of side 4 if NBD4 and IBD4 are not specified.
SMOOTH	Number of smoothing iterations for the free surface mesh (10 is default). The smoothing is not used if SMOOTH = 0.
XU1,YU1	x and y coordinate of the most upstream point on side 2 of the free-surface.
XU2,YU2	x and y coordinate for an additional point to be included in the definition of side 2 of the free-surface upstream of the bow.
XU3,YU3	x and y coordinate for an additional point to be included in the definition of side 4 of the free-surface upstream of the bow.
XU4,YU4	x and y coordinate of the most upstream point on side 4 of the free-surface.
XD1,YD1	x and y coordinate for an additional point to be included in the definition of side 2 of the free-surface downstream of the stern.
XD2,YD2	x and y coordinate of the most downstream point on side 2 of the free-surface.
XD3,YD3	x and y coordinate of the most downstream point on side 4 of the free-surface.
XD4,YD4	x and y coordinate for an additional point to be included in the definition of side 4 of the free-surface downstream of the stern.
XUPS	x-coordinate for the upstream boundary of the upstream part of the free-surface.
XBOW	x-coordinate for the boundary between the upstream and middle parts of the free-surface.
XSTE	x-coordinate for the boundary between the middle and downstream parts of the free-surface.

	XDOW	x-coordinate for the downstream boundary of the downstream part of the free-surface.
(1)	STAU	Number of stations from XUPS to XBOW.
(1)	STAM	Number of stations from XBOW to XSTE.
(1)	STAD	Number of stations from XSTE to XDOW.
	STRU	Stretch function between XUPS and XBOW.
	STRM	Stretch function between XBOW and XSTE.
	STRD	Stretch function between XSTE and XDOW.
	DFU	Size of the first panel in the interval XUPS to XBOW.
	DFM	Size of the first panel in the interval XBOW to XSTE.
	DFD	Size of the first panel in the interval XSTE to XDOW.
	DLU	Size of the last panel in the interval XUPS to XBOW.
	DLM	Size of the last panel in the interval XBOW to XSTE.
	DLD	Size of the last panel in the interval XSTE to XDOW.

Notes

- (1) GRNO, STAU, STAM, STAD and POINT must always be specified in normal input mode.
- (2) All input on the FREE command must be non-dimensionalized by Lpp.
- (3) DF1 and DL1 must be 0.0 or left out if STR1 = 0.

Only one of DF1 and DL1 must be specified if STR1 = 1, 2, 3 or 4. The mesh is clustered towards the first station if DF1 is specified and towards the last station if DL1 is specified.

Both DF1 and DL1 must be specified if IGRID = 5.

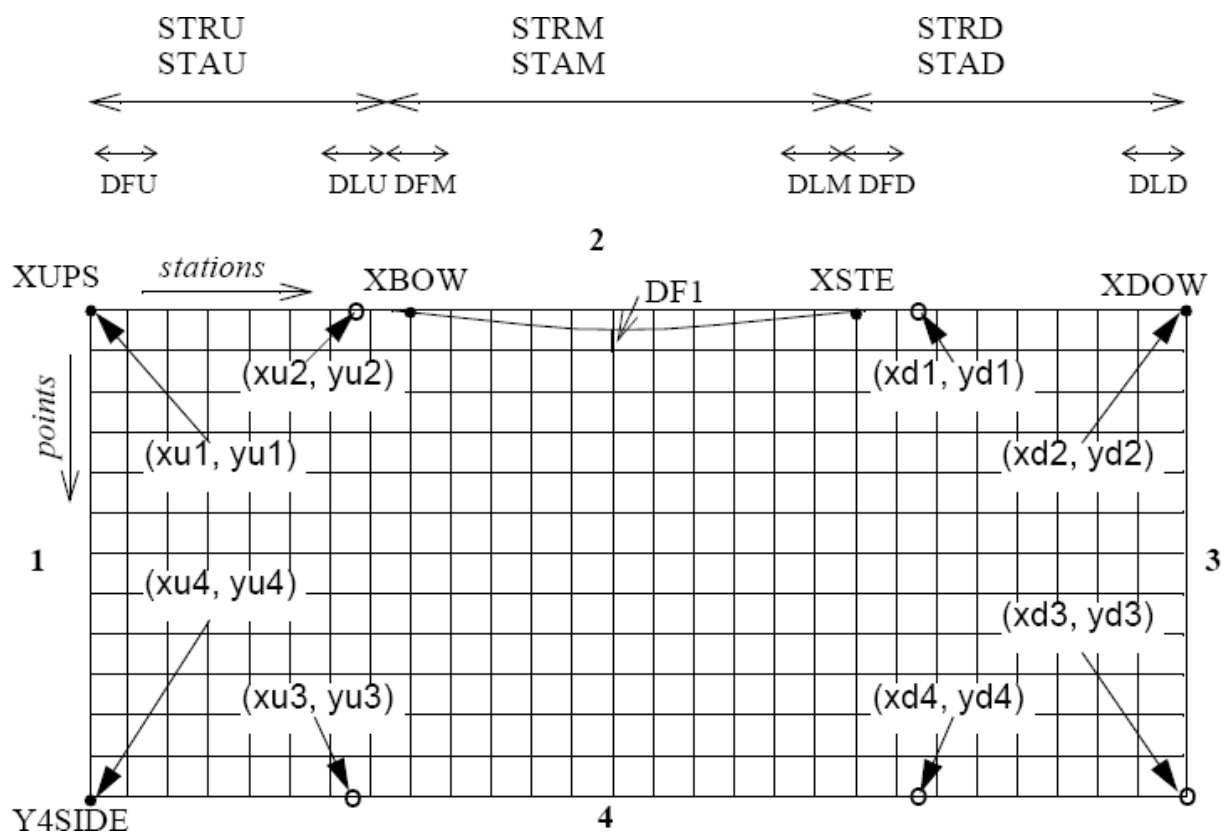
The input for side 1 is used as default for side 3.

The input for side 2 is used as default for side 4.

See the figure below for definitions of some of the keywords.

- (4) The free-surface panels are attached to the intersection between the free-surface and the BODY groups specified on IBD2 and IBD4 unless the keyword NFSINCLUDE is specified on the BODY command. If the BODY groups are modelled above the free-surface level the free-surface panels will be attached to the intersection between

the free-surface and the BODY groups for $x/L_{pp} < 0.6$ and to the vertical projection of the intersection line above the free-surface aft of $x/L_{pp} = 0.6$. This approach will give a gap between the hull and the free-surface for sterns with a flare but this is necessary for the potential-flow method.



3.6. TRANSOM

This command specifies that a transom stern case is to be computed. In the longitudinal direction the hull will be cut at the last offset station that has a point below the free-surface. A TRANSOM group will be generated on the free-surface downstream of the transom. The TRANSOM command can be used for high speed vessels, sailing yachts, tunnel sterns and for flat overhang sterns. For slow and medium speed vessels it is recommended to model the hull also above the free-surface, see EXPANEL on the BODY command in XMESH and ZFACT on the CONTROL command in XPAN. Use one TRANSOM group only downstream of a hull. The TRANSOM group can be connected to several BODY groups.

```
TRANSOM (  GENE      ,  HIGHER      ,  TRXDIR      ,  
           READ      ,  FIRST      ,  TRXYZDIR     ,  
                                           TRHBUT  
  
           GRNO = n   ,  OFFSETGROUP = "c" ,  POINT = n     ,  
  
           STR1 = n   ,  DF1 = v           ,  DL1 = v       ,  
  
                                           IBD1 = [n1, ... ,nn] ,  STAD = n     ,  
  
           DFD = v    ,  DLD = v           ,  STRD = n      )
```

General form of command

Default values

```
TRANSOM (  GENE, FIRST, STR1 = 0, DF1 = 0.0, DL1 = 0.0, IBD1 = 0,  
  
           STRD = 0, DFD = 0.0, DLD = 0.0, TRXDIR)
```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	GRNO	Group number. The panel groups must be numbered from 1 to the maximum number of groups, starting with the propeller group followed by the environment groups, the lift groups, the body groups, the free-surface groups, the transom free-surface groups, the strip groups, the offbody point group and last the plot groups. The group number must always be specified.
	GENE	[GENMODE] The panels are generated according to the information on the present transom command.

	READ	[GENMODE] The offset points are directly used as panel corner points.
	HIGHER	[PTYPE] Parabolic panels with a linearly varying source strength are generated.
	FIRST	[PTYPE] Flat panels with constant source strength are generated.
	TRXDIR	[WEDMODE] The wave profile will continue horizontally after the transom. Only used here in Manual mode. In Automatic mode this option can be specified in the HULLTYPE command.
	TRXYZDIR	[WEDMODE] The wave profile will follow the slope of the wedge after the transom. Only used here in Manual mode. In Automatic mode this option can be specified in the HULLTYPE command.
	TRHBUT	[WEDMODE] The wave profile can take any direction after the transom. Only used here in Manual mode. In Automatic mode this option can be specified in the HULLTYPE command.
	OFFSETGR	Label name for the points in the offset file. Specified in manual input mode, MANU.
(1)	POINT	Number of points on a station.
	STR1	Stretch function for side 1 of the transom groups. = 0 uniform distribution = 1 hyperbolic tangent stretching, one end specified = 2 exponential stretching, one end specified = 3 hyperbolic sine stretching, one end specified = 4 geometric stretching, one end specified = 5 hyperbolic tangent stretching with spacing specified at two ends.
(2)	DF1	Size of the first panel on side 1 in percent of Lpp.
(2)	DL1	Size of the last panel on side 1 in percent of Lpp.
(1)	IBD1	The group number for the body groups connected to side 1 of this transom free-surface group.
(1), (3)	STAD	Number of stations from XSTE to XDOW.
	STRD	Stretch function between XSTE and XDOW.
	DFD	Size of the first panel in the interval XSTE to XDOW.
	DLD	Size of the last panel in the interval XSTE to XDOW.

Notes

- (1) GRNO, STAD, POINT, NBD1 and IBD1 must always be specified in normal input mode. Use one TRANSOM group only behind each hull.

- (2) DF1 and DL1 must be less than the inverse of the number of panels along side 1.

DF1 and DL1 must be 0.0 or left out if STR1 = 0.

Only one of DF1 and DL1 must be specified if STR1 = 1, 2, 3 or 4. The mesh is clustered towards the first station if DF1 is specified and towards the last station if DL1 is specified.

Both DF1 and DL1 must be specified if IGRID = 5.

The input for side 1 is used as default for side 3.

The input for side 2 is used as default for side 4.

- (3) STAD must be set to the same number as STAD for the FREE group aside of the present TRANSOM group.

3.7. FSFAR

This command is used to define the far-field part of the free-surface and to specify the number of panels to display far-field waves. The XPAN command TWCUT must also be included in order to compute the far-field waves.

General form of command

```
FSFAR ( GRNO = n          , POINT = n          , STR1 = n          ,
        DF1 = v            , DL1 = v            , Y2SIDE = v          ,
        Y4SIDE = v         , XU1 = v            , XU2 = v            ,
        XD1 = v            , XD2 = v            , STAU = n            ,
        STAM = n           , STAD = n           , STRU = n           ,
        STRM = n           , STRD = n           , DFU = v         ,
        DFM = v            , DFD = v            , DLU = v         ,
        DLM = v            , DLD = v            )
```

Default values

```
FSFAR (Y2SIDE = 0.0, Y4SIDE = -1.4, XU1 = 2.5, XU2 = 3.5, XD1 = 4.5,
        XD2 = 5.5, STR1 = 0, DF1 = 0.0, DL1 = 0.0, STRU = 0, DFU = 0.0,
        DLU = 0.0, STRM = 0, DFM = 0.0, DLM = 0.0, STRD = 0, DFD = 0.0,
        DLD = 0.0)
```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1), (2)	GRNO	Group number. The panel groups must be numbered from 1 to the maximum number of groups, starting with the propeller group followed by the environment groups, the lift groups, the body groups, the free-surface groups, the transom free-surface groups, the far-field group, the strip groups, the offbody point group and last the plot groups. The group number must always be specified.
(1)	POINT	Number of points on a station.
(3)	STR1	Stretch function for side 1 of the free groups. = 0 uniform distribution = 1 hyperbolic tangent stretching, one end specified = 2 exponential stretching, one end specified

		= 3 hyperbolic sine stretching, one end specified = 4 geometric stretching, one end specified = 5 hyperbolic tangent stretching with spacing specified at two ends.
(3)	DF1	Size of the first panel on side 1 in percent of Lpp.
(3)	DL1	Size of the last panel on side 1 in percent of Lpp.
	Y2SIDE	y-coordinate of side 2
	Y4SIDE	y-coordinate of side 4
	XU1	x coordinate for the start of the upstream part of the far-field group.
	XU2	x coordinate for the start of the mid part of the far-field group.
	XD1	x coordinate for the start of the downstream part of the far-field group.
	XD2	x coordinate for the end of the downstream part of the far-field group.
(1)	STAU	Number of stations from XU1 to XU2.
(1)	STAM	Number of stations from XU2 to XD1.
(1)	STAD	Number of stations from XD1 to XD2.
(3)	STRU	Stretch function between XU1 and XU2.
(3)	STRM	Stretch function between XU2 and XD1.
(3)	STRD	Stretch function between XD1 and XD2.
(3)	DFU	Size of the first panel in the interval XU1 to XU2.
(3)	DFM	Size of the first panel in the interval XU2 to XD1.
(3)	DFD	Size of the first panel in the interval XD1 to XD2.
(3)	DLU	Size of the last panel in the interval XU1 to XU2.
(3)	DLM	Size of the last panel in the interval XU2 to XD1.
(3)	DLD	Size of the last panel in the interval XD1 to XD2.

Notes

- (1) GRNO, STAU, STAM, STAD and POINT must always be specified.

(2) All input on the FSFAR command must be non-dimensionalized by L_{pp} .

(3) DF1 and DL1 must be 0.0 or left out if STR1 = 0.

Only one of DF1 and DL1 must be specified if STR1 = 1, 2, 3 or 4. The mesh is clustered towards the first station if DF1 is specified and towards the last station if DL1 is specified.

Both DF1 and DL1 must be specified if IGRID = 5.

The input for side 1 is used as default for side 3.

The input for side 2 is used as default for side 4.

Similar for the streamwise stretch functions STRU, STRM, STRD,

3.8. STRIP

This command is used to specify the strips for the dipole distribution on a lifting surface and its trailing wake. The lifting surface is specified by two LIFT groups and one STRIP group. The STRIP group introduces the dipole distribution representing the lift force on its associated two LIFT groups, and on the trailing wake.

General form of command

```
STRIP ( GENE          , FIRS          , FINS          ,
        READ          NFIR          NFIN          ,

        GRNO = n      ,              , POINT = n      ,

        STATION = n    , OFFSETGROUP = "c" , POIW = n      ,

        STRW = n      , DFW = v          , DLW = v      ,

                          IBDS = [1, nn] , EFIRST = v    ,

        EFINAL = v    , WASTART = n      , WEXTENS = v   ,

        PKUTTA = v    , XCMIN = v        , XCMAX = v     ,

        YCMIN = v     , YCMAX = v        , ZCMIN = v     ,

        ZCMAX = v     , XROT = v          , YROT = v      ,

        ZROT = v      , TEXROT          , TEYROT        ,

        TEZROT        )
```

Default values

```
STRIP (GENE, NFIR, NFIN, STRW = 0.0, DFW = 0.0, DLW = 0.0, EFIRST = 0.0,
        EFINAL = 0.0, WASTART = 1, WEXTEN = 2.0, PKUTTA = 0.005,
        XCMIN = -1.0e6, XCMAX = 1.0e6, YCMIN = -1.0e6, YCMAX = 1.0e6,
        ZCMIN = -1.0e6, ZCMAX = 0.0, XROT = 0.0, YROT = 0.0, ZROT = 0.0,
        TEXROT = 0.0, TEYROT = 0.0, TEZROT = 0.0)
```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1), (2)	GRNO	Group number. The panel groups must be numbered from 1 to the maximum number of groups, starting with the propeller group followed by the environment groups, the lift groups, the body groups, the free-surface groups, the transom free-surface groups, the strip groups, the offbody

		point group and last the plot groups. The group number must always be specified.
	GENE	[GENMODE] The panels are generated according to the information on the present strip command.
	READ	[GENMODE] The offset points are directly used as panel corner points.
	OFFSETGR	Label name for the points in the offset file. Specified for manual input mode, MANU.
(1)	STATION	Number of stations on the attached LIFT groups.
(1)	POINT	Number of points on a station on the attached LIFT groups.
(1)	POIW	Number of points on the trailing wake.
	STRW	Stretch function for the wake downstream of the trailing edge.
	DFW	Size of the first wake panel at the trailing edge in percent of Lpp.
	DLW	Size of the last wake panel far downstream in percent of Lpp.
(1), (5)	IBDS	The group numbers of the two LIFT groups that defines the lifting surface.
(4)	EFIRST	Size in the spanwise direction of the extra-strip on the first side of the lifting surface.
(4)	EFINAL	Size in the spanwise direction of the extra-strip on the final side of the lifting surface.
	WASTART	The number of the wake point where the wake angle starts.
	WEXTENS	The downstream extension of the wake.
	PKUTTA	The distance from the trailing edge to the Kutta-point in fractions of the chord length.
(3)	XCMIN	The extra strips are not generated outside XCMIN in the x- direction.
(3)	XCMAX	The extra strips are not generated outside XCMAX in the x- direction.
(3)	YCMIN	The extra strips are not generated outside YCMIN in the y- direction.

(3)	YCMAX	The extra strips are not generated outside YCMAX in the y- direction.
(3)	ZCMIN	The extra strips are not generated outside ZCMIN in the z- direction.
(3)	ZCMAX	The extra strips are not generated outside ZCMAX in the z- direction.
	XROT	The wake panels are rotated XROT, YROT, ZROT degrees around the x, y and z axis respectively. XROT, YROT and ZROT are applied from wake station WASTART.
	YROT	“-----”-----”-----”
	ZROT	“-----”-----”-----”
	TEXROT	The wake panels are rotated TEXROT, TEYROT, TEZROT degrees around the x, y and z axis respectively. The rotations are applied at the trailing edge.
	TEYROT	“-----”-----”-----”
	TEZROT	“-----”-----”-----”
	FIRS	[VTXFIR] A symmetry condition is used for the bound vortex at the first side of the lifting surface when the lift force and induced drag is computed.
	NFIR	[VTXFIR] No symmetry condition is used for the bound vortex at the first side of the lifting surface when the lift force and induced drag is computed.
	FINS	[VTXFIN] A symmetry condition is used for the bound vortex at the final side of the lifting surface when the lift force and induced drag is computed.
	NFIN	[VTXFIN] No symmetry condition is used for the bound vortex at the final side of the lifting surface when the lift force and induced drag is computed.

Notes

- (1) GRNO, STATION, POINT, POIW and IBDS must always be specified.
- (2) All input on the STRIP command must be non-dimensionalized by Lpp.
- (3) XCMIN, XCMAX, YCMIN, YCMAX, ZCMIN and ZCMAX are all specified in the offset coordinate system.
- (4) The extra-strips are used to move the tip-vortex of a lifting surface to a position other

than the tip. A typical example is from a keel tip to the centre of the keel bulb or from the keel root to the undisturbed free-surface level. EFIRST and EFINAL must be specified also in manual mode, MANU.

- (5) IBDS must be specified also in manual input mode, MANU.

3.9. OBPOINT

This command is used to specify additional points in the flow field where velocity vectors and pressure contour lines are to be plotted. The input must be specified in the offset coordinate system. Only one OBPOINT group is allowed.

General form of command

```
OBPOINT ( BOX          , GENE          , FOLLOW  ,
          CYL          , READ          , FIXED   ,

          GRNO = n    , OFFSETGROUP = "c" , IOFF = n  ,

          JOFF = n    , KOFF = n          , X1OF = v  ,

          Y1OF = v    , Z1OF = v          , X2OF = v  ,

          Y2OF = v    , Z2OF = v          , R1OF = v  ,

          R2OF = v    , ANGLE = v        , LENG = v  ,

          XDIR = v    , YDIR = v        , ZDIR = v  )
```

Default values

```
OBPOINT (GENE, BOX, FIXED, IOFF = 0, JOFF = 0, KOFF = 0,
          X1OF = 0.0, Y1OF = 0.0, Z1OF = 0.0, X2OF = 0.0, Y2OF = 0.0,
          Z2OF = 0.0, R1OF = 0.0, R2OF = 0.0, ANGLE = 0.0, LENG = 0.0,
          XDIR = 0.0, YDIR = 0.0, ZDIR = 0.0)
```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(2)	GRNO	Group number. The panel groups must be numbered from 1 to the maximum number of groups, starting with the propeller group followed by the environment groups, the lift groups, the body groups, the free-surface groups, the transom free-surface groups, the strip groups, the offbody point group and last the plot groups. The group number must always be specified. Only one off-body group is allowed.
	GENE	[GENMODE] The off-body points are generated according to the information on the present OBPOINT command.
(3)	READ	[GENMODE] The offset points are directly used as

		offbody points.
	OFFSETGR	Label name for the off-body points in the offset file when the MANUAL input mode is used.
(3)	BOX	[OBPTYP] Instructs the program to generate off-body points in a box defined by the keywords X1OF, Y1OF, Z1OF and X2OF, Y2OF, Z2OF.
(3)	CYL	[OBPTYP] CYL instructs the program to generate offbody points in a cylinder defined by the keywords X1OF, Y1OF, Z1OF, R1OF, R2OF, ANGL, LENG, XDIR, YDIR and ZDIR.
	FIXED	[FIXFRE] The off-body points are kept in a fixed position during sinkage and trim.
	FOLLOW	[FIXFRE] The offbody points moves with the hull if sinkage and trim is allowed.
(1),(3)	IOFF	Number of offbody points in the <i>i</i> -direction. <i>i</i> is the axial direction for a cylinder.
(1),(3)	JOFF	Number of offbody points in the <i>j</i> -direction. <i>j</i> is the circumferential direction for a cylinder.
(1),(3)	KOFF	Number of offbody points in the <i>k</i> -direction. <i>k</i> is the radial direction for a cylinder.
(1)	X1OF,	<i>x</i> , <i>y</i> , and <i>z</i> coordinate for the first corner
(1)	Y1OF,	(the origin of the <i>i</i> , <i>j</i> , <i>k</i> system) when the
(1)	Z1OF	offbody points are generated as a BOX.
(1)	X2OF,	<i>x</i> , <i>y</i> and <i>z</i> coordinate for the second (diagonal)
(1)	Y2OF,	corner point when the offbody points are
(1)	Z2OF	generated as a BOX.
(1)	X1OF,	<i>x</i> , <i>y</i> , and <i>z</i> coordinate for the centre of the
(1)	Y1OF,	base plane when the offbody points are
(1)	Z1OF	generated as a CYLINDER.
	R1OF	Inner radius when the offbody points are generated as a CYLINDER.

(1)	R2OF	Outer radius when the offbody points are generated as a CYLINDER.
(1)	ANGLE	The circumferential extension in degrees when the offbody points are generated as a CYLINDER.
(1)	LENG	The length of the cylinder when the offbody points are generated as a CYLINDER.
(1)	XDIR,	Direction cosines for the cylinder axis when
(1)	YDIR,	the offbody points are generated as a CYLINDER.
(1)	ZDIR	

Notes

- (1) The following keywords are mandatory if the off-body points are generated as a BOX: GRNO, IOFF, JOFF, KOFF, X1OF, Y1OF, Z1OF, X2OF, Y2OF and Z2OF.
- (1) The following keywords are mandatory if the off-body points are generated as a CYLinder: GRNO, IOFF, JOFF, KOFF, X1OF, Y1OF, Z1OF, R2OF, ANGLE, LENG, XDIR, YDIR and ZDIR.
- (1) The following keywords are mandatory if the off-body points are input in MANUAL mode: GRNO, IOFF, JOFF and KOFF.
- (2) All input on the OBPOINT command must be in the offset coordinate system.
- (3) The group type OBPOINT is specified when the velocities and pressure are to be displayed in specified points in the flowfield. The offbody points must always form a topological box. The *i*-, *j*- and *k*-directions must form a right handed system in space.

The off-body points are displayed in the post-processor as a block of $(n_i-1)*(n_j-1)*(n_k-1)$ elements. Since at least one element must be formed, n_i , n_j , n_k must be at least two.

The offbody points may be input in manual mode or be generated automatically from the input on the group command. The offbody points must be included in the offset file if the manual mode is selected.

In manual model the lattice of off-body points must be given in the order indicated by the following pseudo program:

```

do i = 1 .. ni
  do j = 1 .. nj
    do k = 1 .. nk
      (read off-body point coordinates:
       xijk, yijk, zijk {and status} )
    enddo
  enddo
enddo

```

3.10. PLOT

This command is used to specify panels that are included to plot parts of the geometry that may be of interest. The PLOT panels are not active in the computation. A typical example is to plot the shape of the hull including the part above the free-surface.

General form of command

```
PLOT (  FIXED      ,  GENE      ,  HIGHER      ,
        FOLLOW      READ      FIRST
        GRNO = n    ,  OFFSETGROUP = "c" ,  XTRA = v    ,
        YTRA = v    ,  ZTRA = v          ,  ZROT = v    ,
        XROT = v    ,  YROT = v          ,  XSCA = v    ,
        YSCA = v    ,  ZSCA = v          ,  YMIR      ,
        STATION = n ,  POINT = n        ,              ,
        STR1 = n    ,  DF1 = v           ,  DL1 = v    ,
        STR2 = n    ,  DF2 = v           ,  DL2 = v    ,
        STR3 = n    ,  DF3 = v           ,  DL3 = v    ,
        STR4 = n    ,  DF4 = v           ,  DL4 = v    )
```

Default values

```
ENVI (GENE, HIGHER, FIXED, XTRA = 0.0, YTRA = 0.0, ZTRA = 0.0,
      XROT = 0.0, YROT = 0.0, ZROT = 0.0, STR1 = 0, DF1 = 0.0, DL1 = 0.0,
      STR2 = 0, DF2 = 0.0, DL2 = 0.0, STR3 = STR1, DF3 = DF1, DL3 = DL1,
      STR4 = STR2, DF4 = DF2, DL4 = DL2, XSCA = 1.0, YSCA = 1.0,
      ZSCA = 1.0)
```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1),(2)	GRNO	Group number. The panel groups must be numbered from 1 to the maximum number of groups, starting with the propeller group followed by the environment groups, the lift groups, the body groups, the free-surface groups, the transom free-surface groups, the strip groups, the offbody point group and last the plot groups. The group number must always be specified.
	GENE	[GENMODE] The panels are generated according to the

		information on the present plot command.
	READ	[GENMODE] The offset points are directly used as panel corner points.
	HIGHER	[PTYPE] Parabolic panels with a linearly varying source strength are generated.
	FIRST	[PTYPE] Flat panels with constant source strength are generated.
	OFFSETGR	Label name for the points in the offset file.
	FIXED	[FIXFRE] The plot panels are kept in a fixed position during sinkage and trim.
	FOLLOW	[FIXFRE] The plot panels moves with the hull if sinkage and trim is computed.
	XTRA	The offset points are moved the distance XTRA in the x-direction for this group. XTRA is given in the offset scale.
	YTRA	The offset points are moved the distance YTRA in the y-direction for this group. YTRA is given in the offset scale.
	ZTRA	The offset points are moved the distance ZTRA in the z-direction for this group. ZTRA is given in the offset scale.
	XROT	The offset points are rotated XROT degrees about the x-axis for this group.
	YROT	The offset points are rotated YROT degrees about the y-axis for this group.
	ZROT	The offset points are rotated ZROT degrees about the z-axis for this group.
	XSCA	The x-coordinates in the offset file are scaled a factor XSCA.
	YSCA	The y-coordinates in the offset file are scaled a factor YSCA.
	ZSCA	The z-coordinates in the offset file are scaled a factor ZSCA.
	YMIR	The offset points are mirrored in the plane $y = 0.0$.
(1)	STATION	Number of stations.
(1)	POINT	Number of points on a station.
	STR1	Stretch function for side 1 of the plot groups.

	STR2	Stretch function for side 2 of the plot groups.
	STR3	Stretch function for side 3 of the plot groups.
	STR4	Stretch function for side 4 of the plot groups. = 0 uniform distribution = 1 hyperbolic tangent stretching, one end specified = 2 exponential stretching, one end specified = 3 hyperbolic sine stretching, one end specified = 4 geometric stretching, one end specified = 5 hyperbolic tangent stretching with spacing specified at two ends.
(3)	DF1,DF2, DF3,DF4	Size of the first panel on side 1, 2, 3 and 4 in percent of Lpp.
(3)	DL1,DL2, DL3,DL4	Size of the last panel on side 1, 2, 3 and 4 in percent of Lpp.

Notes

- (1) GRNO, STATION and POINT must always be specified in normal input mode.
- (2) The PLOT groups are not active in the computation, they are only included to plot parts of the geometry that may be of interest. The PLOT groups are not cut at the free-surface i.e. all offset points are used in the panelization of the PLOT group. It may be necessary to use the Panel Category menu in the SHIPFLOW post-processor to select or deselect panel groups to be plotted. The GROUP command in FIPOST can be used to select the groups to be plotted.
- (3) DF1 and DL1 must be 0.0 or left out if STR1 = 0.

Only one of DF1 and DL1 must be specified if STR1 = 1, 2, 3 or 4. The mesh is clustered towards the first station if DF1 is specified and towards the last station if DL1 is specified.

Both DF1 and DL1 must be specified if IGRID = 5.

The above is valid also along side 2, 3, 4.

The input for side 1 is used as default for side 3.

The input for side 2 is used as default for side 4.

4. XPAN commands

This section details the commands necessary for applying the potential flow module.

4.1. CONTROL

General form of command

```
CONTROL ( SAVE          , LINEAR      , FIXED          ,  
          NOSAVE        , NONLINEAR  , FREE           ,  
                                     BERNOU  
                                     ITSOLV      , DISKSAVE       ,  
                                     GESOLV      , NODISKSAVE  
  
          EQSING = v    , EQAVFA = v    , EQCONV = v    ,  
  
          SHALLOW = v  , ZRAISE = v    , XSHIFT = v    ,  
  
          ZFACT = v    ,  
  
          DOUBLE        , AFSSHIFT      , TWO  
          NEUMAN        , UFSSHIFT      , THREE  
          SINGLE        ,                , FOUR  
                                     ANALYT  
  
          NOLAST        , CWPI          ,  
          LAST
```

Default values

CONTROL (SAVE, FIXED or FREE, LINEAR or NONLINEAR, ITSOLV, NODISKSAVE, EQSING=1.0e-5, EQAVFA=5.0e-3, EQCONV=1.0e-3, DOUBLE, FOUR, ZRAISE=FZR, ZFACT = 0.75 or 0.0, XSHIFT=0.3, AFSS, NOLAST)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	NOSAVE	[XPDB] The data base file XPDB will not be created.
(1)	SAVE	[XPDB] XPAN will save all the solution quantities that XBOUND and XCHAP need on the id_XPDB file.
(2)	FIXED	[MODEL] The ship is kept in a fixed position during the computation. Default for double model flow computations.
(2)	FREE	[MODEL] The ship is allowed to sink and trim during the computation. Default for free surface flow computations.
	SHALLOW	Shallow water effects are taken into account. The distance from the undisturbed free-surface to the bottom is

		specified by the SHALLOW keyword. The distance is input in the offset scale.
(3)	ZRAISE	The raised panel method is used as default. The distance from the free-surface to the source panels is specified by the ZRAISE keyword. ZRAISE must be specified as a factor (FZR) which is then multiplied by a typical panel size internally in the program to give the distance.
(13)	ZFACT	The BODY groups are panelized up to a distance ZFACT*ZRAISE above the wavy free-surface. See also the keyword EXPANEL on the BODY command.
(14)	BERNOU	[BC] The Bernoulli (double-model) wave is computed. No free- surface boundary conditions are applied.
(4)	LINEAR	[BC] A linearized free surface boundary condition is used. Default for double model flow computations.
(5)	NONLINEAR	[BC] The non-linear free surface boundary condition is used. Default for free surface flow computations.
(6)	DOUBLE	[BMMOD] A double model is used as the first basic model solution. The xy-plane is automatically treated as a symmetry plane for the body groups.
(6)	NEUMAN	[BMMOD] An undisturbed flow is used as the first basic model flow for the free-surface computation.
(6)	SINGLE	[BMMOD] A single model is used as the first basic model solution.
(7)	TWO	[OPERATOR] A two point operator is used to compute the velocity derivative in the streamwise direction on the free surface.
(7)	THREE	[OPERATOR] A three point operator is used to compute the velocity derivative in the streamwise direction on the free surface.
(7)	FOUR	[OPERATOR] A four point operator is used to compute the velocity derivative in the streamwise direction on the free surface.
(7)	ANALYT	The velocity derivatives are computed from analytical expressions and the collocation points are shifted one panel length upstream.
(8)	XSHIFT	The free-surface collocation points are shifted a distance XSHIFT times a typical panel size in the x-direction. XSHIFT must be specified as a positive factor.
(8)	AFSSHIFT	[SHIFT] All collocation points on the free surface is shifted a distance XSHIFT times a typical panel size.

(8)	UFSSHIFT	[SHIFT] Only collocation points upstream of FP are shifted a distance XSHIFT times a typical panel size.
(10)	ITSOLV	[SOLVER] An iterative solution procedure is used to solve the system of equations.
(10)	GESOLV	[SOLVER] A Gauss-elimination is used to solve the system of equations.
(11)	DISKSAVE	[DISK] The influence coefficients are stored on temporary disk during the execution.
(11)	NODISKSAVE	[DISK] The influence coefficients are re-computed during the execution. No temporary disk is used for the influence coefficients during the execution.
	EQSING	Parameter used to determine if the system of equation is singular when the iterative solver is used.
(12)	EQAVFA	Parameter that governs the pre-conditioning step before the iterative solver.
	EQCONV	Convergence criteria for the iterative solver.
	NOLAST	[CGIT] Save all iterations to CGNS file.
	LAST	[CGIT] Save only the last iteration to CGNS file.
	CWPI	Print CW based on the pressure integration to id_OUTPUT file.

Notes

- (1) Once the XPDB file has been created, there is no need to perform a new potential flow computation for each new run of XBOUND or XCHAP. (XPAN can thus be removed from the PROGRAM command.
- (2) The program integrates the hydrodynamic forces and adjusts the sinkage and trim to balance these forces if FREE is specified. The number of sinkage and trim iterations to be performed is controlled by the ITERATION and CONVERGENCE commands. Some cases shows an oscillatory or divergent behaviour. It is then necessary to introduce an under-relaxation for the sinkage, trim and/or wave height change. The under-relaxation is introduced on the RELAXATION command.
- (3) The raised panel method is used as default for both linear and non-linear computations. The distance between the free-surface and the source panels is computed as $FZR * (1 / (STAM - 1))$. The default value of FZR is $1.0 - 1.25 * F_n$ and the minimum value of FZR is 0.2. The default value can be used in most cases, but if the panel aspect ratio (dx/dy) is large as often occurs for high speed transom stern cases it may be necessary to reduce FZR to smaller values than 0.2. The value of FZR is

then specified by the ZRAISE keyword.

- (4) The non-linear terms in the boundary conditions are assumed to be small and can therefore be neglected. In this case the free surface boundary conditions are applied on the flat free surface. The basic model solution is used as the known solution in the linearization.
- (5) A linear solution is first performed and the free-surface boundary conditions are then moved to the computed wavy free surface. The linear solution is used as the known base solution in the next iteration. This procedure is repeated until convergence is achieved. It is sometimes difficult to get a converged solution if there is a large change in the water-plane area between the iterations or if the computation is performed for a high speed. Under-relaxation must be introduced if the solution shows a divergent behavior. The iterative process is controlled with the commands ITERATION, CONVERGENCE and RELAXATION.

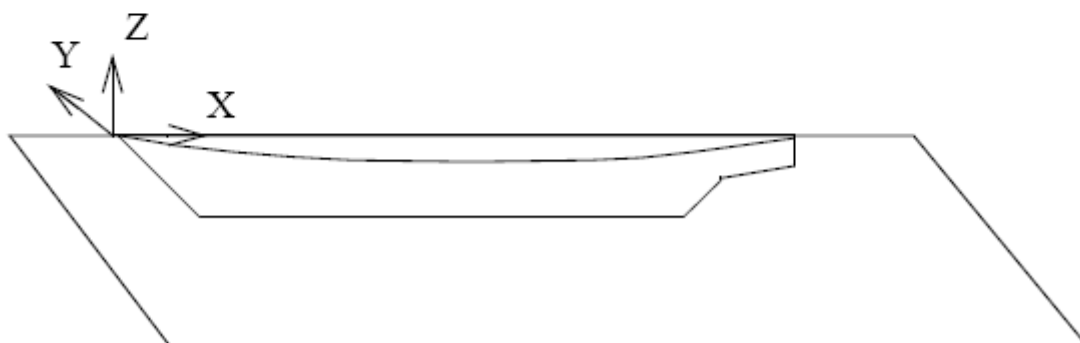
The program automatically finds the intersection between the hull and the new wavy free surface. The non-linear boundary conditions are exactly satisfied on the wavy free- surface when convergence is achieved.

- (6) The computations can be performed with or without a free surface. A flow without a free surface is called “basic model flow”. A flow with a free surface is called a “free surface flow”. The basic-model solution is also used for the first linearization of the free surface boundary conditions. There are three types of basic model computations. double-model, single-model and Neumann-Kelvin.

Double-model



Single-model



In the double-model computation a mirror image of the hull is taken into account to ensure zero flow through the water plane. Zero flow condition is exactly fulfilled for the double-model. The double-model solution is used for cases without a free surface and if the linear free surface boundary condition is used.

In the single-model computation a region of the water plane is covered with panels. A Neumann boundary condition is applied on these panels to prevent a flow through the water plane. The zero flow condition is only approximately fulfilled for the single-model. The single-model is used for the non-linear, the shallow water and the submarine option.

- (7) The operator FOUR works well in most cases and should not be changed unless there are very strong reasons to do so. The operators TWO, THREE and FOUR all have a central-difference operator in the transverse direction. The analytical method ANALYT requires an aspect ratio (dx/dy) smaller one for the free-surface panels.
- (8) A collocation point shift in the x-direction on the free-surface can be introduced to further enforce the radiation condition. The collocation points are shifted a distance $XSHIFT*(1/(STAM-1))$ in the x-direction. This type of shift is used together with the numerical operators TWO, THREE and FOUR. There are two possibilities to apply the collocation point shift. AFSSHIFT (All Free Surface Shift) specifies that all collocation points on the free surface are shifted. UFSSHIFT (Upstream Free Surface Shift) specifies that only collocation points upstream of the ship are shifted. The program default option is to use the FOUR point operator, AFSSHIFT and $XSHIFT=0.3$. The default should not be changed unless there are very strong reasons to do so.
- (10) The iterative solver is normally much faster than the Gauss-elimination solver, but it requires that the system matrix is stored in core. An estimate of the memory size required is printed in the OUTPUT file when XMESH is executed. The Gauss-elimination solver is a block solver and it can be used with a much smaller memory size available. The system matrix is on the other hand stored on temporary disk. An estimate of the disk space required is printed in the OUTPUT file when XMESH is executed.
- (11) There are two possibilities to handle the influence coefficients. They are either computed once and stored on temporary disk or they are computed two times without being stored on disk. The computing time for the two options is similar for non-linear computations while linear computations are faster if writing to disk is allowed. The disk space required for storing the influence coefficients is $24*N*N$. Where N is the total number of panels.
- (12) It is sometimes necessary to reduce the value of EQAVFA for “difficult” cases like very high Froude numbers or complicated geometries like a ship in a canal. A better but more time consuming pre-conditioning will then be performed before the iterative solver.
- (13) The BODY groups may be modelled to a distance above the free-surface or they may be cut at the intersection between the group and the free-surface. The BODY groups are modelled to a distance $ZFACT*ZRAISE$ above the free-surface. The default value for ZFACT is 0.75. The BODY groups will be cut at the free-surface level if $ZFACT*ZRAISE$ is smaller than 0.001. The number of panels in the girthwise direction above the free-surface is specified by EXPANEL on the BODY command in XMESH.
- (14) A FREE group must be defined in order to plot the Bernoulli wave in the post-processor.

4.2. CONVERGENCE

This command is used to specify convergence criteria for sinkage and trim and non-linear computations.

General form of command

CONVERGENCE (EPTRIM = v , EPSINK = v , EPWAVE = v ,
WCHMAX = v

Default values

CONVERGENCE (EPTRIM = 1.e-2, EPSINK = 1.e-5, EPWAVE = 5.e-5, WCHMAX=1.0)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	EPTRIM	Convergence tolerance for trim angle change in degrees.
(1)	EPSINK	Convergence tolerance for sinkage change.
(1)	EPWAVE	Convergence tolerance for wave height change in non-linear computations.
(2)	WCHMAX	Maximum allowed wave height change between two non-linear iterations. The computation is stopped if a larger wave height change is found.

Notes

- (1) The iteration procedure is stopped when the convergence criteria is satisfied for both the sinkage and the trim. The convergence criteria on the wave height change is added for non-linear computations. The solution continues MAXIT iterations if no converged solution is found. The maximum number of iterations, MAXIT, is specified on the ITERATION command.

The residuals of the free-surface boundary conditions are printed for each iteration in the OUTPUT file for non-linear computations. The residuals must diminish towards small values for the converged solution.

- (2) The non-linear computation is considered to be divergent if the wave height change is larger then WCHMAX. The value of WCHMAX must be non-dimensionalized by Lpp.

4.3. EXFORCE

This command is used to specify external forces acting on the hull surface.

General form of command

EXFORCE (CVFORM = v , CVLIFT = v , CVBODY = v ,
CVERT = v)

Default values

EXFORCE (CVFORM = 0.0, CVLIFT = 0.0, CVBODY = 0.0, CVERT = 0.0)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	CVFORM	Form factor for the hull.
(1)	CVLIFT	Viscous resistance coefficient for the LIFT groups.
(1)	CVBODY	Viscous resistance coefficient for the BODY groups.
(2)	CVERT	Force coefficient for a vertical external force

Notes

- (1) The influence of the viscous forces can be included in the sinkage and trim potential-flow computations if the viscous resistance coefficients are known or can be estimated. See also the EXMOMENT command.

The viscous resistance coefficients must be computed as

$$CVLIFT = \frac{Force}{0,5 * \rho * U_{\infty}^2 * S_L} \quad \text{where } S_L \text{ is the total wetted surface for the LIFT groups.}$$

$$CVBODY = \frac{Force}{0,5 * \rho * U_{\infty}^2 * S_H} \quad \text{where } S_H \text{ is the total wetted surface for the BODY groups.}$$

- (2) The influence of a vertical external force can be included in the sinkage and trim potential-flow computations. The force coefficient is defined as:

$$CVERT = \frac{Force}{0,5 * \rho * U_{\infty}^2 * S} \quad \text{where } S \text{ is the total wetted surface of the ship.}$$

See also the EXMOMENT command.

4.4. EXMOMENT

This command is used to specify the position of external forces.

General form of command

EXMOMENT (TOWX = v , TOWZ = v , TOWAGL = v ,
ZMLIFT = v , ZMBODY = v , XVERT = v)

Default values

EXMOMENT (TOWX = LCB (longitudinal centre of buoyancy as computed by the program),
TOWZ = VPoR (vertical position of the resistance as computed by the program),
TOWAGL = 0.0)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	TOWX	Longitudinal position of the towing force.
(1)	TOWZ	Vertical position of the towing force.
(1)	TOWAGL	Angle of the towing force direction with respect to the horizontal plane in degrees.
(1)	ZMLIFT	Vertical position of the viscous force from LIFT groups.
(1)	ZMBODY	Vertical position of the viscous force from BODY groups.
(2)	XVERT	Longitudinal position of the external vertical force.

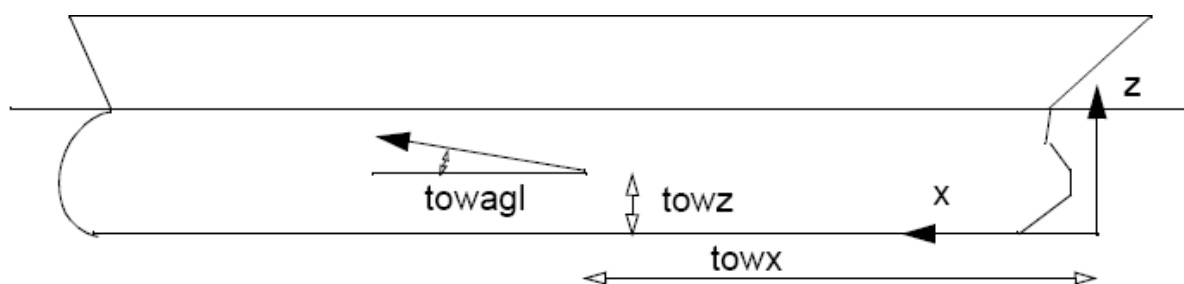
Notes

- (1) The influence of the viscous and towing forces can be included in the sinkage and trim potential-flow computations if the position of these forces are known or can be estimated. See also the EXFORCE command.

The position of the forces must be specified in the offset coordinate system.

- (2) The influence of an external vertical force can be included in the sinkage and trim potential-flow computations if the position of the forces is known. See also the EXFORCE command. The input is given in the offset coordinate system but the

output is written in the computational coordinate system.



Towing point position and direction of towing force.

The position of the forces must be specified in the offset coordinate system.

4.5. ITERATION

This command is used to specify the maximum number of iterations to be carried out for sinkage and trim and non-linear computations.

General form of command

ITERATION (MAXIT = n , IUPD = [n₁, n₂, ..., n_n])

Default values

ITERATION (MAXIT = 20, IUPD = 100*1)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	MAXIT	Maximum iteration number for sinkage and trim and/or non-linear calculations.
(2)	IUPD	Specifies when the geometry is updated in a non-linear computation.

Notes

- (1) The MAXIT keyword is active for non-linear and sinkage and trim cases. The sinkage and trim computation is performed in every non-linear iteration if FREE is specified in the CONTROL command.
- (2) It is not necessary to update the position of the source panels in each non-linear iteration when the raised panel method is used. The keywords NUMB and IUPD are used to specify if the position is to be updated or not. NUMB = 8, IUPD = [1, 1, 0, 0, 1, 0, 0, 0] means that the position is updated in the first, second and fifth iteration. The update of body geometry in non-linear computations is stopped if sinkage and trim has converged and IUPD is 0 for the present iteration.

4.6. PARALLEL

This command controls parallel execution of XPAN. The technique supported in the current version is multithreading. It means that the program still runs in one process with one memory space, but with more than one program pointer. If the hardware supports it, the different threads may be executed concurrently on different processors. The kind of hardware that can support this kind of execution is Shared Memory Processor (SMP) machines. This includes multi-processor machines and multi-core processors. Parallel computers without shared memory, like clusters, cannot use multiple threads.

If other processes are running, the maximum number of threads for a new process is $M-N$, where M is the maximum number of processes allowed by the license, and N the number in use. Here a process running with n threads counts as n processes.

General form of command

PARALLEL (NTHREADS = n)

Default values

PARALLEL (NTHREADS = adapted to your CPU and license)

Keywords

<i>keyword</i>	<i>description</i>
NTHREADS	Requested number of threads in parallel execution.

4.7. RELAXATION

This command is used to specify relaxation factors for sinkage, trim and for non-linear computations also for source strength and wave height change.

General form of command

RELAXATION (RFTRIM = v , RFSINK = v , RFSOUR = v ,
RFWAVE = v)

Default values

RELAXATION (RFTRIM = 1.0, RFSINK = 1.0, RFSOUR = 0.7, RFWAVE = 1.0)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	RFTRIM	Relaxation factor for trim angle in sinkage and trim calculations.
(1)	RFSINK	Relaxation factor for sinkage in sinkage and trim calculations.
(1)	RFSOUR	Relaxation factor for source singularity for non-linear computations.
(1)	RFWAVE	Relaxation factor for wave height change in non-linear computations.

Notes

- (1) A divergent or oscillatory behaviour is sometimes found for sinkage and trim or non-linear computations. The convergence history can be improved if relaxation is introduced. The relaxation forces the solution to move slower towards the final solution. The trim, sinkage, source strength and wave height are in every step updated according to the following formula:

$$var(k) = var(k-1) + RFvar * (var(k) - var(k-1))$$

where k is the iteration step and $RFvar$ is the value of RFTRIM, RFSINK, RFSOUR and RFWAVE.

Relaxation on the wave height, RFWAVE, is normally only used if severe oscillations occurs for the shape of the waterline in the non-linear iterations.

4.8. TWCUT

A transverse wave cut computation of wave resistance is added when TWCUT is included. This command must be used if the FSFAR group for far-field waves is included in XMESH. TWCUT can only be used for cases having a symmetry plane at the centre plane.

General form of command

```
TWCUT      (  ON                ,  XSTTWC = v   ,  XENTWC = v   ,
              STATWC = n   ,  STRTWC = n   ,  DFTWC = v   ,
              DLTWC = v   ,  NVALTW = n   ,  NWAVNU = n   ,
              YTWC = v                                     )
```

Default values

TWCUT(XSTTWC = 1.4, XENTWC = computed by the program, STATWC = 8, STRTWC= 1, DFTWC = (XENTWC-XSTTWC)/STATWC/2, NVALTW = 100, NWAVNU = 100, YTWC = min(-0.45*XD2, 1.1*Y4SI)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	ON	The wave cut computation is included using default values for all keywords
(1)	XSTTWC	Longitudinal position of the first transverse wave cut
(1)	XENTWC	Longitudinal position of the last transverse wave cut
(1)	STATWC	Number of transverse wave cuts
(1)	STRTWC	Stretch function for distributing the transverse wave cuts between XSTTWC and XENTWC (See the BODY command on XMESH.
(1)	DFTWC	Distance between the first and second transverse wave cut.
(1)	DLTWC	Distance between the last and second last transverse wave cut.
(1)	NVALTW	Number of data points generated on each transverse wave cut.
(1)	NWAVNU	The number of wave numbers included in the computation.

- | | | |
|-----|------|---|
| (1) | YTWC | Transverse extension of the transverse wave cuts. |
|-----|------|---|

Notes

- (1) The wave resistance is computed from multiple transverse wave cuts downstream of the ship. The wave cuts are distributed between XSTTWC and XENTWC. By default XENTWC is located one fundamental wave length downstream of XSTTWC. It is therefore important to use a free-surface that is large enough to capture this distance. If the free-surface is too small XENTWC will be adjusted to stay upstream of the downstream boundary. It is also important to have a free-surface that is wide enough to ensure that the wave system leaves the free-surface at the downstream boundary, not at the side of the free-surface.

The default values for the keywords should normally not be changed.

The input must be non-dimensionalized by L_{pp} .

4.9. WAVECUT

This command is used to print the wave profile along longitudinal and transverse wave cuts on the free-surface.

General form of command

WAVECUT (LWAVEC= [v₁,...,v_{nl}] , DXWAVE = v ,
TWAVEC = [v₁,...,v_{nt}] , DYWAVE = v)

Default values

No default values

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	LWAVEC	Transverse position(s) of the longitudinal wave cut
(1)	DXWAVE	Distance between data points for longitudinal cuts.
(1)	TWAVEC	Longitudinal position(s) of the transverse wave cuts
(1)	DYWAVE	Distance between data points for transverse cuts.

Notes

- (1) The longitudinal wave profiles will be printed in the file id_LWAVECUT and the transverse wave profiles will be printed in id_TWAVECUT.

LWAVEC, DXWAVE, TWAVEC and DYWAVE must be non-dimensionalized by Lpp.

5. XBOUND commands

The following commands will be interpreted by XBOUND if found in the input file between the SHIPFLOW module delimiter commands XBOUND and END.

5.1. CONTROL

This command is used to specify the mode of execution of XBOUND.

General form of command

```
CONTROL ( STREAMLINE , SAVE , FREESURFACE ,  
          NOSAVE  
  
          FILE = "c" )
```

Default values

CONTROL (SAVE)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	STREAMLINE	Streamline tracing only. Potential flow streamlines are traced as specified by the TRACE commands.
	SAVE	[XBDB] Create the data base file id_XBDB and save solution quantities for Navier-Stokes computation.
	NOSAVE	[XBDB] The data base file id_XBDB will not be created.
	FREESURFACE	XBOUND will use the linear free-surface solution instead of the double-model solution in id_XPDB. No id_XBDB file will be created. This option should only be used in combination with the linear free-surface option in XPAN.
	FILE	Name of file to store the limiting streamlines for plotting in FIPOST. Default name is XBLIMIT.

5.2. INICON

This command is used to specify initial conditions for the boundary layer computation. One command is needed for each potential flow streamline group.

General form of command

```
INICON ( SGROUP = n , TRCR = v , LAMINAR
        TURBULENT
        GIRTH = [V1, V2, ..., VPOINT] , T11 = [V1, V2, ..., VPOINT] ,
        H12 = [V1, V2, ..., VPOINT] , BETA = [V1, V2, ..., VPOINT] )
```

Default values

INICON (TURBULENT, T11 = [1.0*10⁻⁴] (for Standard Case only), H12 = [1.4] (for Standard Case only), GIRTH = [0.0], BETA = [0.0], *see note 2 for H12*)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	SGROUP	The number on the streamline group that the INICON command is referring to.
(2)	GIRTH	Girth length from the lower edge of the section to data point. GIRTH equals zero at the lower edge and one at the upper edge.
(2)	T11	Momentum thickness at data point ($T11 = \theta_{11}/L_{pp}$).
(2)	H12	Form factor at data point.
(2)	BETA	Wall cross flow angle at data point. Must be specified in degrees.
(3)	LAMINAR	[FLOW] Streamline starts at a stagnation point. The flow is assumed to be laminar up to a point of transition which is computed by the e^n -method.
(3)	TURBULENT	[FLOW] The boundary layer is assumed to be turbulent from the start.
(3)	TRCR	Transition criteria.

Notes

- (1) The program assigns a "group number" to each streamline group. This number corresponds to the order of the appearance of the equivalent TRACE command in the input file. The value of SGROUP specifies what streamline group the INICON command is referring to. Boundary layer computations are only carried out for the streamline groups that have been assigned initial conditions.
- (2) Initial conditions i.e. T11, H12 and BETA are specified at a maximum of 10 data points from the start position of the streamline group. The number of data points is specified with POINT. The transverse position of each point is specified by GIRTH. GIRTH is zero at the lower edge of the section and one at the upper side. Thus the direction of GIRTH is the same as that for the corresponding frames of the section in the offset file. SGROUP and T11 must be specified. Linear interpolation is used between the data points and the starting points of the streamlines. Constant start values can be specified as shown in the following example:

INICON (SGROUP=1, T11=[1.0*10⁻⁴], H12=[1.4])

H12 is estimated from flat plate correlations if it is not specified:

$$H12 = \frac{1,0}{1,0 - 0,5118 * (RT11)^{\frac{-1}{12}}}, \text{ where } RT11 = \frac{Rn * \theta}{Lpp} = Rn * T11$$

where RT11 is the Reynolds number based on the momentum thickness. See, *White F. M., "Viscous Fluid Flow", ch. 6-6, McGraw-Hill.*

The default value of BETA is zero.

- (3) The boundary layer calculation can be started from a point where the flow is turbulent. The momentum thickness must then be given. It can also start from a stagnation point. XBOUND then assumes a laminar flow and computes the transition point with the n^+ -method. The calculation continues, after the location of the transition point is determined, with a computation of the turbulent boundary layer. The laminar option requires a well defined stagnation point. The transition criteria TRCR is 9.0 by default.

5.3. LIMIT

The limiting streamline tracing is controlled with this command. Each LIMIT command in the command file will produce a group of limiting streamlines on the specified panel group. A maximum of 30 limiting streamline groups can be traced at the same time.

The limiting streamline tracing is carried out immediately after the boundary layer calculation. The velocities close to the hull are transformed according to the computed cross-flow angle.

All key words are the same as for the TRACE command for tracing the potential flow streamlines.

General form of command

```
LIMIT  (  GROUP = n   ,  STATION = n   ,  STREAM = n   ,
          ISTART = n   ,  PARAMETRIC   ,  IDISTR = n   ,
          S1 = v       ,  DS1 = v       ,  SN = v       ,
          DSN = v      ,  JDISTR = n    ,  P1 = v       ,
          DP1 = v      ,  PN = v       ,  DPN = v       )
```

Default values

LIMIT (no default values exist for GROUP, STATION, STREAM or ISTART, but: IDISTR = 0, S1 = 0, DS1 = 0, SN = 1, DSN = 0, JDISTR = 0, P1 = 0, DP1 = 0, PN = 1, DPN = 0)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	GROUP	Section number of the panel section on which this group of streamlines will be traced.
	STATION	Number of streamline stations along the hull for streamline tracing. (That is, the number of points per streamline.)
	STREAM	Number of streamlines in the group.
	ISTART	Index of the streamline station from which the tracing will be done. The streamline stations are counted from the bow, so downstream tracing is accomplished with ISTART = 1 and upstream tracing with ISTART = STATION. Values in between will give rise to streamlines that are traced in two directions from the streamline

station specified by ISTART.

PARAMETRIC This keyword specifies whether the start and end point of the streamlines is given in a computational coordinate system or in the parametric space. It thus determines the meaning of S1 and SN.

The values given to the keywords IDISTR ... DSN specifies the distribution of the streamline stations along the section. The values of the keywords JDISTR ... DPN controls the girthwise distribution of the streamlines at the ISTART's streamline station.

IDISTR Type of stretching function that should be used:

0 linear DS1 and DSN ignored

1 hyperbolic tangents DS1 or DSN required

2 exponential function ----- " -----

3 hyperbolic sine ----- " -----

4 geometric", partial sums of a geometric progression DS1 or DSN required

5 hyperbolic tangents DS1 and DSN required.

(1) **S1** Location of the first streamline station given in the computational coordinate system or the parametric space. See note 1.

DS1 Spacing of the streamline stations at the upstream end of the streamlines.

(1) **SN** Location of the last streamline station given in the computational coordinate system or the parametric space. See note 1.

DSN spacing of the streamline stations at downstream end of the streamline group.

JDISTR Same as IDISTR but concerning the girthwise distribution of the streamlines.

(2) **P1** Girthwise location of the keelmost streamline at the ISTART's streamline station.

(2) **DP1** Spacing of streamlines at the keelmost streamline.

(2) **PN** Girthwise location of the topmost streamline at the ISTART's station.

(2) **DPN** Spacing of streamlines at the topmost streamline.

Notes

- (1) S1 and SN must be given as absolute coordinates in the computational coordinate system or in the parametric space if the keyword PARAMETRIC is used. The value of S is 1 at the first station and equal the number of panels minus one at the last station of a panel group in the parametric space. Tracing the streamlines all the way to the end of a section that ends in a point is not a good idea. The mapping to the girth-wise arc-length coordinate becomes singular and some of the boundary layer assumptions cease to be valid.
- (2) P1, DP1, PN and DPN should be specified using a normalized arc-length coordinate that is zero at the keel most edge of the panel section and one at the topmost edge.

5.4. RESISTANCE

This command limits the region used in the computation of the total skin friction coefficient.

General form of command

RESISTANCE (XL1 = v, XL2 = v)

Default values

RESISTANCE (XL1 = -1.0e12, XL2 = 1.0e12)

Keywords

<i>keyword</i>	<i>description</i>
XL1	Lower limit of x for the total skin friction integration.
XL2	Upper limit of x for the total skin friction integration.

Notes

The total skin friction coefficient CF is integrated using the domain limited by $XL1 < x < XL2$. All hull sections are involved in the integration and every panel has been assigned a local skin friction value from the nearest streamline. This means that panels on sections where no streamlines have been started will contribute to CF. Streamlines that have been traced but not been assigned initial conditions for boundary layer computations will contribute with a local skin friction coefficient of zero.

5.5. ROUGHNESS

This command is used to specify the hull surface roughness.

General form of command

ROUGHNESS(C = v, H = v)

Default values

ROUGHNESS (C = 0.0, H = 0.0)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	C	Surface efficiency parameter
(2)	H	Dimensionless surface roughness height. Dimensionless by the reference length, i.e. $H=h/REFLEN$, where h is the average full scale roughness height. See REFLEN in the VSHIP command in the XFLOW section.

Notes

- (1) The surface roughness is defined by two parameters, h and C. h is simply the average roughness height of the roughness elements. The efficiency parameter C is obtained experimentally, and is a measure on how effective a surface roughness geometry is in terms of hydrodynamic friction. If the parameters h and C has been obtained experimentally, it is possible to scale the roughness to some extent, by changing h (i.e. barnacle surfaces for example). No universal value for C exists, though.

Some different artificial and marine fouling surfaces has been measured in [1], and C values from this paper can be used as reference.

[1] Michael Leer-Andersen, Lars Larsson. "An experimental/numerical approach for evaluating skin friction on full-scale ships with surface roughness", J Mar Sci Technol (2003) 8:26-36.

5.6. TRACE

The streamline tracing is controlled with this command. Each TRACE command in the input file will produce a group of streamlines on one of the panel groups.

General form of command

```
TRACE ( GROUP = n , STATION = n , STREAM = n ,  
        ISTART = n , PARAMETRIC , IDISTR = n ,  
        S1 = v , DS1 = v , SN = v ,  
        DSN = v , JDISTR = n , P1 = v ,  
        DP1 = v , PN = v , DPN = v )
```

Default values

TRACE (no default values exist for GROUP, STREAM or ISTART, but: STATION=100, IDISTR = 0, S1 = 0, DS1 = 0, SN = 1, DSN = 0, JDISTR = 0, P1 = 0, DP1 = 0, PN = 1, DPN = 0)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	GROUP	Section number of the panel section on which this group of streamlines will be traced.
	STATION	Number of streamline stations along the hull for streamline tracing. (That is, the number of points per streamline.)
	STREAM	Number of streamlines in the group.
	ISTART	Index of the streamline station from which the tracing will be done. The streamline stations are counted from the bow, so downstream tracing is accomplished with ISTART = 1 and upstream tracing with ISTART = STATION. Values in between will give rise to streamlines that are traced in two directions from the streamline station specified by ISTART.
	PARAMETRIC	This keyword specifies whether the start and end point of the streamlines is given in a computational coordinate system or in the parametric space. It thus determines the meaning of S1 and SN.

		The values given to the keywords IDISTR ... DSN specifies the distribution of the streamline stations along the section. The values of the keywords JDISTR ... DPN controls the girthwise distribution of the streamlines at the ISTART' s streamline station.
	IDISTR	Type of stretching function that should be used: 0 linear DS1 and DSN ignored 1 hyperbolic tangents DS1 or DSN required 2 exponential function ----- " ----- 3 hyperbolic sine ----- " ----- 4 geometric", partial sums of a geometric progression DS1 or DSN required 5 hyperbolic tangents DS1 and DSN required.
(1)	S1	Location of the first streamline station given in the computational coordinate system or the parametric space. See note 1.
	DS1	Spacing of the streamline stations at the upstream end of the streamlines.
(1)	SN	Location of the last streamline station given in the computational coordinate system or the parametric space. See note 1.
	DSN	spacing of the streamline stations at downstream end of the streamline group.
	JDISTR	Same as IDISTR but concerning the girthwise distribution of the streamlines.
(2)	P1	Girthwise location of the keel most streamline at the ISTART's streamline station.
(2)	DP1	Spacing of streamlines at the keel most streamline.
(2)	PN	Girthwise location of the topmost streamline at the ISTART's station.
(2)	DPN	Spacing of streamlines at the topmost streamline.

Notes

- (1) S1 and SN must be given as absolute coordinates in the computational coordinate system or in the parametric space if the keyword PARAMETRIC is used. The value

of S is 1 at the first station and equal the number of panels minus one at the last station of a panel group in the parametric space. Tracing the streamlines all the way to the end of a section that ends in a point is not a good idea. The mapping to the girthwise arc-length coordinate becomes singular and some of the boundary layer assumptions cease to be valid.

- (2) P1, DP1, PN and DPN should be specified using a normalized arc-length coordinate that is zero at the keel most edge of the panel section and one at the topmost edge.

6. XGRID commands

The following commands will be interpreted by XGRID if found in the input file between the SHIPFLOW module delimiter commands XGRID and END. XGRID only uses the first four characters of a command or a keyword.

Note that some options are by default turned off. So even if you don't intend to change any of the default values, you will at least have to give its command to activate the option.

Note also that there are two exceptions to the rule that all lengths and coordinates given must suit the dimensionless coordinate system, namely the commands SINGUL and RADIUS.

There are a few commands in the XFLOW section that will effect the way XGRID will design the grid.

The command HULL and the keywords WSING and NOWSING will control the default behaviour of XGRID for the waterline singularity. The free-surface is normally treated as a symmetry plane by XGRID, but can be fitted to the free-surface computed by XPAN when the keyword VFSFLOW is given.

Tuning commands

Some of the XGRID commands will be referred to as tuning commands in the following. These commands are not likely to be used by the user. The tuning commands have been included only to give the user access to some "constants" that really shouldn't have to be changed and some options that are not likely to be used. Since we haven't had the chance to try XGRID on all possible hull shapes (there are quite a few of them), we cannot guarantee that the tuning command default values will be right for all of them. These commands should thus be used restrictively!

Multiple XGRID modules

From version 6.0 it is possible to use the XGRID module more than once in a command file in order to more conveniently specify the geometry of a multihull vessel for the viscous solver XCHAP. In order to comply with the default monohull behavior, the multihull behavior is achieved by pre-processing the command file and running XGRID separately to create grids and then in the final run import them. This is done for all occurrences of XGRID except the last one, which is run together with the rest of the system as usual.

By default, XGRID takes the geometry from the XFLOW module, it places the hull centerline at $y=0$, the forward perpendicular at $x=0$ and the aft perpendicular at $y=0$ and it also sets the domain size and boundary conditions to represent the whole fluid domain in the computation. When adding hulls this will not work, so the defaults have to be overridden.

The domain size and other aspects of the grid details can be set directly in the XGRID

module. To reduce the domain size, use XGRID|XDISTR|XSTART, XGRID|XDISTR|XEND and XGRID|RADIUS|RADIUS. See other sections in the XGRID chapter for details.

The geometry of the hull is given in the XFLOW|OFFSET and XFLOW|HULL commands. If those commands are given in an XGRID module (except the last one), they will override the commands given in the XFLOW module. It is also possible to override the XFLOW|SYMMETRY command in the same way. This is useful in order to specify for example a symmetric trimaran configuration where the center hull is XFLOW|SYMMETRY|SYMM and the side hull(s) are XFLOW|SYMMETRY|NOSYMM.

The location, orientation and boundary conditions for an imported grid are specified with the XCHAP|IMPORT command. Since having multiple hulls in the same position is not sensible, an IMPORT command must be given in all XGRID modules except the last one. See the XCHAP chapter for details on the IMPORT command.

The generated command files are found in <id>_RUN_DIR and are named xgrid_<import_ID>, where import_id is the ID parameter given in the IMPORT command.

To demonstrate the new functionality, “trimaran_xchap” has been added to the example collection in the distribution.

Since catamarans do not have a hull on the centerline, the functionality described above does not apply. Like in earlier versions the demi-hull has to be embedded in a box grid using the XFLOW|ENVBOX or XFLOW|BOX commands. The location, orientation and boundary conditions are specified in the XCHAP|XGRID command.

For symmetric ships with an even number of hulls in general, one demi-hull has to be specified according to the previous paragraph, while the other hulls can be added with extra XGRID modules.

For unsymmetric configurations like a proa, one hull can use the default XGRID behaviour, with XFLOW|SYMMETRY|SYMM, and the rest of the hulls are added with extra XGRID modules.

6.1. COARSE

This command specifies the relation between the preliminary grid and the final grid. The use of a coarser grid gives the user a chance to

- (i) avoid the problems that arise due to the extreme resolution close to the hull that XCHAP (or any other code) demands at high Reynolds numbers. The preliminary grid needs only to be fine enough to resolve the curves that we want the ζ -lines to be able to do close to the hull.
- (ii) save some time thanks to the sheer reduction of the number of points that have to be processed (see also OUTPUT).

General form of command

COARSE (FATTEN = v, KSICOPY = g, ZETAPRE = g)

Default values

COARSE (FATTEN = 1.0, KSICOPY = ksimax + 0.5, ZETAPRE = zetamax)

ksimax and zetamax are entered with the "SIZE" command

Keywords

<i>keyword</i>	<i>description</i>
FATTEN	The first ζ -layer outside the hull in the preliminary grid will lie a factor FATTEN further out from the hull than the distance specified in the SKIN command.
KSICOPY	The ξ -coordinate of the last plane that will be produced through solving the Poisson equations. The planes after this plane will be copies of it. To make XGRID use the Poisson solver on the whole domain, KSICOPY must be KSIMAX + ..5.
ZETAPRE	The number of ζ -planes used in the preliminary grid.

6.2. CONTROL

This command is used to specify whether the program should use original or post processed offset file and also if the quality test should be performed.

General form of command

CONTROL (XGR8, OFFTEST, OVHG, OVFW, OVAFT, NODECK, BALA)

Default values

None

Keywords

<i>keyword</i>	<i>description</i>
OFFTEST	The offset file quality test is performed to search for possible errors. In total there is seven different criteria used to judge the correctness of the data: ● Length change large - checks if the section girth change is not too large ● Low point number - checks number of points at each section ● First point y.ne.0 - checks if the first point is on the symmetry plane ● Section non-planar - checks if the section is planar ● Double point - check the occurrence of double points ● Small distance - checks the distance between consecutive points on sections ● Mixed point order - checks if the points are sorted properly All the above messages are printed to the screen together with the information in which offset group and what location the problem was encountered.
XGR8	Switches on new version of XGRID with contour line following
OVHG	Switches on special treatment for stern overhang (VOF only)
BALA	Switches on special treatment for ballast conditions (VOF only)
OVFW	Offset of the hnd parameter from keel line at its intersection
OVAFT	Offset of the hnd from keel line at the transom
NODECK	Prevent XGRID to create deck for VOF cases.

6.3. OFFSET

This command can be used to specify the names of the offset groups in the offset file that will be used in the construction of the grid. The specified offset groups will override those entered in the command “HULLTYPE” on page 51

General form of command

```
OFFSET ( H1GR = “c” , H2GR = “c” , H3GR = “c” , OGRP = “c” ,
        ABGR = “c” , FBGR = “c” , H1GI = “c” , H2GI = “c” ,
        H3GI = “c” , OGRI = “c” , ABGI = “c” , FBGI = “c” ,
        H1GO = “c” , H2GO = “c” , H3GO = “c” , OGRO = “c” ,
        ABGO = “c” , FBGO = “c” )
```

Default values

OFFSET (No default values)

Keywords

<i>note</i>	<i>key</i>	<i>description</i>
(1)	H1GR	Name of an offset group that describes the main hull.
(1)	H2GR	Second offset group that describes the main hull.
(1)	H3GR	Third offset group that describes the main hull.
(1)	OGRP	Name of offset group that describes the overhang in the stern.
(1)	ABGR	Name of offset group that describes the stern bulb.
	FBGR	Name of offset group that describes the bulb.
(2)	H1GI	Name of an offset group that describes the inner part of the main hull.
(2)	H2GI	Second offset group that describes the inner part of the main hull.
(2)	H3GI	Third offset group that describes the inner part of the main hull.
(2)	OGRI	Name of offset group that describes the inner part of the

		overhang in the stern.
(2)	ABGI	Name of offset group that describes the inner part of the stern bulb.
(2)	FBGI	Name of offset group that describes the inner part of the bulb.
(2)	H1GO	Name of an offset group that describes the outer part of the main hull.
(2)	H2GO	Second offset group that describes the outer part of the main hull.
(2)	H3GO	Third offset group that describes the outer part of the main hull.
(2)	OGRO	Name of offset group that describes the outer part of the overhang in the stern.
(2)	ABGO	Name of offset group that describes the outer part of the stern bulb.
(2)	FBGO	Name of offset group that describes the outer part of the bulb.

Notes

- (1) At most three offset groups can be used to describe the main hull. The frames in the groups H1GR, H2GR and H3GR must be in global order from the foreship to the stern. ABGR and OGRP shall be used when there is a stern bulb and an overhang. See “HULLTYPE” on page 51 and “Mono-hull” on page 326 for more information.
- (2) Offset groups used to describe inner and outer parts of the twin skeg hull. More informations on grid generation for this type of ships can be found in “Offset file format for twin skeg hulls” on page 313.

6.4. OUTPUT

This command gives the user control over what grids will be written to files and if any iteration history should be given.

General form of command

```
OUTPUT ( NONE      , PNONE      , FULL  , HISTON )
          COARSE    PCOARSE
          FINE      PFINE
```

Default values

```
OUTPUT ( FINE, PCOARSE, HISTON )
```

Keywords

<i>keyword</i>	<i>description</i>
NONE	Nothing will be sent to the id_XVGRID file.
COARSE	The preliminary, uninterpolated, grid will be sent to the id_XVGRID file.
FINE	The interpolated grid will be written to id_XVGRID.
PNONE	[POST] No data for post-processing is saved.
PCOARSE	[POST] The preliminary uninterpolated grid is stored for post-processing.
PFINE	[POST] The interpolated grid is stored for post-processing.
FULL	This keyword can be used in combination with PCOARSE and PFINE. All points used by XGRID will be saved for post-processing, not just those that correspond to the p-points. The pictures of the grid from such a file will undoubtedly look more impressive to the eye of a novice but this option should be used with some care since the resulting data files will become approximately 8 times larger.
HISTON	The presence of HISTON in the command turns on the iteration history that will be written to the output file.

Notes

The interpolation itself is controlled by the command COARSE.

6.5. POISSON

This command controls the Poisson solver.

General form of command

```
POISSON ( MAXIT = n      , YCRIT = v      , ZCRIT = v      , ORFY = v      ,  
          ORFZ = v      , OFF              , OUTER          , FREEZE      ,  
                               INNER  
          JUFREEZE = n, JAFREEZE = n, ANGFREEZE = v      )
```

Default values

```
POISSON ( MAXIT = 60, YCRIT = 1.0D-09, ZCRIT = 1.0D-09, ORFY = 1.0,  
          ORFZ = 1.0, OFF*, JUFREEZE = UETA, JAFREEZE = AETA,  
          ANGFREEZE = 25 )
```

Twin skeg inner: POISSON (MAXIT = 150, ...)

Twin skeg outer: POISSON (MAXIT = 100, ...)

* for submarines

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	MAXIT	The maximum number of iterations allowed for solving the Poisson equations.
(1)	YCRIT	Convergence limit for the y-component changes.
(1)	ZCRIT	Same as YCRIT but for the z-component.
	ORFY	Over relaxation factor for the y-equation.
	ORFZ	----- " ----- z-equation.
	OFF	[POISOFF] Its presence in the command turns off the poisson solver. The initial guess as produced by transfinite interpolation will be used for output. This is useful when a quick check of the boundary point distribution is wanted.
(2)	OUTER	The information in the command will concern the outer block of the grid for twin skeg hull.
(2)	INNER	The information in the command will concern the outer block of the grid for twin skeg hull.
(3)	FREEZE	Turn off the poisson solver for VOF grids in certain problematic areas.
(3)	JUFREEZE	From this eta position start searching (in increasing eta direction) for a chine. Freeze the poison solver from the

- chine and for larger etas if keyword FREEZE is present. This is done separately for each ksi-plane.
- | | | |
|-----|-----------|--|
| (3) | JAFREEZE | Freeze the poisson solver from this eta position and for larger etas if keyword FREEZE is present. |
| (3) | ANGFREEZE | Chine detection grid angle for JUFREEZE. |

Notes

- (1) Convergence is achieved³ when $y_{chang} < YCRIT$ at the same time as $z_{chang} < ZCRIT$. y_{chang} and z_{chang} are the maximum changes made to the grid between two consecutive iterations. Note that the largest change in the y-direction (y_{chang}) may have taken place at a another point than the largest z-direction change. The iteration history in terms of y_{chang} and z_{chang} is printed out in the `id_OUTPUT` file of the y and z-coordinate for any grid point.
- (2) Keywords that can be used only in connection with the twin skeg hull. More informations on grid generation for this type of ships can be found in “Offset file format for twin skeg hulls” on page 313
- (3) This functionality is typically used when you have a hull with a chine which you will run with VOF. It is hard for the poisson solver to produce a good grid around the chine and for some cases some cells might be twisted inside out if FREEZE is not used.

When the freezing functionality is used, it might be necessary to reduce the number of poisson iterations.

³ We have to remember that convergence of the Poisson equations is not the ultimate goal of our efforts. If we can produce an acceptable grid after say three sweeps, that's excellent and we don't care that it's a poor solution to our grid generation equations.

6.6. RADIUS

The radius of the domain is entered in this command. Note that centre demands input given in the offset file coordinate system. (The only other XGRID command that needs input in that system is SINGUL.)

General form of command

RADIUS (RADIUS = v, CENTRE = v, STING = v)

Default value

RADIUS (RADIUS = 3.0, CENTRE = 0.0, STING = 0.0)

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
(1)	RADIUS	Radius of the grid.
(2)	CENTRE	Z-coordinate of the center line of the grid, this must be given in the offset file system!
(2)	STING	Thickness of a STING that will replace the polar singularity in the wake behind a submarine.

Notes

- (1) The radius will have to be chosen large enough to avoid blockage effects. If a potential flow solution is used as a boundary condition, the radius can be chosen much smaller as long as the outer boundary doesn't enter the viscous part of the flow.
- (2) This keyword is supposed to be used to inform XGRID where the sting of a submarine is located. Centre should be the z-coordinate of the centre of the grid, given in the hull offset coordinate system. This keyword will be ignored for anything but submarines.

6.7. SINGUL

This command gives the location of the two parametric edges that detach from the keel and water line. They are often called singularity lines since the mapping of a 90° edge to a flat surface results in a singularity in the mapping. This is one of the only two XGRID commands that needs input given in the offset file coordinate system. (The other one is RADIUS.)

General form of command

```
SINGUL ( KEEL           , OFF           ,
          WATER
          BOW

          XYZFWD = [V, V, V] , XYZAFT = [V, V, V] )
```

Default values

SINGUL (OFF)

Keywords

<i>keyword</i>	<i>description</i>
KEEL	[WHICH] The information in the command will concern the water edge leaving the keel.
WATER	[WHICH] The information in the command will concern the water edge leaving the waterline.
BOW	[WHICH] The information in the command will concern the water edge leaving the keel at the bow.
XYZFWD	The x, y and z coordinate of the point where the singularity line leaves the hull. It should coincide with the hull. This point will, in the WATER case, follow the water line if the ship is subject to sinkage and trim.
XYZAFT	The x, y and z coordinate of the point where the singularity line should change direction to become parallel to the x-axis. XGRID assign values to XYZAFT that makes the singularity line continue aft of XYZFWD with constant y and z-coordinates if not given.
OFF	The presence of OFF turns off the user defined parametric edges. The default behaviour is used instead which means that the parametric edge will follow the x-axis aft of the hull. The η -boundary smoothing is turned

6.8. SIZE

The number of planes in the grid is specified with this command.

If the refinement is changed with VCOARSE, COARSE or MEDIUM, the refinement of all other grids is changed accordingly. Se XCHAP|OVERLAP for more details.

General form of command

```
SIZE ( GLOBAL      , ETAMAX = n  , ZETAMAX = n  , IETAMAX = n  ,  
      OETAMAX = n  , VCOARSE      , UETAMAX = n  , AETAMAX = n  ,  
                        COARSE  
                        MEDIUM  
                        FINE  
      JMAXF = n    , JMAXP = n    ,  
      OEBO = n     , OEBA = n     , OEAF = n     , OEAB = n     ,  
      IEAF = n                                           )
```

Default values

SIZE (FINE or inherited from XFLOW::HULL)

Keywords

<i>keyword</i>	<i>description</i>
ETAMAX	Number of planes in the circumferential direction. The planes are uniformly distributed. Use IETAMAX and OETAMAX instead for twin skeg.
ZETAMAX	Number of planes in the radial direction.
IETAMAX	Number of planes in the circumferential direction in the inner block of the grid for twin skeg hull. Keyword can only be used in connection with a twin skeg hull. More informations on grid generation for this type of ships can be found in “Offset file format for twin skeg hulls” on page 313
OETAMAX	Number of planes in the circumferential direction in the outer block of the grid for twin skeg hull. Keyword can only be used in connection with a twin skeg hull. More information on grid generation for this type of ships can be found in “Offset file format for twin skeg hulls” on page 313
UETAMAX	Available only for VOF. Number of planes in the circumferential direction up to HUND location. See section XDISTR for more information.

AETAMAX	Available only for VOF. Number of planes in the circumferential direction up to HABO location. See section XDISTR for more information.
JMAXF	Available only for VOF. Number of planes in the circumferential direction up to the deck location. See section XDISTR for more information.
JMAXP	Available only for VOF. Number of planes in the transverse of the deck plug grids.
OEBO	[XGRID7] Number of points in transverse directions on the boss.
OEBA	[XGRID7] Number of points in transverse directions above the boss.
OEAF	[XGRID7] Number of points in transverse directions on the outer part of the aft overhang.
OEAB	[XGRID7] Number of points in transverse directions below the aft overhang
IEAF	[XGRID7] Number of points in transverse directions on the inner part of the overhang.
VCOARSE	[XGDENS] Available only for VOF. Will change the default values to: SIZE (ETAMAX=74, ZETAMAX=59) XDISTR (NU=27, NF=65, NM=77, NA=27, NW=30) YPLUS (YTARGET=1.18)
COARSE	[XGDENS] Will change the default values to: SIZE (ETAMAX=49, ZETAMAX=71) XDISTR (NM=52, NA=32, NW=28) YPLUS (YTARGET=0.99) For twin skeg: SIZE (IETAMAX=25, OETAMAX=25, ZETAMAX=71) XDISTR (NM=52, NA=32, NW=28) YPLUS (YTARGET=0.99)
MEDIUM	[XGDENS] Will change the default values to: SIZE (ETAMAX=59, ZETAMAX=84) XDISTR (NM=62, NA=38, NW=34) YPLUS (YTARGET=0.83) For twin skeg: SIZE (IETAMAX=30, OETAMAX=30, ZETAMAX=84) XDISTR (NM=62, NA=38, NW=34) YPLUS (YTARGET=0.83)
FINE	[XGDENS] Will change the default values to:

SIZE (ETAMAX=70, ZETAMAX=100)
XDISTR (NM=74, NA=44, NW=40)
YPLUS (YTARGET=0.7)

For twin skeg:

SIZE (IETAMAX=35, OETAMAX=35, ZETAMAX=100)
XDISTR (NM=74, NA=44, NW=40)
YPLUS (YTARGET=0.7)

GLOBAL

Will change the default values to:

XDISTR (XSTART=-0.8, XEND=1.8, XFPU=-0.05,
 XFPD=0.225, XAPU=0.88, XAPD=0.99)

Together with the COARSE keyword it will also set:

SIZE (ETAMAX=49, ZETAMAX=71)
XDISTR (NU=32, NF=78, NM=92, NA=32, NW=35)
YPLUS (YTARGET=0.99)

For twin skeg:

SIZE (IETAMAX=25, OETAMAX=25, ZETAMAX=71)
XDISTR (NU=32, NF=78, NM=92, NA=32, NW=35)
YPLUS (YTARGET=0.99)

Together with the MEDIUM keyword it will also set:

SIZE (ETAMAX=59, ZETAMAX=84)
XDISTR (NU=38, NF=92, NM=109, NA=38, NW=42)
YPLUS (YTARGET=0.83)

For twin skeg:

SIZE (IETAMAX=30, OETAMAX=30, ZETAMAX=84)
XDISTR (NU=38, NF=92, NM=109, NA=38, NW=42)
YPLUS (YTARGET=0.83)

Together with the FINE keyword it will also set:

SIZE (ETAMAX=70, ZETAMAX=100)
XDISTR (NU=45, NF=110, NM=130, NA=45, NW=50)
YPLUS (YTARGET=0.7)

For twin skeg:

SIZE (IETAMAX=35, OETAMAX=35, ZETAMAX=100)
XDISTR (NU=45, NF=110, NM=130, NA=45, NW=50)
YPLUS (YTARGET=0.7)

6.9. XDISTR

This command controls the x-plane distribution.

General form of command

```
XDISTR ( XSTART = v , NU = g , XFPU = v , NF = g ,
        XFPD = v , NM = g , XAPU = v , NA = g ,
        XAPD = v , NW = g , XEND = v , UNIFORM ,
        HUND = v , HABO = v )
```

Default values (for a MEDIUM grid)

```
XDISTR ( XSTART=0.5, XEND=1.8, XAPU=0.88, XAPD=0.99,
        NM=62, NA=38, NW=34 )
```

Keywords

<i>keyword</i>	<i>description</i>
XSTART	x-coordinate of the inflow end of the computational domain.
NU	number of planes (of clusters) in the longitudinal (ξ) direction between coordinates XSTART and XFPU.
XFPU	The x-coordinate of the change in stretching of the bow region upstream of the FP.
NF	number of planes (of clusters) in the longitudinal (ξ) direction between coordinates XFPU and XFPD.
XFPD	The x-coordinate of the change in stretching of the bow region downstream of the FP.
NM	number of planes (of clusters) in the longitudinal (ξ) direction between coordinates XFPD and XAPU.
XAPU	The x-coordinate of the change in stretching of the stern region upstream of the AP.
NA	number of planes (of clusters) in the longitudinal (ξ) direction between coordinates XAPU and XAPD.
XAPD	The x-coordinate of the change in stretching of the stern region

downstream of the AP.

NW	number of planes (of clusters) in the longitudinal (ξ) direction between coordinates XAPD and XEND.
XEND	The x-coordinate of the end of the grid.
UNIFORM	A uniform distribution is made. Only XSTART and XEND needs to be specified.
HUND	The non-dimensional z-coordinate of the lower grid plane of the refined free surface interface. Available only for VOF. Default value is -0.005.
HABO	The non-dimensional z-coordinate of the higher grid plane of the refined free surface interface. Available only for VOF. Default value is 0.015.

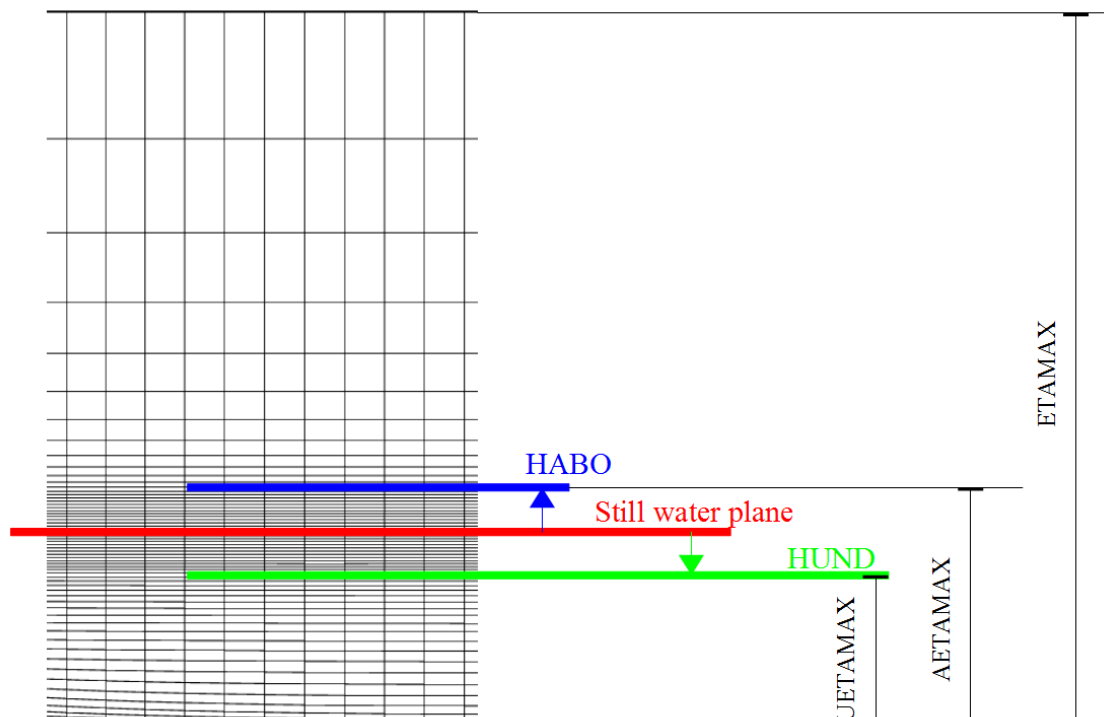
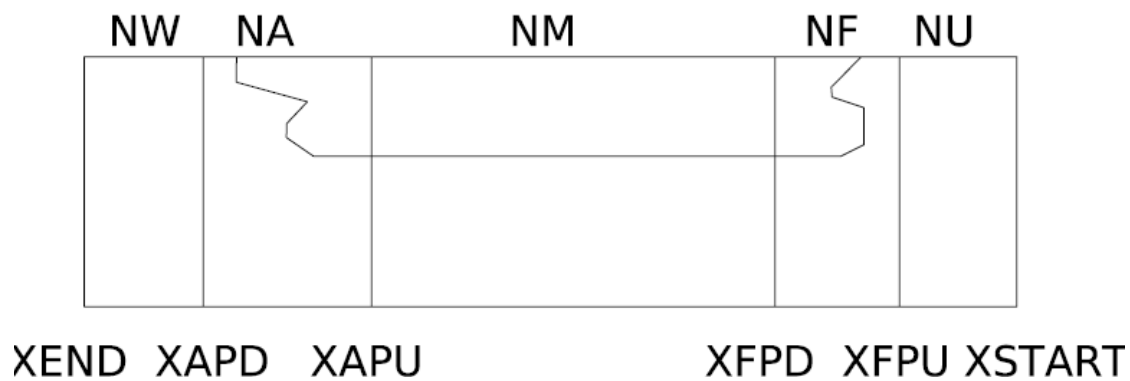


Illustration 1: Side view on the hull surface mesh for VOF computations.

6.10. ETASMOOTH

TUNING COMMAND

This command controls the fourth order smoothing of the longitudinal lines on the η -boundaries.

General form of command

ETASMOOTH (TIMESM = v, ZETASM = n, OFF)

Default values

ETASMOOTH (OFF)

ZETAPRE is truncated to an integer.

Keywords

<i>keyword</i>	<i>description</i>
TIMESM	The smoothing of the longitudinal lines on the η boundaries is governed by fourth order diffusion. TIMESM is the time that the diffusion will act on the lines (see theoretical manual).
ZETASM	The time of diffusion used will increase gradually from zero at the hull to TIMESM at the parametric grid coordinate $\zeta = \text{ZETASM}$. The lines outside $\zeta = \text{ZETASM}$ will be smoothed using a time of diffusion = TIMESM. The reason for this is that the mechanism that has been introduced to prevent the lines from moving inside each other or the hull isn't elegant enough to produce as good lines as a lower time of TUNING COMMAND. This command allows the user to tunediffusion will.
OFF	The η -boundary smoothing is turned off if OFF is found in the command.

6.11. FEEDBACK

TUNING COMMAND

This command allows the user to tune the improvement of the Poisson equation source terms (see "Source term improvement" in the theoretical manual).

General form of command

```
FEEDBACK ( HULL          , TRANS = v , NORMAL = v ,  
           KEEL  
           WATER  
  
           MAXDIFF = v )
```

Default values

FEEDBACK (TRANS = 1.0, NORMAL = 0.7, MAXDIFF = 0.5)

Keywords

<i>keyword</i>	<i>description</i>
HULL,	[WHICH] Feedback coefficients are chosen separately for the hull, water and centre
WATER,	plane surface. The presence of the keyword HULL in this command tells
KEEL	XGRID that the quantities given in the command should be used for the hull surface. WATER and KEEL works in the same way but for the water surface and the centre plane below the keel. This command can thus appear three times in the input file.
TRANS	This is the feedback amplification factor K_t . It affects the feedback of the "transversal" source term, i.e. the source term that influences the direction of the coordinate lines leaving the surface of interest.
NORMAL	Same as TRANS but concerning the other feedback amplification factor, K_n . It effects the source term that influences the spacing of the coordinate planes "parallel" to the surface of interest.
MAXDIFF	This parameter puts a limit on the "error" that enters the feedback loop. It is an adjustable level of saturation for the feedback loop.

6.12. IMPROVE

TUNING COMMAND

This command controls when the source terms of the Poisson equations should be improved (see "Source term improvement" in the theoretical manual).

General form of command

```
IMPROVE (  IMPW8 = n      ,  IMPHOP = n      ,  IMPMAX = n      ,
           ANGSEEP = v    ,  CONFUNBLEED = g    ,  OFF           ,
           OUTER                                     )
           INNER
```

Default values

IMPROVE (IMPW8 = 4, IMPHOP = 1, IMPMAX = 40, ANGSEEP = 0.001,
CONFUNBLEED = 0.40*ZETAMAX)

Twin skeg inner: IMPROVE (IMPW8 = 100, IMPHOP = 2, IMPMAX = 200,
ANGSEEP = -0.005, CONFUNBLEED = 0.59*ZETAMAX)

Twin skeg outer: IMPROVE (IMPW8 = 100, IMPHOP = 2, IMPMAX = 200,
ANGSEEP = 0.001, CONFUNBLEED = 0.59*ZETAMAX)

Keywords

<i>keyword</i>	<i>description</i>
IMPW8	The number of Poisson solver iterations before the first manipulation of the sources.
IMPHOP	The number of Poisson solver iterations between each change.
IMPMAX	The maximum number of changes that should be made.
ANGSEEP	Non-orthogonal edges of the domain will infinitely frustrate the source term feedback process unless the grid is allowed to be slightly non-orthogonal close to these corners. ANGSEEP is the non dimensional extent of this region.
CONFUNBLEED	This is the <u>parametric</u> distance of influence in the preliminary grid of the source (control function) changes.
OFF	The presence of OFF in the command turns off the iterative improvement of the source terms.
OUTER	The information in the command will concern the outer block of the grid

for twin skeg hull. Keyword can only be used in connection with a twin skeg hull. More informations on grid generation for this type of ships can be found in “Offset file format for twin skeg hulls” on page 313

INNER

The information in the command will concern the inner block of the grid for twin skeg hull. Keyword can only be used in connection with a twin skeg hull. More informations on grid generation for this type of ships can be found in “Offset file format for twin skeg hulls” on page 313

6.13. NEUMANN

TUNING COMMAND

A Neumann boundary condition at the η -boundaries can be explicitly implemented by movement of the boundary points in between sweeps in the Poisson solver.

General form of command

NEUMANN (NEUW8 = n, NEUHOP = n, NEUMAX = n, OFF)

Default values

NEUMANN (OFF)

Keywords

<i>keyword</i>	<i>description</i>
NEUW8	The number of Poisson solver iterations before the first boundary point movement.
NEUHOP	The number of Poisson solver iterations between each movement.
NEUMAX	The maximum number of changes that should be made. The actual number of changes made will be limited by convergence of the grid generation process or by the value of MAXIT (see "POISSON").
OFF	The presence of OFF in the command turns off the migration of the boundary points.

6.14. SKIN

TUNING COMMAND

The distance between the first and second ζ plane is specified with this command.

General form of command

SKIN (KEEL , X = [v₁, ..., v_{number}] ,
WATER
THICKNESS = [v₁, ..., v_{number}])

Default values

SKIN (Default values depend on Reynolds number and ship type.)

Keywords

<i>keyword</i>	<i>description</i>
KEEL	[WHICH] The distances given will be the distances between the first and second ζ -plane at the keel edge of the parametric grid.
WATER	[WHICH] Same as KEEL but this time at the waterline. The absence of any default values will force the use of two SKIN commands.
X	The x-coordinates of the thickness values. The sequence x ₁ .. x _{number} has to be monotonously increasing.
THICKNESS	The distances that XGRID will try to achieve. Linear interpolation is used between the values used. The first and last values given are used in front of x ₁ and after x _{number} respectively.

Notes

The values given are only what XGRID will try to aim for. Unwise use of η boundary smoothing (see ETASMOOTH) and a very small first distance in the preliminary grid (see COARSE) can make the final grid look a lot different from what we specified.

The thickness values have to be chosen so that the second p-layer outside the hull does not end up outside the logarithmic overlap region of the boundary layer. Consult the theoretical manual for more details on how the SKIN thickness values should be chosen.

6.15. YPLUS

TUNING COMMAND

Effects how the distance between the first and second ζ plane is computed.

General form of command

```
YPLUS ( NOWALLAW , YTARGET = v , EXPAND = v )
      WALLAW
```

Default values

```
YPLUS ( NOWALLAW, YTARGET=0.83, EXPAND=1)
```

Keywords

<i>keyword</i>	<i>description</i>
NOWALLAW	[WALL] Yields suitable y^+ values for XCHAP that use no-slip boundary condition at the hull surface.
WALLAW	[WALL] Yields suitable y^+ values for the use of wallaw boundary condition at the hull surface. XCHAP does not use wallaws so do not use this option.
YTARGET	Overrides the target value for y^+ .
EXPAND	Expands linearly the YTARGET value to EXPAND*YTARGET at XCH2 or XEND.

6.16. TUNE

TUNING COMMAND

The parameters changed by this command are of minor importance.

General form of command

TUNE (CORKY = v, PENDEP = v, SMALLP = v, EXPLICIT)

Default values

TUNE (CORKY = 0.6, PENDEP = 0.25, SMALLP = 0.05)

Keywords

<i>keyword</i>	<i>description</i>
CORKY	In the vicinity of the parametric longitudinal edges at the keel and waterline, the source terms will have to be adjusted (if improvement of source terms is turned on) to satisfy the boundary conditions of two surfaces. This can simply become too much for the feedback process in some cases unless we diminish the feedback amplification in the corners slightly. This extra multiplicative factor is named CORKY. The default value will make the feedback 40% weaker in these corners than otherwise.
PENDEP	The iterative "improvement" of the source terms will take place in a region of parametric thickness CONFUNBLEED (see "IMPROVE"). At the edge of this region, the changes will have dropped to zero. At the parametric distance PENDEP*CONFUNBLEED from the boundary, the influence will have dropped to one half of that at the surface.
SMALLP	The equations that govern the feedback contain boundary values of the source terms. If these turn out to be zero, no change will ever be made. To avoid this, these terms are never allowed to become smaller than SMALLP.
EXPLICIT	The presence of this word in the command tells XGRID to satisfy the Neumann boundary condition at the η -boundaries explicitly. This option is "old" but has been included until the implicit approach has been thoroughly tested.

7. XGR8 commands

The following commands will be interpreted by XGR8 if found in the input file between the SHIPFLOW module delimiter commands XGR8 and END. XGR8 only uses the first four characters of a command or a keyword.

At the moment XGR8 is a surface mesh generator for the hull surface. This mesh is then read in by XGRID and a volume grid is created.

XGRID uses transfinite interpolation to create a starting guess for the volume grid, but if XGR8 has been used to mesh the hull surface then it can also create a starting guess for the XGRID volume grid. The command XGR8:IGUE controls the starting guess grid generation in XGR8.

In order to make your grid generation as efficient as possible, it is crucial to have a rough understanding of how XGR8 works:

- The hull is rotated into position. The unused parts of the hull geometries are cut away. For double models this is at $z=0$ and for VOF this is at $z=FRBO$
- The hull is initially in one **STRIP**, but can be subdivided into two or more strips by defining additional **SPLITS**. Each split defines the upper side of one strip and the lower side of the next strip (see illustration below). The splits cut every station from the most upstream position to the most downstream position.

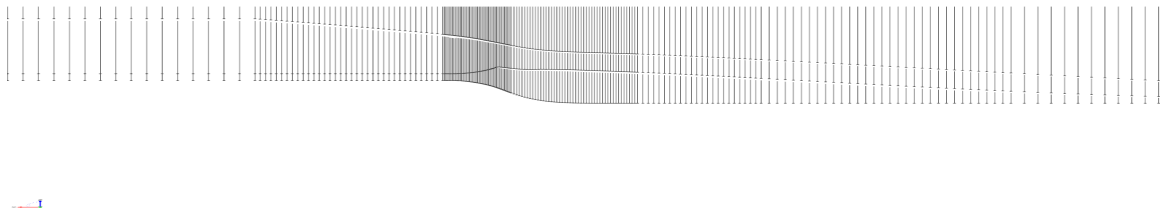


Illustration 2: Side view of the aft of KVLCC2 and the trailing symmetry plane. In this example there are 4 splits and therefore 3 strips. The first split defines the keel singularity position. The second split follows the upper contour of the boss. The third split follows the contour of the aft overhang. The fourth split does not do anything since there is no waterline singularity defined.

- Each strip contains a user controlled number of cells in the girthwise direction. This means that by adjusting where a split cuts through the stations or by adding more splits, the user gains control of the girthwise cell-distribution.
- You can have as many splitting lines as you want, just make sure they are ordered from keel to deck. (It not the order in the config file that needs to be ordered, but the

identifying number XGR8:SPLIT:NUMB.) Between split number 1 and split number 2 is strip number 1. And between split number 2 and split number 3 is strip number 2. And etc.

- The path of each individual split can be controlled in some different ways:
 1. XGR8:SPLIT:LFOLLOW – make the split follow a straight line between two points as closely as possible.
 2. XGR8:SPLIT:ZFOLLOW – follow a z-coordinate in computational space.
 3. XGR8:SPLIT:FOLLOW – follow another splitting line at a certain distance.
 4. XGR8:SPLIT:CFOLLOW – follow the hull contour (the beginning or end of stations) between two points.
 5. XGR8:SPLIT:SING – create a singularity line from a certain point.
 6. XGR8:RANGE & XGR8:RPOS – define a range distribution of the splitting line on a certain station.
- The above order is also the priority order if there are multiple SPLIT commands in the same region for the same splitting line (LFOLLOW has the highest priority and RANGE the lowest priority)

Multiple XGR8 modules

In SHIPFLOW offset files for TWINSKEGs are split into a forward part, an aft outer part and an aft inner part. XGR8 uses these offset groups and because of that TWINSKEGs require three XGR8 modules. However, the surface mesh and the starting guess exported from XGR8 is always in one block.

XGRID 8 DEFAULTS

The strips will be setup using a DEFAULTS program, which accesses the offset file, identifies points of interest and applies appropriate controls to achieve a uniform girth-wise distribution of the cells. DEFAULTS are automatically triggered when XGRID8 is used. The number of cells girth-wise are computed from XGRI:SIZE based on ETAMAX value. It will also inherit values from XGRI:SIZE for IETA, OETA, UETA, AETA, and JMAX.

The set up strips and controls are written in the configuration file inside RUN_DIR, which can be used to run with manual control of the strips. DEFAULTS will read sinkage and trim from XFLOW:IPOS and will be corrected after XPAN computation.

7.1. CONTROL

This command is used for specifying the overall hull shape; for instance mono vs twinskeg or wet transom vs dry transom.

General form of command

```
CONTROL ( ALL          , WTRAN      , FRBO = v      , PORT          ,
           INNER        DRYTRAN
           OUTER        NOTRAN
           FWD

           POFF          , DUCK        , MSTR          , BSING = v      ,
           BCF=v         , ACF=v       , WSAN=v        , WSIN=v        ,
           BLUFF         , BALA        )
           NOBA
```

Default values

```
CONTROL ( FRBO = 0.0, PORT,BSING=10.0,BCF=170.0,ACF=155.0,WSAN=45.0 )
```

Keywords

<i>keyword</i>	<i>description</i>
ALL	The entire hull geometry will be given in this XGR8 module. Used for mono hulls.
INNER	The inner stern hull geometry will be given in this XGR8 module. Used for twinskeg hulls and used together with other XGR8 modules that specify the rest of the hull geometry.
OUTER	The outer stern hull geometry will be given in this XGR8 module. Used for twinskeg hulls and used together with other XGR8 modules that specify the rest of the hull geometry.
FWD	The bow hull geometry will be given in this XGR8 module. Used for twinskeg hulls and used together with other XGR8 modules that specify the rest of the hull geometry.
WTRAN	The hull has a wet transom. The last hull station is copied backwards all the way to the outlet boundary and the hole is filled with a transom block grid.
DRYTRAN	The hull has a dry transom. The grid lines at the end of the overhang snaps to $z = 0$ (DM) or $z = \text{FRBO}$ (VOF).
NOTRAN	The hull has no transom. The grid lines at the end of the overhang snaps to $y = 0$.

FRBO	The freeboard height in the offset scale. Used for VOF.
PORT	The geometry is on the port side.
STAR	The geometry is on the starboard side.
POFF	Print intermediate offset files for debugging.
DUCK	WTRAN has a duck tail shape.
MSTR	Run with manually set up strips, will not run the DEFAULTS for XGRID8.
BSING	Angle control for the bow singularity line for the DEFAULTS.
BCF	Angle control for the contour follow above the bulb for the DEFAULTS.
ACF	Angle control for the contour follow on the aft overhang for the DEFAULTS.
WSAN	Angle control for the water-line singularity line for the DEFAULTS.
WSIN	Beam controlled water-line singularity for the DEFAULTS. WSIN is a fraction of the maximum half beam. The water-line singularity is controlled by angle by default.
BLUFF	Bluff bow correction – post correction on XGR8 to improve skewed cells on the bow, works on the bottom parts of the bow. Can be an option for wide bows.
BALA	(MONO only) Running the DEFAULTS with special treatment of the free surface refinement for VOF cases for ballast drafts. Without this ballast treatment the free surface refinement band will continue towards the transom, this will rather make the band continue at constant horizontal plane. DEFAULTS will detect ballast drafts automatically when the aft overhang is significantly above the free surface, this BALA keyword forces it to be treated as a ballast draft.
NOBA	(MONO only) for DEFAULTS - Overrides ballast treatment of VOF cases to no ballast.

7.2. OFFSET

This command can be used to specify the names of the offset groups in the offset file that will be used in the construction of the grid. The OFFSET command imports one offset group. To import multiple offset groups, one needs to have multiple OFFSET commands.

NOTE: One of the offset groups has to be defined as the MAIN group.

General form of command

```
OFFSET ( GRNA = "c" , MAIN      , EXTEND
        BOSS      , NOEXTEND
        AFT
        BULB
        FWD
        KEXT
```

Default values

OFFSET (EXTEND (For MAIN, AFT and FWD) | NOEXTEND (Otherwise))

Keywords

<i>keyword</i>	<i>description</i>
GRNA	Name of the offset group.
MAIN	The main hull group. This group is mandatory.
BOSS	The boss group.
AFT	The aft overhang group.
BULB	The bulb group.
FWD	The bow overhang group.
KEXT	The aft keel extension group.
EXTEND	Extend all stations that does not reach the cutting line.
NOEXTEND	Do nothing.

Notes

Right now XGR8 does not use the hull geometry defined in XFLOW, so all the hull geometry has to be defined by XGR8:OFFSET commands.

7.3. STRIP

This command gives the user control of the girthwise cell distribution within each strip. Since there is no default value for NCELL, the STRIP command is mandatory for each strip.

General form of command

```
STRIP    (  NUMBER = n ,  NCELL = n   ,  UNIFORM   ,  DF1L = v   ,
           DF1H = v       ,  DFNL = v    ,  DFNH = v    ,  NS2I      ,
           BSYM           ,  SSYM
           BFOLD          ,  SFOLD
           BINTERIOR      ,  SINTERIOR
           BWPL           ,  SWPL
           )
```

Default values

STRIP (DF1L = 0.95, DF1H = 1.05, DFNL = 0.95, DFNH = 1.05)

Keywords

<i>keyword</i>	<i>description</i>
NUMBER	Strip number. Ordered from keel to deck.
NCELL	Number of cells in the girthwise direction on this strip.
UNIFORM	This strip gets a uniform cell distribution in girthwise direction.
DF1L	The lowest allowed cell size at the low side of the strip.*
DF1H	The highest allowed cell size at the low side of the strip.*
DFNL	The lowest allowed cell size at the high side of the strip.*
DFNH	The highest allowed cell size at the high side of the strip.*
NS2I	Change BC from NOSLIP to INTERIOR.
BSYM	Upstream of the bow this strip is snapped to y=0. BC is SLIP or INTERIOR.
BFOLD	Upstream of the bow the nodes are snapped together from both sides of the fold.
BINTERIOR	Upstream of the bow the BC's will be INTERIOR.
BWPL	Upstream of the bow this strip is snapped to z=0. BC is SLIP.

SSYM	Downstream of the stern this strip is snapped to $y=0$. BC is SLIP or INTERIOR.
SFOLD	Downstream of the stern the nodes are snapped together from both sides of the fold.
SINTERIOR	Downstream of the stern the BC's will be INTERIOR.
SWPL	Downstream of the stern this strip is snapped to $z=0$. BC is SLIP.

Notes

* When UNIFORM is not set for a strip, then the girthwise cell distribution is stretched to get a smooth variation of the cell sizes. This is done for each station individually:

1. First an average cell size ($\Delta s_{ave,i}$) for the sub-station of each strip (i) is computed.
2. At the intersection between strips i and i+1 these averages are combined into new averages ($\Delta s_{ave,i+1/2} = (\Delta s_{ave,i} + \Delta s_{ave,i+1})/2$)
3. The first cell size in girthwise direction for strip i is then calculated according to
$$\begin{aligned}\Delta s_{first,i} &= \min(DF1L * \Delta s_{ave,i-1/2}, \Delta s_{ave,i}) && \text{if } \Delta s_{ave,i} > \Delta s_{ave,i-1/2} \\ \Delta s_{first,i} &= \max(DF1H * \Delta s_{ave,i-1/2}, \Delta s_{ave,i}) && \text{if } \Delta s_{ave,i} < \Delta s_{ave,i-1/2}\end{aligned}$$
4. The last cell size in girthwise direction for strip i is then calculated according to
$$\begin{aligned}\Delta s_{last,i} &= \min(DFNL * \Delta s_{ave,i+1/2}, \Delta s_{ave,i}) && \text{if } \Delta s_{ave,i} > \Delta s_{ave,i+1/2} \\ \Delta s_{last,i} &= \max(DFNH * \Delta s_{ave,i+1/2}, \Delta s_{ave,i}) && \text{if } \Delta s_{ave,i} < \Delta s_{ave,i+1/2}\end{aligned}$$
5. $\Delta s_{first,i}$ and $\Delta s_{last,i}$ is then used as boundary values for stretching number 5 described in this manual in section XMESH:BODY

7.4. SPLIT

The lower and upper bounds of STRIPS in girthwise direction are defined by SPLITS. There are 5 different ways to control the position of a SPLIT with the SPLIT command:

1. SPLIT:LFOLLOW – make the split follow a straight line between two points as closely as possible.
2. SPLIT:ZFOLLOW – follow a z-coordinate in computational space.
3. SPLIT:FOLLOW – follow another splitting line at a certain distance.
4. SPLIT:CFOLLOW – follow the hull contour (the beginning or end of stations) between two points.
5. SPLIT:SING – create a singularity line from a certain point.

These commands are described separately below.

7.4.1. SPLIT:LFOLLOW

This command sets a line between points FROM and TO, and makes the SPLIT number NUMBER follow that line as closely as possible.

General form of command

```
SPLIT    (  LFOLLOW = n          ,  NUMBER = n          ,  
            FROM = [v, v, v]      ,  TO = [v, v, v]      ,  
            SMIN = v              ,  SMAX = v              ,  
            DSDZ = v              ,  DZDX = v              )
```

Default values

SPLIT (SMIN = 0.01, SMAX = 0.99, DSDZ = 3.0, DZDX = 3.0)

Keywords

<i>keyword</i>	<i>description</i>
LFOLLOW	Follow a line defined by two points. If you also want to restrict this split compared to say split number i, then set LFOLLOW=i. If you don't want this functionality, then set LFOLLOW=0
NUMBER	This SPLIT command concerns split number NUMBER. Splits are ordered from keel to deck.
FROM	The x, y and z coordinate in offset file coordinate system of the point defining the start of the line.
TO	The x, y and z coordinate in offset file coordinate system of the point defining the end of the line.
SMIN	Minimum distance to the splitting line that is being followed (ratio of the sub-station length to the station length). *
SMAX	Maximum distance to the splitting line that is being followed (ratio of the sub-station length to the station length). *
DSDZ	Maximum allowed sub-station length, (ds) to dz ratio between this splitting line and the splitting line that is being followed. If the maximum value is exceeded, the sub-station length is set to $ds = DSDZ * dz$. *
DZDX	Maximum allowed dz to dx ratio, where dx and dz is the difference in x-direction and z-direction of splitting line of two consecutive offset

stations. If the maximum value is exceeded, the z-position for the split of the downstream station (of these two stations) is set to $z_{\text{downstream}} = z_{\text{upstream}} + dz$ where $dz = DZDX * dx$. *

Notes

* When SMIN, SMAX, DSDZ and DZDX is applied, it is done in the following order: DSDZ, SMIN, SMAX & DZDX.

7.4.2. SPLIT:ZFOLLOW

This command sets a splitting line at $z = Z_C$ in computational space. The command is especially useful for VOF computations.

General form of command

```
SPLIT    (  ZFOLLOW = n          ,  NUMBER = n          ,  
            XMIN = v              ,  XMIN = v              ,  
            SMIN = v              ,  SMAX = v              ,  
            DSDZ = v              ,  DZDX = v              ,  
            Z_C = v                )
```

Default values

```
SPLIT ( SMIN = 0.01, SMAX = 0.99, DSDZ = 3.0, DZDX = 3.0 )
```

Keywords

<i>keyword</i>	<i>description</i>
ZFOLLOW	Follow a z-value between XMIN and XMAX. If you also want to restrict this split compared to say split number i, then set ZFOLLOW=i. If you don't want this functionality, then set ZFOLLOW=0
NUMBER	This SPLIT command concerns split number NUMBER. Splits are ordered from keel to deck.
XMIN	The x coordinate in computational coordinate system of the point defining the start of this ZFOLLOW.
XMAX	The x coordinate in computational coordinate system of the point defining the end of this ZFOLLOW.
Z_C	The z coordinate in computational coordinate system of the splitting line between XMIN and XMAX.
SMIN	Minimum distance to the splitting line that is being followed (ratio of the sub-station length to the station length). *
SMAX	Maximum distance to the splitting line that is being followed (ratio of the sub-station length to the station length). *
DSDZ	Maximum allowed sub-station length, (ds) to dz ratio between this

splitting line and the splitting line that is being followed. If the maximum value is exceeded, the sub-station length is set to $ds = DSDZ * dz$. *

DZDX

Maximum allowed dz to dx ratio, where dx and dz is the difference in x-direction and z-direction of splitting line of two consecutive offset stations. If the maximum value is exceeded, the z-position for the split of the downstream station (of these two stations) is set to $z_{\text{downstream}} = z_{\text{upstream}} + dz$ where $dz = DZDX * dx$. *

Notes

* When SMIN, SMAX, DSDZ and DZDX is applied, it is done in the following order: DSDZ, SMIN, SMAX & DZDX.

7.4.3. SPLIT:FOLLOW

If you do not want to follow a line (SPLIT:LFOLLOW) or a z-coordinate (SPLIT:ZFOLLOW), but you still want to restrict the route this splitting line takes, you can use this command.

General form of command

```
SPLIT    (  FOLLOW = n          ,  NUMBER = n          ,  
           XMIN = v            ,  XMIN = v            ,  
           SMIN = v            ,  SMAX = v            ,  
           DSDZ = v            ,  DZDX = v            )
```

Default values

SPLIT (SMIN = 0.01, SMAX = 0.99, DSDZ = 3.0, DZDX = 3.0)

Keywords

<i>keyword</i>	<i>description</i>
FOLLOW	If you also want to restrict this split compared to say split number i, then set FOLLOW=i.
NUMBER	This SPLIT command concerns split number NUMBER. Splits are ordered from keel to deck.
XMIN	The x coordinate in computational coordinate system of the point defining the start of this FOLLOW.
XMAX	The x coordinate in computational coordinate system of the point defining the end of this FOLLOW.
SMIN	Minimum distance to the splitting line that is being followed (ratio of the sub-station length to the station length). *
SMAX	Maximum distance to the splitting line that is being followed (ratio of the sub-station length to the station length). *
DSDZ	Maximum allowed sub-station length, (ds) to dz ratio between this splitting line and the splitting line that is being followed. If the maximum value is exceeded, the sub-station length is set to $ds = DSDZ * dz$. *
DZDX	Maximum allowed dz to dx ratio, where dx and dz is the difference in x-direction and z-direction of splitting line of two consecutive offset

stations. If the maximum value is exceeded, the z-position for the split of the downstream station (of these two stations) is set to $z_{\text{downstream}} = z_{\text{upstream}} + dz$ where $dz = DZDX * dx$. *

Notes

* When SMIN, SMAX, DSDZ and DZDX is applied, it is done in the following order: DSDZ, SMIN, SMAX & DZDX.

7.4.4. SPLIT:CFOLLOW

This command sets a hull contour line for SPLIT number NUMBER to follow.

General form of command

```
SPLIT    (  CFOLLOW          ,  NUMBER = n          ,  
          FROM = [V, V, V]    ,  TO = [V, V, V]      )
```

Default values

SPLIT ()

Keywords

<i>keyword</i>	<i>description</i>
CFOLLOW	This splitting line follows the hull contour between FROM and TO.
NUMBER	This SPLIT command concerns split number NUMBER. Splits are ordered from keel to deck.
FROM	The x, y and z coordinate in offset file coordinate system of the point where the contour line starts following the hull contour. The point closest to FROM on a beginning or end of a station will be used.
TO	The x, y and z coordinate in offset file coordinate system of the point where the contour line stops following the hull contour. The point closest to TO on a beginning or end of a station will be used.

7.4.5. SPLIT:SING

This command sets a line from the hull, upstream or downstream, for SPLIT number NUMBER.

General form of command

```
SPLIT  (  SING          ,  NUMBER = n      ,  FROM = [V, V, V]    ,
        BOW            ,  YSNAP = v      ,  ZSNAP = v          )
        STERN
```

Default values

SPLIT ()

Keywords

<i>keyword</i>	<i>description</i>
SING	A “singularity” line exits the hull.
NUMBER	This SPLIT command concerns split number NUMBER. Splits are ordered from keel to deck.
FROM	The x, y and z coordinate in offset file coordinate system of the point where the singularity line leaves the hull. The point closest to FROM on a beginning or end of a station will be used.
BOW	Bow singularity line.
STERN	Stern singularity line.
YSNAP	Snap singularity line to this y-value in comp coord.
ZSNAP	Snap singularity line to this z-value in comp coord.

7.5. RANGE

This command defines a range that can be applied at a station with the command XGR8:RPOS.

General form of command

RANGE (ID = "c" , RNGS = [v₁, ..., v_n])

Default values

RANGE ()

Keywords

<i>keyword</i>	<i>description</i>
ID	Id of this RANGE. Use the same id in RPOS to apply it.
RNGS	This vector needs to contain the range for each splitting line. The value of element i is the ratio of station length up to split i and the total station length. Thus, element values should be between 0.0 and 1.0. Setting the element i to a negative value, will have the effect that this range will not be applied to split i.

7.6. RPOS

This command where a range should be applied.

General form of command

```
RPOS      (  ID = "c"                ,  POS = [v, v, v]          )
           FIRST
           LAST
```

Default values

RPOS ()

Keywords

<i>keyword</i>	<i>description</i>
ID	Id of the range to apply.
POS	Station closest to POS (in offset coordinate system) will get this range applied.
FIRST	Range will be applied to the most upstream station.
LAST	Range will be applied to the most downstream station.

7.7. IGUIESS

Create a starting guess for the volume grid generator in XGRID.

General form of command

```
IGUESS (  ON
          OFF
          ,  FGRO = v
          ,  FANG = v
          ,
          FNDI = v
          ,  FTAN = v
          ,  RAMP = n
          ,
          NSTE = n
          ,  NSKI = n
          ,  JFIX = n
          ,
          YZJF = v
          ,  JFS0 = v
          ,  JFS1 = v
          ,
          JFIM = v
          )
```

Default values

RPOS (ON (for TWIN), OFF (for MONO), FGRO = 1.0, FANG = 1.0, FNDI = 1.0,
FTAN = 1.0, RAMP = 40, NSTE = 8000, NSKI = 100, YZJF = 3.0, JFIM=0)

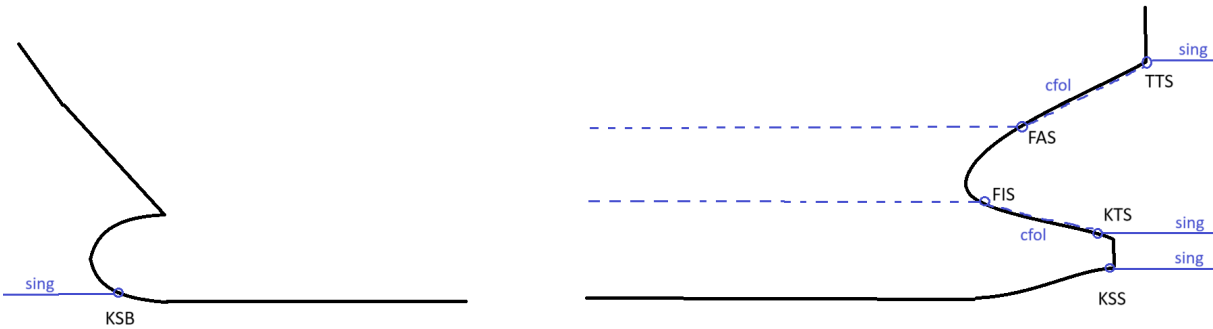
Keywords

<i>keyword</i>	<i>description</i>
ON	Starting guess for XGRID will come from XGR8.
OFF	XGR8 will not produce a starting guess.
FGRO	Global growth factor for step size control.
FANG	Angle factor to multiply the normal step size control.
FNDI	Girthwise node distance factor to multiply the normal step size control.
FTAN	Girthwise node distribution equalizer factor.
RAMP	Ramp up of girthwise node distribution equalizer factor.
NSTE	Number of steps to generate in the normal direction.
NSKI	Number of steps to skip when writing to VTK file.
JFIX	From plane jfix and upwards in the girthwise direction, create horizontal planes.
YZJF	Max ratio dy/dz where JFIX is allowed.

JFS0	Parameter value [0.0-1.0] of the grid radius where the girthwise distribution below jfix is smoothened out. Smoothing factor is 0% below JFS0, increases linearly until 100% at JFS1 and above.
JFS1	Parameter value [0.0-1.0] of the grid radius where the girthwise distribution below jfix is smoothened out. Smoothing factor is 0% below JFS0, increases linearly until 100% at JFS1 and above.
JFIM	Grid plane in girth-wise direction below which the girth-wise distribution is not smoothened out.

7.8. FTPT

This is an optional way to add more control to the XGR8 DEFAULTS and running it in a semi automatic way. Instead of manually adjusting the points in the splits and running from the _RUNDIR config file, this method provides an easier option to fine tune known points for common hull shapes. DEFAULTS will use the points provided here and will find other required points and adjust cell distributions automatically. Refer to the following figure for the convention on the points, not all points are used for all types of stern and bows. For TWIN the same convention applies on the skegg plane, except TTSC



General form of command

FTPT (TTS = [v, v, v] , TTSC = [v, v, v] , KSS = [v, v, v] ,
KTS = [v, v, v] , FIS = [v, v, v] , FAS = [v, v, v] ,
KSB = [v, v, v])

Default values

FTPT ()

Keywords

<i>keyword</i>	<i>description</i>
TTS	Singularity line from the end of aft overhang
TTSC	(TWIN only) Singularity line from the end of aft overhang at the center line
KSS	Keel singularity line from the lower end of the boss
KTS	Singularity line from the upper end of the boss
FIS	Forward end of the cfollow on the upper side of the boss
FAS	Forward end of the cfollow on the aft overhang

KSB Keel singularity line from the bow

8. XCHAP commands

The following commands will be interpreted by XCHAP if found in the input file between the SHIPFLOW module delimiter commands XCHAP and END. XCHAP only uses the first four characters of a command or a keyword.

8.1. ACTUATOR

This method for modeling a propeller is deprecated. Use the lifting line propeller model (LLINE) instead.

This command is used to control the running of the force actuator disc. Thrust and moment is taken from the PROPELLER command in XFLOW section.

General form of command

```
ACTUATOR ( ID="c" , ON
           OFF
           XUPSTR = v , XDOWNSTR = v , RMIN = v ,
           RMAX = v , DIMENSION = [v1,v2,v3] , BC11 = v ,
           BC12 = v , BC21 = v , BC22 = v ,
           LOW = [v1,v2] , HIGH = [v1,v2] , RUNINT )
```

Default values

```
ACTUATOR ( ID="ID", ON, BC11="INTERIOR", BC12="INTERIOR",
           BC21="INTERIOR", BC22="INTERIOR", LOW= [0,0],
           HIGH= [1,1], RUNINT=(10 for ADI, 1 for KRYLOV) )
```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	ID	A string that is used to identify the actuator disc object
(1)	ON	The propeller model is run.
(1)	OFF	The propeller model is turned off.
	ACTIVE	The body forces from the propeller is transferred to the propeller grid and the flow is solved there. The default behavior when ACTIVE is not specified is to interpolate the body forces directly from the propeller to the background grid.
	XUPSTR	Upstream position of propeller grid in offset coordinate system.
	DOWNSTR	Downstream position of propeller grid in offset coordinate system.

RMIN	Inner radius of propeller grid in offset coordinate system.
RMAX	Outer radius of propeller grid in offset coordinate system.
DIMENSION	Dimension of grid in axial, radial and circumferential direction respectively.
BC11	Specify the boundary condition of the grid. Possible values are “NOSLIP”, “SLIP”, “INFLOW”, “OUTFLOW”, “INOUT” and “INTERIOR”. The INTERIOR condition means that the boundary values are taken by interpolation from other grids.
BC12	Specify the boundary condition of the grid.
BC21	Specify the boundary condition of the grid.
BC22	Specify the boundary condition of the grid.
LOW	Defines the parametric extent in axial and radial direction used for transferring propeller forces. LOW gives start point. HIGH gives end point. Values should be between 0 and 1.
HIGH	Defines the parametric extent in axial and radial direction used for transferring propeller forces. LOW gives start point. HIGH gives end point. Values should be between 0 and 1.
RUNINT	Number of outer solver iteration between running the propeller model.

Notes

- (1) Turning it OFF will still make Chapman create a propeller and a propeller grid, so start-up will take a little longer and more memory will be used. If you want to restart later with the propeller turned ON, this will only be possible if the propeller was present but OFF from the beginning. When the propeller is ON it will be run after every 10 Chapman iterations.

8.2. CONTROL

This command is used to specify if the program shall generate an initial solution or restart from a previous solution. The maximum number of iterations can also be set and some additional output data can be requested.

General form of command

```
CONTROL (  START          ,  MAXITER=n  ,  GRID          ,
           RESTART        ,              OVERLAP        ,
           NEWCONFIG

           KRYITER = n    ,  MAXV = v    ,  DUMP = n      ,
           VERBOSE = n    ,  YTARGET = v ,  K0 = v        ,
           W0 = v        ,

           EASM           ,  ADI          ,  FROMM         ,
           KWSST          ,  KRYLOV     ,  SUPERBEE      ,
           KWBSL          ,  KRAD       ,  SUPBC         ,
                           ADKR          ,  STACS

           STRETCH        ,  RMODEL = n  ,
           NOSTRETCH

           AHR = v        ,  KS = v        ,  AH_KS = v      ,

           FLPO = n      )
```

Default values

```
CONTROL (  START, MAXITER = [depends on grid size],
           EASM(KWSST for VOF), STRETCH, YTARGET=1,
           DUMP=1000, VERBOSE=1, SUPERBEE, RMODEL = 0,
           ADI (for VOF) or KRYLOV (for double model), KRAD,
           K0=1.0e-8, W0=10.0, RMODEL=4, AH_KS=5      )
```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	START	[START] Start new computation.
	RESTART	[START] Restart from old solution.
	NEWCONFIG	[START] Restart from an old solution, but with some change in the configuration. For example if a propeller is turned on or if an appendage is moved.
(1)	MAXITER	Maximum number of iterations.

	FLPO	Number of data points from the _FORCELOG used for calculating the average of forces and moments.
	KRYITER	Run the first KRYITER iterations with KRYLOV, then use ADI up to MAXITER. There is seldom any reason to do this, but if many and/or complicated appendages are used this may improve convergence.
	STRETCH	[STRETCH] All imported and parametric grid components are stretch towards noslip boundaries.
	NOSTRETCH	[STRETCH] The opposite to STRETCH
	EASM	[TURB] Selects the Explicit Algebraic Stress Model. This is the most advanced turbulence model in XCHAP.
	KWSST	[TURB] Selects the k- ω SST turbulence model
	KWBSL	[TURB] Selects the k- ω BSL turbulence model
	K0	Initial and inlet value for the turbulent kinetic energy k. Default value is 1.0e-8, a small value intended to represent a non-turbulent inflow. For zonal approach computations this is not used, instead the turbulent inlet quantities are determined from the boundary layer solution on the forebody. There is in general no reason to adjust this value, it should only be done for certain validation cases.
	W0	Initial and inlet value for the turbulent frequency ω . Default value is 10. For zonal approach computations this is not used, instead the turbulent inlet quantities are determined from the boundary layer solution on the forebody. There is in general no reason to adjust this value, it should only be done for certain validation cases.
(3)	AHR	Sets the Average Hull Roughness value given in μm .
(3)	KS	Sets the equivalent sand grain roughness height in μm .
(3)	AH_KS	Controls the AHR/KS ratio necessary to convert between the different measures.
(3)	RMODEL	Roughness model. 0 – Hellsten-Wilcox, 1 – Knopp, 2 – Knopp-modified, 3 – Aupoix-Nikuradse, 4 – Aupoix-Colebrook
	YTARGET	Control the y^+ target value for first cell center of all grids except the XGRID grid. For the grids with NOSTRETCH defined in the import command this has no effect.
	MAXV	Maximum velocity restriction. The velocity in all cells is restricted to the value after each iteration. If this happens in any cell the progress indicator '.' is changed to '*'.

	DUMP	Control how many iterations should be run between each save to the CGNS file(s). Regardless of what DUMP is set to the results will be saved when the case is finished.
	GRID	[STOP] Only the grids are created in XCHAP. The program will be stopped before calculating overlap information and running iterations. This command should be used for a quick check of grids.
	OVERLAP	[STOP] Only overlap is computed in XCHAP. The program will be stopped before running iterations. This command should be used for a quick check of overlap.
	VERBOSE	Control the level of output to the terminal. 0 = no output 1 = normal progress reporting 2 = full progress reporting 3 = debugging information
	ADI	[SOLVER] Use the Alternating Direction Implicit solver to solve the linear equations.
	KRYLOV	[SOLVER] Use a krylov solver to solve the linear equations. So far it is only implemented for double model XCHAP. In general krylov is faster then ADI, but it uses more memory (about 6kB per cell).
	KRAD	[SOLVER] Hybrid ADI-Krylov scheme. Used by default if KRYITER is specified.
	ADKR	[SOLVER] Hybrid ADI-Krylov scheme. The ADI solver is first run with MAXIT-KRYIT iterations and then the Krylov solver is run with the remaining KRYIT iterations.
(4)	FROMM	[DScheme] Discretization scheme for volume fraction equation. More stable but diffusive.
(4)	SUPERBEE	[DScheme] Discretization scheme for volume fraction equation. Best balance between stability and diffusion.
(4)	SUPBC	[DScheme] Discretization scheme for volume fraction equation. More compressive but less stable.
(4)	STACS	[DScheme] Discretization scheme for volume fraction equation. Blended scheme, very compressive, more unstable.

Notes

- (1) The maximum number of iterations to be performed. Data to the log history is saved every tenth iteration with ADI and every iteration with KRYLOV.

- (2) This is useful only for developers.
- (3) Note that applying roughness requires lowering YTARGET. It is recommended to use YTARGET = 0.5. Recommended model: 4 – Aupoix-Colebrook.
- (4) Command applies to VOF solver only.

8.3. CONVERGENCE

This command is used to specify the solver convergence criteria.

General form of command

```
CONV      (  ON
              OFF
              ,  EPFX=v      ,  FXPSTD=v      ,
              MINIT=n      ,  TRBSTD=v      )
```

Default values

```
CONV      (  ON, EPFX=[0.1 for ADI, 0.025 for KRYLOV],
              FXPSTD=[1.0 for ADI, 0.25 for KRYLOV], MINIT=0,
              TRBSTD=0.01                                     )
```

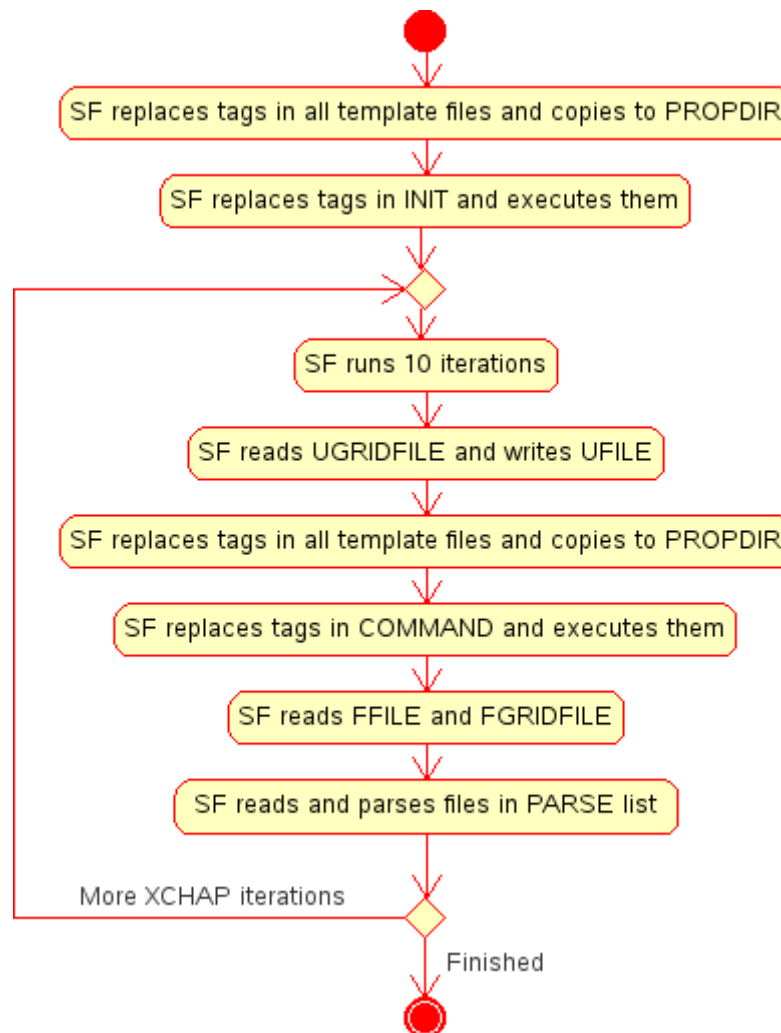
Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	ON/OFF	[CONVON] Switch on or off the automatic solver termination at convergence.
	EPFX	Maximum percentage difference of average Fx value between current and 100 iterations old solutions.
	FXPSTD	Maximum standard deviation of pressure contribution to Fx expressed in percent.
	MINIT	Minimum number of iterations before program stops due to convergence.
	TRBSTD	Maximum standard deviation of Thrust-Resistance balance.

8.4. EXTPROPELLER

The EXTPROPELLER command offers an open interface to external propeller programs. Every 10th iteration in XCHAP the velocities in the propeller grid is written to the UFILE and the propeller program(s) is launched. The propeller program(s) should write the body forces to the FFILE in the same grid.

Commonly an external propeller code requires the running of more than one programs, typically there is the potential flow propeller program itself and one or more programs that adapts the wake coming from SHIFLOW and computes the time averaged force field read by SHIPFLOW. There may also be a need for copying and/or renaming files, so to be able to do all these operations in general, EXTPROP runs a sequence of shell (command window) commands specified by the INIT and COMMAND parameters. The process is outlined in the diagram below.



In the directory where you have your command and offset file you should also have your template command and data file(s) for the propeller program. The user can make SHIPFLOW transfer data to the template file(s) and to the commands by entering certain tags in them. The format of a tag is $\$(ID)$, where ID is the name of the tag. Currently the following tags are available:

- ID = Propeller object ID.
- NRUN = Number of times the external program has been called.

J	=	Current advance ratio.
UGRIDFILE	=	Name of the file containing the wake grid.
UFILE	=	Name of the file containing the absolute wake
FORCEFILE	=	Name of the file containing the propeller force field
XP	=	X-coordinate of propeller, in propeller model coordinates
XEFF	=	X-coordinate of the wake plane, in propeller model coordinates
NI	=	First propeller grid dimension
NJ	=	Second propeller grid dimension
NK	=	Third propeller grid dimension

The propeller program will be run in a subdirectory called RUNID_DIR/PROPID_DIR. The RUNID is the same as the shipflow command file name and the PROPID is the same as the propeller name (or ID in the EXTPROPELLER command).

Most propeller programs behaves different the first time it is run compared to the following times it is run, since for the first run it does not have the values of the sources. This can be flagged in the appropriate template file by inserting the \$(NRUN) tag. The first time the propeller program is called \$(NRUN) will be replaced with 0, the following time \$(NRUN) will be replaced by 1 and so on.

General form of command

```

EXTPROP  ( ID="c"           , INIT=["c", ...]           , COMMAND=["c", ...]   ,
          TYPE="c"         , CTOW=v           ,
          INPUT=["c", ...] , OUTPUT=["c", ...] , UGRID="c"           ,
          UGFORM="c"       , UFILE="c"        , UFORM="c"           ,
          FGRID="c"        , FGFORM="c"       , FFILE="c"           ,
          FFORMAT="c"     , DMODEL=v        , ACTIVE              ,
          ON               , DEBUG           , ROTATION = [v1,v2,v3] ,
          OFF              , NODEBUG         ,
          FIX              ,
          EXTGRID          , FS_1            , US_1                ,
          INTGRID          , FS_KT           , US_V                ,
          UNC              , FNC             , FVOL                ,
          UCC              , FCC             , FCELL               ,
          CTGIVEN=v        , FRACTION=v      , DIME=[v1,v2,v3]    ,
          XUPSTREAM=v     , XDOWNSTREAM=v   , RMIN=v              ,

```

```

RMAX=v      , BC11="c"      , BC12="c"      ,
BC21="c"      , BC22="c"      , STR1="c"      ,
STR2="c"      , RELAX=v      , NLOG = n      ,
SLEEP = v      , RUNINT = n      , STARTAT = n      )

```

Default values

```

EXTPROP ( UGRID="grid3d.dat" , FFILE="f.dat" , UFILE="u.dat" ,
ON , RELAX=1.0 , CTOW=0.0 , XUPST=-0.3 ,
XDOWNST=0.3 , RMAX=1.2 , FRACTION=1.0 ,
DIME=[12,16,31] ,
UNC , US_1 , FCC , FS_1 , FCELL ,
BC11=BC12=BC21=BC22="INTERIOR" , NLOG=1 ,
SLEEP=0.1 , RUNINT=(10 for ADI, 1 for KRYLOV) ,
STARTAT=0 , NODEB

```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	ID	A string that is used to identify the propeller object. The string should correspond to the ID given in the PROPELLER command in the XFLOW section.
	INIT	Vector of strings, each one executed as a system command before XCHAP starts iterating. Before execution tags in the format \$(TAG) are replaced by current values.
	TYPE	Name of a file stored in <your_homedir>/.shipflow/propmodel/<type>. The file should be a Shipflow command file with a single XCHAP:EXTPROP command. This will be read before the actual case command file to set defaults for a particular external propeller model, thus allowing the case command file to skip these and be much shorter.
	COMMAND	Vector of strings, each one executed as a system command after each 10 iterations in XCHAP. Before execution tags in the format \$(TAG) are replaced by current values.
	INPUT	Vector of strings that specify names of template files. These will become the input files for the external propeller program. Before running the commands in INIT or COMMAND, tags in the format \$(TAG) are replaced by current values (file splicing) and the modified files are copied to the propeller directory where the external propeller program is run.

OUTPUT	Vector of strings that specify the names and formats of output files from the external propeller program. These files will be parsed according to the format after each run of the external propeller program. The format of each entry in PARSE is “filename(format)”, for example “mp2208.out(skipline \$(float) \$(KT) \$(KQ))”. This means: in file mp2208.out, skip the first line, skip a floating point value, read KT, readKQ.
DEBUG	Turn on debug output.
NODEBUG	Turn off debug output.
UGRID	Name of the wake grid file. This file should be created by the propeller program and read by SHIPFLOW, which interpolates the cartesian velocity components to each grid node and writes it to UFILE.
UGFORMAT	Specification of the format of UGRID. This is a recipe for telling SHIPFLOW how the propeller program specifies the grid in UGRID, for example “NI NJ{X Y Z}” which means: read grid dimensions NI and NJ, and then for each J=1..NJ, for each I=1..NJ, read x(I,J), read y(I,J), read z(I,J). A format code like “NI NJ{X} {Y} {Z}” means that all x values are read first, then y etc.
UFILE	Name of the wake file. This file is written by SHIPFLOW before calling the external programs(s) and contains the velocity of the current XCHAP solution, interpolated to the grid nodes or cell centers of UGRID.
UFORMAT	Format of the UFILE, specified in the same way as UGFORMAT. The velocity components are designated U, V and W in the format descriptor. The field dimensions are assumed to be the same as for the UGRID. For example “{U V W}” writes U(I,J) V(I,J) W(I,J) separated by spaces for each J,I.
FGRID	Name of the force grid file. This file should be created by the propeller program and read by SHIPFLOW. FGRID may be omitted if the propeller program writes both force and grid to the FFILE, see below.
FGFORMAT	Format of FGRID, specified in the same way as UGFORMAT.
FFILE	Name of the force field file. This file is written by the external propeller program and contains the time-averaged propeller force field. The force field is read by SHIPFLOW and added as a volume force to the flow equations. FFILE may also contain the grid for the force field.
FFORMAT	Format of the FFILE, specified in the same way as

		UGFORMAT. The force components are designated FX, FY and FZ. For example: "NI NJ NK{X Y Z FX FY FZ}".
	DMODEL	Diameter of the propeller in the external model. Normally this is either 1 (normalized by diameter) or 2 (normalized by radius).
	EXTGRID	[GRID] The wake and force field grids are produced by the external program.
	INTGRID	[GRID] A cylindrical grid made by SHIPFLOW is used for the force and wake fields.
	UCC	[ULOC] The values in the VELOCITYFILE are given in the cell centers.
	UNC	[ULOC] The values in the VELOCITYFILE are given in the node centers.
	FCC	[FLOC] The values in the FORCEFILE are given in the cell centers.
	FNC	[FLOC] The values in the FORCEFILE are given in the node centers.
	US_V	[US] Velocities in the wake file are non-dimensionalized by the free stream velocity at infinity, i.e. ship speed.
	US_1	[US] Velocities in the wake file are not scaled.
	FS_KT	[FS] Forces in the force file are non-dimensionalized by KT. This means that all the forces added up should be equal to KT.
	FS_1	[FS] Forces in the force file are raw. This means that all the forces added up should be equal to the force in the computation.
	FVOLUME	[FUNIT] Forces are defined as force per volume.
	FCELL	[FUNIT] Forces are defined as force per cell.
	ACTIVE	[ACTIVE] Run the RANS solver on the propeller grid. For embedded propeller grids this gives essentially the same result as the default mode where the propeller grid is only used for interpolation. With ACTIVE the influence of interpolation errors tends to be smaller, but at the default resolution details in the wake may be lost since the propeller grid is rather coarse close to the symmetry plane. If the propeller grid is used stand-alone, without being embedded in (an)other grid(s), boundary conditions have to be specified and ACTIVE mode used.
(1)	ON	[DISKON] The propeller model is run and the body forces

		are updated every 10 iterations.
(1)	OFF	[DISKON] The propeller model is not run and the body forces are set to zero.
	FIX	[DISKON] The propeller model is not run and the body forces are kept constant.
	ROTATION	Rotation of the grid(s) specified by the Euler angles with the pitch, roll, yaw convention. Given in degrees. Use this transformation to align the external propeller model coordinate system axis with the EXTPROPELLER convention: x-axis pointing downstream, y-axis pointing to starboard looking upstream and z-axis pointing up.
	DIME	Specify number of nodes in the axial, radial and tangential direction.
	XUPST	The position of the propeller grid inlet plane expressed as nondimensional propeller radius. TUNING COMMAND.
	XDOWNST	The position of the propeller grid outlet plane expressed as nondimensional propeller radius. TUNING COMMAND.
	RMIN	The propeller grid inner radius. If set to zero, BC21 is disregarded. If no value is given, RMIN is computed from DHUB in the PROPELLER command. TUNING COMMAND.
	RMAX	The propeller grid outer radius expressed as nondimensional propeller radius, i.e. RMAX=1 will put the outer boundary at the propeller tip. TUNING COMMAND.
	BC11	Specify the boundary condition on the upstream plane. Possible values are "NOSLIP", "SLIP", "INFLOW", "OUTFLOW", "INOUT" and "INTERIOR". The INTERIOR condition means that the boundary values are taken by interpolation from other grids. Only relevant when running the propeller grid alone.
	BC12	Specify the boundary condition on the downstream plane.
	BC21	Specify the boundary condition on the inner radius.
	BC22	Specify the boundary condition on the outer radius.
	STR1	Stretching in axial direction.
	STR2	Stretching in radial direction.
	RELAX	Relaxation factor. The force added to the right hand side of the RANS equations are relaxed by this factor. A low value makes the model more stable but slower to respond.

NLOG	Maximum number of log and error files that are saved from the external propeller model runs.
SLEEP	Sleep time in seconds between checking if the propeller model has finished. Very small values will load the computer with MPI polling, very large values will give latency. Not a critical value, but should not be larger than about 0.1 times the expected propeller model run time.
RUNINT	Number of solver outer iterations between running the propeller model.
STARTAT	Number of solver outer iterations before the proepller model is run for the first time.

Notes

- (1) Turning it OFF will still make XCHAP create a propeller grid, so start- up will take a little longer and more memory will be used. If you want to restart later with the propeller turned ON, this will only be possible if the propeller was present but OFF from the beginning. When the propeller is ON it will be run after every 10 Chapman iterations.

8.5. EXTRACT

The EXTRACT command to interpolates the solution to a set of specified points and writes the result to a Tecplot file. The points can be specified as a two-dimensional Cartesian grid, a two-dimensional circular grid or by a Tecplot file containing a point or block zone.

The extraction points will be transformed from the offset file coordinate system to the SHIPFLOW computational system, including sinkage and trim.

General form of command

```
EXTRACT ( FILE                , INFILE="c"                , OUTFILE="c"    ,
          DISC                ,
          RECT                ,
          LINE                ,

          X=v                  , Y=[v1,v2]          , Z=[v1,v2]      ,
          R=v                  , N=[v1,v2]          , SCALE=v        ,
          YSIGN=v              , ROTATION=[v1,v2,v3] ,
                                TRANSLATION=[v1,v2,v3] ,
          XIN="C"              , YIN="C"            , ZIN="C"        ,
          XVAR="C"             , YVAR="C"            , ZVAR="C"        ,
          UVAR="C"             , VVAR="C"            , WVAR="C"        ,
          PVAR="C"             , KVAR="C"            , OMVAR="C"       ,
          FIBVAR="C"           , LFIBVAR="C"         )
```

Default values

EXTRACT(SCALE=1, YSIGN=-1, ROTATION=[0,0,0], TRANSLATION=[0,0,0])

Keywords

<i>keyword</i>	<i>description</i>
FILE	[TYPE] The extraction points are given as a point zone in a Tecplot file. See the Tecplot manual for a description of the file format. The name of the Tecplot file is given by the OUTFILE parameter.
DISC	[TYPE] The extraction points are arranged in a uniform circular grid. The geometry is specified by the X and R parameters.

RECT	[TYPE] The extraction points are arranged in a Cartesian grid. The geometry is specified by the X, Y and Z parameters.
LINE	[TYPE] The extraction points are distributed on a linear segment. The end points are specified by the X, Y and Z parameters.
X	The x coordinate of the grid given in the offset coordinate system.
R	The inner and outer radius of the circular grid.
Y	The y coordinates of the corners of the Cartesian grid.
Z	The z coordinates of the corners of the Cartesian grid.
N	The dimensions of the circular or Cartesian grid. For a circular grid the first dimension is the number of points in the tangential direction, and for a Cartesian grid it is the number of points in the y direction. For a linear grid only the first value is used.
SCALE	Scale factor for the grid.
YSIGN	Set to 1 if the grid is on the positive side of $y=0$, -1 otherwise. This will place the grid on the negative side of $y=0$ where the XGRID grid is for symmetrical computations.
ROTATION	Rotation of the grid(s) specified by the Euler angles with the pitch, roll, yaw convention. Given in degrees.
TRANSLATION	Translation of the grid origin, specified in offset file coordinates.
XIN	(FILE only) Find which column to use as x-coordinate. If this parameter is omitted the value of X is used for all points.
YIN	(FILE only) Find which column to use as y-coordinate. If this parameter is omitted the value of Y is used for all points.
ZIN	(FILE only) Find which column to use as z-coordinate. If this parameter is omitted the value of Z is used for all points.
DIN	(FILE only) Find which column to use as distance.
XVAR	Extract the x-coordinate. The column will be identified by the value of XVAR. The extracted value is in the computational coordinate system.
YVAR	Extract the y-coordinate. The column will be identified by the value of YVAR. The extracted value is in the computational coordinate system.
ZVAR	Extract the z-coordinate. The column will be identified by the value of ZVAR. The extracted value is in the computational coordinate system.
UVAR	Extract the x-component of the velocity. The column will be identified by the value of UVAR. The extracted velocity is non-dimensionalized by ship speed.

VVAR	Extract the y-component of the velocity. The column will be identified by the value of VVAR. The extracted velocity is non-dimensionalized by ship speed.
WVAR	Extract the z-component of the velocity. The column will be identified by the value of WVAR. The extracted velocity is non-dimensionalized by ship speed.
PVAR	Extract the non-dimensional pressure $p=\frac{1}{2}C_p$. The column will be identified by the value of PVAR.
KVAR	Extract the non-dimensional turbulent kinetic energy. The column will be identified by the value of KVAR.
OMVAR	Extract ω . The column will be identified by the value of OMVAR.
FIBVAR	Extract distance to nearest NOSLIP boundary. The column will be identified by the value of FIBVAR.
LFIBVAR	Extract distance to nearest component grid boundary boundary. The column will be identified by the value of LFIBVAR.

8.6. FREE

The FREE command controls RANS computation with free sinkage and trim. It only applies to VOF free surface computations.

Computations with free sinkage and/or trim is done by heaving and pitching the hull to achieve balance in lift and pitching moment. After every 10 iterations the difference between the hydrodynamic lift force and pitching moment and the gravitational force and pitching moments are computed. Using the waterline area and area moment from XMESH, an equilibrium position is determined and the hull position is updated by deforming the grid.

During each iteration the convergence and number of iterations is controlled with the XCHAP|CONTROL|MAXIT parameters and the XCHAP|CONV. Since the grid is modified, more iterations may be needed than for a fixed hull computation.

If either a propeller position in XFLOW|PROPELLER or a tow force position (TOWX,0,TOWZ) is specified, then the horizontal force that balances the hydrodynamic drag is applied there. If no force position is given it is assumed that the drag is balanced by a force acting in the hydrodynamic center, so that there is no trimming moment. The latter condition can be also be mandated with the keyword NOMOMENT.

The result from each iteration is logged in the file <id>_RUN_DIR/xship_TRIMLOG. The final result is also written to <id>_OUTPUT.

Since XMESH is used to get hydrostatic data and LPP is used, free sinkage and trim calculations require that there is an XFLOW|OFFSET command.

General form of command

FREE	(	ON	,	HORIZON	,	ITSTART = i	,
		TRIM		SHAFT			
		SINK		NOMOMENT			
		OFF					
		ITEND = i		ITINTER = i		VERB = i	
		XTOW = v	,	ZTOW = v	,	TRCRIT = v	,
		SICRIT = v	,	DTRIMMAX = v	,	DSINKMAX = v	,
		TRMAX = v	,	SIMAX = v	,	TRRELAX = v	,
		SIRELAX = v	,	DSOLID = v	,	DFIXED = v	)

Default values

FREE (ON, NOMOMENT, ITSTART=300, ITEND=-1, ITINTER=10, SHAFT,

TRCRIT=1e-3, SICRIT=1e-5, DTRIMMAX=0.002, DSINKMAX=1e-5,
TRIMMAX=2, SIMAX=0.05, TRRELAX=0.02, SIRELAX=0.01, DSOLID=0.2,
DFIXED=0.4, VERBOSE=1)

Keywords

<i>keyword</i>	<i>description</i>
ON	[FREEON] The model is free to sink and trim.
TRIM	[FREEON] The model is free to trim.
SINK	[FREEON] The model is free to sink.
START	[MODE] In each iteration the RANS solver is started from an initial guess.
ITSTART	The hull remains fixed in its initial position until ITSTART iterations have been done. This is to let the initial solution gradients damp out.
ITEND	The hull remains fixed after this iteration. Default is ITEND=-1 which means infinity, i.e. the hull remains free for all iterations. There is usually no reason to change this parameter.
ITINTER	Number of iterations between each hull position update. There is usually no reason to change this parameter.
SHAFT	[TOWDIR] If an XFLOW PROPELLER command is present, the propeller thrust is applied parallel to the shaft.
HORIZONTAL	[TOWDIR] If an XFLOW PROPELLER command is present, the propeller thrust is applied in the horizontal direction.
XTOW	x-position in the offset (design) coordinate system of the centre of the drag balancing force.
ZTOW	z-position in the offset (design) coordinate system of the centre of the drag balancing force.
TRCRIT	Trim angle convergence criterion, in degrees.
SICRIT	Sinkage convergence criterion, expressed as fraction of LPP.
DTRIMMAX	Maximum absolute trim change.
DSINKMAX	Maximum absolute sinkage change.
TRMAX	Maximum allowed absolute trim angle.
SIMAX	Maximum allowed absolute sinkage.
TRRELAX	Relaxation factor for trim change.

SIRELAX	Relaxation factor for sinkage.
DSOLID	Inside this distance from the hull, the grid is deformed as a rigid body. The distance is non-dimensionalised with L_{pp} , so 0.2 means 20% of L_{pp} .
DFIXED	Outside this distance the grid is not deformed. Between DSOLID and DFIXED the deformation reduces linearly from rigid body to zero. DFIXED must be larger than DSOLID and the difference, typically 0.2, must allow for a smooth transition between the rigid body region and the fixed region.
VERBOSITY	VERBOSITY=2 prints diagnostics at each hull position update, the default is silent.

8.7. IMPORT

This command imports internally or externally generated structured grids. The command may be repeated to import grids from other files or re-import from previously used files. If a file is used more than once, the grids will appear as new objects, i.e. the previously imported grids will not be modified or overwritten. Importing a grid twice (or more) is not by itself useful since the grids will cover the same domain, but by applying translation and/or rotation it can be used to represent a recurring geometrical feature.

SHIPFLOW can re-import XCHAP-grids from an CGNS file produced from SHIPFLOW in a previous computation.. Typically, this is useful when setting up a catamaran or trimaran case. In order for SHIPFLOW to recognize that it is such a CGNS file, the keyword XCHAP should be set in the IMPORT command. GRIDNAME also needs to be set to read in the correct grid from the CGNS file. If there is a need to change the boundary conditions the BC11, BC12, BC21, BC22, BC31, BC32 keywords can be used.

Externally created grids in Plot3D ASCII file format can also be imported. The file must have the suffix .fmt or .p3d to be recognized as a Plot3D file. Each file may contain one or more grids. The Plot3D format is

```
N
L1 M1 N1 L2 M2 N2 ... Lnb Mnb Nnb
X11 X12 ... X1N1
Y11 Y12 ... Y1N1
Z11 Z12 ... Z1N1
m11 m12 ... m1N1
X21 X22 ... X2N2
.
.
.
Znb1 Znb2 ... ZnbNnb
mnb1 mnb2 ... mnbNnb
```

where N is the number of grids (blocks), L₁ M₁ N₁ are the dimensions of the first grid, x₁₁, y₁₁, z₁₁ are the x, y, z-coordinates of the first node in the first grid, m₁₁ is an integer value that specifies properties for the first node in the first grid (not used by SHIPFLOW) and N₁ = L₁ x M₁ x N₁ is the number of nodes in the first grid. The ordering of the node data is such that if node number i corresponds to point (l,m,n) in index space, then node i+1 corresponds to (l+1, m,n). The numbers are separated by whitespace.

SHIPFLOW can also import externally generated grids from files in the CGNS format. Such files can be exported from for example ICEM CFD and Gridgen. The advantage with this format is that it can contain the connectivity information for the butt-joined grid faces and the boundary conditions for all faces. If the connectivity is specified in the file then it is important to set the NOSTRETCH parameter, since stretching the component grids is likely to invalidate the connectivities. If boundary conditions are specified in the command file they will override the values in the CGNS file. If no boundary conditions need to be specified in the command file then all grids in the CGNS file can be imported with a single IMPORT command.

Since there are several ways to store a grid in a CGNS file, the following should be noted:

8. Use CGNS version 2.
9. There should be only 1 base in the file.
10. The domains should be located directly under the base.
11. Each domain should be of ZoneType Structured
12. The ZoneGridConnectivity data should have the Node Label GridConnectivity1to1_t.
13. The ZoneBC data should have the Node Label BC_t and the values map to SHIPFLOW boundary conditions according to the table below. Any unspecified boundary conditions will be assigned INTERIOR.

CGNS BC identifier	SHIPFLOW bc identifier
BCInflow	INFLOW
BCOutflow	OUTFLOW
BCSymmetryPlane	SLIP
BCWall	NOSLIP
BCWallViscous	NOSLIP
BCWallInviscid	SLIP
BCTypeNull	INTERIOR

14. Under each BC_t node should be
 1. PointRange with Node Label equal to IndexRange_t.
 2. InwardNormalIndex.
 3. Any other nodes are ignored.

Use a CGNS browser such as adfviewer to check that the grid data is correctly stored in the CGNS file.

The grids are assumed to be specified in the offset file coordinate system, but if it is not it may be translated, rotated and scaled. The rotation is specified by Euler angles using the pitch-roll-yaw conventions. All transformations are done starting from the grid coordinate system. The transformations are made in the order SCALE, YSIGN, ROTATION, TRANSLATION.

General form of command

```
IMPORT      (  ID = "c"                      ,  GROUP = n                      ,
              FILE= "c"                      ,  NUMBER = n                      ,
              TRANSLATION = [v1,v2,v3]      ,  ROTATION = [v1,v2,v3]          ,
              RANGE = [n1,n2,n3,n4,n5,n6]   ,  DIMENSION = [n1,n2,n3]        ,
              YSIGN = n                      ,  SCALE =v                      ,
              BC11 = "c"                    ,  BC12 = "c"                    ,
              BC21 = "c"                    ,  BC22 = "c"                    ,
              BC31 = "c"                    ,  BC32 = "c"                    ,
              ROTCENT = [v1,v2,v3]          ,  STRETCH
                                                NOSTRETCH                      ,
              GRIDNAME = "c"                ,  XCHAP                          ,
              CUT                            ,  TRIM                          ,
              NOCUT                          ,  NOTRIM                        ,
              SCUT=[n1,n2,n3,n4,n5,n6]      ,  STRDIR = [n1,n2,n3]          ,
              COMPUT                        LINEAR                          )
                                                CUBIC
                                                HERMITE
                                                ENO
```

Default values

```
IMPORT(  NUMBER=0, TRANSLATION=[0,0,0], ROTATION=[0,0,0], YSIGN=-1,
        SCALE=1, BC11="INTERIOR", BC12="INTERIOR", BC21="INTERIOR",
        BC22="INTERIOR", BC31="INTERIOR", BC32="INTERIOR", CUT, TRIM,
        ENO)
```

Keywords

<i>keyword</i>	<i>description</i>
----------------	--------------------

ID	Gives the imported grid a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different grids may have the same name, but then only the total force on all of those objects will be calculated and output. For import file formats such as CGNS that supply a name, ID may be omitted.
----	--

FILE	Name of the Plot3D file that specifies the grid. The file name must end with .fmt or .p3d.
NUMBER	If there are more than one grids in the file, import only the NUMBERth one. The first grid in the file is number one. If the default value NUMBER=0 is given, all grids in the file are imported and they get the same name, transformations and boundary conditions.
GROUP	Group number in the overlapping grid algorithm. If not specified the grid will be the only grid in the group.
SCALE	Scale factor for the grid.
YSIGN	Set to 1 if the grid is on the positive side of $y=0$, -1 otherwise. This will place the grid on the negative side of $y=0$ where the XGRID grid is for symmetrical computations.
ROTATION	Rotation of the grid(s) specified by the Euler angles with the pitch, roll, yaw convention. Given in degrees.
ROTCENT	Center of rotation for the euler angles given in ROTATION.
TRANSLATION	Translation of the grid origin, specified in offset file coordinates.
RANGE	Index range to import, [Lmin,Lmax,Mmin,Mmax,Nmin,Nmax]. If not specified, the whole range is imported.
DIMENSION	Set grid dimensions. Default is to use the dimensions in the file. Changing the default means that new node positions are interpolated from the original ones, this may lose features like sharp edges.
BC11	Specify the boundary condition on side $l=1$ of the grid. Possible values are "NOSLIP", "SLIP", "INFLOW", "OUTFLOW", "INOUT", "INTERIOR" and "MIXED_BC" (see section "H-O topology grid import" on page 312). The INTERIOR condition means that the boundary values are taken by interpolation from other grids. For grids imported from SHIPFLOW CGNS files there is an additional option. One boundary condition can be changed to some other boundary condition with the '$\rightarrow$' operator. Examples are "INTERIOR$\rightarrow$SLIP" "SLIP$\rightarrow$INTERIOR" etc.
BC12	Specify the boundary condition on side $l=L$ of the grid.
BC21	Specify the boundary condition on side $m=1$ of the grid.
BC22	Specify the boundary condition on side $m=M$ of the grid.
BC31	Specify the boundary condition on side $n=1$ of the grid.
BC32	Specify the boundary condition on side $n=N$ of the grid.
STRETCH	[STRETCH] Stretch the grid.

NOSTRETCH	[STRETCH] Do not stretch the grid. The grid is used as given in the imported file.
STRDIR	Stretch only in directions j where STRDIR[j] == 1.
XCHAP	File to import from is a SHIPFLOW (XCHAP) CGNS file.
GRIDNAME	Name of the grid in the imported SHIPFLOW CGNS file.
CUT	Include faces with boundary condition=NOSLIP in hole cutting. This is seldom a concern.
NOCUT	Exclude grid faces from hole cutting.
SCUT	Select which grid faces to include in hole cutting. 1=side cuts, 0=side does not cut, [Lmin,Lmax,Mmin,Mmax,Nmin,Nmax].
TRIM	Translate and rotate the grid(s) together with the hull sinkage and trim. This is the default and appropriate for grids that model appendages on the hull.
NOTRIM	Keep the grid(s) fixed, do not follow the hull sinkage and trim. This is appropriate for grids that model objects in the ships environment like a canal bank.
COMPUT	Do not transform the grid(s) to the computational coordinate system. Use this if the imported grids are already defined there, i.e. Lpp=1, origin at FP.
LINEAR	Linear interpolation. Robust.
CUBIC	Cubic interpolation. May oscillate.
HERMITE	Hermite interpolation. May oscillate.
ENO	ENO interpolation. Robust and accurate. Default.

8.8. KRYLOV

This command is used to control the krylov solver. The krylov solver is turned on in the CONTROL command in the XCHAP section.

General form of command

```
KRYLOV ( STARTIT = n      ,  NSTOL = v          ,  KTOL = v      ,  OTOL = v      ,
          MAXIT = n
          RELAX = v        ,  VLIMIT = v          ,  NSBJAC      ,  KBJAC      ,
                               NSASM      ,  KASM
          OBJAC      ,  NSGMRES      ,  KGMRES      ,  OGMRES      )
          OASM      NSBCGS      KBCGS      OBCGS
                   NSCGS      KCGS      OCGS
                   NSBICG      KBICG      OBICG
```

Default values

```
KRYLOV ( STARTIT=30, MAXIT=100, NSTOL=1e-3, KTOL=1e-5, OTOL=1e-5, )
          RELAX=0.8,
          VLIMIT=0.5, NSBJAC, KBJAC, OBJAC, NSGMRES, KGMRES,
          OGMRES
```

Keywords

<i>keyword</i>	<i>description</i>
STARTIT	Starting number of inner iterations per outer iterations. The number of inner iterations will increase as the computation progresses. STARTIT can be increased if problems are encountered.
MAXIT	Maximum number of inner iterations.
NSTOL	Relative residual decrease per outer iteration for the Navier-Stokes and Continuity equations.
KTOL	Relative residual decrease per outer iteration for the k-equation.
OTOL	Relative residual decrease per outer iteration for the omega-equation.
RELAX	Blending control between old and new values. 0 means keep the old value. 1 means use the new value entirely.
VLIMIT	Limit the update of the normalized velocities.
NSBJAC	[NS_PC] Block jacobi preconditioner for Navier-Stokes and the

continuity equations.

NSASM	[NS_PC] Additive Schwarz Method preconditioner for Navier-Stokes and the continuity equations.
KBJAC	[K_PC] Block jacobi preconditioner for k equation.
KASM	[K_PC] Additive Schwarz Method preconditioner for k equation.
OBJAC	[O_PC] Block jacobi preconditioner for omega equation.
OASM	[O_PC] Additive Schwarz Method preconditioner for omega equation.
NSGMRES	[NS_KSP] GMRES solver for Navier-Stokes and the continuity equations.
NSBCGS	[NS_KSP] BiCGStab solver for Navier-Stokes and the continuity equations.
NSCGS	[NS_KSP] CGS solver for Navier-Stokes and the continuity equations.
NSBICG	[NS_KSP] BiCG solver for Navier-Stokes and the continuity equations.
KGMRES	[K_KSP] GMRES solver for k equation.
KBCGS	[K_KSP] BiCGStab solver for k equation.
KCGS	[K_KSP] CGS solver for k equation.
KBICG	[K_KSP] BiCG solver for k equation.
OGMRES	[O_KSP] GMRES solver for omega equation.
OBCGS	[O_KSP] BiCGStab solver for omega equation.
OCGS	[O_KSP] CGS solver for omega equation.
OBICG	[O_KSP] BiCG solver for omega equation.

8.9. LLINE

This command is used to control the running of the lifting line actuator disc. The propeller position, geometry and advance ratio is taken from the PROPELLER command in the XFLOW section.

LLINE in a duct

The potential flow Goldstein Kappa model in LLINE assumes that the propeller is in free flow. A propeller duct with small tip clearance modifies the blade span loading in a way that cannot be reproduced by LLINE making the result inaccurate. To handle this the parameters DUCT, UTRED and UTEXP are used.

The effect of a duct is accelerate/decelerate the axial velocity and to act as an endplate that stops the tip vortex from forming close to the blade. The first effect is captured in the wake, but the second needs to be handled directly in the potential flow model. This is done by reducing the tangential component of induced velocity, which otherwise decreases the blade angle of attack and hence the blade load at the tip. The reduction is done by multiplying the tangential induced velocity by $(1 - UTRED * (r/R)^{UTEXP})$. Adding the keyword DUCT sets $UTRED = 0.9$ and $UTEXP = 1.5$ which is chosen to replicate the Wageningen Ka systematic series. It is expected that ducted propellers with similar blade and duct geometry will be accurately represented with these values, but if open water data is available it should be used to tune the values of UTRED and UTEXP. For geometries that differ significantly from the Ka series, this is mandatory. Giving the UTRED and UTEXP values explicitly overrides the defaults.

See also the XCHAP|POW section on ducted propellers.

General form of command

```
LLINE ( ID="c" , RELAX = v , CF=v , ACTIVE ,
        DJLIMIT = v DIME=[v1,v2,v3] , XUPST=v , XDOWNST=v ,
        RMIN=v , RMAX=v , BC11="c" , BC12="c" ,
        BC21="c" , BC22="c" , STARTAT=n , WPLANE = n
        ON , DKDJF = v , RUNINT=n , VERBOSITY = n ,
        OFF
        FIX
        CTGIVEN = v DUCT , UTRED = v , UTEXP = v )
```

Default values

```
LLINE ( ID from PROP if only one propeller, RELAX=0.2,
        CF=0.004, XUPST=-0.3, WPLANE=5, )
```

XDOWNST=0.3, RMAX=1.2, STARTAT=0,
BC11=BC12=BC21=BC22="INTERIOR", DIME=[12,16,31],
DKDJF=2 (KRYLOV) or 1 (ADI), RUNINT=1 (KRYLOV) or 10 (ADI),
VERBOSITY=1, UTRED=0, UTEXP=1.5

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	ID	A string that is used to identify the propeller and connect it to a PROPELLER. Most files output by MPUF will have names composed with ID as the first part. All terminal output from the last running of MPUF (and e3dummy) will be directed to a file called <ID>_out.log. KT, KQ and VEFF for all runs of MPUF are saved in the file <ID>_PROPLOG.
	RELAX	Relaxation factor. Both the circulation in the lifting line model and the force added to the right hand side of the RANS equations are relaxed by this factor. A low value makes the model more stable but slower to respond.
	CF	Wing section friction drag coefficient for the propeller blades. Setting CF to a positive value increases KQ. CF depends on the Reynolds number, typical values are in the range 0.002 to 0.008.
(1)	ON	[DISKON] The propeller model is run and the body forces are updated every 10 iterations.
(1)	OFF	[DISKON] The propeller model is not run and the body forces are set to zero.
	FIX	[DISKON] The propeller model is not run and the body forces are kept constant.
	ACTIVE	[ACTIVE] Run the RANS solver on the propeller grid. For embedded propeller grids this gives essentially the same result as the default mode where the propeller grid is only used for interpolation. With ACTIVE the influence of interpolation errors tends to be smaller, but at the default resolution details in the wake may be lost since the propeller grid is rather coarse close to the symmetry plane. . If the propeller grid is used stand-alone, without being embedded in (an)other grid(s), boundary conditions have to be specified and ACTIVE mode used.
	STARTAT	Start the propeller model after STARTAT iterations. This can typically be set to 40 for the KRYLOV solver if problems are encountered.
	XUPST	The position of the propeller grid inlet plane expressed as nondimensional propeller radius. TUNING COMMAND.

XDOWNST	The position of the propeller grid outlet plane expressed as nondimensional propeller radius. TUNING COMMAND.
RMIN	The propeller grid inner radius. If set to zero, BC21 is disregarded. If no value is given, RMIN is computed from DHUB in the PROPELLER command. TUNING COMMAND.
RMAX	The propeller grid outer radius expressed as nondimensional propeller radius, i.e. RMAX=1 will put the outer boundary at the propeller tip. TUNING COMMAND.
BC11	Specify the boundary condition on the upstream plane. Possible values are “NOSLIP”, “SLIP”, “INFLOW”, “OUTFLOW”, “INOUT” and “INTERIOR”. The INTERIOR condition means that the boundary values are taken by interpolation from other grids. Only relevant when running the propeller grid alone.
BC12	Specify the boundary condition on the downstream plane.
BC21	Specify the boundary condition on the inner radius.
BC22	Specify the boundary condition on the outer radius.
DIME	Specify number of nodes in the axial, radial and tangential direction.
WPLANE	Propeller matching plane in the cylindrical propeller grid.
DKDJF	Factor controlling the change in J due to the imbalance between thrust and drag for a self-propulsion run.
DJLIMIT	Maximum allowed change of J in self propulsion computation.
RUNINT	Number of solver outer iterations between running the propeller model.
VERBOSITY	VERBOSITY=2 prints diagnostics at each propeller force update, the default is silent.
CTGIVEN	Total drag coefficient, overrides value from computation.
DUCT	Keyword that turns on duct compensation with UTRED=0.9 and utexp=1.5.
UTRED	Tangential velocity reduction factor, see the section on ducted propellers above.
UTEXP	Tangential velocity reduction exponent, see the section on ducted propellers above.

Notes

- (1) Turning it OFF will still make XCHAP create a propeller and a propeller grid, so start- up will take a little longer and more memory will be used. If you want to restart later with the propeller turned ON, this will only be possible if the propeller was present but OFF from the beginning. When the propeller is ON it will be run after every 10 Chapman iterations for ADI and every Chapman iteration for KRYLOV.

8.10. OVERLAP

This is a tuning command that gives the user control of some of the parameters in the overlapping grid algorithm. The normal use of this command is to diagnose problems with non- fluid points “leaking” out into and filling the fluid domain. By giving the keywords BDFLAG or ONEFILL the recursive filling is inhibited and the non-fluid classified points can be inspected in SPOST.

A global refinement factor can be specified for all grids with XGRID|SIZE|VCOARSE, COARSE, MEDIUM or FINE. If a different factor is required for all grids except the hull grid, like rudders, shafts, brackets etc, this is done with the corresponding keywords in the OVERLAP command together with the keyword REFSEP (REFine SEParately). FINE is the default and corresponds to a refinement factor $r=1$.

The refinement factor can also be given explicitly with GREFINE. A value >1 gives more cells and <1 fewer cells. The factor is applied equally in all directions so $\text{GREFINE} = 2$ gives $2^3 = 8$ times more cells. In keyword form the ratio between values is $\sqrt[4]{2}$ so MEDIUM gives a refinement factor of $1/\sqrt[4]{2}$ and VFINE $\sqrt[4]{2}$. Refinement factors that are not an integer power of 2 means that the number of cells is rounded and hence the refinement is only approximately uniform.

The value of YTARGET is multiplied by $1/r$ to keep the refinement uniform.

Specifying the grid dimensions explicitly with the DIMENSION parameter overrides this functionality.

General form of command

```
OVERLAP ( FILL          , TRIM          , RECLASS      , VCOARSE      ,
          NOFILL        , NOTRIM         , NORECLASS     COARSE
          BDFLAG         ,                 ,                MEDIUM
          ONEFILL        ,                 ,                FINE
                                   VFINE

          FIL1           , REFEXGR        , IWEIGHT       ,
          FIL2           , REFSEP         , NOIWEIGHT     ,
          FIL3

          NFILL = n      , VOLDIST=v      , GREFINE = v   NOLEAKCHECK,
          RELAX = v      , NTHREAD = n    , SMALLV = v    ,
          INTSAVE        , GHTRIM         ,                )
```

Default values

```
OVERLAP ( FILL, NFILL=-1, FIL1, TRIM, SMALLV=0.0, VOLDIST=10e15,
          NOIWEIGHT,RELAX=1.0 (0.8 if Krylov solver is used,
          NTHREAD=MAX(PARA|NTHREAD,PARA|NPROC), COSVMIN=0.5,
          FINE, IWEIGHT, REFEXGR)
```

Keywords

<i>keyword</i>	<i>description</i>
FILL	[FILL] Recursively mark cells that are outside of NOSLIP boundaries as non-fluid.
NOFILL	[FILL] Do not mark any cells as non-fluid.
BDFLAG	[FILL] Mark cells that are cut by a NOSLIP boundary as non-fluid. This is done to seed regions that are outside of the fluid domain before recursively filling them. This keywords stops the algorithm after the seeding.
ONEFILL	[FILL] Do the seeding and do the filling step once. Useful for finding “leaks”.
NFILL	Stop the filling of non-fluid volumes after NFILL recursions. This is useful for finding the location of the leak. The technique is to find NFILL such that only a few outside cells leaks out in the fluid volume. The cells classified as outside can be visualized in Shipflow Design/CAESES.
FIL1	Select filling algorithm 1. This is the old algorithm and also the default. It finds initial non-fluid cells by investigating cells that are intersected by a physical boundary and using a geometric criterion to determine if the intersected cells center is inside or outside of the intersecting cell face.
FIL2	Select filling algorithm 2. It finds initial intersected cells like algorithm 1 but then finds the closest point on the total physical boundary. If the scalar product of the vector from the cell center to the closest point and the inward (seen from the fluid side) surface normal vector in the closest point is positive the cell is classified as non-fluid.
FIL3	Select filling algorithm 3. It loops through all cells and finds and classifies the nearest point on the physical boundary like algorithm 2. Since it investigates all cells it is considerably slower than the other two.
IWEIGHT	Use a more accurate calculation of interpolation weights for surface integrals in overlapped regions.
NOIWEIGHT	Use a less accurate calculation of interpolation weights for surface integrals in overlapped regions. This can be used in the rare case that the more accurate method fails.
TRIM	[TRIM] No recursive interpolation between overlapping grids will occur. This is the preferred way to run overlapping grids.
GHTRIM	[TRIM] In an additional step, recursive interpolation involving ghost cells are removed.
NOTRIM	[TRIM] Should be seen as a post processing command. If the plots look strange in SPOST try restart SHIPFLOW with NOTRIM turned on and MAXIT=0 and RESTART set in the CONTROL command in the XCHAP module.

SMALLV	Controls what is considered a small cell volume. Cells smaller than this value will be treated in a more robust but less accurate way. SMALLV can not be set to less than 0.0.
VOLDIST	If distance to the nearest NOSLIP boundary is greater than VOLDIST, then the cell with the larger volume will interpolate from the cell with the smaller volume if they overlap. VOLDIST should be given in the nondimensional coordinate system.
RELAX	Blending control between old and new value in interpolation cells. 0 means keep the old value. 1 means use the new value entirely.
NTHREAD	Number of threads used in the classification of cells.
RECLASS	Reclassify orphan interpolation INTERIOR to NOSLIP.
NORECLASS	Do not reclassify orphan interpolation INTERIOR to NOSLIP.
REFXGR	Refine globally COARSE, MEDIUM etc according to the value in XGRID SIZE.
REFSEP	Refine separately from the specification in XGRID SIZE.
VCOARSE	[REFLEV] Sets the refinement factor to $1/2$.
COARSE	[REFLEV] Sets the refinement factor to $1/\sqrt{2}$.
MEDIUM	[REFLEV] Sets the refinement factor to $1/\sqrt[4]{2}$.
FINE	[REFLEV] Sets the refinement factor to 1.
VFINE	[REFLEV] Sets the refinement factor to $\sqrt[4]{2}$.
GREFINE	Sets the refinement factor. Value must be >0 . Overrides REFLEV.
NOLEAKCHECK	Do not check the overlapping grid for leaks. In rare cases the test gives a false positive, i.e. a component grid has a large proportion of outside and interpolation cells but not due to a leak. In this case NOLEAKCHECK allows the computation to proceed.
INTSAVE	Save interpolation information as fields in the CGNS file, so that it can be visualized in external software like Paraview. This is mainly for code development purposes.

8.11. PARALLEL

This command controls parallel execution of XCHAP. The techniques supported in the current version are message passing and multithreading.

Message passing

In this parallelization technique the problem is split up and worked upon by different processes that communicate by sending messages. Each process runs a program in its own memory space. The programs are usually identical except for the data they work on. The most common standard for message passing is MPI (Message Passing Interface) which has several implementations. Shipflow uses OpenMPI on Linux and Microsoft MPI on Windows.

MPI programs are run on clusters of computers connected through a network. For small clusters a standard gigabyte Ethernet may be enough, but for large clusters specialized hardware is used since the amount of communication grows as the problem is divided into smaller parts.

A requirement of Shipflow is that there is a common file system, i.e. that all involved computers can access the same data directory. This is not an unusual requirement and if you use a large cluster a common file system is probably already there. If you set up your own you will have to do this yourself using for example NFS.

Running on a Linux cluster

The small cluster scenario is a few standard workstations, say up to 8, connected with a standard Ethernet. The OpenMPI default way to establish communication between the processes is using ssh. For this to work you need to set up your computers so that giving a password is not required when communicating between them. The recommended way is using DSA passphrases. The FAQ on the OpenMPI web pages gives these instructions on how to do it on a Linux machine:

The first thing that you need to do is generate an DSA key pair to use with ssh-keygen:

```
shell$ ssh-keygen -t dsa
```

Accept the default value for the file in which to store the key (`[extract_itex]HOME/.ssh/id_dsa`]) and enter a passphrase for your key pair. You may choose to not enter a passphrase and therefore obviate the need for using the ssh-agent. However, this greatly weakens the authentication that is possible, because your secret key is potentially vulnerable to compromise because it is unencrypted. It has been compared to the moral equivalent of leaving a plain text copy of your password in your `[/extract_itex]HOME` directory. See the ssh documentation for more details.

Next, copy the `[/extract_itex]HOME/.ssh/id_dsa.pub` file generated by ssh-keygen to `[/extract_itex]HOME/.ssh/authorized_keys` (or add it to the end of `authorized_keys` if that file already exists):

```
shell$ cd [extract_itex]HOME/.ssh  
shell$ cp id_dsa.pub authorized_keys
```

In order for DSA authentication to work, you need to have the `[/extract_itex]HOME/.ssh` directory in your home directory on all the machines you are running Open MPI. If your home directory is on a common filesystem, this may be already taken care of. If not, you will need to copy the

\$HOME/.ssh directory to your home directory on all Open MPI nodes (be sure to do this in a secure manner -- perhaps using the scp command -- particularly if your secret key is not encrypted).

ssh is very particular about file permissions. Ensure that your home directory on all your machines is set to at least mode 755, your \$HOME/.ssh directory is also set to at least mode 755, and that the following files inside \$HOME/.ssh have at least the following permissions:

```
-rw-r--r-- authorized_keys
-rw----- id_dsa
-rw-r--r-- id_dsa.pub
-rw-r--r-- known_hosts
```

The phrase "at least" in the above paragraph means the following:

The files need to be readable by you

The files should only be writable by you

The files should not be executable

Aside from id_dsa, the files can be readable by others, but do not need to be

Your \$HOME and \$HOME/.ssh directories can be readable by others, but do not need to be

For large clusters you are probably not required to do this since they have an internal network that does not require high security. On the other hand you are probably required to share the nice, big cluster with other users so some kind of queue system is likely.

SHIPFLOW integrates with the PBS/Torque job by looking at the value of the environment variable PBS_HOST_FILE. If this variable is set to a valid file name then a list of hosts is read from that file and neither NPROC nor HOSTFILE need to be specified in the SHIPFLOW command file. This feature can be turned off by the NOPBS keyword.

Running on a Windows cluster

Running on more than 1 node requires that the user has the same user credentials on all nodes. This in turn requires that a Windows Domain with Windows Active Directory is set up. It also requires that a master node running the Windows HPC Server 2008 operating system to manage the cluster. The detailed set up and configuration of the cluster can be done in many ways and we refer to the Microsoft documentation for this information.

SHIPFLOW integrates with the Windows Job Manager by looking at the value of the environment variable CCP_HOST_LIST. If this variable is set then a list of hosts is read from it and neither NPROC nor HOSTFILE need to be specified in the SHIPFLOW command file.

Multithreading

This means that the program still runs in one process with one memory space, but with more than one program pointer. If the hardware supports it, the different threads may be executed concurrently on different processors. The kind of hardware that can support this kind of execution is Shared Memory Processor (SMP) machines. This includes multi-processor machines and multi core processors. Parallel computers without shared memory, like clusters, cannot use multiple threads, but within each node you can still benefit from threading. In XCHAP when you specify NTHREADS>1, each process will run with that number of threads.

The whole XCHAP code is not parallelized yet, only the two most work-demanding where the

flow equations are set up and solved. This means that the pre-processing stage where the grid is set up, and the post-processing stage where data is saved etc. will not run faster.

If other jobs are running, the maximum number of threads for a new job is M-N, where M is the maximum number of processes allowed by the license, and N the number in use. Here a job running with m processes and n threads counts as m x n threads.

General form of command

```
PARALLEL ( NPROCESSES = n      , NTHREADS = n      ,  
           HOSTFILE = "c"      , NPERNODE = n      ,  
           MPIPARAM = "c"      , NOPBS              )  
           ENVP = "c"          , ENVN = "c"  
           ENVC = "c"
```

Default values

```
PARALLEL ( NPROCESSES = [adapted to your CPU and license] , NTHREADS = 1 )
```

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	NPROCESSES	Requested number of processes in parallel execution. Do not set to a higher number than your number of physical CPU cores, since that would only slow down the computation.
	NTHREADS	Requested number of threads in parallel execution for one process. XCHAP does not support running with multi-threading and multi-processes at the same time, so keep NTHREADS=1 if NPROCESSES>1. Do not set to a higher number than your number of CPU threads, since that would only slow down your computation. NOTE FOR LINUX: From SHIPFLOW 6.0, OpenMPI v1.8.3 is being used where the default is to use the same number of CPU cores as the number of processes i.e. 1. That means that only one CPU core will be used no matter how many threads you set. A workaround for this is for example for 4 threads to set MPIPARAM="--map-by slot:pe=4".
	NPERNODE	Maximum number of processes per node. If undefined the MPI implementation distributes the processes.

	HOSTFILE	Name of a text file containing information about the machines the program will run on. The file is a text file with a computer URL on each line, optionally followed by slots=n where n is the number of processors on the machine. The number of slots gives MPI a hint about how many processes it should run on a particular node to achieve load balance. A host file may look like this: <pre>serv1.domain.name slots=1 serv2.domain.name slots=1</pre>
	MPIPARAM	Extra MPI parameters that are passed verbatim to mpiexec.
	NOPBS	Do not use the PBS/Torque host file.
(1)	ENVP	Environment Variable for Number of processes.
(1)	ENVC	Environment Variable for Number of cores per node.
(1)	ENVN	Environment Variable for Number of nodes.

Notes

- (1) Used to override number of processes from the configuration file by the system environment variables. Either use ENVP or ENVC and ENVN. See the queueing system documentation for relevant variables.

8.12. POW

The Propeller Open Water simulations can be controlled with this command.

NOTE: XFLOW:HULLTYPE:POW needs to be set.

Ducted and other close-coupled propeller devices

When doing open water calculations with a ducted propeller, the duct has to be included since the propeller and duct interaction is so close-coupled that analyzing them separately is not accurate. Other types of appendages may also need to be included in a POW computation. To do this, use the APPENDAGE and KEEP parameters. Grids listed in APPENDAGE are included in the POW computation, and their thrust is added to the propeller thrust. Objects listed in KEEP are also copied to the POW command file, this is used to keep things like for instance a 2D section for a ROTBODY forming a duct.

See also LLINE|DUCT, UTRED and UTEXP to learn how to run LLINE in a duct.

General form of command

```

POW      (  J = [v1,v2...]          ,  OUTPUTFILE = "c",  MAXIT = n  ,
          RN = v                    ,  DIME = [n1,n2,n3]  ,  REFINES = v  ,
          VERBOSE = n               ,  COUPLED              ,
                                     UNCOUPLED
          ON                        ,  START                COARSE  ,
          OFF                      ,  RESTART              MEDIUM ,
                                     FINE
          APPENDAGE =               KEEP =                  )
          ["c1","c2",...]          ["c1","c2",...]

```

Default values

```

REFINE  ( ON, J=[0.5,0.6,0.7,0.8,0.9], OUTPUTFILE="id_pow.dat",
          MAXIT=500 (150 with KRYLOV), FINE (inherited from XFLOW:HULL)
          REFINES=1, VERBOSE=1, START, COUPLED)

```

Keywords

<i>keyword</i>	<i>description</i>
----------------	--------------------

J	List of advance coefficients to be calculated.
---	--

OUTPUTFILE	Name of the file where a table of J, KT, KQ and Eta_o is stored.
MAXIT	Number of iterations in XCHAP for each J.
RN	Reynolds number based on the propeller diameter.
DIME	Dimensions of the box grid.
REFINE	Refinement factor for box grid.
VERBOSE	How much diagnostics to print. [0..4]
START	[START] Start from initial guess.
RESTART	[START] Restart from the previous run.
COARSE	[GDENS] Coarse box grid.
MEDIUM	[GDENS] Medium box grid.
FINE	[GDENS] Fine box grid.
ON	[POWON] Run the POW calculations.
OFF	[POWON] Do no run the POW calculations.
COUPLED	[COUPLED] Run the propeller model coupled to the RANS solver. This includes the effects of coupling as in the self propulsion calculation and generally gives better results.
UNCOUPLED	[COUPLED] Run the propeller model without coupling to the RANS solver. Much faster but only useful to test the propeller model.
APPENDAGE	List of strings that match the ID (name) of 3D component grids that make up a close-coupled propeller appendage that needs to be included in the POW computation. The thrust (negative drag) on these grids are added to the propeller thrust. Three columns are added to the OUTPUTFILE: Ktd which is the K_T contribution from the appendages (typically a duct), Kta which is the total K_T and eta_a which is the efficiency of the propeller and appendage(s).
KEEP	XFLOW commands/objects used in the APPENDAGE grids, e.g. a 2D WSECT used by a ROTBODY to make a duct.

8.13. PRIORITY

Sometimes when overlapping grids are used non-fluid points can leak out and fill entire grids. If there are at least three grids that overlap each other where the leak starts then it might be possible to fix this by assigning different priorities to different grids.

A grid with a certain priority x can cut holes, and thereby define which cells are non-fluid, in any grid with priority $x-1$ or higher. By default all grids have the same priority. The priority command can be used up to 100 times.

One possible scenario: There are three grids that overlap. Grid G1 is a coarse background grid. Grid G2 is a bodyfitted grid. Grid G3 is a very small body fitted grid around some small detail. Defining non-fluid points in G1 from G2 and in G2 from G3 gives no problems. The leak occurs when non-fluid points in G1 is defined from G3. Since G1 in this case has one cell center in the region that is considered to be a non-fluid by G3 (inside the G3 grid) but no cell center in the region that is considered to be fluid by G3 (in the G3 grid) the algorithm will not know where the non-fluid domain ends and will fill all of G1.

The fix is to give G1 priority 1, G2 priority 2 and G3 priority 3 so that G3 will not cut any holes in G1.

General form of command

PRIORITY (PRIO = n , GROUP = [n1,n2...nn])

Default values

No default values.

Keywords

<i>keyword</i>	<i>description</i>
----------------	--------------------

PRIO	Set the priority.
------	-------------------

GROUP	Control which grids should be assigned the priority.
-------	--

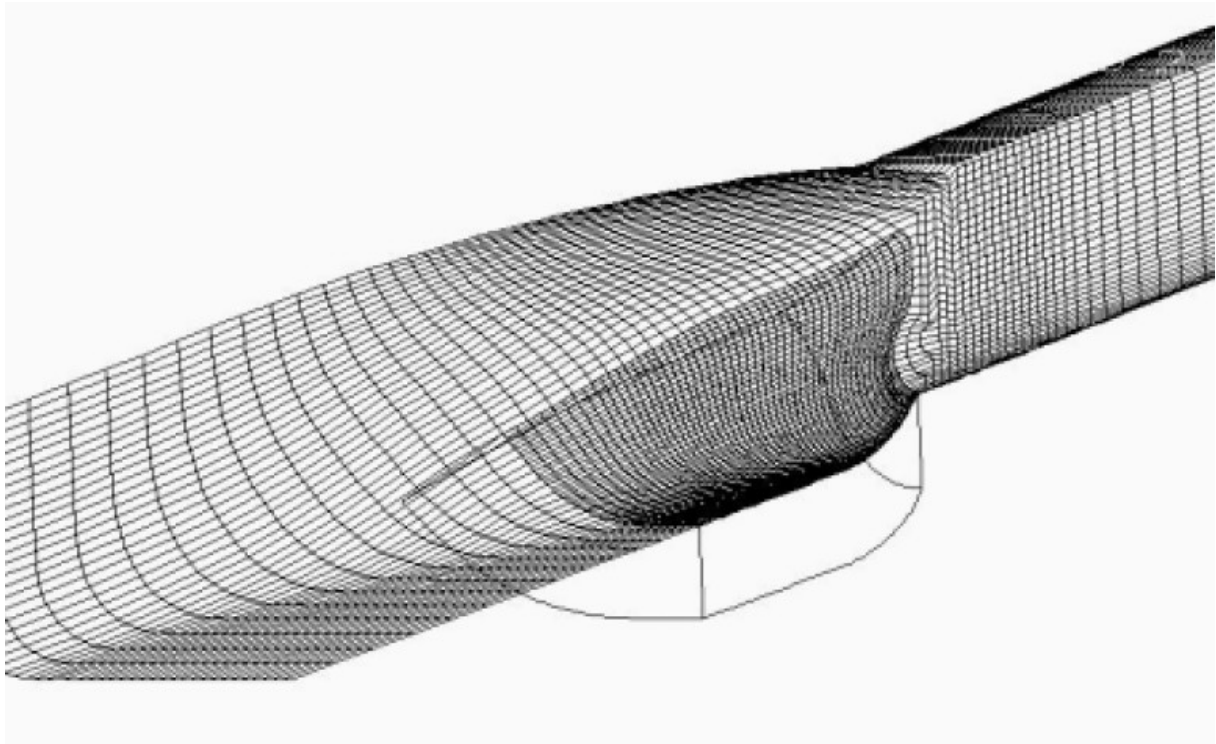
8.14. REFINE

This command refines a region in the grid made by XGRID. The command may be repeated to specify several refinement regions. The refinement is done by cubic interpolation in the original grid, so oscillations may occur in regions where it is not smooth, for example around the aft tip of an aft bulb where the propeller shaft exits. The local refinement is intended to be used to resolve flow features, not geometry. To improve resolution of the hull geometry, use the SIZE and XDISTR commands in XGRID.

Example: The following command

```
refine( level=[1,1,1], low=[0,0,0.2], high=[0.65,0.5,0.55])
```

refines the region where the bilge vortex is formed, see figure below. If the size (i.e. number of nodes) is changed in XGRID, the refinement region will stay approximately the same, but if the point distribution is changed with XDISTR or by changing the ship speed the refinement region will change.



General form of command

```
REFINE ( LOW = [v1,v2,v3] , HIGH = [v1,v2,v3] , ENO ,  
          LINEAR  
          CUBIC  
          HERMITE  
  
          LEVEL = [v1,v2,v3] , GRID = "c" )
```

Default values

REFINE(GRID="xgrid", LEVEL=[0,0,0], LOW=[0,0,0], HIGH=[1,1,1], ENO)

Keywords

<i>keyword</i>	<i>description</i>
LEVEL	An integer value that specifies the refinement level. A level of 0 means no refinement in the direction and level 1 means that each cell is divided in two in that direction. Setting LEVEL=[1,1,1] means that every cell will be divided into 8 new cells.
LOW	The low corner of the refinement region. The values are given in the range [0,1], 0 being the first node and 1 the last. The actual region used is adjusted so that its corners coincide with grid nodes in the coarser grids. For every component it must be true that low<=high.
HIGH	The high corner of the refinement region. The values are given in the range [0,1], 0 being the first node and 1 the last. The actual region used is adjusted so that its corners coincide with grid nodes in the coarser grids.
ENO	[INTERP] Essentially Non-Oscillatory interpolation from the default grid to obtain the new grid. Uses a high order interpolation scheme to give a smooth representation of the geometry while still keeping sharp corners without creating oscillations. This is the preferred interpolation scheme.
CUBIC	[INTERP] Cubic interpolation from the default grid to obtain the new grid. This is more accurate than linear, but can create wiggles on a grid with sharp turns from one cell to the next.
HERMITE	[INTERP] Hermite polynomial interpolation from the default grid to obtain the new grid. This is more accurate than linear, but can create wiggles on a grid with sharp turns from one cell to the next.
LINEAR	[INTERP] Linear interpolation from the default grid to obtain the new grid. This should only be used when the ENO and cubic interpolations does not work.
GRID	Define at which grid the local refinement should be created by referring to the ID of the grid. Default is to use the XGRID grid.

8.15. UNION

When using overlapping grids the total volume occupied by fluid is the intersection of the fluid volume of each grid. If all component grids only have physical boundaries (NOSLIP, SLIP, INFLOW, OUTFLOW), forming the intersection is easy, it is all points that are inside all of the grids. If a component grid has the INTERIOR boundary condition on part of its boundary then its fluid volume extends to infinity but the total volume is still bounded since one or more grids will have to form the physical boundary of the fluid volume.

This default behavior means that the fluid volume cannot be extended if one or more grids together form a closed boundary with no INTERIOR boundary, such as for example the grid produced by XGRID. To extend the fluid volume with for instance a thruster tunnel, the UNION command is used. This command splits the component grid in two sets. Within each set the total fluid volume is the intersection of the grids in the set, but the total fluid volume is the union of the fluid volumes of the two sets.

For the example with a thruster tunnel we model the tunnel with a CYLINDER:

```
xflow
...
  cylinder ( id="bowtunnel", r=[0,2], from=[146,0,4], to=[146,6,4], dime=[30,25,30],
            bc11="INTERIOR", bc12="NOSLIP", bc31="SLIP", bc32="noslip")
...
end
```

and then form the union

```
xchap
...
  union    ( set1=["Xgrid"], set2=["bowtunnel"] )
...
end
```

The grids in the sets are identified by name. Local refinements are treated as part of the enclosing grid so the suffix `_1`, `_2` etc that can be seen in SPOST should not be specified. The default name of the XGRID grid is "Xgrid".

General form of command

```
UNION      ( SET1=["c1", ...] , SET2=["c1", ...] )
```

Default values

```
UNION      ( )
```

Keywords

<i>keyword</i>	<i>description</i>
----------------	--------------------

SET1	Names of the grids in the first set.
------	--------------------------------------

SET2	Names of the grids in the second set.
------	---------------------------------------

8.16. VOLUME

Create a volume grid from an imported surface grid. A hyperbolic volume grid generator is used.

The imported surface grid is specified in the Plot3D ASCII file format. The file must have the suffix .p3d to be recognized as a Plot3D file. Each file may contain one or more grids. The Plot3D format is

```

N
L1 M1 N1 L2 M2 N2 ... LN MN NN
X11 X12 ... X1N1
Y11 Y12 ... Y1N1
Z11 Z12 ... Z1N1
m11 m12 ... m1N1
X21 X22 ... X2N2
.
.
.
Znb1 Znb2 ... ZnbNnb
mnb1 mnb2 ... mnbNnb

```

where N is the number of grids (blocks), L₁ M₁ N₁ are the dimensions of the first grid, x₁₁, y₁₁, z₁₁ are the x, y, z-coordinates of the first node in the first grid, m₁₁ is an integer value that specifies properties for the first node in the first grid (not used by SHIPFLOW) and N₁ = L₁ x M₁ x N₁ is the number of nodes in the first grid. The ordering of the node data is such that if node number i corresponds to point (l,m,n) in index space, then node i+1 corresponds to (l+1, m,n). The numbers are separated by a white space.

The grids are assumed to be specified in the offset file coordinate system, but if it is not it may be translated, rotated and scaled. The rotation is specified by Euler angles using the pitch-roll-yaw conventions. All transformations are done starting from the grid coordinate system. The transformations are made in the order SCALE, YSIGN, ROTATION, TRANSLATION.

General form of command

```

VOLUME ( ID = "c" , GROUP = n ,
        FILE= "c" , NUMBER = n
        TRANSLATION = [v1,v2,v3] , ROTATION = [v1,v2,v3] ,
        YSIGN = n , SCALE =v ,
        BC11 = "c" , BC12 = "c" ,
        BC21 = "c" , BC22 = "c" ,
        BC31 = "c" , BC32 = "c" ,

```

ROTCENT = [v1,v2,v3]	,	STRETCH	,
		NOSTRETCH	
KMAX = n	,	STEP = v	,
FAC = v	,	VOLSM = v	,
MAXVOLSM = n	,	KTRVOLSM = n	,
ITETA = v	,	JTETA = v	,
EPSE = v	,	PRIO = n	,
HB11 = n	,	HB12 = n	,
HB21 = n	,	HB22 = n	,
CUT	,	NOTRIM	)
NOCUT			

Default values

VOLUME(PRIORITY=1, NUMBER=0, TRANSLATION=[0,0,0], ROTATION=[0,0,0],
ROTCENT=[0,0,0], YSIGN=-1, SCALE=1, BC11="INTERIOR",
BC12="INTERIOR", BC21="INTERIOR", BC22="INTERIOR",
BC31="INTERIOR", BC32="INTERIOR", CUT)

Keywords

<i>keyword</i>	<i>description</i>
ID	Gives the imported grid a name. The name is used in general to identify the object and to associate results like integrated forces with it. Different grids may have the same name, but then only the total force on all of those objects will be calculated and output.
FILE	Name of the Plot3D file that specifies the grid. The file name must end with .p3d.
NUMBER	If there are more than one grids in the file, import only the NUMBERth one. The first grid in the file is number one. If the default value NUMBER=0 is given, all grids in the file are imported and they get the same name, transformations and boundary conditions.
GROUP	Group number in the overlapping grid algorithm. If not specified the grid will be the only grid in the group.
SCALE	Scale factor for the grid.
YSIGN	Set to 1 if the grid is on the positive side of y=0, -1 otherwise. This will place the grid on the negative side of y=0 where the XGRID grid is for symmetrical computations.

ROTATION	Rotation of the grid(s) specified by the Euler angles with the pitch, roll, yaw convention. Given in degrees.
ROTCENT	Center of rotation for the euler angles given in ROTATION.
TRANSLATION	Translation of the grid origin, specified in offset file coordinates.
BC11	Specify the boundary condition on side $l=1$ of the grid. Possible values are “NOSLIP”, “SLIP”, “INFLOW”, “OUTFLOW”, “INOUT”, “INTERIOR” and “MIXED_BC” (see section “H-O topology grid import” on page 312). The INTERIOR condition means that the boundary values are taken by interpolation from other grids.
BC12	Specify the boundary condition on side $l=L$ of the grid.
BC21	Specify the boundary condition on side $m=1$ of the grid.
BC22	Specify the boundary condition on side $m=M$ of the grid.
BC31	Specify the boundary condition on side $n=1$ of the grid.
BC32	Specify the boundary condition on side $n=N$ of the grid.
STRETCH	[STRETCH] Stretch the grid.
NOSTRETCH	[STRETCH] Do not stretch the grid. The grid is used as given in the imported file.
KMAX	Number of planes to make in normal direction.
STEP	First step size. The length is given in the computational coordinate system (where the hull has length 1).
FAC	Expansion factor.
VOLSM	Volume smoothing parameter.
MAXVOLSM	Maximum number of volume smoothings.
KTRVOLSM	Reduce the number of smoothings when $k < KTRVOLSM$, where k is the plane number in the normal direction.
ITETA	Diffusion parameter in i-direction.
JTETA	Diffusion parameter in j-direction.
EPSE	Added diffusion.
HB11	Boundary condition for the grid generator. First parameter direction, low side. 0 = free 1 = constant x 2 = constant y 3 = constant z

HB12	Boundary condition for the grid generator. First parameter direction, high side.
HB21	Boundary condition for the grid generator. Second parameter direction, low side.
HB22	Boundary condition for the grid generator. Second parameter direction, high side.
CUT	The created grid should cut holes in other grids.
NOCUT	The created grid should not cut holes in other grids.
NOTRIM	Do not apply hull sinkage and trim to the generated grid.

8.17. VOF

This command sets the parameters of the VOF free surface solver. The physical properties are Z0, free surface z-coordinate at rest, RHOR, the density ratio between water and the fluid above, and NYR, the dynamic viscosity ratio between water and the fluid above. The defaults are set to represent the lighter fluid as being air. The method expects a heavier fluid as reference so do not set RHOR>1.

Reflection of transient waves at the boundaries of the computational region can cause the solution to not reach a steady state for a long time. To avoid this the z-component of velocity is damped outside the region $XDU2 < x < XDD1$, $YD1 < |y| < YD2$. The default values are set to work with the default dimensions of the grid made by XGRID. If these are changed or other grids used, the dimensions of the damped region may need to be changed too.

General form of command

```
VOF ( Z0=v          , RHOR=v          , NYR=v          , LVCDIST=v      ,
      PINTERP       , HSCORR          , TRWALL          , ICFL=i          ,
      NOPINTERP      NOHSCORR        NOTRWALL
      XDU1=v         , XDU2=v         , XDD1=v         , XDD2=v         ,
      YD1=v         , YD2=v         , DAMP=v         )
```

Default values

VOF (Z0=0, RHOR=0.00127, NYR=15, LVCDIST=0.001, PINTERP, HSCORR, ICFL=10, DAMP=0.2, XDU1=-1.0, XDU2=-0.5, XDD1=1.75, XDD2=2.5, YD1=1.0, YD2=1.75, TRWALL)

Keywords

<i>keyword</i>	<i>description</i>
Z0	Free surface level.
RHOR	Fluid density ratio.
NYR	Fluid viscosity ratio.
LVCDIST	VOF extrapolation depth. Expressed as the non-dimensional distance to the nearest NOSLIP boundary, i.e. hull, appendages etc. Used to avoid air suction under the hull.
PINTERP	Switch on hydrostatic pressure gradient compensation in interpolation. This is important if overlapping grids are used. It improves interpolation of the pressure by subtracting the hydrostatic pressure gradient in each cell that is supplying the interpolation data and then add the hydrostatic pressure gradient of the cell that the receiver point is located in.

NOPINTERP	Switch of hydrostatic pressure gradient compensation in interpolation. Usually a bad idea but if no overlapping grids are used it will save a small amount of computation time.
HSCORR	Switch on hydrostatic pressure drag correction. If appendage grids are used the computation of integration weights is not exact so the projected hull area as seen from the front and rear are not exactly equal. Even a very small difference will affect the pressure drag at low Froude numbers, so to correct for this the pressure drag for the hydrostatic case (initial solution), which should be zero, is saved and subtracted from the final result.
NOHSCORR	Switch off hydrostatic pressure drag correction.
ICFL	CFL number in the implicit direction of the ADI sweep. Increasing this parameter will make the boundary layer converge faster but also destabilizes the solution around the free surface.
DAMP	Strength of damping of the z-component of velocity in the outer parts of the computational region. This is needed to damp out transient waves and reach a steady state in a reasonable number of iterations. Setting DAMP=0 turns off damping and DAMP=1 is the maximum value. Depending on grid stretching, running without damping may or may not work. Too strong damping can cause long waves to bounce back and forth a long time.
XDU1	Start of the upstream damping ramp. For $x < XDU1$ the damping value is DAMP.
XDU2	End of the upstream damping ramp. For $XDU2 < x < XDD1$ the damping value is 0.
XDD1	Start of the downstream damping ramp.
XDD2	End of the downstream damping ramp. For $x > XDD2$ the damping value is DAMP.
YD1	Start of the transverse damping ramp. For $ y < YD1$ the damping value is 0.
YD2	End of the transverse damping ramp. For $ y > YD2$ the damping value is DAMP.
TRWALL	Use wall-law for turbulent quantities boundary conditions on the transom. This may be important for high Reynolds number (full scale) cases due to the relatively thick cells there compared to on other NOSLIP boundaries.
NOTRWALL	Use the same near-wall boundary condition for the turbulent quantities on the transom as on other NOSLIP boundaries.

8.18. WAKE

This command is used to calculate the nominal wake for a propeller, i.e. the volumetric mean of one minus the velocity through the propeller disc. The geometry is taken from the PROPELLER command in the XFLOW input section. Diameter and shaft position is required. The wake computations are always active whenever the necessary data in the PROPELLER command is given.

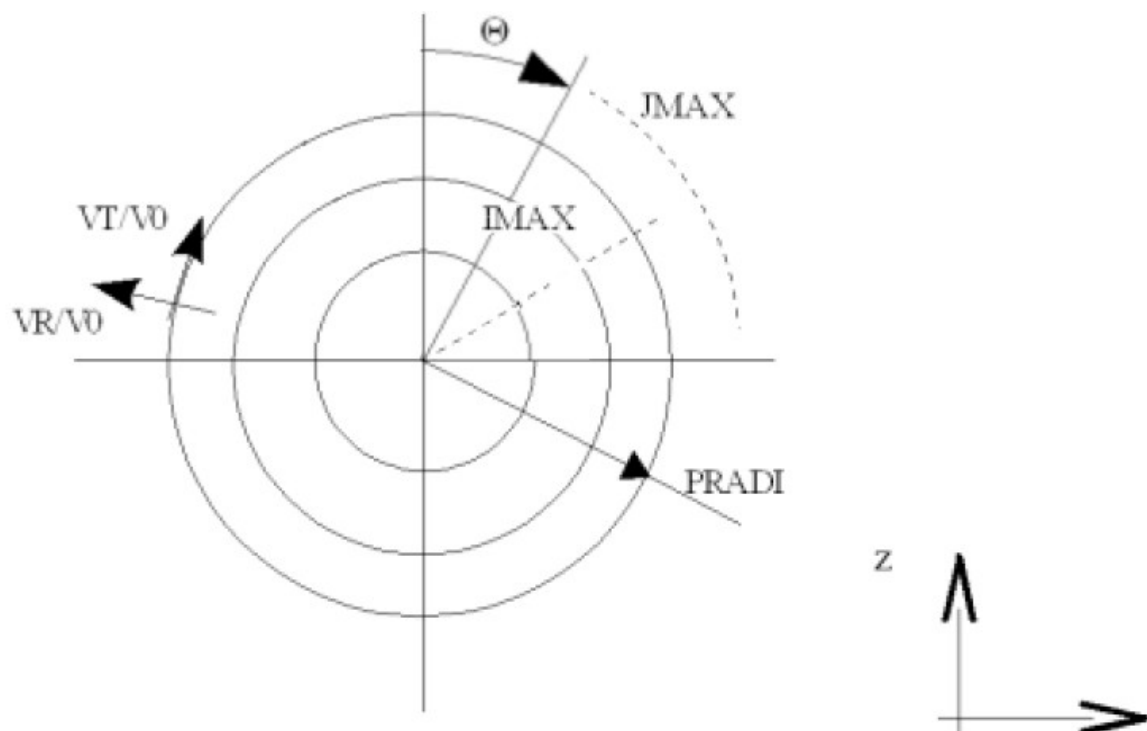
The nominal wake is printed in the OUTPUT file if the propeller plane is defined. If there is an active propeller in the propeller plane the thrust and torque coefficients will also be written to the OUTPUT file. Additionally a history of the thrust and torque coefficients are logged every 10 iterations in the propellerid_PROPLOG file.

If the propeller plane is defined in the PROPELLER command then wake distribution data will also be written to the file id_propellerid.html. The following data is written to the html file:

- V/V_0 , The total velocity
- V_a/V_0 , The axial velocity component. Positive direction is downstream.
- V_{tr}/V_0 , The transversal velocity component.
- V_t/V_0 , The tangential velocity component. Positive direction is clockwise.
- V_r/V_0 , The radial velocity component. Positive direction is outwards.
- $w = 1 - V/V_0$, Total wake fraction
- $w_a = 1 - V_a/V_0$, Axial wake fraction

The angular position is 0 degrees at the top and increases in clockwise direction looking from behind. In case the propeller is rotated the axial, radial, tangential components will follow the shaft direction.

The points for extracting the wake at the propeller disc are organized in the following way, looking from behind of the ship:



General form of command

WAKE (ON
OFF , IMAX=n , JMAX=n , CSV ,
PRADI = [V₁, V₂, .., V_{NRAD}] CSV , DAT)

Default values

WAKE (ON, IMAX = 25, JMAX = 25, PRADI = [0.1, 0.2, ..., 1.0])

Keywords

<i>note</i>	<i>keyword</i>	<i>description</i>
	ON	[ONOFF] WAKE computation is on.
	OFF	[ONOFF] Disable the WAKE computations.
(1)	MAX	Number of integration points in the radial direction of the propeller.
(1)	JMAX	Number of integration points in the circumferential direction of the propeller. The circumferential wake distribution is printed at JMAX points.
	PRADI	Radii where the local wake is calculated. PRADI must be non-dimensional with respect to the propeller radius, i.e. PRADI equals one at the propeller tip. Uniform distribution if not given.
	CSV	The extracted wake data will also be exported in Comma-Separated Values format.
	DAT	The extracted wake data will also be exported in Grace format.

Notes

- (1) The longitudinal velocity component is interpolated from the solution to IMAX*JMAX integration points on the propeller disc. The average wake is thereafter calculated using the assumption of a constant velocity on IMAX*JMAX area elements.

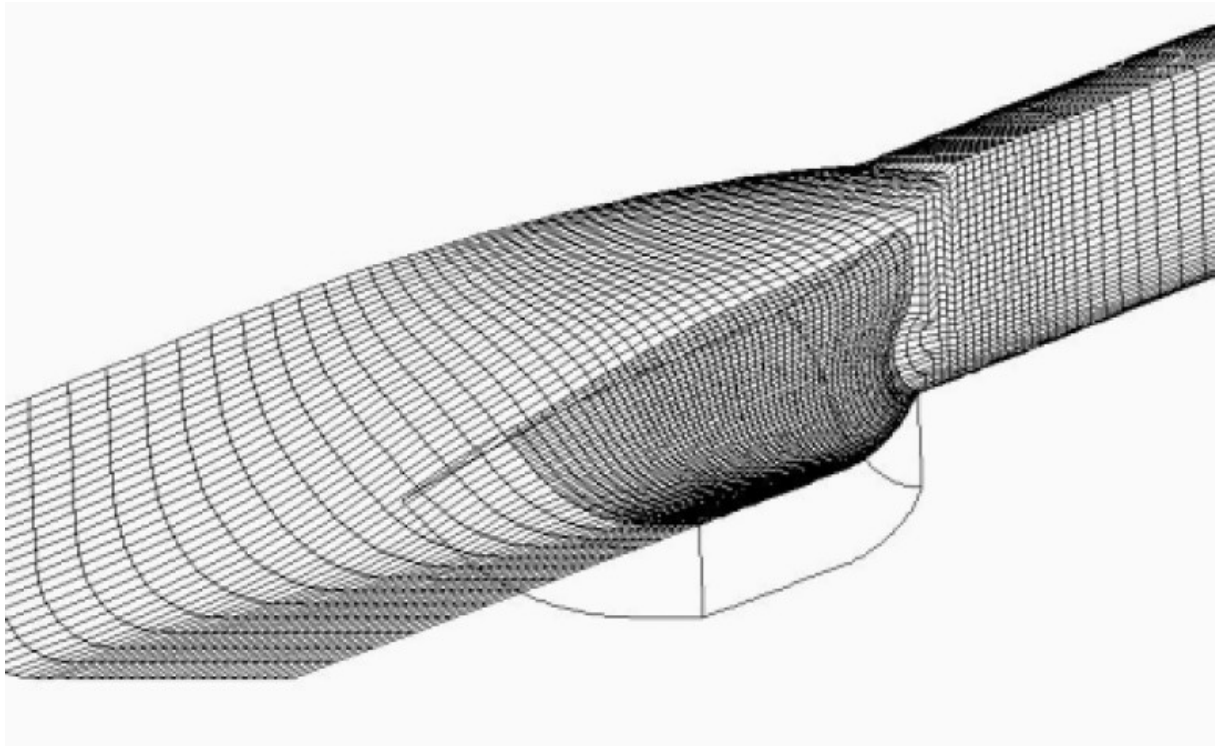
8.19. XGREFINE

This commands refines a region in the grid made by XGRID. The refined grid will be the same that can be created with the REFINE command, but this input format is more convenient for the user.

If propeller and/or appendage grids are used, default values for XSTART, XEND, ZETAEND, ETASTART and ETAEND will be automatically set to cover these grids with the refinement. Any of these values can be overridden by setting them manually.

If only an XGRID grid is used, it is mandatory to set XEND, ZETAEND, ETASTART and ETAEND

The local refinement is intended to be used to resolve flow features, not geometry. To improve resolution of the hull geometry, use the SIZE and XDISTR commands in XGRID.



General form of command

```
XGREFINE( XSTART = v      , ON
          , OFF           , ENO
                              LINEAR
                              CUBIC
          ,
          ZETAEND = g      , ETASTART = g      , ETAEND = g      ,
          LEVEL = [v1,v2,v3] , EXPROPCELLS = g , EXCELLS = g      ,
          XEND = v                                     )
```

Default values

XGREFINE(LEVEL=[1,1,1], XSTART=0.85, ENO, EXPROPCELLS=5, EXCELLS=3)

NOTE: It is mandatory to specify ON or OFF.

Keywords

<i>keyword</i>	<i>description</i>
LEVEL	An integer value that specifies the refinement level. A level of 0 means no refinement in the direction and level 1 means that each cell is divided in two in that direction. Setting LEVEL=[1,1,1] means that every cell will be divided into 8 new cells.
ON	[ONOFF] The refinement is created.
OFF	[ONOFF] The refinement is not created.
XSTART	The x-coordinate of where the refinement starts.
XEND	The x-coordinate of where the refinement ends.
ZETAEND	Number of cells in the normal direction in the XGRID grid covered by the refinement.
ETASTART	Starting cell number in the girth-wise direction in the XGRID grid covered by the refinement.
ETAEND	Ending cell number in the girth-wise direction in the XGRID grid covered by the refinement.
EXPROPCELLS	Used when calculating the default extent of the refinement for a case with propeller grids (the cylinder grid used by lifting line). The propeller grids will be fully covered by the refinement plus EXPROPCELLS in all directions.
EXCELLS	Used when calculating the default extent of the refinement for a case with appendage grids. The solid boundaries (NOSLIP) of the appendage grids will be fully covered by the refinement plus EXCELLS in all directions.
ENO	[INTERP] Essentially Non-Oscillatory interpolation from the default grid to obtain the new grid. Uses a high order interpolation scheme to give a smooth representation of the geometry while still keeping sharp corners without creating oscillations. This is the preferred interpolation scheme.
CUBIC	[INTERP] Cubic interpolation from the default grid to obtain the new grid. This is more accurate than linear, but can create wiggles on a grid with sharp turns from one cell to the next.
LINEAR	[INTERP] Linear interpolation from the default grid to obtain the new grid. This should only be used when the ENO and cubic interpolations

does not

8.20. XGRID

Command to control and modify the XGRID grid. The rotation is specified by Euler angles using the pitch-roll- yaw conventions. All transformations are done starting from the grid coordinate system. The transformations are made in the order SCALE, YSIGN, ROTATION, TRANSLATION.

General form of command

```
XGRID      (  TRANSLATION = [v1,v2,v3]   ,  ROTATION = [v1,v2,v3]      ,  
              YSIGN = n                  ,  SCALE =v                    ,  
              BC11 = "c"                 ,  BC12 = "c"                 ,  
              BC21 = "c"                 ,  BC22 = "c"                 ,  
              BC31 = "c"                 ,  BC32 = "c"                 ,  
              TR12 = "c"                 ,  TR22 = "c"                 ,  
              TR32 = "c"                 ,  
              NBCC                        ,  STRETCH                      ,  
              NBGHOST                     NOSTRETCH                     ,  
              YTAR = v                    ,  GROUP = n                    ,  
              TRSTRETCH                    )
```

Default values

```
XGRID  (  TRANSLATION=[0,0,0], ROTATION=[0,0,0], SCALE=1, NBCC, YTAR=1)
```

Keywords

<i>keyword</i>	<i>description</i>
GROUP	Group number in the overlapping grid algorithm. If not specified the grid will be the only grid in the group.
SCALE	Scale factor for the grid.
YSIGN	Mirror XGRID grid around y-axis.
ROTATION	Rotation of the grid(s) specified by the Euler angles with the pitch, roll, yaw convention. Given in degrees.

TRANSLATION	Translation of the grid origin, specified in offset file coordinates.
BC11	Specify the boundary condition on side $l=1$ of the grid. Possible values are “NOSLIP”, “SLIP”, “INFLOW”, “OUTFLOW”, “INOUT”, “INTERIOR” and “MIXED_BC” (see section “H-O topology grid import” on page 312). The INTERIOR condition means that the boundary values are taken by interpolation from other grids.
BC12	Specify the boundary condition on side $l=L$ of the grid.
BC21	Specify the boundary condition on side $m=1$ of the grid.
BC22	Specify the boundary condition on side $m=M$ of the grid.
BC31	Specify the boundary condition on side $n=1$ of the grid.
BC32	Specify the boundary condition on side $n=N$ of the grid.
TR12	Specify the boundary condition on side $l=L$ for the transom grid.
TR22	Specify the boundary condition on side $m=M$ for the transom grid.
TR32	Specify the boundary condition on side $n=N$ for the transom grid.
STRETCH	Switches on ytarget stretching (Default for $R_n \geq 1e8$).
NOSTRETCH	Switches off ytarget stretching (Default for $R_n < 1e8$).
TRSTRETCH	Stretch the transom grid towards the transom. Gives better resolution of the boundary layer but breaks the cells matching on the interior boundary with the hull grid.
NBCC	Use neighbouring cell center locations for interpolation to interior ghost cells.
NBGHOST	Use ghost cell centers for interpolation to interior ghost cells.

9. XPOST

This section describes available controls for automatic reporting tool.

9.1. CONTROL

This command can be used to control automatic report generation.

General form of command

```
CONTROL ( ON  
        OFF
```

Default values

```
CONTROL ( ON )
```

Keywords

<i>keyword</i>	<i>description</i>
ON	[ONOFF] Enable the report generation.
OFF	[ONOFF] Disable the report generation.

10. Offset file format

NOTE: For standard hulls it is also possible to have the hull geometry described in IGES or STL format.

The purpose of this file is to describe the geometry of the vessel. This is done by a list of coordinate values of points lying on the hull surface.

The following concepts are extensively used when the geometry is described in the manual.

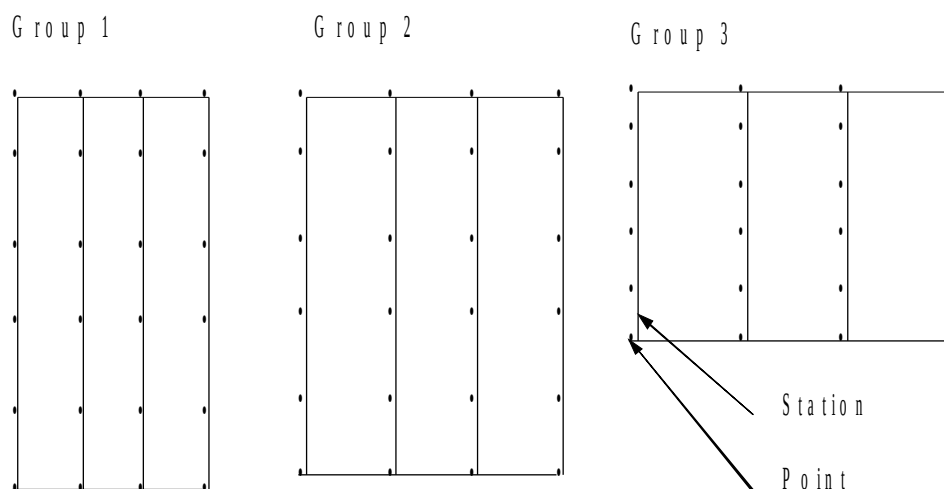
Point: A point is defined by its x, y and z coordinates.

Status flag: A status flag can be assigned to a point. The status flag tells the program how to treat the point.

Station: A station is a set of points that describe the intersection between the hull and a constant x plane. The XPAN module also allows for varying x.

Group or offset group: A group describes the shape of a part of the hull. Typically one group is used to describe the bulb, the keel or the rudder. One or more groups can be used to describe the main part of the hull.

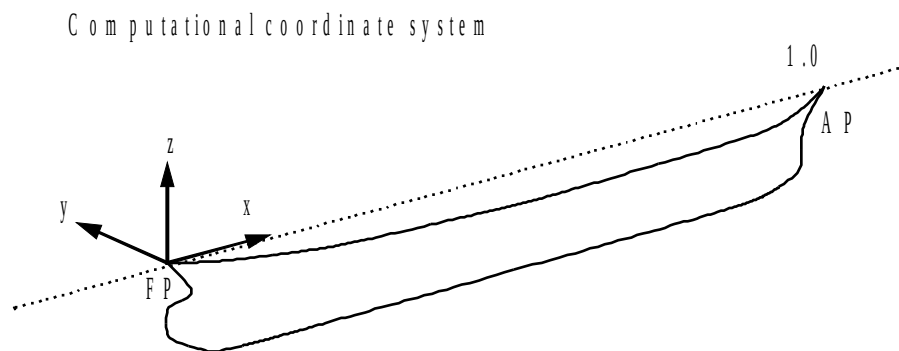
Group label: Label name of an offset group used to identify the group.



10.1. Syntax

10.1.1. Coordinate systems

This section contains a description of the two coordinate systems used in SHIPFLOW. The computational coordinate system is used internally during the computation, in the output files and for post-processing. The offset coordinate system is used for defining the offset points in the offset file. The relation between the two systems is defined with the command "OFFSETFILE" on page 71



- Non-dimensionalized coordinates are used internally in SHIPFLOW.
- The ship length LPP is used for non-dimensionalization.
- The ship is positioned as in the figure during the computations.
- The offset points can either be input directly in full scale coordinates together with a reference length LPP or in non-dimensionalized form.
- The coordinate system used for the offset points can be located anywhere in the center- plane.
- Only one side has to be specified if the ship is symmetric.

10.1.2. Line syntax

Each non-empty line of this *ascii* file should contain either a group label such as "XYZHULL" or the x-, y- and z-coordinate of one point and its status flag, e.g.

```
0.213900      0.012500      0.108500      0
```

The coordinates and their status flag are read by a "free format" read statement. The y-coordinate of the example could also have been written as 0.0125, 1.25e-2, 1.25E-02, 1.25d-02 or perhaps 12.500D-03. Integers will be converted to real numbers and real numbers to integers if necessary:

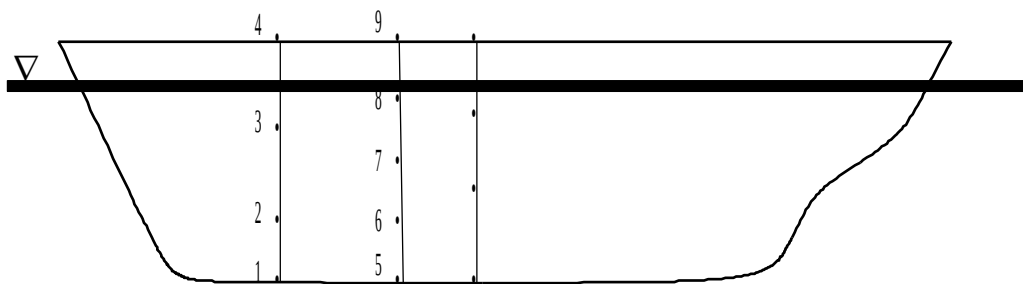
```
2            2            2            3.2e+0  will be interpreted as
2.0          2.0          2.0          3
```

The file must start with a group label and end with the end of file label "END".

10.1.3. Order of points and stations

Body groups

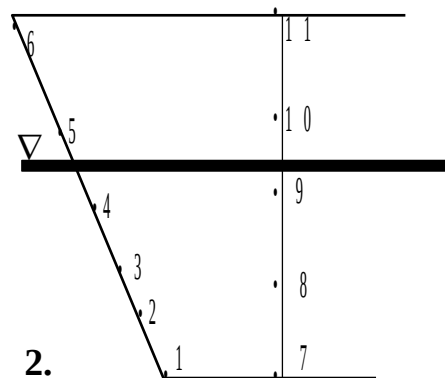
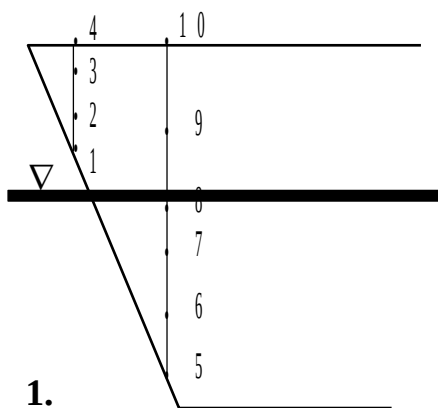
- The points within each group must be given as stations, i.e. as sets of points lying on the intersection of the hull surface and constant x planes. The mesh generator for the potential flow part XPAN can handle points of varying x.
- The stations should be given from the bow towards the stern and the points within each station should be given from the keel and up.
- The hull must be defined also above the water plane.



The first station

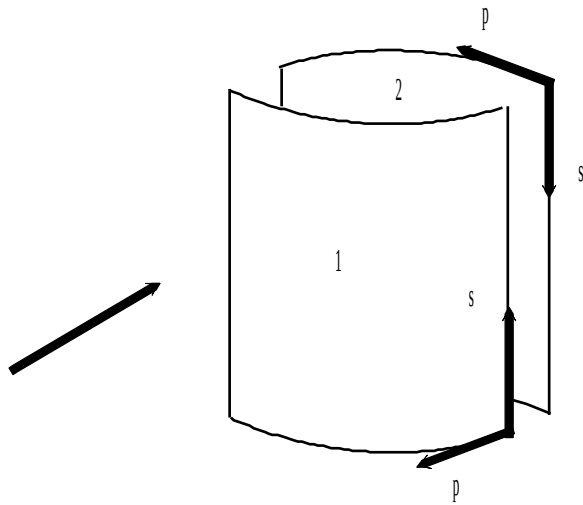
The first station can be defined in two ways:

1. As a constant x-station
2. As a set of points defining the shape of the bow in the centre plane.



Lift groups

Lift groups are defined with an other ordering compared to body groups. The points must be given in the chord-wise direction. The ordering is illustrated in the figure below.

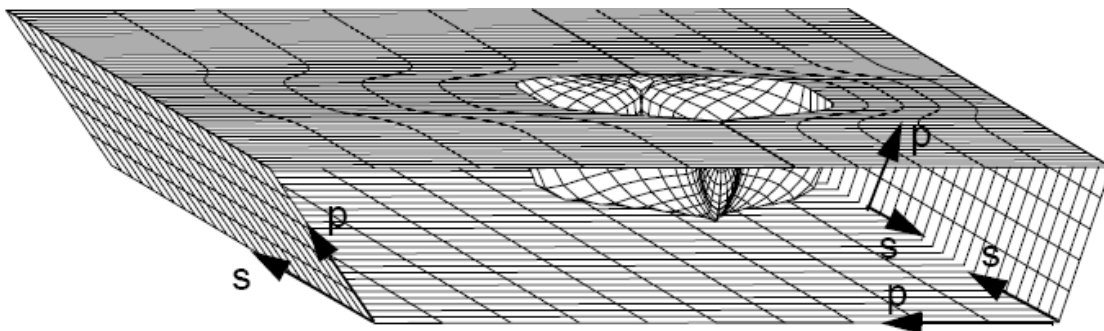


p means the direction of increasing point index and s the direction of increasing station index. The point should then be given so that the p index varies faster than the s index in the offset file. Note the different ordering on the pressure and suction sides.

Environment groups

The points and stations must be ordered as shown in the figure below in order to find the intersection between the free-surface and the sides of the canal.

Ship in a canal. ENVIRONMENT groups, order of offset points (p) and stations (s).



Ship in a canal. ENVIRONMENT groups, order of offset points (p) and stations (s).

10.1.4. Groups

- A hull can be divided into several groups.
- Stations defining a boundary between two groups must be defined two times in the offset file.

10.1.5. Usage of group labels and status flags

- Each group must be preceded by a group label.
- The first point of each of the stations within a group should have status flag 1.
- The last point of each group must have the status flag 9. (This turns out to be the points immediately preceding the labels.)
- The vast majority of the points, those that are not the first of the frame they belong to or the last of a group, must have the status flag 0 (zero).
- The offbody group must be preceded by a group label. The last offbody point must have the status flag 9. All other offbody points must have the status flag 0 (zero).

10.2. XGRID requirements

Since XGRID will make a grid where the hull surface (+ a part of the wake centre plane) will be covered with one continuous surface and for a number of fundamental differences between XPAN and XCHAP (and hence XGRID), XGRID users will have to get used to different / more restrictions than where imposed on them when they were using XMESH. The current grid topology cannot handle appendices such as keels and rudders.

Note also that the hull representation used by XGRID differs from that of XMESH. XGRID can only handle frames with constant x-coordinates. Parametric cubic splines will only be used in the circumferential direction. Linear interpolation will be used in between frames.

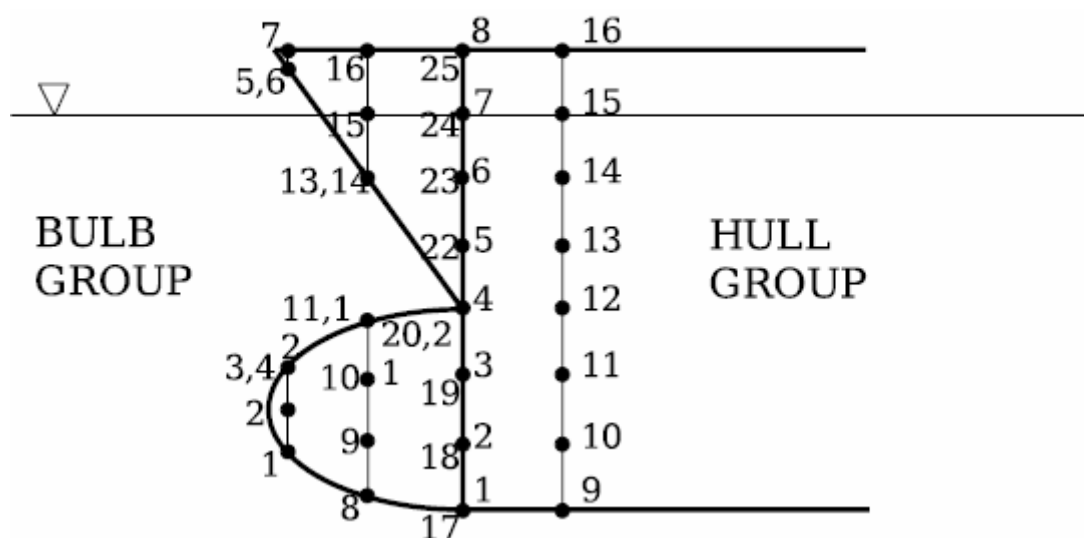
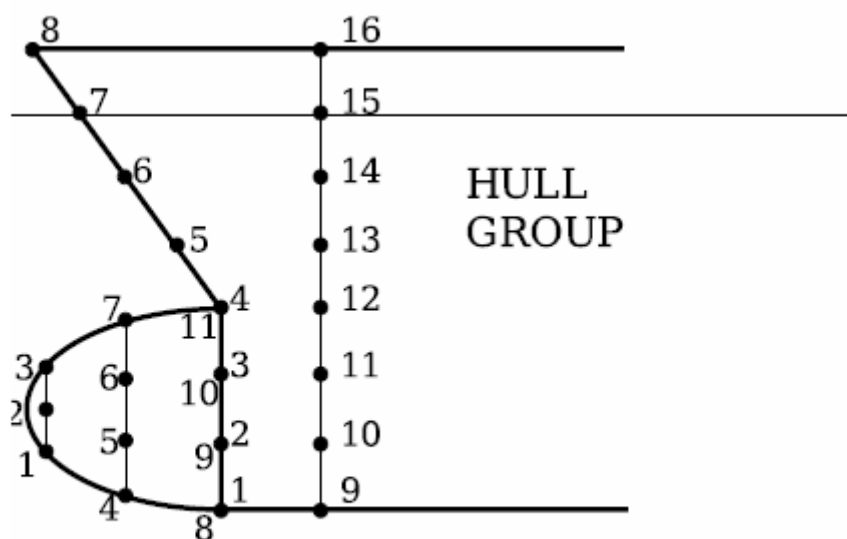
The main hull can be described by at most three offset groups. The names of these groups are assigned to H1GR, H2GR and H3GR in the OFFSET command for XGRID. The names of a stern bulb and an overhang in the offset file is assigned to ABGR and OGRP, respectively.

The frames in H1GR, H2GR, H3GR and OGRP must be in global order.

The frames given in the ABGR and OGRP groups must lie in the same x-planes where they overlap for SHIPFLOW version 3.2 or earlier. For SHIPFLOW 3.3 this is only true if keyword RAWOFFSET is set.

10.3. Hull with bulbous bow

To ensure the best hull shape representation for both potential flow as well as RANS computations it is necessary to pay special attention to the body groups and the points order in the offset file. For the XMESH it is more suitable to describe the stem together with the first station in the hull group creating a three dimensional curve as shown in the upper figure. However, the XGRID module is prepared to use the two dimensional stations lying on the x-planes. Therefore, in the case when one wants to extend the computational domain for the RANS solver to cover the whole hull it is advised to use a separate hull group for this module. The bulbous bow should be described in an additional offset group for both solvers, however the one for the XGRID should consist also the points belonging to the bow overhang, see lower figure.

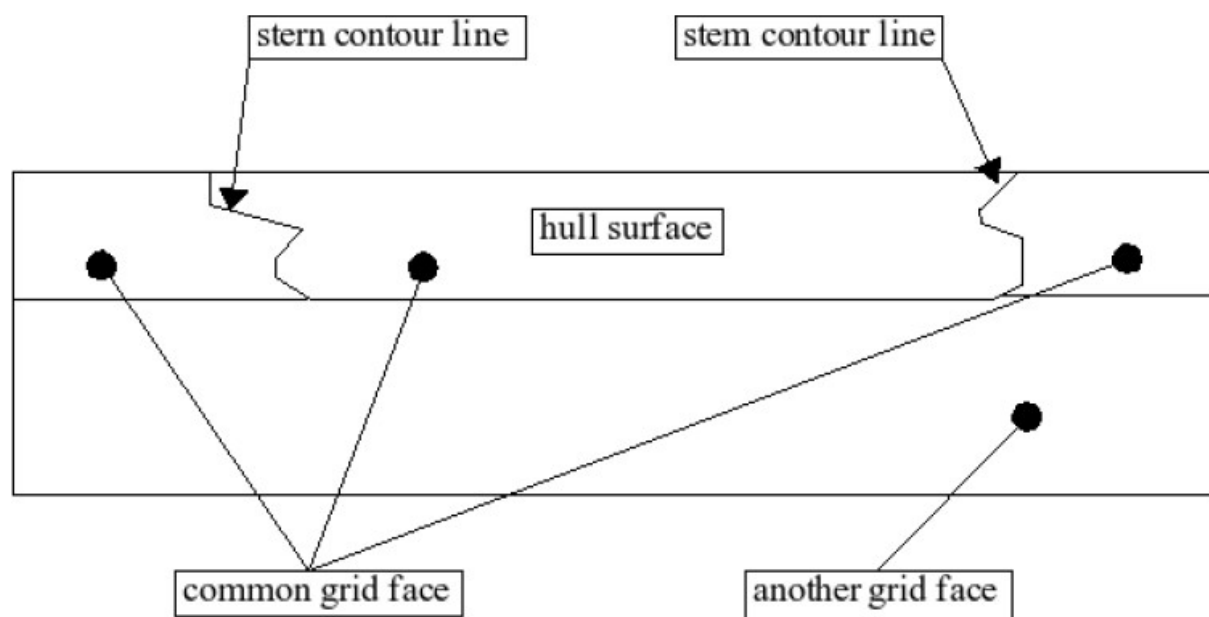


The offset groups for XMESH should be specified in the command file in the xflow section in the hulltype command. The groups for XGRID can be respecified in the command file in the xgrid section in the offset command

10.4. H-O topology grid import

When H-O topology is imported, it is not possible to set the boundary conditions unambiguously for the face that consists of the hull surface. The face is continuous along the whole domain span and covers not only the hull but also parts of the center plane in front and behind the ship. Because of that the position of the hull can not be known exactly from just looking at the grid.

A remedy to this is to describe the hull location explicitly at the stem and/or the stern. In order to do that a special function has been developed that enables a manual specification of the contour lines and additionally gives a possibility to define a specific boundary condition for particular cell faces.



The function is triggered by specifying a bcXX in the import command to MIXED_BC. XCHAP then expects a file that contains the necessary information. The file should be named id.bc, where the id is the id of the grid that is being imported. The file may contain keywords: stern and stem followed by the contour lines in the grid coordinate system and keywords: SLIP, NOSLIP and INTERIOR followed by the cell face indices. After each keyword a number of point coordinates or cell indices has to be given. An example id.bc file is given below

```
stern 2
141      10.000000   -12.700000
144.5    8.952560   -8.737370

stem 3
-142.35  0.000000   -11.700000
-144.84  0.000000   -11.485041
-158.31  0.000000   19.715700

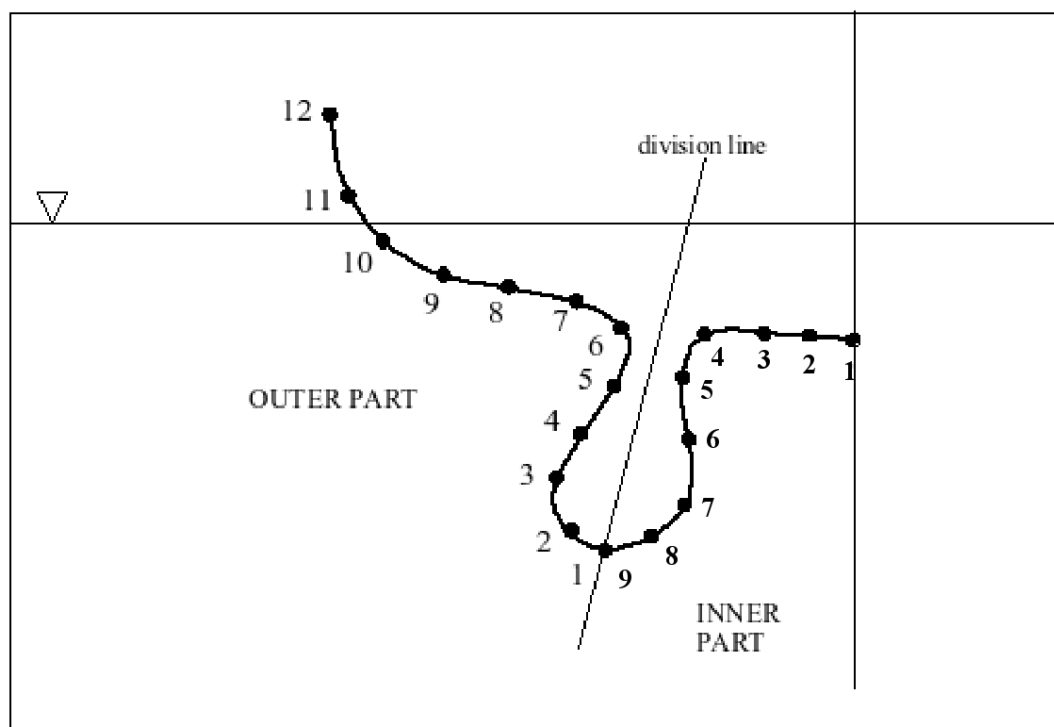
NOSLIP 2
138 14
139 26

INTERIOR 3
100 5
101 5
100 6

SLIP 1
140 15
```

10.5. Offset file format for twin skeg hulls

For the XCHAP calculations the twin skeg hull has to be split into body groups in a similar way as a standard case hull i.e. bulb, bossing, stern overhang and up to three main groups. However an additional longitudinal split has to be added thus creating inner and outer parts. The split should correspond approximately with the lower edge of the skeg (gondola) as shown in the figure below. The offset points should be arranged in such a way that the first point of each section is at the longitudinal split edge.



For global computations the H1GR should overlap with H1GI and H1GO offset groups. The H1GI and H1GO should extend about 5% of L_{pp} upstream of the midship while H1GR about 5% downstream of the midship.

11. System practicalities

The following section gives an overview of the program file structure and tabulates the i/o units used by SHIPFLOW. A brief description of memory allocation is also given.

11.1. Files

The following table shows all the i/o units used by SHIPFLOW.

<i>unit</i>	<i>name</i>	<i>description</i>
6		Terminal output.
5		Terminal input.
3		(not used)
4	(user spec.)	Geometry file containing the offset coordinates.
1	(user spec.)	Command file. Its name "id" will be used as prefix for all files that SHIPFLOW creates. This naming convention can be overruled with the FILES command.
2	id_OUTPUT	Detailed output.
7	id_SUMMARY	Summarized output.
	id.cgns	Data for visulaization of results from all modules with SPOST.
8	id_XPDB	XPAN results needed by XBOUND and XCHAP.
9	id_XBDB	XBOUND results needed by XCHAP.
17	id_XVGRID	Grid coordinate file for XCHAP.
19	id_XVPROF	Inflow boundary velocity profiles, generated and used by XCHAP.
22	id_XPRES	Restart file for XPAN.
26	id_LWAVECUT	Contains wave profiles along longitudinal wave cuts. Generated when thenWAVECUT command is used.
27	id_TWAVECUT	Contains wave profiles along transverse wave cuts. Generated when the WAVECUT commandn is used.
52	id_XGPOST.plt	XGRID mesh for TECPLOT.
53	id_XPPOST.plt	XPAN results for TECPLOT.
54	id_XBPOST.plt	XBOUND results for TECPLOT.
55	id_XBSTRL.plt	XBOUND streamlines for TECPLOT.
55	id_XBLIMIT.plt	XBOUND limiting streamlines for TECPLOT.
58	id_FPXYPL.plt	Wave profile data for TECPLOT.

61	optres.dat	Contains resistance, displacement and LCB. To be used when SHIPFLOW is included in an optimization loop. Generated when the OPTIM command is used.
67	id_FPXYPL	Wave profile data for FIDAP. Moved to id_XPOST/XPAN.FPXYPL.
68	id_WAVECP	Wave profile from hull surface pressure for FIDAP. Moved to id_XPOST/WAVECP.

Extra data output from XPAN and XBOUND, (see description of the PRTOPT command).

unit	name	description
20	id_XYPLOT	Speed, onset-flow and iteration history of XPAN results for xy-plot in FIDAP.
28	id_PGEOM	Panel geometry information.
29	id_FDOP	Finite difference operator terms from XPAN.
30	id_PCON	Panel connectivities from XPAN.
31	id_STRLRES	Streamline coordinates from XBOUND.
32	id_PANCO	Panel corner coordinates from XPAN.
33	id_NLTERM	Non-linear terms from XPAN.
34	id_BMRES	Double or single model information: control point coordinates, velocities and pressures, panel source strengths and normal vectors from XPAN.
35	id_FSRES	Free surface calculation information: control point coordinates, velocities and pressures, panel source strengths and normal vectors, wave heights from XPAN.
36	id_OBPRES	Offbody point calculation information: offbody point coordinates, velocities and pressures from XPAN.
37	id_NAPA	Output of XPAN results for NAPA interface.
57	id_XYPLOT.plt	Speed, onset-flow and iteration history of XPAN results for xy-plot in TECPLOT.
62	id_OPTAIJ	Special purpose optimization output. Not for general use.
63	id_OPTXYZ	Special purpose optimization output. Not for general use.
64	id_OPTRES	Special purpose optimization output. Not for general use.
66	id_TRACE	TRACE file, a runtime log of all actions taken by XPAN, useful if anything goes wrong.

Additional files used by XCHAP.

<i>unit</i>	<i>name</i>	<i>description</i>
	id_chp.cmd	Command file for CHAPMAN. Automatically produced by SHIPFLOW.
	id_chp.xgrd	Grid file produced by XGRID for CHAPMAN
	id_chp.data	Solution exported by CHAPMAN.
	id_chp.cgrd	Grid file exported by CHAPMAN.
	id_chp.resi	Accumulated iteration history of integrated forces.
	id_chp.log	Accumulated norms of residuals from CHAPMAN.
	id_FORCELOG	Iteration history of integrated forces from the last run.
	id_CONV	Norms of residuals from last run of CHAPMAN.
	id_chp.log	Iteration history of the integrated forces from CHAPMAN.
	id_Propeller.html	Extracted wake data in html format.

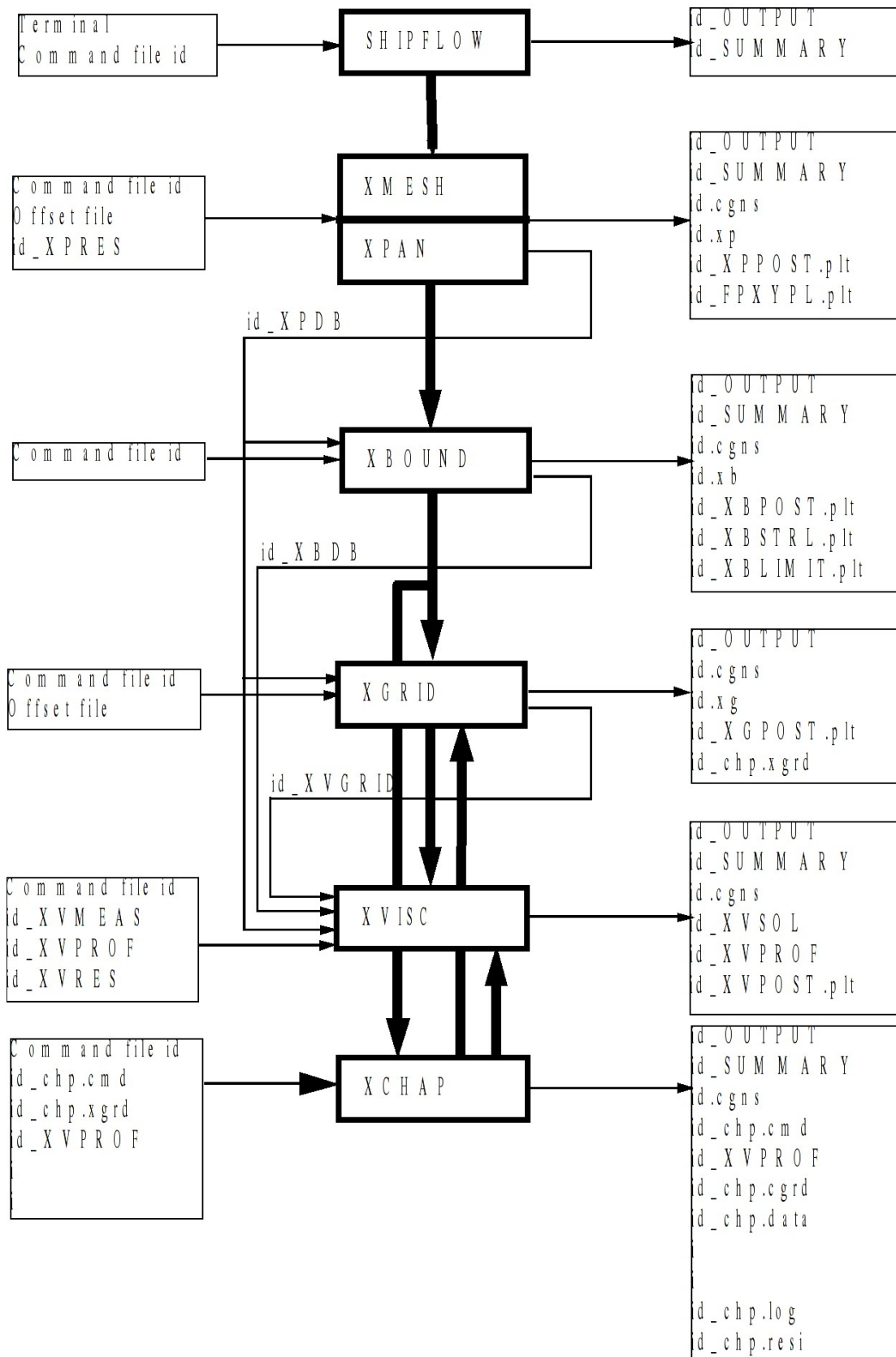
SHIPFLOW (XPAN) also uses a number of scratch files unit 40, 42, 43, 44, 45, 71, 72, 80, 81 and 82 which should not be removed until SHIPFLOW has finished. They disappear by themselves unless SHIPFLOW is terminated abnormally.

The location of scratch files can normally be specified by the system environment variable TMPDIR. This variable effects all scratch files.

In addition it is also possible to specify the location of the scratch files for the influence matrices and the system matrix in XPAN with the environment variable SHIPFLOWTMPDIR, which should be set to the path of the directory that shall contain the files.

The program and file structure is shown in the following section.

11.2. SHIPFLOW program and file structure



11.3. Memory utilization

SHIPFLOW has been written mainly in FORTRAN 77 for maximum speed and portability. One drawback of the language is that the size of vectors and arrays must be known when the program is written. This has been circumvented by storing all the problem size dependent vectors and arrays as parts of a very large vector named "**ia**". The size of "**ia**" can be controlled by the user. This enables the user to adjust the size of SHIPFLOW to achieve any of the following goals:

- Make the problem fit into the program
- Make the program fit into the computer
- Leave some of the computer capacity for other tasks
- Gain certain performance benefits

The "performance benefits" depend mainly on the following mechanism.

XPAN divides the system matrix in blocks which are stored on disc. The number and size of blocks depends on the available memory in "**ia**".

The size of the workspace "**ia**" can be change by the user. The technique for this is dependent on the platform. SHIPFLOW running under UNIX uses C-routines for memory allocation and the size can be specified through the environment variable SHIPFLOWMEM.

SHIPFLOW stops the execution and prints an error message on the terminal and in `id_OUTPUT` whenever the problem size requires more work space. The required value of SHIPFLOWMEM is also printed when possible.

XCHAP is mainly written in C++ and memory is allocated and deallocated dynamically when required and the work space "**ia**" is not used. Due to this difference XCHAP deallocates the work space "**ia**" before starting the CHAPMAN process so it can be reused.

12. Remote computations

12.1. Instructions for remote computations

By remote computation we mean execution of a SHIPFLOW computation on a host that must be a Linux server from a client that can be either a Windows or Linux platform. The computation is initiated from the client. The command and offset files are transferred over to the remote host where the computation is carried out. The output and CGNS files are then transferred back to the client.

Data files are synchronized instead of copied to avoid unnecessary copying. Synchronization is done with the program rsync using SSH (Secure Shell protocol). SSH is also used for controlling the execution of SHIPFLOW on the host. A password free connection is needed to enable SHIPFLOW Design to use the remote computation.

The rshipflow script in the bin directory contains a few variables that must be given the right values for the client / host set up. These are

RHOST	User and host address of the remote host, e.g. Claus@northpole.earth
RWDIR	Name of the working directory on the remote host.
RSHIPFLOW	Location of the shipflow script on the remote host, e.g. /opt/shipflow/bin/shipflow

Files are copied from the client to a location under the RWDIR directory on the Linux host using the full directory path to avoid name conflicts. The data files are located under the directory cygdrive in the RWDIR for Windows clients. It is recommended to use unique names of RWDIR if the RHOST is accessed from several clients.

To start a remote computation run the rshipflow command in the working directory

```
>cd working_dir  
>rshipflow id
```

To use rshipflow in SHIPFLOW Design replace the shipflow by rshipflow in the Computation editor.

12.1.1. Windows specifics

The rshipflow.bat script assumes that cwRsync has been installed. This program installs also a version of SSH. See, <http://www.itefix.no/i2/cwrsync>

The Linux host needs the dos2unix tools in order to convert the command and offset files from Dos to Unix format.

The SHIPFLOW script for remote computations is a batch file located in the bin directory, rshipflow.bat.

12.1.2. Linux specifics

Client.

The SHIPFLOW script for remote computations is a bash script located in the bin directory, rshipflow.sh.

Host.

The host must have SSH service set up and made accessible for the clients. It needs also a authorization mechanism set up to allow password free access. Typically this is achieved with public key authentication.

12.1.3. SSH public key authentication

This paragraph outline the basic step to set up a password free connection from the client to the server using public keys. More details are found in the documentation of the Linux distribution.

1.Generate a key pair on the client. Run “ssh-keygen -t rsa” and do not enter any password (just type Enter on all questions. The program ssh-keygen is found in the installation directory of cwrsync.

2.Copy the public key id_rsa.pub to the Linux host and append it to the file .ssh/authorized_keys in the remote users home directory.

Example for Windows:

Open a Command Prompt, then type:

```
> cd C:\Program Files\cwRsync\bin  
> ssh-keygen -t rsa
```

12.1.4. Known limitations

The rshipflow script must be start in the directory where the command file is located (current working directory)

Don't use space or “.” in the path or name of the command and offset files

13. Standard Cases

The idea behind the Standard Case concept is to simplify the input for commonly used ship types and offset group configurations. The program will automatically choose suitable default values for numerical parameters and mesh generation based on ship speed, geometry etc. The default values for the mesh generation selected by the program must be considered as a starting point for a preliminary computation. The accuracy of the computed results can in most cases be improved further if the mesh is refined compared to the Standard Case. It is important to check carefully the computational model suggested by the Standard Case procedure since it cannot be guaranteed that the procedure will work for all possible cases.

13.1. Mono-hull

This Standard Case covers the most common case for a SHIPFLOW user. It can handle a mono- hull consisting of one to three offset groups for the main hull and one offset group each for the fore bulb, the stern bulb and the overhang. See figure.



A command file for the case above would typically look like:

```
XFLOW
  TITLE      ( TITLE="MONO-HULL CASE" )
  PROGRAM    ( XPAN, XBOUND, XGRID, XCHAP)
  HULLTYPE   ( MONO, FSFLOW, TRANSOM, WSING, BDENS= , FDENS = ,
                H1GR="..." , H2GR="..." , H3GR="..." ,
                OGRP="..." , FBGR="..." , ABGR="..." )
  OFFSET     ( FILE="..." )
  VSHIP      ( FN=[] , RN =[] )
END

XGRID
  SIZE       ( FINE )
END

XCHAP
  PARALLEL   ( NPROC= )
  CONTROL    ( MAXITER=300, KRYLOV )
END
```

Comments

H1GR, H2GR and H3GR should be assigned the label of the three offset groups in the offset file that describe the main hull. They must be in the order as shown in the figure above. At least H1GR must be specified. The mono-hull Standard Case is triggered by the presence of H1GR.

OGRP should be assigned the label of the offset group in the offset file that describes the stern overhang. Not mandatory.

FBGR should be assigned the label of the offset group in the offset file that describes the fore

bulb. Not mandatory.

ABGR should be assigned the label of the offset group in the offset file that describe the aft bulb. Not mandatory.

FSFLOW specifies that a free-surface is included in the potential flow computation.

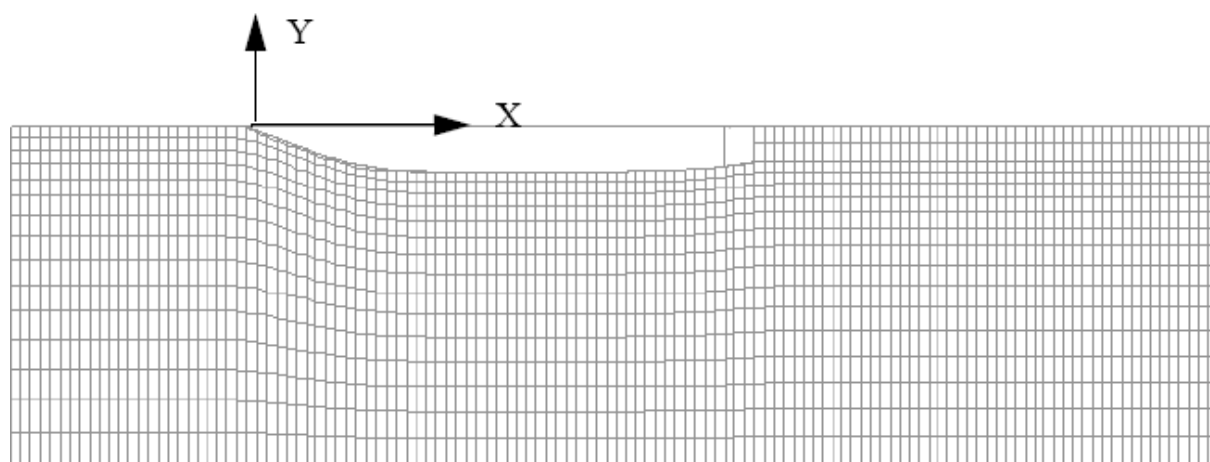
TRANSOM specifies that the free-surface panelization is generated for a transom stern case. A special TRANSOM group is attached behind the transom stern.

BDENS, FSDENS effects the number of panels on the hull and free-surface, respectively.

WSING specifies that the grid generated by XGRID has a grid singularity line in the water-line plane. The line leaves the hull at the point where the y-coordinate is 95% of the half-breadth. A singularity line in the vertical symmetry plane is always generated. The line starts near the first point of the last frame of panel groups H1GR, H2GR or H3GR.

XMESH automatically generates panel groups for the specified offset groups as well as on the free-surface. The numbering of these panel groups start with one for H1GR, then follows H2GR, H3GR, OGRD, FBGR, ABGR, free-surface and transom. Thus, if a hull consist of H1GR, OGRD, FBGR and ABGR, the corresponding panel group number (

GRNO) in XMESH is 1, 2, 3 and 4, respectively. The number of panels on the free-surface determined by the Froude number, F_n .



- A uniform panel distribution is always used in the x-direction. The number of panels is a function of F_n .
- The panels are stretched towards the hull in the transverse (y) direction. The number of panels is a function of F_n , but the number is maximized to 15 to reduce the total number of free-surface panels. The size of the first panel at the hull Δy is $0.02 \cdot L_{pp}$.
- The same free-surface panelization is used for linear and non-linear computations.
- The hull is extended a small distance above the free-surface in non-linear computations. Two panels are used in the grid wise direction to represent the part above the free- surface.

The three figures below show the number of panels per wave length, the free-surface extension and the total number of free-surface panels as function of the Froude number.

The panel density can be changed individually for the hull and the free-surface.

XBOUND starts a group of streamlines and computes the turbulent boundary layer on H1GR.

XGRID creates the grid for XCHAP. The keyword FINE in the size command changes the default values in XGRID so that in many cases a sufficiently good grid is obtained. Keywords COARSE and MEDIUM can also be used.

XCHAP solves the viscous equations. For a fine grid XCHAP usually runs around 100 iterations when run with KRYLOV (a gmres solver is being used), but it is wise to set MAXIT to a higher number and let XCHAP automatically stop when the convergence criterias are reached. In order to speed up the calculation multi-processes with MPI can be turned on with the PARALLEL command. NOTE: Multe-process or multi-threading is only possible on multi-core computers. Never use more processes or threads then there are cores in the computer, since this will only slow down XCHAP.

All default values may be changed by adding commands in the command file.

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